AAP Series on Digital Signal Processing, Computer Vision and Image Processing

Artificial Intelligence and Machine Learning Techniques in Image Processing and Computer Vision

Karm Veer Arya
Ciro Rodriguez Rodriguez
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Editors

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Digital signal processing, computer vision and image processing as a whole is considered to be one of the most rapidly evolving areas of research and technology today with growing applications in almost all disciplines of engineering. Medical imaging, computer vision, healthcare, medical applications, remote sensing, agriculture, robotics, communication systems and space exploration are some of the applications of digital signal processing, computer vision and image processing, to name a few. The present day curriculum covers many aspects of digital signal processing, computer vision and image processing, addressing the theoretical aspects in particular. This book series is intended to supplement the theoretical knowledge with special emphasis on the practical side. The series content has been specifically chosen to give a thorough understanding of the fundamental aspects to advanced applications of digital signal processing, computer vision and image processing.

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ABBREVIATIONS

A	adenosis
ABC	artificial bee colony
ADFV	absolute difference of frame variance
AI	artificial intelligence
AMD	age-related macular degeneration
ART	artificial reproductive techniques
ASPP	atrous spatial pyramid pooling
AUC	area under the curve
BGR	blue-green-red
CAD	computer-aided diagnosis
CASA	computer-aided sperm analysis
CNF	context-wise network fusion
CNN	convolutional neural networks
CRFs	conditional random fields
CRP	C-reactive protein
CXR	chest X-ray
DC	ductal carcinoma
DC-GAN	deep convolutional generative adversarial networks
DCNN	deep convolutional neural network
DR	diabetic retinopathy
EX	exudates
FA	fibroadenoma
FCL	fully connected layer
FCNs	fully convolutional networks
FN	false negatives
FODPSO	fractional order Darwinian particle swarm optimization
FoV	field-of-view
FP	false positives
GA	genetic algorithm
GANs	generative adversarial networks
GLCM	grey level co-occurrence matrix
GPU	graphics processing unit
HAR	human activity recognition

H&E	hematoxylin and eosin
HBMO	honey-bee mating optimization
HEMs	hemorrhages
HOG	histogram of oriented gradients
HR	high-resolution
HSN	hourglass-shaped network
HuSHeM	human sperm head morphology dataset
iDT	improved density trajectory
ILSVRC	ImageNet Large Scale Visual Recognition Challenge
IR	infrared
IS	image segmentation
IVF	in vitro fertilization
JPU	Joint Pyramid Upsampling
LBD	lung boundary detection
LBP	local binary pattern
LC	lobular carcinoma
LCF	local convergence filters
LR	low-resolution
LSTM	long short-term memory
LTD	lung tumor detection
MA	microaneurysms
MAP	maximum a posteriori
MC	mucinous carcinoma
MCIS	multi-resolution color image segmentation
MCNN	multiscale CNN
MIR	mid-IR
MLP	multilayer perceptron
MRF	Markov random fields
MRIs	magnetic resonance imaging
MSE	means squared error
NIR	near-IR
NPA	natural protected areas
OSA	obstructive sleep apnea
PC	papillary carcinoma
PCOS	polycystic ovary syndrome
PDE	partial differential equations
PNN	probabilistic neural network
PSNR	peak signal-to-noise ratios

PSO	particle swarm optimization
PT	phyllodes tumor
ReLU	rectified linear unit
RMSE	root-mean-square error
ROC	receiver operating characteristics
SA	simulated annealing
SIFT	scale invariant feature transform
SLAM	simultaneous localization and mapping
SP	spatial pyramids
SR	super-resolution
SSF	scale-space filter
SSGAN	semi-supervised GAN
SSIM	structural similarity index
SURF	speeded up robust features
SVM	support vector machine
TA	tubular adenoma
TN	true negatives
TP	true positives
VDSR	very deep super-resolution
VHR	very high resolution
VLAD	vector of linearly aggregated descriptors
WHO	World Health Organization

PREFACE

Machine learning (ML) algorithms learn from data by building adaptive mathematical models which can detect the pattern in the training data. One of the applications of machine learning is image processing and computer vision. Image processing (IP) is the manipulation of an image to process, analyze, and obtain the information provided for human interpretation. The interpretation of images by machine learning algorithms is the same as is done by our brains.

ML algorithms typically have steps to learn from data which include data preparation, data extraction, training the model, and making predictions by testing the model. High-quality data are the primary requirement of ML algorithms for fair level of predictions.

This book is a road map for the improvement of computer vision and IP field. The book will play a significant role in providing information about the ML algorithms and models used in IP and computer vision to new researchers and practitioners who wish to start working in this field. They would be able to differentiate between the various algorithms available and which would get the most precise result for a specific task involving certain constraints. This book will be useful because it will provide a vast amount of information and clarification on topics researched by a number of professionals working in the field. The researchers will come across the future scope of these technologies and get to know about the potential application of the same in solving real-life problems.

Chapter 1 covers semen analysis which is one of the key components of clinical pathology. The sperm morphology is analyzed for possible structural abnormalities of the sperm manually done by an embryologist. Fatigue due to increased workload and interpersonal bias creep in when diagnosing sperm morphology abnormalities. These factors may lead to error-prone results. The research objective is to develop an automated clinical tool which would yield rapid and accurate results. The sperm image dataset is aggregated from Mendeley data. Various ML models are trained and tested with this dataset in a training:testing ratio of 80:20. This model would be suggested for clinical deployment in patient care.

This automated tool could aid physicians by providing an accurate, rapid assessment of the human sperm head morphology.

Chapter 2 covers the strategic processing of data used for diagnosing polycystic ovary syndrome (PCOS) and a model to predict its imminent occurrence. PCOS is exceedingly widespread and represents the most confronted endocrinopathy in women of reproductive age. The characteristics that confirm its presence in an individual are not very accurate due to its clinical heterogeneity. Hence, employing ML concepts can play a beneficial role in detecting the presence of this condition. Therefore, this chapter presents an ML-based classification system built on various features like age, weight, etc. The experimental results show that the proposed method for PCOS diagnosis is decently high in terms of accuracy and low in error rate, making it a suitable tool to be used.

Chapter 3 covers the disadvantages of annotated data and traditional deep learning methods. The semisupervised learning approach, using generative adversarial networks (GANs), can solve the lack of labeled input data when training a deep learning classifier. We demonstrated that semisupervised GAN using both labeled and unlabeled datasets can achieve a significant level of performance as traditional convolutional neural networks. This method reduces the cost and time related to annotating cell images for deep learning classifier development.

Chapter 4 covers the issues in video copyright using local features of frames and talks about keyframes. Keyframes are effectively identified by using entropy and the absolute mean difference of the video frames. The Haar wavelet is adapted for decomposing the frames into two levels; the LH band is chosen for watermarking process. For aiming at improving efficiency of the robustness and imperceptibility, the artificial bee-colony optimization algorithm is applied to choose the optimal scaling factor.

Chapter 5 covers textures analysis which plays a vital role in classifying the image type or the region of interest. Texture analysis based on extracted features is widely used in classification of images; it could be medical, photomicrograph, satellite, or aerial images. This chapter gives a clear explanation of one state-of-art method, the "grey level co-occurrence matrix," its advantages, disadvantages, examples, and applications of this feature extraction method.

Chapter 6 talks about computer vision which can be used to identify the changes in Amazon Rainforest in Brazil using satellite imagery at two different points in time using the metrics of structural similarity index

(SSIM). This study demonstrates how using a simple implementation of principles of computer vision and efficient open-source libraries can detect the difference in two images in the most efficient way. The application in monitoring the deforestation as demonstrated in this chapter alone has multiple utilities including, but not limited to, avoiding/controlling wildfires, identifying the areas for new plantation, avoiding some serious climate risks, and saving some wildlife from extinction.

Chapter 7 covers a new biometric recognition model which is based on a proposed convolutional neural network using biometric information from sea turtle images. The results generated for the recognition of sea turtles with the proposed biometric model are presented in the final part of the chapter. The proposed model outperforms the artificial intelligence technique that is implemented with RGB neural network architecture—AlexNet—which has an accuracy of 73% in validation and 74% in testing.

Chapter 8 covers computer vision and remote sensing and its applications. Satellite imagery is a vast source of information about the Earth and its environment however, is inherently challenging to draw useful conclusions from it owing to complex spectral and spatial details. This chapter presents the state-of-the-art techniques for semantic segmentation based on deep learning and their applications in satellite imagery. The various challenges faced by the researchers in the application of these techniques on satellite imagery are also discussed.

Chapter 9 covers CNN for visual recognition of objects in an image and using Resnet34 pretrained on ImageNet to separate these objects into individual entities and train our model to guess the color of the objects. This also ends up restoring the picture to a certain extent. The proposed method will be much easier to use than using photoshopping softwares which also require certain technical skills.

Chapter 10 Deep-learning-based convolutional neural network has achieved remarkable performance in the field of image and computer vision tasks. In this chapter, we explore how these models are related to image super-resolution (SR) algorithms. Furthermore, we discuss the basic terminology that is used in the image SR reconstruction. Finally, we present the quantitative results in terms of PSNR/SSIM.

Chapter 11 covers the research endeavor in which, proposition is to find image scale (size in real-world unit per pixel) using the common size of the objects existing in the visual description. These regular objects could be people, cars, bikes, signposts, etc., depending upon the location, traffic, and time when the image was taken. The dataset which is used for the investigation is attained from KITTI, and the RGB-D images of the dataset have been taken into account. It is then wielded in derivation of the mathematical function to corelate depth of the concerned entity with object in real world unit per pixel. In this way, the desired outcome is achieved.

Chapter 12 covers metaheuristic methods which are global optimizations that find global solutions to avoid stagnation. These algorithms are mostly nature-inspired, and they are turning out to be exceptionally incredible in solving global optimization problems. Various metaheuristic algorithms are reviewed in this chapter. Further, the freely accessible benchmark datasets for segmentation are also reviewed.

Chapter 13 covers the diabetic retinopathy which is an eye disease occuring in diabetic patients that damage the retina. Accurate detection of diabetic retinopathy lesions from fundus images is a difficult task that necessitates the development of automated computer-aided diagnostic techniques. This review identifies the changes that must be made in the future to create the best automated diagnostic tool that performs better and avoids all of the pitfalls found in current literature.

Chapter 14 covers a study which is being conducted on how to construct accurate data representations in the human action recognition process. This research looks at approaches to basic human actions as well as approaches to abnormal activity levels. The interest and constraints of each system are observed and used to classify these methods.

Chapter 15 covers a research whose objective is to provide cost-effective and more accurate results in less time by less human intervention. The main challenge here is the preprocessing of the images, as the images are in different formats and may have different resolutions. This chapter talks about determining diseases from a standard image dataset using supervised ML algorithms and compares the accuracy obtained by different algorithms.

PART I
Health Care Systems

CHAPTER 1

MACHINE LEARNING MODEL-BASED DETECTION OF SPERM HEAD ABNORMALITIES FROM STAINED MICROSCOPIC IMAGES

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ABSTRACT

Semen analysis is one of the key components of clinical pathology. The stained sample is analyzed under a microscope for eliciting the morphology of the sperm. The sperm morphology is analyzed for possible structural abnormalities of the sperm. This procedure is one of the mainstays in infertility treatment procedures. This examination is manually done by an embryologist. Fatigue due to increased workload and inter-personal bias creep in when diagnosing sperm morphology abnormalities. These factors may lead to error-prone results. There is a dire need to develop a system for accurate and rapid automated diagnosis. The objective of this work is to develop an automated clinical tool which would yield rapid and accurate results. The various existing literatures in this domain have been elaborately studied. The sperm image dataset is aggregated from Mendeley Data. Various machine learning models are trained and tested with this dataset in training: testing ratio of 80:20. From the experiments, it is found that the support vector

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machine (polynomial) model performs the best with accuracy of 75.76%, sensitivity of 75.84%, specificity of 91.83%, precision of 80%, recall of 76%, and F1-Score of 76%. This model would be suggested for clinical deployment in patient care. This automated tool could aid physicians by providing an accurate, rapid assessment of the human sperm head morphology.

1.1 INTRODUCTION

Semen analysis is one of the key components of clinical pathology. This procedure involves analyzing the semen sample collected from the patient in the laboratory. The semen sample is collected with the help of a container. It is also very imperative that these samples be transported immediately to the laboratory or at most within an hour. Semen analysis is carried out principally for the following indications: in infertility treatments, to evaluate the success of vasectomy procedure, and in medicolegal cases.

The applications of semen analysis in infertility treatments are of immense importance. It is one of the routine procedures carried out in in-vitro fertilization (IVF) treatment. The semen analysis is principally carried out to study the various properties of sperm, which is the male gamete. The semen sample is subjected gross, microscopic, microbiological assay, immunologic assay, and chemical examination. The microscopic examination is carried out to study various properties like motility, viability, morphology, and count. The motility of the sperm is examined after by examining it under the microscope. If the motility is below the threshold, the viability of the sperms is studied. This is done by staining procedures with stains like methylene blue. The count of sperms is examined to diagnose conditions like oligozoospermia or azoospermia. The morphology of the sperm is analyzed to examine any structural defects associated with the head or the tail of the sperm.

This analysis is also useful in studying the success of the vasectomy. Vasectomy is a procedure in which the vas deferens of the male genital tract is cut and sealed, thereby blocking the passage of sperms from the testis. The semen sample is collected post-vasectomy procedure after some weeks and analyzed for the presence of sperms. The absence of any sperms in the semen sample indicates a successful vasectomy procedure. This procedure is repeated after certain years to ensure the patency of the procedure. This semen analysis technique is also valuable in terms of its medico-legal significance. This technique can be used to prove the sterility of a male, which can be used to solve paternity disputes.

1.2 RELATED WORKS

The standard technique of evaluating the morphology of the sperm using the Krueger grading criteria has been critically evaluated. This method is found to be time-consuming and provides error-prone results at times. Hence, the need to automate this process has arised. An artificial neural network model has been built which was trained and tested with a dataset of 3500 sperm image dataset. The model was able to provide excellent results. This automated model could be easily interfaced with the existing optical instruments. This novel model provides an accurate and time-effective tool for semen analysis.[1] Applications of modern technologies like artificial intelligence and machine learning have been rapidly evolving and are increasingly being used in all fields. The field of medicine is no exception. The various fundamental principles of artificial intelligence and machine learning, their functioning, and various pros and cons have been detailed. Also, the scope and challenges for the implementation of these technologies in the reproductive medicine domain has been briefed.[2] The assessment of ideal sperm morphology has always been a controversial topic. Applications of computer vision techniques in eliciting sperm biometrics are quite an interesting problem. The various perspectives on this utilization and its pros and cons have been discussed.[3] Sperm motility is a parameter of paramount importance in normal fertilization. This is being done through computer-aided sperm analysis (CASA). A decision tree model based on support vector machine (SVM) was built. This model analyzed 2817 CASA sperm tracks and was used to classify the sperm motility. The model has been able to classify the sperm motility into various distinctive classes.[4] The essentiality of vigorous sperm motility of male gametes in female reproductive tract for the fertilization has been iterated. An automated model to classify the sperm motility into five distinct classes has been proposed. A supervised learning model, SVM is used for this automation process. This model is trained and tested upon 2000 sperm tracks captured using CASA. Sperm motility profiles have been computed with this model.[5]

Sperm quality in fish can be evaluated with sperm motility as a biomarker. These are used in various applications like evaluation of the effect of various pollutants on sperm motility, in cryopreservation, and in studying the optimized conditions for fertilization. The computer-assisted sperm analysis can be used as an effective tool to determine the quality of the sperm of fish, which is in turn used to evaluate the probability of successful fertilization.[6] The worldwide increase in infertility rates and subsequent demand for fertility care is growing every day. Automated machine-learning techniques have been employed in fertility care. This has been used in a range of domains like in embryo selection, embryo gradation, and sperm selection. These techniques powered with powerful data processing algorithms could prove to be a game-changer in reproductive medicine.[7] The Deep Convolutional Neural Network (DCNN) model, VGG-16 has been trained on ImageNet. This model classifies the sperm morphology into various World Health Organization (WHO) defined categories. This model proves better in various performance metrics. This model exhibits the potential of artificial intelligence technology in the classification of sperm head morphology even better than human experts.[8] A dictionary learning model has been proposed for sperm morphology classification. The principal motive behind this is to build a dictionary of several sperm morphology. Sperm morphology analysis has been one of the mainstays in the male infertility diagnosis and treatment. The process of analyzing sperm morphology has predominantly been manual and hence, prone for subjective bias. The DL-based model has been proposed to serve as automated process. Various performance metrics like accuracy, precision, recall, and F1-Score have been used to evaluate the model. This model has proved to be much better than classifiers in the classification of the morphology of human sperm.[9] Sperm motility identification from videos has been proposed. Machine learning models were tested with 85 videos of semen samples. Automatic analysis with machine learning techniques yields a rapid and consistent result of sperm motility prediction from human semen sample analysis. This would provide an automatic tool for clinical diagnosis of male infertility and further treatment.[10]

Natural sperm selection is the process by which the highest quality sperm is destined to reach the oocyte and fertilize it to form the zygote. The various artificial reproductive techniques (ART) have been described. The existing and upcoming newer techniques used in selecting the best quality of sperm pool have been detailed.[11] Traditionally, the best quality sperm has been selected based on motility and morphology. The DNA content of the sperm has not been accounted in selecting the sperms. A deep learning model has been proposed for accounting the DNA integrity as a parameter for choosing human sperms. A dataset of 1000 sperms with known DNA content has been used to train and validate this model. This provides a rapid tool for sperm selection with the best DNA quality.[12] A two-stage classification scheme has been proposed to identify the human sperm head morphology. Accordingly, these have been classified into five classes according to the classification given by the WHO. Out of the five classes, one class denoted the normal human sperm head morphology whereas the other four denote classes of abnormal morphology. The performance of this model has been at par with the human experts. This paves the way for the labeling of the new unknown sperm head image.[13] The various ART Techniques could be undertaken based on a range of other reproductive parameters apart from the sperm morphology.[14] The smartphone-based tool for analyzing sperm morphology has been proposed. The proposed model consists of two sequential steps. Firstly, the segmentation step in which the segmentation of the sperm is carried out. Secondly, the segmented sperm images are subjected to classification, following which the normal and abnormal sperms are identified. The automated tool based on the mobile net has been developed.[15]

A R-CNN-based model has been proposed which serves the dual functionality of identification of the sperms as well as calculation of the velocity of the sperm movement. This model achieved an accuracy of 91.77% when analyzed with the VISEM dataset. The results are quite promising and hence, could be deployed in artificial insemination procedures.[16] A faster R-CNN model with elliptic scanning algorithm was proposed to automate the analysis of sperm morphology and motility. The proposed model outweighed all the existing models with superior performance metrics with an accuracy of 97.37%.[17] A potential model which could surpass embryologists has been proposed. A specialized CNN architecture was proposed to analyze the sperm head morphology. This model outputs a recall of 88 and 95% on SCIAN and HuSHeM datasets, respectively.[18] Utilization of SVM classifiers to classify gray scale images of sperms can be used to develop automated clinical tools which could be deployed in patient care.[19] The applications of computer-aided sperm analysis in selecting the best quality of sperms which exhibit a range of desirable features like superior motility and good mucus penetrating property have been discussed.[20]

The difficulties of assessing sperm head morphology manually have been discussed. The various intricate structures of the sperm head, the acrosome and nucleus, and the framework for its automated detection from microscopic images have been detailed.[21] The imperativeness of the evaluation of sperm head morphology in fertility treatment has been discussed. The practical difficulty in evaluating this sperm head morphology has been briefly described. A novel framework has been puts forth an automated segmentation of sperm head has yielded extremely impressive results.[22] The state of digital pathology and its various aspects have been described. The existing literature about models available to evaluate microscopic images captured from a variety of microscopes are been explained. Moreover, the scope and challenges in the field of digital pathology have been briefed.[23] A "minimum-model" approach which aids in the detection of cell nuclei has been described. It has produced good results in terms of distinguishing between normal and diseased cells.[24]

1.3 MATERIALS AND METHODS

The aim of this work is to develop a superior diagnostic tool suitable for clinical deployment which would accurately classify sperm head morphology. The architecture of the proposed model is shown in Figure 1.1. The dataset used in this work is the Human Sperm Head Morphology Dataset (HuSHeM) which is acquired from https://data.mendeley.com/datasets/tt3yj2pf38/3. This dataset consists of stained images of sperms from the semen samples collected from 15 patients at the Isfahan Fertility and Infertility Center. The method used for fixing and staining the sperms was the Diff-Quick method. The microscope used in aggregating this dataset was Olympus CX21 which provided a magnification of 1000× with an objective of 100× and an eyepiece of 100×. The images were captured using a Sony SSC-DC58AP. These images were then analyzed by embryologists and labeled into four classes according to the sperm head morphology. These were namely normal, tapered, pyriform, and amorphous.[25]

Various machine learning models like Decision Tree (Gini), Decision Tree (Entropy), Logistic Regression, SGD Classifier, Gaussian NB (Naïve Bayes), K-Neighbours Classifier, Random Forest Classifier, Support Vector Machine (Sigmoid), Support Vector Machine (Linear), Support Vector Machine (RBF), Support Vector Machine (Polynomial), and XGB Classifier (XGBoost) are used for this purpose. The total number of images is 216. It comprises of images from all four classes. The composition of the various classes is depicted in Table 1.1. The sample images from various classes of the dataset are shown in Figure 1.2. This dataset is utilized in the training: testing ratio of 80:20. Various performance metrics like accuracy, sensitivity, specificity, precision, recall, and F1-Score are calculated.

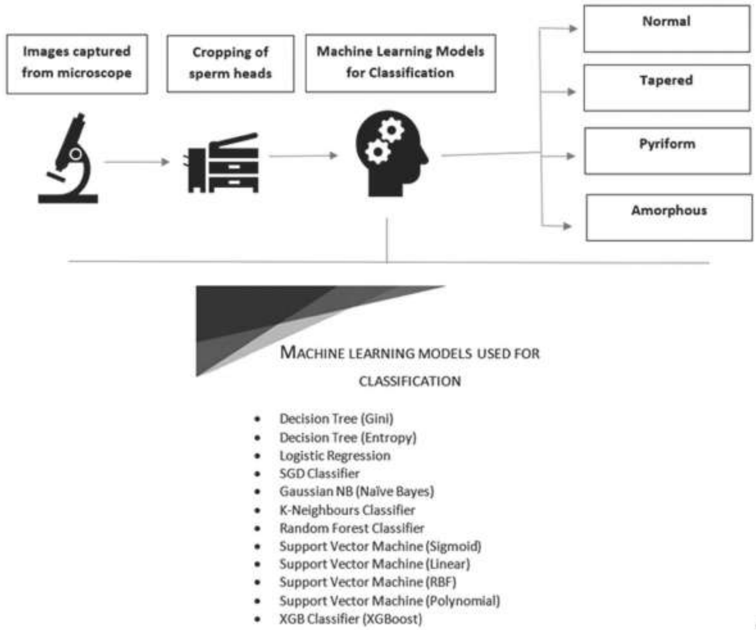

FIGURE 1.1 Architecture of the proposed model.

TABLE 1.1 Composition of the Various Classes of the Human Sperm Head Morphology Dataset.

Class	Absolute number of images	Percentage of images
Normal	54	25
Tapered	53	24.53
Pyriform	57	26.38
Amorphous	52	24.07

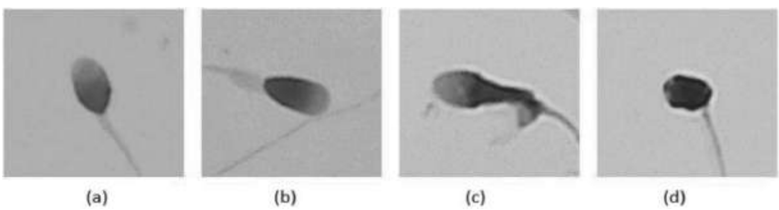

FIGURE 1.2 Sample images from the four different classes of the dataset: (a) normal, (b) tapered, (c) pyriform, (d) amorphous.

Source: Reprinted from dataset contributed by Fariba Shaker, Mendeley Data, 2018. https://creativecommons.org/licenses/by/4.0/ (Ref. [25]).

1.4 EXPERIMENTS AND RESULTS

Performance metrics like accuracy, sensitivity, specificity, precision, recall, and F1-Score are calculated as given in eq (1.1), eq (1.2), eq (1.3), eq (1.4), eq (1.5), and eq (1.6).

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{1.1}$$

$$Sensitivity = \frac{TP}{TP + FN} \tag{1.2}$$

$$Specificity = \frac{TN}{TN + FP} \tag{1.3}$$

$$Precision = \frac{TP}{TP + FP} \tag{1.4}$$

$$Recall = \frac{TP}{TP + FN} \tag{1.5}$$

$$F1 - Score = 2.\frac{(Precision * Recall)}{(Precision + Recall)} \tag{1.6}$$

where

TP denotes True Positive,

TN denotes True Negative,

FP denotes False Positive,

FN denotes False Negative.

The performance metrics of the various models are listed in Table 1.2. The comparison of Accuracy across the various models is depicted in Figure 1.3. The comparison of Specificity across the various models is depicted in Figure 1.4. The comparison of Sensitivity across the various models is depicted in Figure 1.5. The comparison of Precision across the various models is depicted in Figure 1.6. The comparison of Recall across the various models is depicted in Figure 1.7. The comparison of F1-Score across the various models is depicted in Figure 1.8.

TABLE 1.2 Performance Metrics of the Various Machine Learning Models.

Machine Learning Models	Accuracy	Specificity	Sensitivity	Precision	Recall	F1-score
Decision Tree (Gini)	42.42	82.06	45.53	48	45.53	45
Decision Tree (Entropy)	27.28	81.05	42.75	43	42.75	41
Logistic Regression	57.58	85.94	57.68	56	57.68	56
SGD Classifier	48.48	75.88	26.77	18	26.77	14
Gaussian NB (Naïve Bayes)	57.58	85.49	57.68	66	57.68	59
K-Neighbours Classifier	57.58	85.70	55.31	59	55.31	52
Random Forest Classifier	66.66	88.98	66.12	69	66.12	65
Support Vector Machine (Sigmoid)	21.21	75	25	5	25	9
Support Vector Machine (Linear)	69.69	89.83	69.49	73	69.49	68
Support Vector Machine (RBF)	66.67	88.54	67.16	77	67.16	68
Support Vector Machine (Polynomial)	**75.76**	**91.83**	**75.84**	**80**	**75.84**	**76**
XGB Classifier (XGBoost)	63.64	87.86	63.69	64	63.69	63

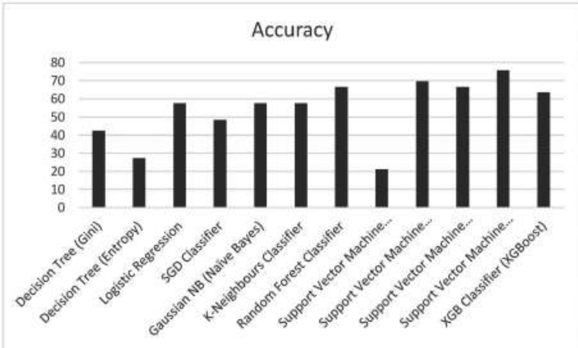

FIGURE 1.3 Comparison of accuracy across the various models.

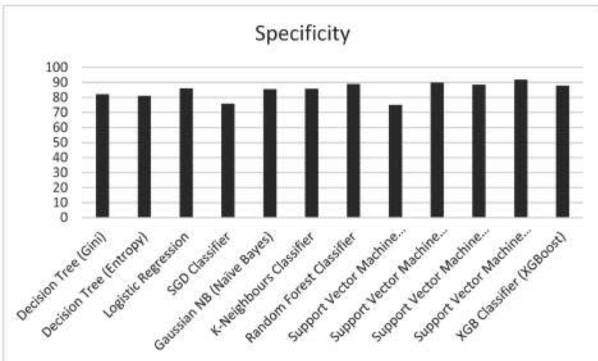

FIGURE 1.4 Comparison of specificity across the various models.

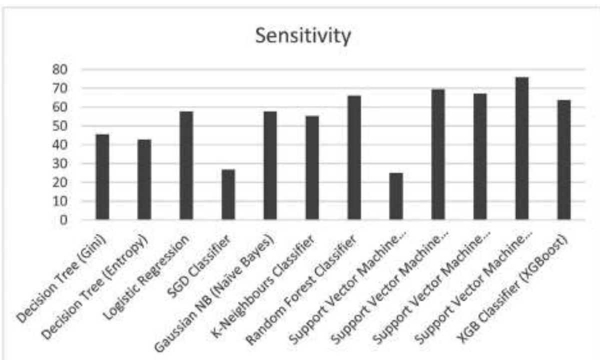

FIGURE 1.5 Comparison of sensitivity across the various models.

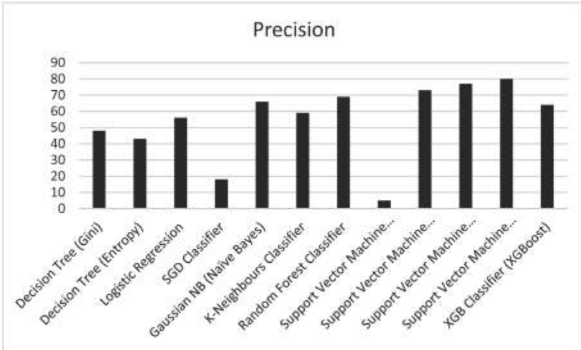

FIGURE 1.6 Comparison of precision across the various models.

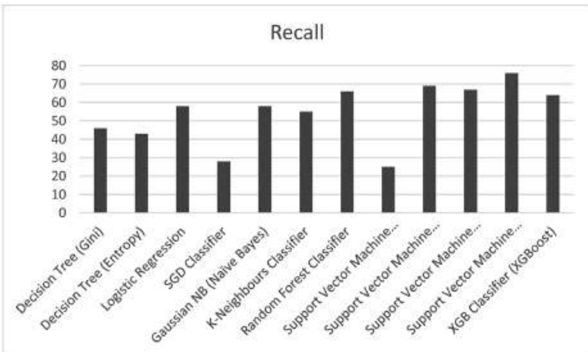

FIGURE 1.7 Comparison of recall across the various models.

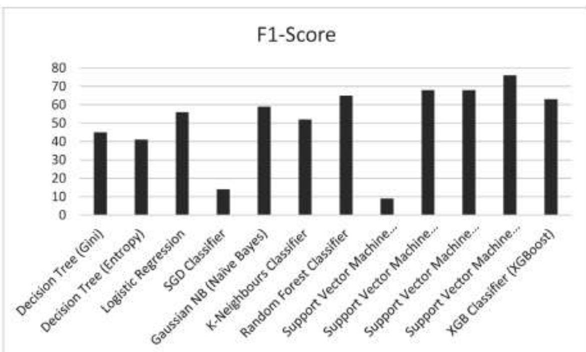

FIGURE 1.8 Comparison of F1-score across the various models.

1.5 CONCLUSION

From the above results, it can be concluded that the support vector machine (Polynomial) model proves to be the better model. It exhibits superior performance metrics with an accuracy of 75.76%, sensitivity of 75.84%, specificity of 91.83%, precision of 80%, recall of 76%, and F1-Score of 76%. The comparison of the manual examination and automated examination with ML-based tools has been depicted in Table 1.3. This model would be suggested for clinical deployment and patient care. Usage of this improved technique in fertility treatments would result in an effective diagnosis of sperm head abnormalities. This procedure would be more accurate and hence, would eliminate inter-personal bias. This tool would be able to aid physicians in the diagnosis of sperm head abnormalities.

TABLE 1.3 Comparison of Manual Examination and Automated Examination with ML-Based Tool.

Parameters	Manual examination	Automated examination with ML-based tool
Accuracy	Lower	Significantly higher
Miss-outs	Present	Absent
Time consumption	Increased	Decreased; rapid
Effect of fatigue	Present	Absent; eliminated
Inter-personal bias	Present	Absent; eliminated
Requirement of advanced computing facilities	Absent	Present
Requirement of trained laboratory staff	Present	Absent
Pros	Low-cost	High accuracy, rapid, useful in low-resource setting with minimal laboratory staff
Cons	Low accuracy, time consuming, mandatory requirement of laboratory staff	High initial cost

KEYWORDS

- **andrology**
- **automated diagnostic tool**
- **human sperm head morphology**
- **machine learning**
- **semen analysis**

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CHAPTER 2

SMART HEALTHCARE SYSTEM FOR RELIABLE DIAGNOSIS OF POLYCYSTIC OVARY SYNDROME

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ABSTRACT

The female body's well-being is about precaution, protection, diagnosis, and treatment. Polycystic ovary syndrome (PCOS) is exceedingly widespread and represents the most confronted endocrinopathy in women of reproductive age. The characteristics that confirm its presence in an individual are not very promising due to its clinical heterogeneity; hence employing machine learning concepts can play a beneficial role to detect the presence of this condition. Therefore, this paper presents a machine learning-based classification system built on various features like age, weight, etc. The proposed methodology entails the strategic processing of data used for diagnosing PCOS and a model to predict its imminent occurrence which yielded a high testing accuracy of 96.36% and an error rate of 3.64%. The experimental results show the proposed method for PCOS diagnosis are decently high in terms of accuracy and low in error rate making it a suitable tool to be used in real-time applications.

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2.1 INTRODUCTION

Polycystic ovary syndrome (PCOS) is an exceedingly growing disease that is very popular among young females. To recognize PCOS, one can observe indicators like irregularities or a pause in menstrual cycles, excessive growth of body hair, acne, etc., but these features are not necessarily always present in patients. This disease needs to be recognized and treated as it is highly likely to cause much bigger complications like Anovulation, premenopausal endometrial cancer, Cardiac-related diseases, and type 2 diabetes.[1] A few other symptoms include obesity, hirsutism, cystic growth in ovaries, and increased levels of testosterone in the body that can be alarming.

Clinically, a fertile healthy woman at her reproductive age tends to develop cyst-like sacs in ovaries every month for producing essential hormones that are termed as follicles. It is responsible for releasing an egg in the ovulation period but a disturbance in hormone levels can affect its growth. In such cases, accumulation of the immature follicles starts which later turns into a bigger cyst. According to the World Health Organization (WHO), people suffering from PCOS are normogonadotropic due to which they experience prolonged or infrequent cycles of menstruation and an imbalance in hormone levels like excess of male hormone in the female body (androgen). Doctors are still unaware of the exact cause of PCOS but most of the professionals tend to rely on the explanation that excess of male hormones in the body can cause this condition preventing ovaries to perform normal functions like hormone production or formation of eggs. As per doctors' recommendations, patients diagnosed by PCOS may opt for treatments like birth control medications and lifestyle alterations.[2] Depending on the size and number of follicles, one can decide whether its presence is harmful or not. Similarly, there are multiple such features that can be used to diagnose PCOS.

The main contribution of this paper is that it proposes an automated machine learning-based smart healthcare system that can be well utilized by doctors and medical professionals for diagnosing PCOS. This work will present a deep artificial neural network model that can diagnose PCOS with great accuracy. It also provides the reader with relevant information about the disease like the prevalent symptoms, common preventions, and treatment.

Another contribution of this research paper involves a comparative analysis of various machine learning models like deep neural network model, non-linear support vector machine model, multinomial Naïve Bayes model, decision tree, random forest, and logistic regression on a dataset collected from around 10 different hospitals of Kerala, India. It also provides a comparative analysis of pre-existing work present in this domain to the best of the author's knowledge.

For better evaluation and understanding of the proposed models, this paper briefly gives various evaluation metrics of the model being used and their corresponding significance. Another noteworthy contribution of the proposed work is its low-time complexity and low-error rate in the machine learning model being used.

In the upcoming sections, the given structure is followed: Section 2 discusses some relevant pre-existing work which is followed by Section 3 which will be providing the reader with information about the dataset along with the necessary steps of cleaning and processing the data. In Section 4, the proposed methodology and the chronology that have been followed while building the model is demonstrated. Section 5 will discuss the results retrieved from the process followed and Section 6 will conclude the findings and future work in this field.

2.2 RELATED WORKS

With reference to PCOS detection, some relevant pre-existing work has been discussed in this section. The performance of the Bayesian classifier on a PCOS recognition problem applied on a dataset of 250 women collected from GDIFR, Kolkata is shown in a work by Ref. [3]. It explains the benefits of using the Bayesian classifier over logistic regression. It shows the analysis of parameters used to determine the model and posterior probability of diagnosing PCOS patients correctly with an accuracy of 93.93%. Later in research,[4] a patient survey of around 541 women was used to analyze the nature of various models and to finally predict the presence of PCOS in patients using the models.

The classification with transformed features using PCA was done using several techniques like CART, Naïve Bayes Algorithm, Logistic Regression Algorithm, K-Nearest Neighbours, Support Vector Machine Algorithm, and Random Forest on a dataset collected from an online repository Kaggle. Results received were reflecting how Random Forest was the best model to be applied with an accuracy of 89.02% whereas the work proposed in present research shows a much better accuracy along with a performance comparison of powerful machine learning models. A technique of image segmentation on ultrasound images[5] was implemented prior to applying a CNN model and tried to attain good accuracy by removing redundant data. As per the proposed work, image segmentation was included between CNN and feature extraction to eradicate unwanted data and to improve accuracy. However, the paper did not mention the accuracy that was finally achieved, and the content was not well written. A new algorithm based on a combination of artificial neural network and fuzzy neural logic was proposed to simplify the job of selecting features and classifying patients.[6] Classification techniques that have been used in this paper are Naïve Bayesian classification algorithm, Artificial neural network, etc., but the accuracy received could have been improved using automatic or hand-tuning of important parameters that were used.

Implementation of a new algorithm based on a combination of artificial neural network and fuzzy neural logic which was an extension of their previous work discussed in a work.[7] The algorithm in the proposed research was employed using MATLAB and has been used to compare with the pre-existing medical details on the disease PCOS. Another work executing particle swarm optimization has been done for the follicle segmentation clustering problem in the proposed work of Ref. [8].

Several computer-aided techniques for detecting follicles and finally diagnosing PCOS in patients using the ultrasound images of the ovary have been surveyed in one of the mentioned works[9] which identified and compared the existing work on PCOS using Transvaginal ultrasound images and concepts of machine learning. Software for detecting the presence of cystic growths in ovaries of the female human body and their classification using the ultrasound imageries with the help of support vector machine algorithm and image processing has been offered.[10] It successfully gained an accuracy of 90%, specificity of 95%, and sensitivity of 88.33%. The classification and detection of ovarian cysts in the research were restricted to only endometrioma cysts based on its geometry.

The conventional VGG-16 model consisting of 16-connected layers which were finely tuned with a dataset collected by the authors of ultrasound images has been used in the research.[11] It was detecting different types of cysts including endometrioid cysts, dermoid cyst, and hemorrhagic ovarian cysts with an accuracy of 92.11%. This work did not acknowledge the reader with any future work.

The research[12] shown in the proposed work can diagnose PCOS using three classification scenarios involving Neural Network-LVQ method, Euclidean distance using K-NN, and RBF Kernel SVM algorithms. As per results, the best accuracy was achieved using SVM-RBF Kernel keeping the regularization parameter, C = 40. Results show that the "A" dataset reached the accuracy of 82.55% and meanwhile the "B" dataset which was obtained from Euclidean distance-KNN classification with K = 5 were able to reach the accuracy 78.81%. For a real-time application, the presented software is not efficient enough as the accuracy can be further improved.

System built in Ref. [13] showcased a relatively different approach mentioning how different patients can be classified in different phenotypic groups. The BorutaShap method and the random forest algorithm were used for making predictions and clustering the PCOS patients. The study shows classified two different clusters considering the relevant features; however, they were not able to provide a system generalizing well on people as the predictions were made considering only an internal validation with a very small sample size.

In Ref. [14], the work presented an analysis of various machine learning algorithms on PCOS diagnosis was released. The work shows all the machine learning models but did not consider the feature extraction and transformation problem that this task suffers from due to the nature of this disease. They did not consider using the capabilities of a neural network for this task.

The present work proposed an automated smart healthcare system based on a deep neural network model that can be well utilized by paramedics for diagnosing PCOS with great accuracy of 96.36%. Various supervised ways of solving the problem and a study showcasing how they yield a different result have been discussed in this work. It can be really challenging to diversify the symptoms of PCOS from the variety of other prospective causes. The etiology of the presence of PCOS can be dubious as the symptoms can be confusing and the state of health of a person may vary from one individual to another individual. Therefore, the problem mentioned in this work required working on several subtasks like feature extraction and transformation. The main motivation behind using a deep neural network model is that it not only transmits data as input and outputs through various connections but also associates well with data transformation and feature extraction. The application of a deep neural network in contrast to a shallow network provides a much more efficient system in terms of a number of parameters and the computational power. This work provides the reader with essential facts about the disease. Also, it provides a comparative analysis of enlightening pre-existing work in this domain which is very closely linked to the work presented here.

2.3 DATASET

In the proposed methodology, the experiments are accomplished on the dataset with 41 features including various clinical and physical parameters like age, weight, height, marital status, etc. Dataset gathering is one of the most crucial steps as it decides the behavior of the model. The one which was used to train the model proposed in this research was collected from 10 different hospitals situated in Kerala, India (Kaggle). It consists of information of about 540 women. The dataset has been analyzed by examining the correlation between all the features to eliminate redundant data. The dataset that has been used in this research contains multiple attributes and all those attributes have been analyzed by calculating their linear interdependency. If a feature is linearly dependent on another feature then it signifies that it is correlated with that feature, hence they are called the collinear features. If two columns have a high correlation, then it suggests that they are having a high covariance so one of the columns should be removed as it brings no new information.

$$\text{Correlation Coefficient}(\rho) = \frac{\sum_{i=1}^{n}\left(yi-\overline{y}\right)(\mathrm{zi}-\overline{\mathrm{z}})}{\sqrt{\sum_{i=1}^{n}(yi-\overline{y})^2}\,\sqrt{\sum_{i=1}^{n}(zi-\overline{z})^2}} \tag{2.1}$$

Correlation can be positive or negative. A positive correlation states that if one feature increases, the target value also increases whereas in the case of negative correlation, the value of the target decreases as the value of a feature increases. Equation 2.1 is used for finding the correlation between various features used in the dataset. In Figure 2.1, a subset of the correlation matrix of the data has been plotted which displays the behavior of the dataset.

Data preparation is a crucial step to improve the learning of models. This step generally involves cleaning,[15] curating, and removal process of unnecessary features. Preparation and cleaning of data were done before using the collected data; one of the most time-consuming and important steps is to clean the raw and unprocessed data. It often deals with challenges like missing data streams, identifying outliers, etc.

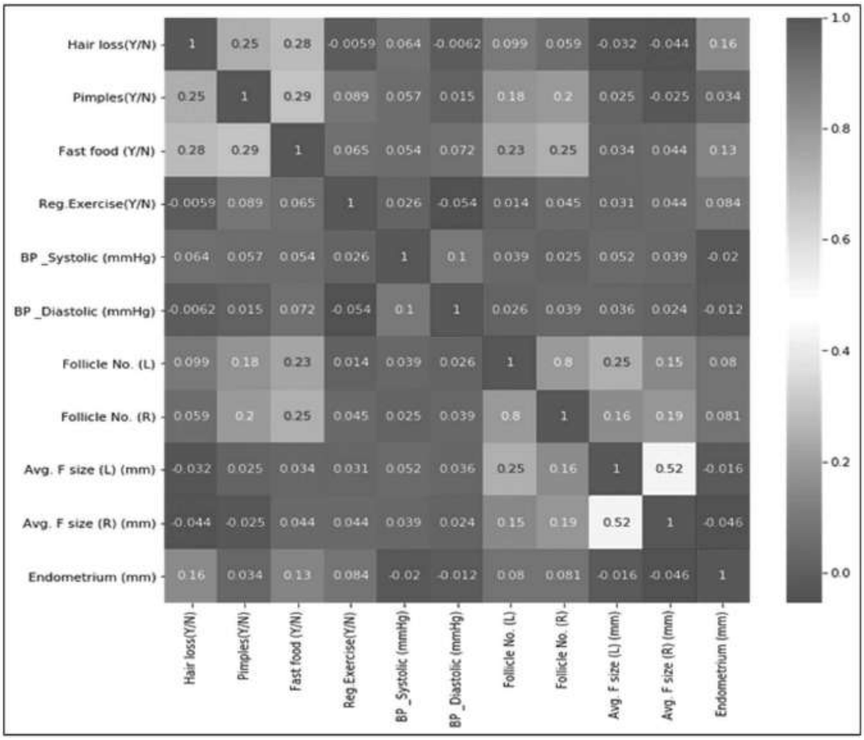

FIGURE 2.1 Heat map of dataset.

In the proposed work, missing data were handled by checking for null values in the data using a function "is null" provided by pandas and dropping those values. Additionally, the research employed a method wherein a limit for outliers was set to identify the set of data that was behaving strangely. Before feeding data to the proposed model, data were split into two sets, namely training and testing data. Non-overlapping sets of data were chosen for training and evaluation. This step is important for ensuring proper testing of the model. Furthermore, this work uses a manual approach for feature selection for reducing the complexity of the model. That would help in removing unwanted data and in improving the model's performance. Hence, the multi-variate technique employing Pearson's Correlation was used with a hand-tuned method with a threshold of 0.85 to find the correlated features with a correlation of 85% and more to remove them. Finally, the function calculated and concluded that the features with a high correlation which were finally removed were as follows: {"Patient File No.," "Waist(inch)," "BMI," "FSH/LH"}

2.4 EXPERIMENTAL METHODS

The main goal of the proposed work is to develop a supervised machine learning-based system that can automate the diagnosis of PCOS in women with the help of certain physical and clinical recorded features of the convalescent.

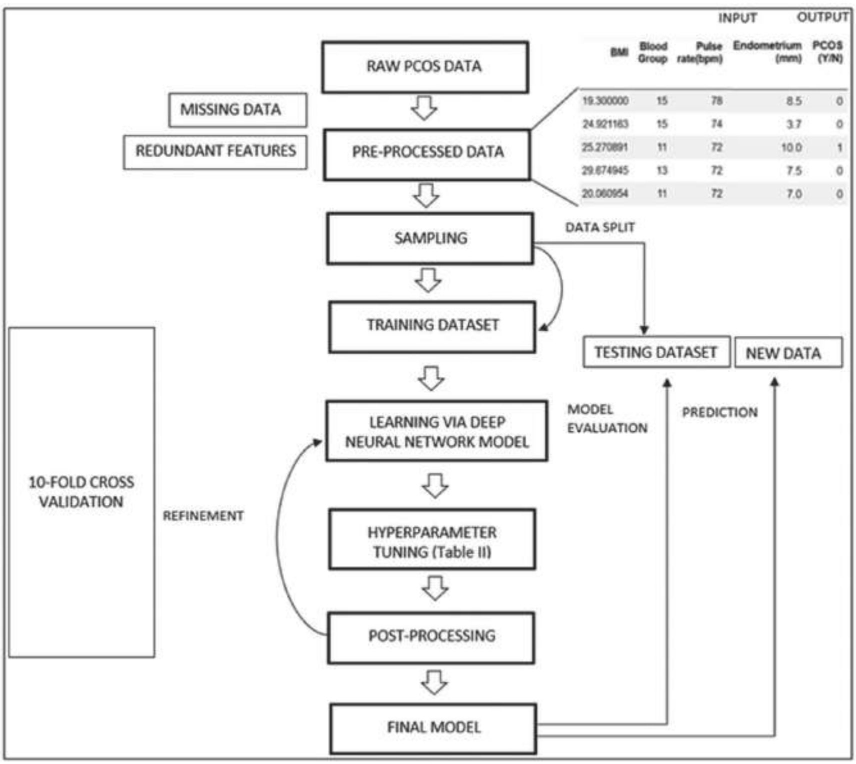

FIGURE 2.2 Flowchart of the proposed machine learning process.

The main goal of the proposed work is to develop a supervised machine learning-based system that can automate the diagnosis of PCOS in women with the help of certain physical and clinical recorded features of the convalescent.

2.4.1 DATASET GATHERING AND PRE-PROCESSING

To build an efficient model, a proper and rigorous process of cleaning, curing, and splitting the data is necessary as generally the input raw data and the output targets are not in a state to train a highly analytical model. A stream of raw and unprocessed data was fed in the system which has the information of multiple patients with dominant symptoms called the features. This set of data consisted of many missing values and redundant features. The pre-processed data were split into two sets such as training data and testing data.

2.4.2 APPLYING A MACHINE LEARNING MODEL

Without a properly labeled target, the supervised machine learning algorithms would be unable to learn anything from the data and will eventually fail to map the given data to its corresponding outcome. So, the target variable was transformed into an understandable form. The training set is used to give the machine learning model a learning ability. Consequently, the construction of more predictive input representations or elements from the raw variables was attempted. The resulting features were fed to the learning algorithms to build a generalized model. Then, the quality of models was evaluated on data that were held out from training.

2.4.3 TUNING THE HYPERPARAMETERS AND TESTING

Theoretically, a hyperparameter can be considered as an orthogonal unit to the learning model in a way that, even though they exist outside the model. There exists a direct relationship between them that tends to affect the model significantly. After the model was trained, hyperparameters were tuned followed by a step of post-processing of data. This further goes for refinement and is again tuned for better performance. The K-fold cross-validation technique which is a statistical tool used for approximation of the ability of machine learning models has been used in this research. In the cross-validation technique, a set of data is divided into "k" number of subsets and in the present work value of k was "10." Once it comes to the selection and optimization of hyperparameters, there are majorly two ways to achieve that: manual selection and automatic selection. In the proposed work, manual testing and tuning of parameters have been performed and the set of parameters that yielded the best accuracy was finally chosen.[16]

Values of various parameters were altered and the behavior of the model for each value was observed. For instance, the number of hidden layers in the model was initially 1 but after a deeper analysis and consecutively observing the results, 5 layers generated the best possible accuracy while training the model. It was seen that the performance of the model started to deteriorate on increasing the number of layers further. Similarly, other parameters were also judged, and based on the analysis, a final set of corresponding values were employed in the deep neural network model as presented in Table 2.1.

TABLE 2.1 Analysis Table Showing Important Parameters and Value.

Serial number	Hyperparameters	Values
1	Number of hidden layers	5
2	Neurons per layer	200
3	Epochs	100
4	Optimizer	ADAGRAD
5	Activation function	RELU
6	Kernel initializer	He-uniform

To avoid the overfitting problem,[17] the process of batch normalization was also implemented. The final model with tuned parameters is tested on the section of testing data[18] which consists of unseen values and provided the final accuracy of the model.

2.5 RESULTS AND DISCUSSION

In the proposed methodology, the experiments are accomplished on the data of healthy patients as well as PCOS-diagnosed patients. In this research, multiple machine learning models[19] were implied, and their accuracy scores were calculated and presented. In Figure 2.3, the comparison of the accuracy scores of all the models has been demonstrated.

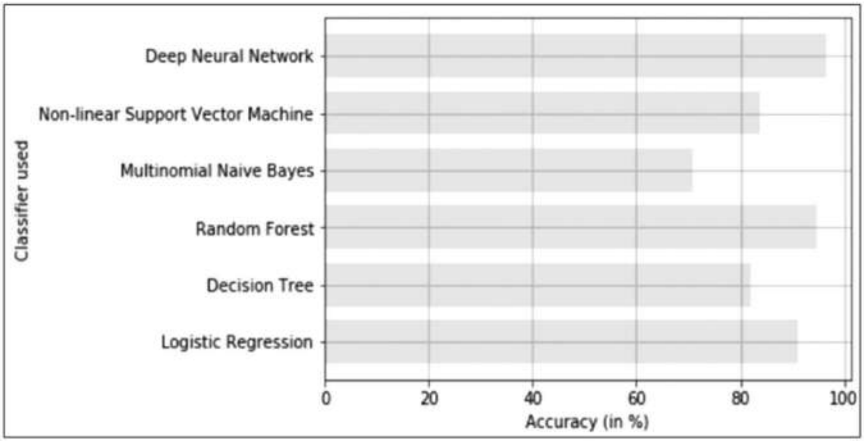

FIGURE 2.3 Comparison of accuracy scores of various classifiers.

As shown in Figure 2.3, the proposed research exemplifies that deep neural networks achieved the best accuracy score of 96.36%, followed by the Random Forest classifier which attained an accuracy of 94.54%. Additionally, logistic regression also performed surprisingly well with an accuracy of 90.09% on this problem whereas the non-linear support vector machine model reached an accuracy of 83.63%. decision tree algorithm was also assessed in this research, and it shows an accuracy of 81.81% and multinomial Naïve Bayes yielded an accuracy score of 70.90% with maximum loss. To further analyze the performance of the star model, a few more metrics were used to test the all-round performance of the model. The final built model yielded a testing accuracy of 96.36%. Figure 2.4 reveals a learning graph for the proposed deep neural network model showing a learning graph in both the phases of training and testing.

Since the deep neural network model showcased the most promising performance, the training and testing accuracies were evaluated in order to understand the model in depth. The training accuracy of the model comes out to be 97.3% and the testing accuracy of the finely tuned model is 96.36%. It shows minor fluctuations while reaching the final precision but since the bias and variance of the given model are well balanced, there is no scope of cases involving overfitting or underfitting.

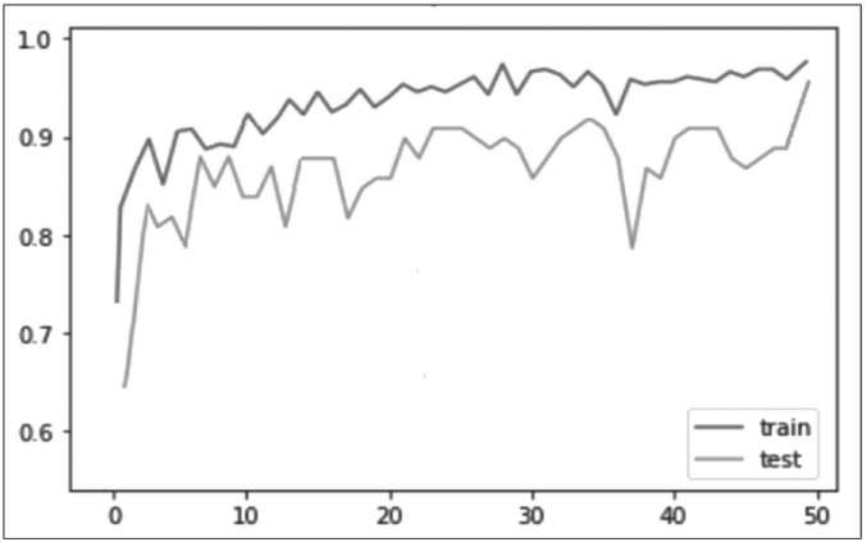

FIGURE 2.4 Learning graph for proposed deep neural network.

For better understanding and proper representation of the results received, confusion matrices have also been presented showing the predictions, more specifically True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN). Table 2.2 brings clarifications for the predictions of metrics in classifications. Further metrics like the Recall, and Precision of the model were calculated using the values of true positive, true negatives, false positives, and false negative.

Using these formulas, the value of all the parameters were computed and evaluated to comprehend the performance of every model. Model evaluation metrics like Recall, F1 score, and Precision, are very vital to evaluate a given model.[20]

In Table 2.3, a detailed report of the output of all the models being used has been discussed. It can clearly be seen from Table 2.3 that the deep neural network model yielded the highest accuracy of 96.36% and a high recall of 96%. Therefore, the F1 measure of the model came out to be 96%. The testing accuracy and training accuracy were also high comparatively to other supervised machine learning models.

TABLE 2.2 True/False Predictions by Different Machine Learning Algorithms.

Model	True positive	True negative	False positive	False negative
Deep Neural Network	38	15	1	1
Non-Linear SVM	35	11	4	5
Multinomial Naïve Bayes	39	0	0	16
Decision Tree	35	10	4	6
Random Forest	38	13	1	3
Logistic Regression	37	13	2	3

TABLE 2.3 Table to Compare the Classifiers.

Model	Precision	Recall	F1-score	Testing accuracy (%)
Deep Neural Network	0.96	0.96	0.96	96.36
Non-Linear SVM	0.83	0.84	0.83	83.63
Multinomial Naïve Bayes	0.50	0.71	0.59	70.90
Decision Tree	0.81	0.82	0. 81	81.81
Random Forest	0.95	0.95	0.94	94.54
Logistic Regression	0.91	0.91	0.91	90.09

A confusion matrix plays an important role in problems involving statistical classification. It is sometimes also referred to as an error matrix. For the current problem, the deep neural network shows a promising performance when compared on different metrics. The confusion matrix of the proposed deep learning model is presented in Figure 2.5.

Each column in the Figure 2.5 confusion matrix representing the result of the deep neural network is representing the instances of an actual class and each row is representing the occurrences in a predicted class. The experiments show that the intended machine learning-based system required an average wall time of around 25 milliseconds per sample and the experimental outcome can achieve good accuracy which implies that it is suitable for real-time applications making it a significant contribution.

Table 2.4 tabulates a comparative study of some existing work in the field of machine learning for diagnosing PCOS. The datasets used in these studies are generally derived locally; hence the datasets which are being used are by different works with different datasets. To compete with different methodologies, a similar dataset should have been employed for better scientific and research standards. Considering how difficult it would be to gain access to these local datasets, this task also becomes challenging.

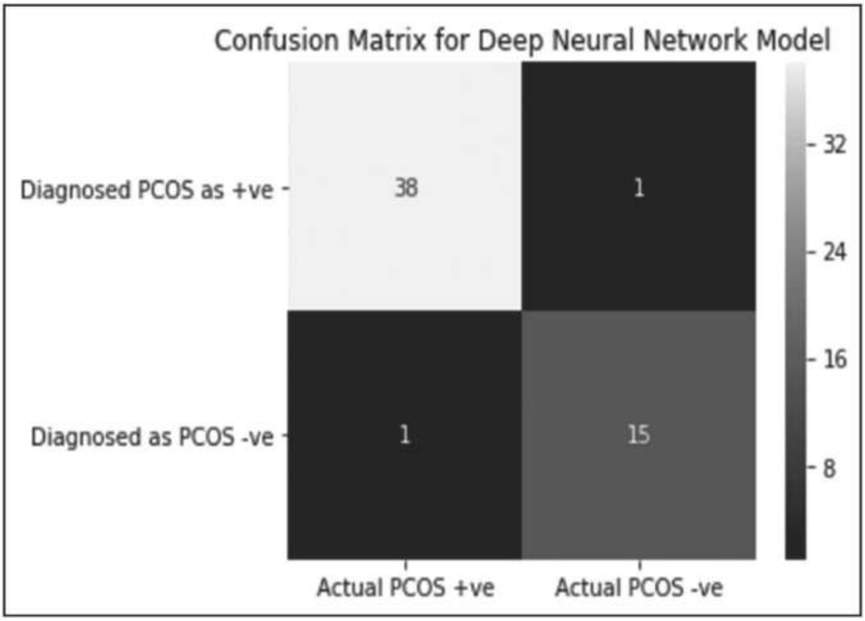

FIGURE 2.5 Confusion matrix for deep neural network model.

2.6 CONCLUSIONS

In this work, a supervised machine learning-based classification technique has been applied to identify the presence of PCOS in females as positive or negative. This paper successfully provides an automated machine learning-based smart healthcare system that can be well utilized by doctors and medical professionals for diagnosing PCOS. Various stages involved in the diagnosis of PCOS were successfully analyzed and an efficient model after proper parameter tuning was also proposed to predict the imminent occurrence of PCOS in this research. The finalized model produces appreciable testing accuracy and prevents overfitting by performing regularization using the batch normalization technique while training the network. It provides a better evaluation and understanding

TABLE 2.4 Comparative Table of Pre-existing Works with the Proposed Method.

Reference titles	Comparison indicators			
	Aim	**Database source**	**Model used**	**Accuracy**
Mehrotra et al. [3]	Bayesian classifier vs logistic regression on a PCOS recognition problem	GDIFR, Kolkata, India	Bayesian classifier	93.93%
Denny et al. [4]	Analyze the nature of various models to diagnose PCOS	Kaggle	CART, Naïve Bayes, Logistic Regression, KNN, SVM, and Random Forest	89.02% Random Forest classifier (best of all)
Soni andVashisht [5]	Technique of image segmentation on ultrasound images before applying CNN model	Ultrasound images from anonymous source	CNN, Support Vector Machine and (Unsupervised learning) K-Nearest Neighbor	82.55% Support Vector Machine (best of all)
Meena et al. [6]	A combination of artificial neural network and fuzzy neural logic	ftp://ftp.ncbi.nlm.nih.gov/geo/datasets/GDS4nnn/GDS4987/	Support Vector Machine, Artificial Neural Network, Classification Tree, Naïve Bayes	83.70% Artificial Neural Network (best of all)
Meena et al. [7]	A combination of artificial neural network and fuzzy neural logic	ftp://ftp.ncbi.nlm.nih.gov/geo/datasets/GDS4nnn/GDS4987/	Neural Fuzzy Rough Set and Neural Network	89%
Rabiu et al. [9]	Surveying computer aided techniques for follicles detection PCOS diagnosis	Two different datasets. Source not mentioned	Not mentioned	97.61% and 98.18%
Chauhan et al. [14]	Early prediction of PCOS	Conducted survey with 267 women	KNN, Naive Bayes, SVM, Decision tree, Logistic Regression	81% decision tree (best of all)

TABLE 2.4 *(Continued)*

Reference titles	Comparison indicators			
	Aim	**Database source**	**Model used**	**Accuracy**
Deepika [21]	Mentions objectives of AI PCOS diagnosis	Not mentioned	Bayesian and Logistic Regression	91.04% for Bayesian and 93.93% for Logistic regression
Present proposed work	Analyze the problem of PCOS, get a reliable machine learning model applied to predict PCOS in patients	10 different hospitals of Kerala, India.	Deep Neural Networks, Non-Linear SVM, Multinomial NB, Decision Tree, Random Forest, and Logistic Regression	96.36%—Deep Neural Networks, 83.63%—Non-Linear Support Vector Machine, 70.90%—Multinomial Naïve Bayes, 94.54%—Decision Tree, 81.81%—Random Forest classifier, and 90.9%—Logistic Regression classifier

of the proposed models by calculating various evaluation metrics and by stating their corresponding significances.

Our future work intends to use methods like transfer learning on complex neural network architecture to further improve the model's performance and make an application out of it for a better user experience. It is believed that developing a software application that uses the deep neural network at its core can be successfully employed by paramedics and by other potential users for performing real-time diagnosis of PCOS.

KEYWORDS

- **artificial intelligence**
- **polycystic ovary syndrome**
- **E-health system**
- **follicles**
- **normogonadotropic**
- **smart healthcare systems**

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CHAPTER 3

CLASSIFICATION OF BREAST HISTOPATHOLOGICAL IMAGES USING SEMI-SUPERVISED GENERATIVE ADVERSARIAL NETWORKS

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ABSTRACT

Women are most at risk for developing breast cancer, leading to a major area of medical research. A biopsy test involves removing tissue and studying it under a microscope to determine the presence of cancer. Often, malignant tumors are missed by histopathologists who are not well trained. Since the automatic analysis of histopathology images can help pathologists to identify cancer subtypes and malignant tumors, this can benefit pathologists and facilitate better diagnosis. Research on deep learning has increased recently, but large amounts of data are needed for its application is the crucial task for extracting performance from them. As a result of the limited number and the high cost of annotating data, pathologists struggle to develop annotated data. Traditional deep learning algorithms

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are ineffective when there are large numbers of unannotated data, which limit the amount of annotated data available. The semi-supervised learning approach, using generative adversarial networks (GANs), can solve the lack of labeled input data when training a deep learning classifier. We demonstrated that semi-supervised GAN using both labeled and unlabeled datasets can achieve a significant level of performance as traditional convolutional neural networks. This method reduces the cost and time related to annotating cell images for deep learning classifier development.

3.1 INTRODUCTION

Every year millions of women are affected by breast cancer and cause many deaths among women. According to the breast cancer survey, in North America, breast cancer is more than 80%, and in Japan and Sweden, nearly 60% of people are affected. The influence of breast cancer is below 40% in low-income countries.[1] According to the data gathered by the American cancer society, around 2,68,600 new invasive cases were diagnosed in 2019, and the expected death case will be roughly about 41,760.[2] Early diagnosis is significant to raise the number of survival people. In recent years, many researchers explore precise models and clarify solutions in the field of medical imaging.

The two major types of cancer are benign (cancer) and malignant (non-cancer). The cell features and properties determine the class of cancer. In situ, tumor cells reside inside ducts (lobules) in the breast. In other cases (invasive), the tumor cells are spread over ducts.[3] Spanhol et al., introduced different types of subclass with various magnifications under benign and malignant.[4] In the beginning, the breast cancer diagnosis is based on self-assessment by utilizing mammography, thermal imaging, and ultrasound imaging techniques with periodic supervision. The needle biopsy is the most dependable technique to clarify malignant growth.[5] The microscopic image gives the structure and elements of tissue accessed by the pathologist. Before analysis, the tissue cells are imposed into the staining process using Hematoxylin and Eosin (H&E). The staining process boosts the study of the tissue structural elements like the shape of the cell and nuclei.[6–8]

The pathologist's workload is reduced, and efficiency improved, using automated processes through computer-aided diagnosis (CAD) systems. In the machine learning technique, the classes are identified using handcrafted features by utilizing various classifiers. The classifiers are trained and predicted the specific label.[9–11] However, the handcrafted features required extensive domain knowledge, and the computational burden needs new advances in machine learning. The advanced machine learning technique is called deep learning makes more attention in the research field. Deep learning is used with a deep convolution neural network, which automatically learns features from the raw input image.[12] The hyperparameters are adjusted manually to improve the results. Recently, researchers focus on convolutional neural network (CNN)-based CAD techniques to improve the quality of medical image analysis results. Deep learning requires more time for training, and the features are learning from experiences by using a large amount of dataset.[13]

During training, the data imbalance causes overfitting may occur rapidly and overcome by using the Generative adversarial network (GAN) network. The GAN network comprises a generator, and the discriminator both performs min–max game which was introduced in 2014. The generator tries to create synthetic data that are closely related to real data.[14] In our case, very few pathologists are specialized in histopathology image analysis in laboratories. The labeling of the image makes slower because of analysts scarcity and more data. The annotation needs more time and does not complete the task at a particular time. In our proposed work, we consider both labeled and unlabeled data for training to improve classification accuracy.

3.2 RELATED WORKS

Spanhol et al.[4] extracted pixel patches from the histopathology image and utilized to train CNN. The 32 × 32 and 64 × 64 patches were clipped out using the sliding window method. The accuracy is between 80.8 and 89.6%. Bayramoglu et al.[15] established single- and multi-task CNN. The single-task CNN performs overall cancer classification accuracy, and the multi-task CNN results in the simultaneous prediction of cancer class and magnification. The authors reported the results in the single task was 82.10 and 84.63%, and for multi-task was between 80.69 and 83.39%.

Feng et al.[16] used annotated and unannotated samples to increase the training accuracy. Unannotated data play a vital role to improve accuracy.

In Ref. [17], the authors utilized the cascaded methods with auto encode for the classification of breast histopathology images. The authors achieve results is between 96.8 and 98.2%. In Ref. [18], the authors used CNN with long-short-term memory (LSTM) network. This model accomplishes the accuracy is between 90 and 91% by utilizing softmax and support vector machine (SVM) classifiers. In Ref. [19], the authors used the GAN network with 70 labeled dermoscopy images. The authors reported that the classifier performance is better than using the auto encoder and hand-crafted features.

Johnson[20] introduced a novel semi-supervised conditional GAN to detect IDC. The framework of GAN accomplished the accuracies are 86.68, 87.45, and 88.33%. In Ref. [21], the authors introduced contextual GAN in mammogram breast cancer images. However, the CNN accuracy is getting inaccurate results due to over-fitting because of using imbalanced data. The author used U-net-based architecture and achieved an AUC of 0.846. In Ref. [22], the authors utilized a three-stage semi-supervised learning method for mammogram images. The reported accuracy increases while increasing the labeled and unlabeled classes. In Ref. [23], the authors introduced a novel supervised and unsupervised network for lung and pancreatic tumors and achieved favorable results. The literature mentioned above showed the deep CNN plays a significant role in medical imaging diagnosis and provide more incredible performance.

3.3 MATERIALS AND METHODS

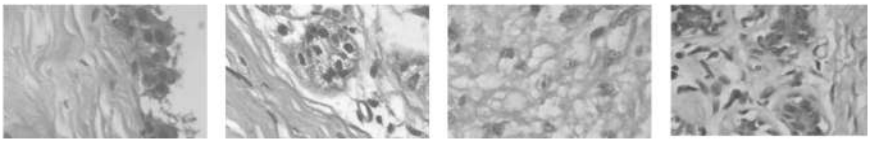
3.3.1 DATASET

Fabio et al., introduced a breast cancer dataset for histopathology images (BreakHis), acquired from 82 patients.[26] The BreakHis is the large scale dataset that comprises 7909 images with benign and malignant, which includes four subclasses for each. The images are categorized under four magnification factors (40×, 100×, 200×, and 400×).[27] The collected images are eight-bit depth RGB images with the size of 700 × 460. The four subclasses for benign are Fibroadenoma (FA), Adenosis (A), tubular adenoma (TA), and phyllodes tumor (PT); and malignant subclass is ductal carcinoma (DC), mucinous carcinoma (MC), lobular carcinoma (LC), and papillary carcinoma (PC). The following Table 3.1 shows the list of BreakHis dataset images.

TABLE 3.1 BreakHis Image Collections in Subclasses with Various Magnification Factors.

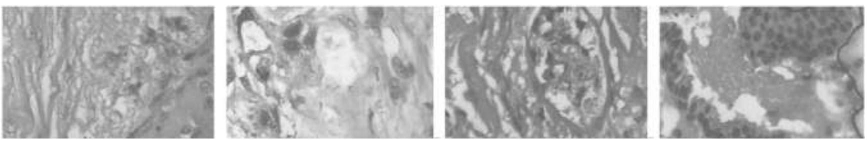
Magnifications	Benign cases				Malignant cases			
	A	FA	TA	PT	DC	LC	MC	PC
40×	114	253	109	149	864	156	205	145
100×	113	260	121	150	903	170	222	142
200×	111	264	108	140	896	163	196	135
400×	106	237	115	130	788	137	169	138

Example histopathology images for benign (400×) and malignant (400×) are shown by following Figures 3.1 and 3.2.



(a) Adenosis (b) Fibro Adenoma (c) Phyllodes Tumor (d) Tubular Adenoma

FIGURE 3.1 Histopathology benign images.



(a) Ductal Carcinoma (b) Lobular Carcinoma (c) Mucinous Carcinoma (d) Papillary Carcinoma

FIGURE 3.2 Histopathology malignant images.

3.3.2 GAN

A GAN is a generative model comprises of two neural networks, the generator and the discriminator. The GAN network based on game theory,[1] having two networks that play vital role to solve min–max game. The fake samples are generated using random noise vector in generator. The discriminator is trained to determine the real and fake samples. The simultaneous training is carried out by two networks. The min–max value function is given by $V(D,G)$,

$$min_G max_D V(D,G) = E_{r\sim P_{data}}\left[logD(r)\right] + E_{f\sim P_f}\left[log(1-(G(f)))\right] \quad (3.1)$$

Where G is generator; D represents discriminator; r denotes real sample; f indicates fake sample; P_{Data} and P_F represents the distribution of real and fake data samples.

3.3.3 SEMI-SUPERVISED GAN

The semi-supervised GAN (SSGAN) is developed from basic GAN network model shown in Figure 3.3. In SSGAN model, the structure of loss function integrates both labeled and unlabeled real data based on deep convolution neural network.[24–26] The real-labeled images perform supervised classification task. The real-unlabeled images are utilized for training discriminator. Raford et al., used the following steps to train GAN network,

- For all generator and discriminator layers, they utilized Relu activation function.
- In generator, tanh activation function and Relu activation function are eliminated.
- Generator used fractional-strided convolution layer while in discriminator pooling layer is replaced with strided convolution layer.
- Batch normalization is utilized for training.

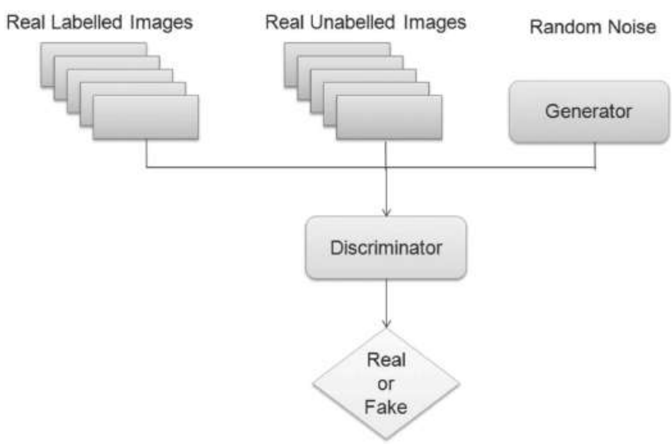


FIGURE 3.3 Semi-supervised GAN network schematic representation.

The combination of labeled (L_L), unlabeled (L_U), and generated (L_G) loss function is used for optimization process.[4] The equations are given below,

$$L = L_L + L_U + L_G \tag{3.2}$$

$$L_L = E_{r,s\sim P_{data}} logp_{model}(s \mid \mathrm{r}, \mathrm{s} < K+1) \tag{3.3}$$

$$L_U = E_{r\sim P_{data}} log(1 - p_{model}(s = \mathrm{K}+1 \mid \mathrm{r}) \tag{3.4}$$

$$L_G = E_{r\sim G} logp_{model}(s = \mathrm{K}+1 \mid \mathrm{r}) \tag{3.5}$$

From the above equations, the notion r defines real image, s indicates label, and P_{data} corresponds to real-data distribution. The SSGAN used K class for generated image while K + 1 class represented for fake image class. The following Figures 3.4 and 3.5 show the schematic representation of generator and discriminator layers.

The SSGAN was designed to handle images of 32 × 32 pixels. At initial stage testing, the BreakHis histopathology image was converted into 32 × 32 pixels and analyzed using conventional algorithms. Due to the resizing, the images suffered from information loss which causes poor performance. Hence, the SSGAN is modified to use the original size of dataset with 700 × 460. The discriminator input layer size was adjusted to 224 × 224 × 3. In the subsequent layers, the reshape layer of the generator was adjusted to 28 × 28 × 256, followed by three convolutional transpose layers, each of which produced images with 224 × 224 pixels of color. The training was performed by dividing the entire training process into mini-batches. The weight has been updated in each stage of dividing the training samples. The input data were normalized by dividing each range by 1. The activation function of Relu was used in the last generator layer.

In eq (3.3), the supervised loss function is calculated after training the discriminator with labeled data and holding the generator weights. In eq (3.4), the unsupervised loss function using discriminator trained on unlabeled data and fake data. As a result, in eq (3.5), the discriminator loss function was calculated by adding both losses together. The backpropagation procedure was utilized to update the discriminator weights. After updating the discriminator, the evaluated weight is fixed froze then the generator training was performed. The mini-batch samples included half-real data and half-fictitious data. Using the discriminator, mini-batch data were fed into the network for analysis. The features are extracted from the real and fictitious data in an intermediate layer of discriminator. At the final stage, the generator weight was updated using back propagation. The evaluation cycle is repeated until the epochs were completed to generate significant result. In this work, our modified model is tabulated in Tables 3.2 and 3.3. The number of 600 noise samples is added to produce the new image.

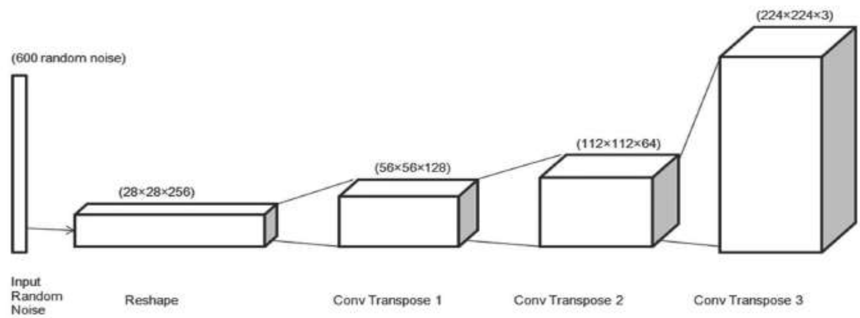


FIGURE 3.4 Generator layers.

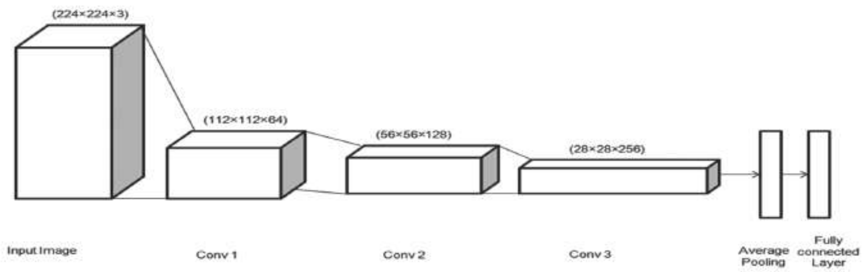


FIGURE 3.5 Discriminator layers.

TABLE 3.2 Generator Analysis.

Block	Kernal	Stride	Feature maps	Dropout	Nonlinearity
Reshape	–	–	28 × 28 × 256	–	–
ConvTrans1	3 × 3	2	128	No	Relu
ConvTrans1	3 × 3	2	64	No	Relu
ConvTrans1	3 × 3	2	8	No	Relu

TABLE 3.3 Discriminator Analysis.

Block	Kernal	Stride	Feature maps	Dropout	Nonlinearity
Conv2D	3 × 3	2	32	No	Relu
Conv2D	3 × 3	2	64	No	Relu
Conv2D	3 × 3	2	128	0.4	Relu
Conv2D	3 × 3	1	256	0.25	Relu

3.4 CONVOLUTION NEURAL NETWORK

As depicted in Figure 3.6, the proposed CNN architecture is comprised with several layers similar to discriminator. The image has three layers of convolution, each with two filter sizes, which are used to extract various attributes, including corners, edges, and color. The convolution layers reduce the size of the input image and increase its depth. In this network, three max-pooling and one average pooling are employed with a 2 × 2 kernel size. The fully connected layer is used to flatten the features and the final output layer predicts whether the results.

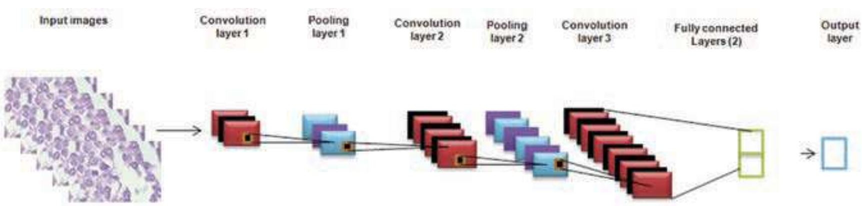


FIGURE 3.6 The proposed CNN layers.

3.5 RESULTS AND DISCUSSIONS

In this work, our modified SSGAN performs the evaluation on breast histopathology classification task. The random selection of data was utilized for training (80%) and testing (20%). Figure 3.7 illustrates the data splitting process. The training data contain labeled and unlabeled data in SSGAN. The hyperparameter settings were given below: Epoch = 20; Batch size = 100; Adam optimizer with learning rate 0.0001.

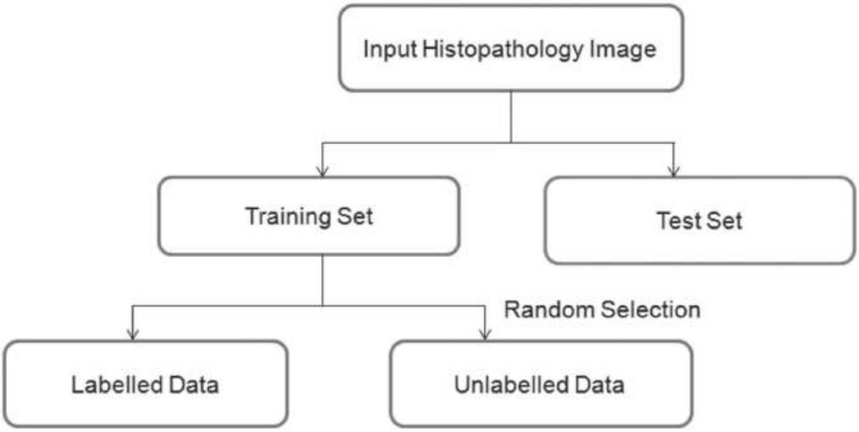


FIGURE 3.7 Data dividing process for experiment.

The performance evaluation of our method is calculated in terms of AUC and accuracy. The image level accuracy can be calculated as follows,

$$Accuracy = \frac{Number\ of\ correctly\ classified\ cancer\ images}{Total\ number\ of\ images} \tag{3.6}$$

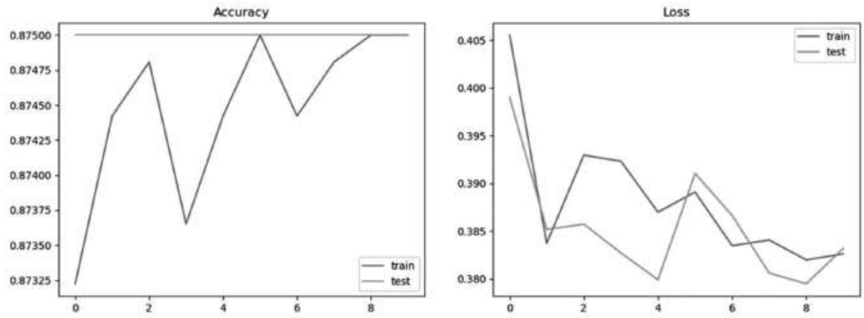
In the field of medical imaging, during cancer diagnosis, positive case is identified as malignant while negative case is identified as benign. The following Figures 3.8, 3.9, 3.10, and 3.11 shows that the accuracy and loss obtained using different magnification factors given in BreakHis dataset. The sensitivity or recall is the important factor in diagnosis. Also, we calculate the other metrics such as precision and F1 score. The following equations represent the calculation of metrics,

$$Sensitivity(recall) = \frac{True\ positive\ rate}{True\ positive\ rate + False\ negative\ rate} \tag{3.7}$$

$$Precision = \frac{True\ positive\ rate}{True\ positive\ rate + False\ positive\ rate} \tag{3.8}$$

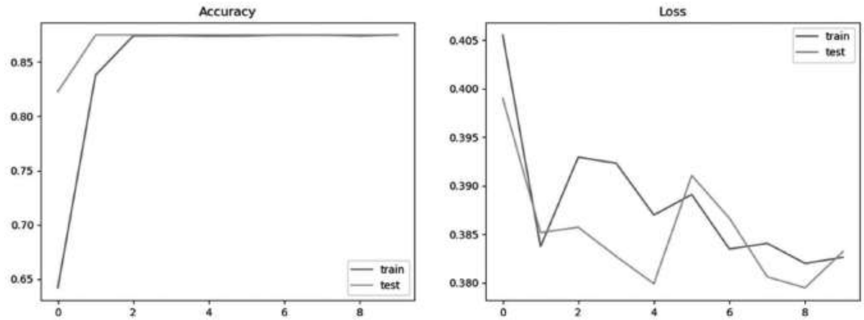
$$F1\,score = \frac{2 \times Recall \times Precision}{Recall + Precision} \tag{3.9}$$

To perform semi-supervised training, different proportions of labeled images are used, such as 30, 40, 50, and 60% and the rest of the images are unlabeled. The labeled data results are illustrated in Table 3.4. The



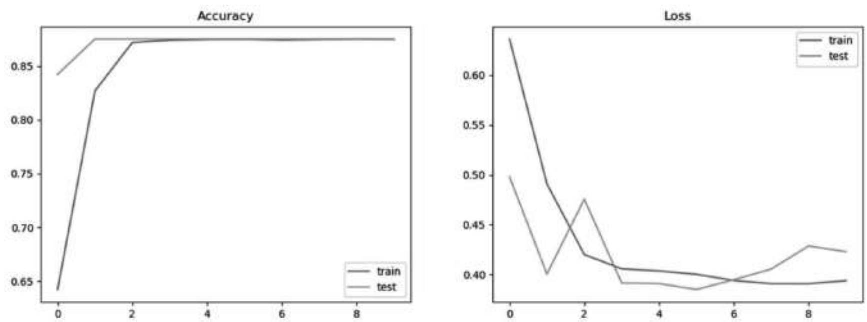
(a) (b)

FIGURE 3.8 SSGAN results for 40× magnification: (a) accuracy plot and (b) loss plot.



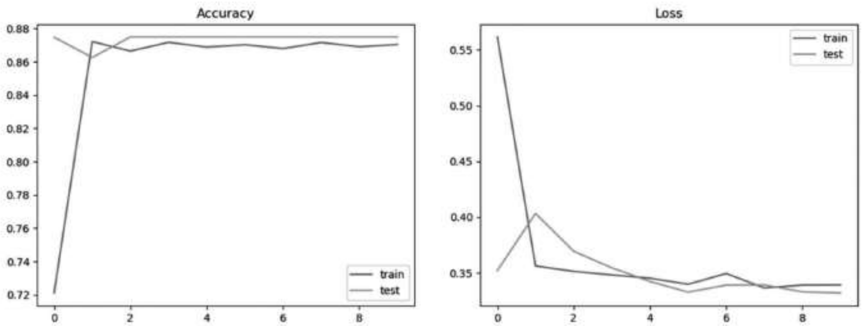
(a) (b)

FIGURE 3.9 SSGAN for 100× magnification: (a) accuracy plot and (b) loss plot.



(a) (b)

FIGURE 3.10 SSGAN results for 200× magnification: (a) accuracy plot and (b) loss plot.



(a) (b)

FIGURE 3.11 SSGAN results for 400× magnification: (a) accuracy plot and (b) loss plot.

experiment was then stopped at 50% of labeled images since there was no improvement. For all experiments, this work utilized random 20% of data for testing. This evaluation is performed using various magnification factors and metrics listed in Table 3.5. This study proposes a method for breast cancer image classification using compact CNN. Our study compared SSGAN to CNN in order to demonstrate the minimally labeled image achieves good performance compared to the largest number of images labeled in CNN. CNN architecture used in this study was similar to that of the SSGAN discriminator.

TABLE 3.4 Accuracy Comparisons for Labeled Data using SSGAN Approach.

Magnifications	Labeled data			
	30%	40%	50%	60%
40×	0.79	0.83	0.87	0.87
100×	0.78	0.83	0.88	0.88
200×	0.82	0.86	0.87	0.87
400×	0.81	0.88	0.88	0.89

TABLE 3.5 SSGAN Performance Metrics for Multiclass Classification.

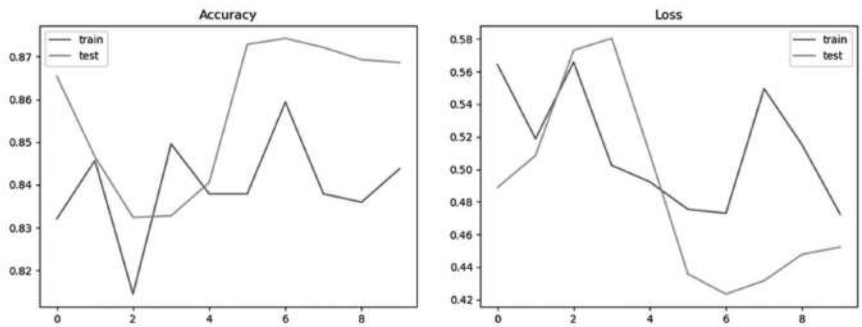
Magnification factors	Accuracy	AUC	Specificity	F1-score
40×	0.87	0.92	0.86	0.87
100×	0.88	0.94	0.88	0.86
200×	0.87	0.92	0.87	0.86
400×	0.88	0.95	0.87	0.87

Moreover, the trail approaches are done using 50, 60, and 80% of training data, and a random set of 20% is used to test the performance of the classifier. The same hyperparameters in SSGAN are utilized for CNN-based classification. As shown in Table 3.6, 80% of the CNN data is used for training while 20% is used for testing to reach the SSGAN results. SSGAN requires fewer labeled images and performs better than CNN. These results are presented in Figures 3.12, 3.13, 3.14, and 3.15. In contrast, CNN cannot use unlabeled images and need more labeled images in order to achieve the same result. The SSGAN was, therefore, found to be valuable when there is a lack of labeled data and was still able to achieve a significant result when compared to conventional CNN.

The proposed models could achieve comparable or better performance than conventional models. Recent years have seen an increase in interest in applying machine learning technology, especially deep learning, to medical research. The software can reduce the workload of pathologists, improve diagnosis quality, and support improved patient care. By utilizing our proposed scheme, we can address breast cancer diagnostic scenarios, and increase diagnostic quality while reducing workload.

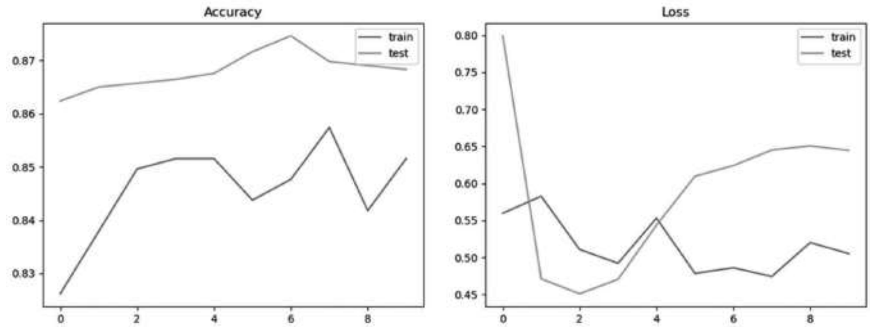
TABLE 3.6 Accuracy Comparisons for Labeled Data using CNN Approach.

Magnifications	**Labeled data**		
	50%	**60%**	**80%**
40×	0.65	0.74	0.88
100×	0.69	0.77	0.87
200×	0.66	0.76	0.86
400×	0.68	0.78	0.87



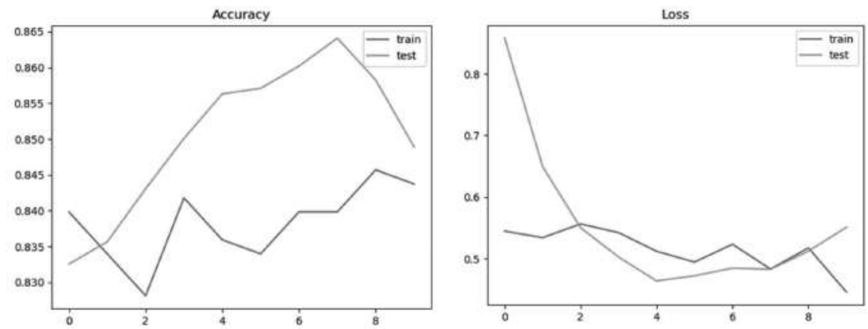
(a) (b)

FIGURE 3.12 CNN results for 40× magnification: (a) accuracy plot and (b) loss plot.



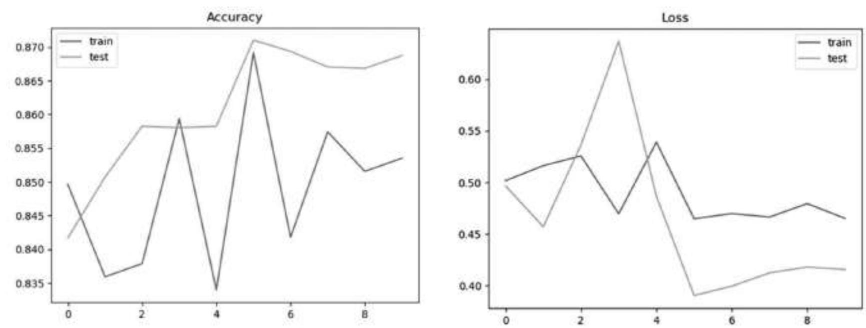
(a) (b)

FIGURE 3.13 CNN results for 100× magnification: (a) accuracy plot and (b) loss plot.



(a) (b)

FIGURE 3.14 CNN results for 200× magnification: (a) accuracy plot and (b) loss plot.



(a) (b)

FIGURE 3.15 CNN results for 400× magnification: (a) accuracy plot and (b) loss plot.

Our hypothesis is, in the future, to incorporate the experience of pathologists when constructing our model. By analyzing deep neural network decisions, we will try to identify regions in an input breast cancer image that indicate a particular type of cancer. By evaluating the diagnostic experience as a priori, we can determine the differences in the support areas when the choice is made between pathologists and algorithms. Additionally, we can construct an attention-based network model that will help to improve the model's accuracy in the future.

3.6 CONCLUSIONS

An adversarial network-based approach provides unique advantages in medical image processing. In this study, we developed a semi-supervised GAN capable of learning both labeled and unlabeled images. Our research has experimentally evaluated this proposed model for multi-classification by using semi-supervised GAN on the BreaKHis dataset over four magnification factors (40×, 100×, 200×, and 400×). For the magnification factor 100× and 400×, the best accuracy of 88% was obtained. Also, the AUC of 0.95 was accomplished in the magnification factor 400×. Furthermore, this study showed that SSGAN performed better with a smaller number of labeled images than CNN. In the future, we will explore multi-class classification using various architectures and data-balancing techniques. The proposed scheme achieved promising results for the task of categorizing breast cancer images. This method can reduce pathologists' workload and improve diagnosis quality when used in breast cancer diagnostic scenarios.

KEYWORDS

- **histopathology**
- **generative adversarial networks**
- **semi-supervised generative adversarial networks (SSGAN)**
- **convolution neural network**
- **breast cancer**

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CHAPTER 4

A SYSTEMATIC REVIEW FOR THE CLASSIFICATION AND SEGMENTATION OF DIABETIC RETINOPATHY LESION FROM FUNDUS

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ABSTRACT

Diabetic retinopathy (DR) is an eye disease caused by diabetic patients that damage the retina. The severity of the disease is found by using the symptoms like hemorrhages, micro aneurysms, exudates, etc. These are the early stage symptoms of DR. The detection of lesion algorithm is used to diagnosis the disease easily by an ophthalmologist. Accurate detection of diabetic retinopathy lesions from fundus images is a difficult task that necessitates the development of automated computer-aided diagnostic techniques. This paper reviewed the state-of-the-art methods currently available for achieving the aforementioned objectives, as well as their benefits and limitations. This review identifies the changes that must be made in the future to create the best automated diagnostic tool that performs better and avoids all of the pitfalls found in current literature.

Artificial Intelligence and Machine Learning Techniques in Image Processing and Computer Vision.
Karm Veer Arya, Ciro Rodriguez Rodriguez, Saurabh Singh, & Abhishek Singhal (Eds.)

4.1 INTRODUCTION

Diabetic retinopathy (DR) affects 80% of diabetic individuals today. Real area and therapy of eye disorders thwarts visual adversity in over portion of patients. Retinal pictures have been displayed to be amazingly successful instruments for affliction assurance. Clinical professionals, for example, evaluate the eye and vital contaminations using ophthalmology photos acquired from the retina. DR and age-related macular degeneration (AMD), for example, manifest themselves on the retina. They are visible through the evaluation of retinal images and may be curable at any stage detected early on. DR is a condition that affects the retina of the eye.

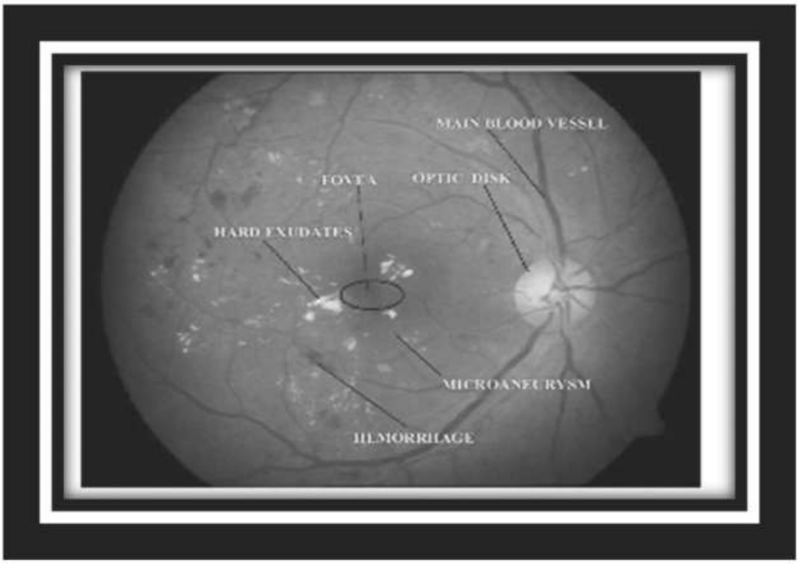

FIGURE 4.1 Diabetic retinopathy-affected image.

It might be reparable when the affliction is found toward the starting stage. Diabetes could be an affliction wherein the body doesn't make or appropriately organizations affront, a substance that is expected to change over sugar, starches, and other food into energy required for technique for life.[1] Tall blood weight is additionally called hypertension raises your chance for coronary episode, stroke, eye issues, and kidney illness. The presence of various wounds insinuated as faint/red bruises, comparable to smaller-than normal aneurysms (MAs) and hemorrhages (HEMs), as well as mind-boggling traumas, similar to exudates, characterizes DR (EXs).

MAs, which are the most common symptoms of DR, appear as bronzed, unassuming, and indirect patches.

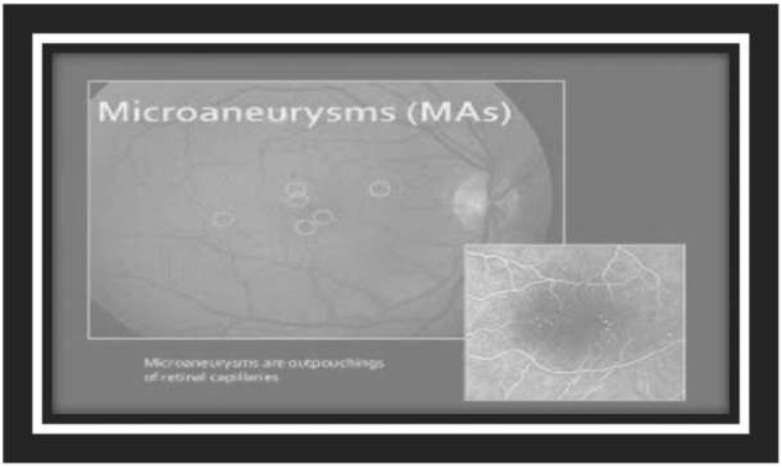

FIGURE 4.2 Microaneurysms lesion.

HEMs are produced by retinal ischemia and the bursting of abnormally fragile retinal veins. They usually appear as bright red spots/patches with a wide range of shapes and looks.

EXs, on the other hand, are yellowish intra-retinal liquid reservoirs that store protein, lipid, cell debris, and other materials. They usually appear as yellowish, dazzling patches with sharp lines in a variety of forms and sizes.

Stages of Diabetic Retinopathy

- **Mild nonproliferative retinopathy:** It is portrayed by expand like growing inside the retina's veins. These are called microaneurysms, and these vessels can spill into the eye.
- **Moderate nonproliferative retinopathy:** Amid this stage, the blood vessels feeding the retina swell and may indeed get to be blocked.
- **Severe nonproliferative retinopathy:** At this stage, the extending number of veins taking care of the eye has been able to be obstructed. Subsequently, the retina is mentioned to foster the present day veins.

- **Proliferative retinopathy:** Usually, the ultimate organization of DR, retinal separation may cause spotty vision, flashes of lights, or serious vision loss.

FIGURE 4.3 Hemorrhages lesion.

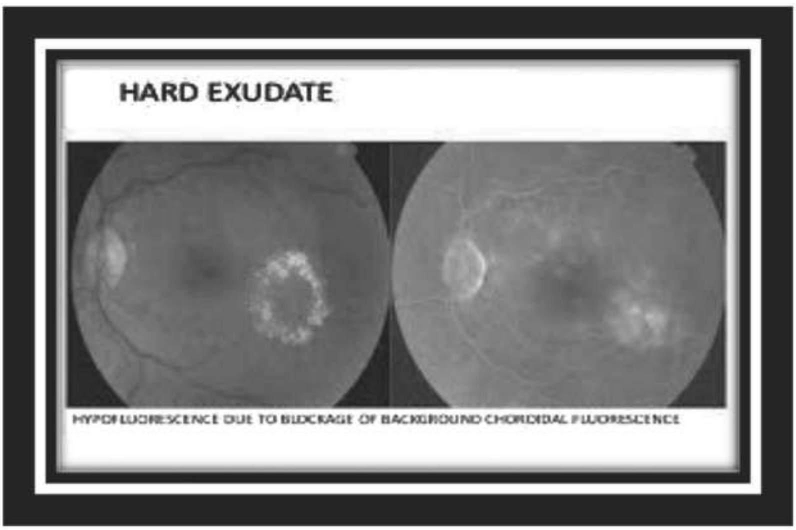

FIGURE 4.4 Exudates lesion.

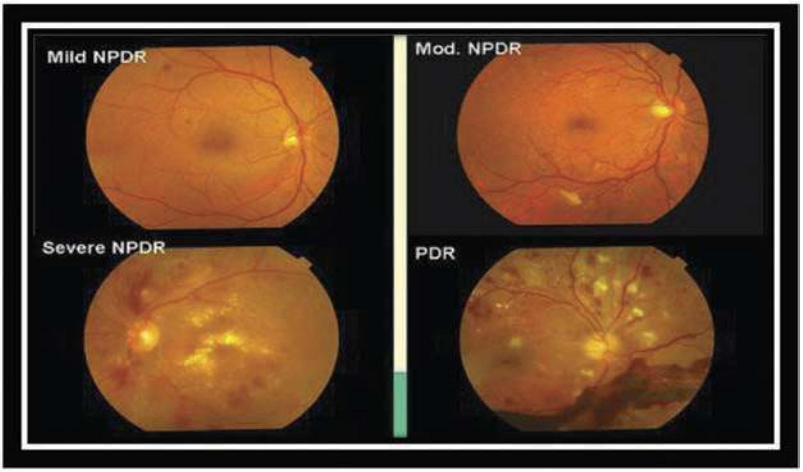

FIGURE 4.5 Stages of diabetic retinopathy.

In any case, as the illness advances, it causes mutilated and clouded vision, requiring an early end to stop the find defilement for visual hindrance expectation. DR is characterized in Ref. [2] by the vicinity of unmistakable wounds alluded to as dim/red wounds, like micro aneurysms (MAs) and hemorrhages (Trims), and sparkling wounds, like Exudates (EXs). MAs, the most well-known side effect of DR, show up as bronzed, more modest, and round dots. A micro aneurysm is the small red spots or swelling that occurs on the inside of a blood vessel. MAs in the retina of the eye are occasionally found in people with diabetes. These smaller aneurysms can rupture and spill blood. As per Refs. [3,4], some exploration proposes that these miniature aneurysms can foresee the movement of diabetic retinopathy, a condition wherein diabetes harms the veins of the retina, which can prompt visual deficiency.[5] Now every day, picture getting ready is among rapidly creating headways. It shapes a center exploration zone inside planning and computer programming disciplines too. Ref. [6] uses a slope weighting approach, as well as another set of highlights based on local convergence filters (LCF) and an irregular underdamping boosting classifier, to identify MAs in retinal images. In Ref. [7], employments of the fluffy C imply calculation is utilized to distinguish the real DR. In Ref. [8], super pixel multi-feature strategy classification is utilized for the programmed location of exudates is created, proposes an optic circle discovery method is outlined to advance progress the execution of classification precision. According to Ref. [9], the SVM could be a unique and noninvasive area system for the early and precise treatment of DR. To energize help setting up, the optic circle and veins are covered first in Ref. [10]. Bend let-based edge upgrade is utilized to isolate the faint wounds from the ineffectually enlightened retinal establishment, while an ideally planned wideband band pass channel improves the differentiation between the shinning wounds and the establishment. In Ref. [11] in any case, early screening through computer-assisted conclusion (CAD) instruments and appropriate treatment have the capacity to control the predominance of DR. A later audit of state-of-the-art CAD frameworks for conclusion of DR is displayed. In Ref. [12], a proposed technique for modified revelation of both miniature aneurysms and hemorrhages in shading fundus pictures is depicted and supported. The procedure defeats various best in class approaches at both per-injury and per-picture levels. DSFs have shown to be solid features, significantly capable of isolating among wounds and vessel partitions. In Ref. [13], proposes a significant MIL procedure for DR revelation by taking the integral central focuses from MIL and significant learning so to speak the picture level clarification is needed to acknowledge both area of DR pictures and DR wounds, in the meantime, features and classifiers are together gained from data. In Ref. [14], the authors present a novel and comprehensive method for halting the progression of DR, which includes two distinct approaches based on the turnover of MAs and hypochondriac risk segments. One methodology, in particular, is modeled after the standard image investigation-based manual for determining MA turnover. The other looks at seven psychotic characteristics linked to MA turnover in order to classify the unmodified, unused, and settled MAs using quantitative assessment and plan characterization procedures, as well as hypochondriac peril components to break down the growth of DR. One methodology, in particular, it is modeled after the standard image investigation-based manual for calculating MAs turnover. The other looks into seven psychiatric characteristics linked to MA turnover in order to categories unaltered, underused, and settled MAs using quantifiable assessment and plan order methodologies. In Refs. [15–17], a directed multivariable classification calculation is additionally presented to recognize the genuine exudates from the spurious candidates. This proposes an programmed seriousness level appraisal of the DR utilizing imaginative image preparing procedures combined with a multilayered manufactured neural arrange demonstrate for classification of retina pictures.

4.2 REVIEW FOR SELECTIVE LITERATURES

Several steps have been taken to detect DR in color fundus images. Figure 4.6 depicts the various steps for detecting DR.

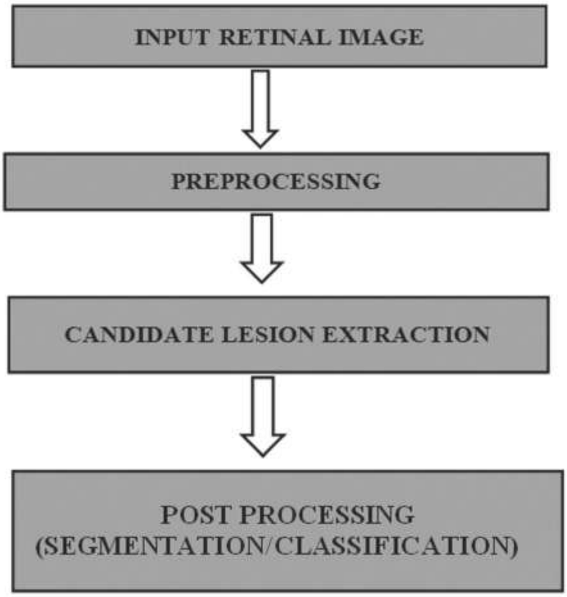

FIGURE 4.6 Various steps for detecting diabetic retinopathy.

4.2.1 PREPROCESSING

Color fundus images are used as a source of information for revealing periods of DR. These photos are simply hiding images that depict sensations related to the retina of the eye. These photographs are preprocessed in order to improve the image's concept, and the preprocessed image is then used in the subsequent steps.

4.2.1.1 MEDIAN FILTER

The separating handling utilizing the middle channel, the undesirable commotion present in the picture has been taken out for the further cycle.[35]

The filtration is the way toward supplanting every pixel power esteem with a new worth assumed control over a neighborhood of fixed size. Middle channels are compelling in eliminating salt and pepper clamor and/or motivation commotion.

4.2.1.2 RANDOM CLIFF OPERATOR

Luca Giancardo et al.[20] present the discovery of small aneurysms using the radon cliff chairman. The study of discrete fixed wavelets that acted on the green plane of the image is included in this work's pre-treatment. A stamped picture is obtained by revamping the image by overriding first scale plane with zeros. This eliminates the need for a picture to be established.[36] After that, the negative pixels are removed and the image is normalized. The optic nerve and amazing bruising are wiped out, and the small aneurysms and vessels are updated. The radon cliff director is used on the grayscale image, and pixels with values greater than 215 are considered up-and-comers.

4.2.1.3 CLAHE

CLAHE (differentiation restricted versatile histogram balancing)[28] is a well-known strategy in biological image preparation because it is particularly effective at making the overwhelmingly compelling surprise areas more evident. The image is divided into discrete locales, with a local histogram evening out applied to each. Then, using a bilinear interposition, the cut-off points between the sections are removed.

4.2.1.4 MORPHOLOGICAL TRANSFORMATION

Preparation is completed in order to separate proof of dark injuries that have a shockingly low content in the green concealing plane.[18,19,37] In addition, the image in the green plane has a lower establishment power assortment. As a result, an image's green channel is obtained from the start, and an area is perceived as focused on each pixel p in the image. The pixel is considered hazier than the including pixels if the faint level of the pixel is less than the modest quantity of the mean. The hazier districts than the standard retina are perceived. Finally, a two-dimensional image with non-zero features for selected faint pixels is obtained.

4.2.2 MICROANEURYSMS DETECTION

It's portrayed by expand like growing inside the retina's veins. These are called MAs, and these vessels can spill into the eye.

4.2.2.1 CIRCULAR HOUGH-TRANSFORMATION

We devised a method based on the identification of small round dots in the image.[20] Candidates are identified by applying indirect Hough change to detect circles on pictures. Many indirect elements can be deleted from the image using this method.

4.2.2.2 LOCAL MAXIMUM REGION EXTRACTION

A grayscale (power) pictures closest by most limit region (LMR) is a related piece of pixels with a constant force respect, such that each connecting pixel of the area has a strictly reduced power.[29] As a result, only the LMRs of the preprocessed image should be considered as probable MA up-and-comer regions. The image's pixels are constantly monitored and diverged from their eight neighbors. The genuine pixel is an LMR if all of its neighbors have a lower power. If there is a higher-powered neighboring pixel, the present pixel may not be as outlandish. If all linked pixels have lesser or equivalent power, a pixel is viewed as a prospective generally outrageous, in which case pixels with a comparative force are taken care of in a line, and attempted as well. The relevant connected component is an LMR if the line is finally depleted to the point where all of the pixels it contained are feasible maxima.

4.2.2.3 AM–FM METHOD

The IF's degree potential increases are negligible at smaller repeat scales, and the isolated AM–FM features depict a constantly shifting picture surface. For veins, for example, the most suited scale is one that obtains frequencies with a period proportional to their width. Higher repeat scales, on the other hand, capture tiny subtleties within certain bruises, such as the small vessels in neovascular formations. We employ a multi-scale redirect breaking down spread out in Ref. [27] to examine the image at multiple scales. Higher repeat scales, on the other hand, capture tiny subtleties within certain bruises, such as the small vessels in neovascular formations. We employ a multi-scale redirect breaking down spread out in Ref. [27] to examine the image at multiple scales. The MAs' and exudates' distances across are on the solicitation for 8 pixels, which address a size of 0.04 mm in the images that were reduced down for this examination. Hemorrhages and cotton downy patches with a width of 25 pixels and a diameter of 0.12 mm are on the request list. To obtain these characteristics of varying sizes, various scales are used. The bearing of the component being encoded is also considered by the various channels (within some irregular scale).

4.2.2.4 DOUBLE RING FILTER

A twofold ring channel was used to detect contender areas for MAs after image pre-treatment.[25] For detecting small aneurysms in the image's containing retinal spaces, estimates of 5 and 13 pixels for the internal and outer rings are utilized separately. Through tailored vein extraction from the pictures, any potential fake positives arranged in the zones associated with veins were removed. Using head portion assessment, a total of 126 picture features were settled on, and 28 portions were chosen. Using standard-based techniques and phoney neural association, the contender wounds were grouped into MAs or sham positives.

4.2.2.5 WAVELET TRANSFORM

Wavelet change is extremely adaptable, and we have the complete control over the limit premise utilized to disintegrate images. As a result, we offer a wavelet premise voyage that may make aided isolating in the wavelet space easier and more useful. In any case, arranging constructed using wavelet bases is cumbersome because several objectives must be met simultaneously (such as biorthogonality, commonness, vanishing minutes, and so on).[21,37]

4.2.3 EXUDATES DETECTION

EXs, on the other hand, are yellowish intra-retinal fluid reservoirs containing protein, lipid, cell detritus, and other materials. They usually appear as yellowish, dazzling patches with sharp lines in a variety of forms and sizes.

4.2.3.1 FCM CLUSTERING

In cushy batching, it is a strategy where huge data are collected into bunches in which each point has a degree of having a spot with gatherings, as indicated by its enlistment instead of having a spot absolutely with just one gathering.[24,26,35] The data centers which are nearer to the center have genuine degree of enlistment than the spotlights on the edge of a bundle have a lesser degree.

4.2.3.2 K-MEANS CLUSTERING

Utilizing the K-implies calculation, it enjoys a benefit of less figuring time. At the end of the day, the partitional grouping is quicker than the progressive bunching. Nonetheless, the diverse introductory centroids will achieve the various outcomes which imply the K-means calculation has an underlying issue. To take care of the underlying issue, one starting point is chosen or utilized the particle swarm optimization (PSO). The k-implies calculation is as follows:

- Decide the quantities of the group N and pick haphazardly N information focuses (N pixels or picture) in the entire picture as the N centroids in N bunches.
- Find out closest centroid of each and every information point (pixel or picture) and arrange the information point into that group the centroid found. In the wake of doing stage 2, all information focuses are grouped in some bunch.
- Calculate the centroid of each bunch. Rehash stage 2 and stage 3 until it isn't changed.

Utilizing the K-implies calculation, it enjoys a benefit of less figuring time. At the end of the day, the partitional grouping is quicker than the progressive clustering.[35] However, the distinctive beginning centroids will achieve the various outcomes which imply the K-means calculation has an underlying issue. To take care of the underlying issue, one introductory point is chosen or utilized the particle swarm optimization (PSO). The K-implies bunching calculation is utilized for the division of the influenced input picture. In this interaction, three levels of division were utilized to find the DR disease and the stage it caused in the given input image. The reformed fragmented image with three degrees of split is obtained.

4.2.3.3 BoVW MODEL

A BoVW model is used to assess individual DR damage.[38–40] These models work by extracting a massive number of feature vectors in photographs [typically around central foci (POIs)] and assigning these vectors to "visual words" based on a visual appearances word reference. Hard exudates, red wounds, small hemorrhages, major hemorrhages, cotton-downy patches, and drusen are all considered injury finders.

4.2.4 CLASSIFIERS USED

4.2.4.1 GAUSSIAN NAÏVE BAYES (GNB) CLASSIFIER

The request task is carried out by artless Bayes classifiers, which use the "Naive's" assumption of self-sufficiency between each pair of features.[32] In the Gaussian Naive Bayes depiction challenge, the likelihood of features is assumed to be Gaussian.

$$P\left(\frac{Xi}{y}\right)=\frac{1}{\sqrt{2\pi\sigma_y^2}}exp-\left(\frac{\left(x_i-\mu_y\right)^2}{\sigma_y^2}\right)$$

Here, x_i is a highlight vector, and y is the element vector's dependent class variable.

_y² = variance of probability work _y = mean

GNB classifiers need a small amount of data preparation, are boundary-free, and are ready quickly.

4.2.4.2 K-NEAREST NEIGHBORS (KNN) CLASSIFIER

The oversaw KNN classifier[34,37] gathers data by moving a bigger chunk of the KNNs' characteristic to the related test. KNN from the pixel of endorsement is processed using Euclidean distance. This estimate essentially saves the planned information. The value of the chosen K worth has a significant role in depiction evaluation. The higher the value of K, the fewer the upheavals, and the lower the value of K, the more clear the class limitations.

4.2.4.3 GENERALIZED LINEAR MODEL (GLM)

The chance of tissue-restricted rot is tended in GLM by a defined limit utilized for distinctive applications.

$$F(t) = e^\wedge t/(e^\wedge t + 1)$$

The t-linear limit of data features x is t = 0 + 1 x 1+..+ n x n, where t = 0 + 1 x 1+..+ n x n.

The benefits of calculation include simplicity, quickness in the design and testing stages, and the veracity of assessing the various effects of the information restriction on the probability of breaking through the limit.[42,43] However, for non-direct difficulties of basic type, this core backslide-based classifier is largely unsuitable.

4.2.4.4 SUPPORT VECTOR MACHINE

The data centers are designed to have a high-dimensional component space with a disengaging hyper-plane. The estimate is chosen in order to improve the partition based on the closest plans, a sum known as the edge. SVMs are learning systems that are used to generally trade precision for unpredictability by limiting the theory goof.[30,31] SVMs have demonstrated in a range of request situations that they can reduce planning and testing blunders, resulting in greater affirmation precision. SVMs may be used on high-dimensional data without having to change their parameters.

Hyper plane articulation

$$w * x + b = 0$$

Where,

x = Set of preparing vectors

w = Vectors opposite to the isolating hyper plane

b = balance boundary which permits the increment of the edge

SVM transforms input vectors into a higher dimensional vector space, from which an ideal hyper plane is constructed. There is only one hyper plane of the several available that increases the distance between itself and the closest information vectors of each class.

4.2.4.5 CONVOLUTIONAL NEURAL NETWORK

The best exactness was acquired with an ANN classifier comprising of two secret layers with 20 and 15 neurons in them individually.[41,44,45] The explicitness and affectability accomplished were 98 and 98.76%, respectively, separately by this classifier. To distinguish dull sores from an unnoticed fundus picture, first districts of interest were separated from the picture following strategies referenced in Section 3 and afterward the extricated locales were grouped to be sores or amiable areas by the prepared ANN classifier.

4.2.4.6 NEURAL NETWORK

At this point, the CNN is applied to a "concealed" fundus image to create a two-dimensional image that distinguishes veins from retinal foundation: pixels' numerical representations are processed separately by the NN. In our case, the NN input units get a standardized version of the highlights provided by Ref. [23,33]. Because a strategic sigmoidal actuation task was chosen for the yield layer's solitary neuron, the NN choice determines a grouping esteem between 0 and 1.[22] As a result, a vessel likelihood map is constructed, which shows the likelihood of a pixel being essential for a vessel. The brilliant pixels in this image have a higher chance of becoming vessel pixels. A thresholding plan on the likelihood map is used to determine whether or not a given pixel is required for a vessel in order to obtain a vessel paired division.

TABLE 4.1 Comparison of Different Type of Lesion Performance Metrics.

Ref no	Feature extracted	Methodologies systems utilized for preprocessing	Methodologies utilized for highlight extraction	Classifier utilized	Performance metrics
1	**Micro aneurysms, Exudates**	Thresholding	Double ring filter	Not used	Sensitivity-68%
2	**Exudates**	Alternating sequential filters (ASF)	SMFC	Not used	AUC of 0.9655
6	**Microaneurysms**	Kernel linear filter	Pathological risk factors	GDBICP	Sensitivity of 94% and a specificity of 93%
14	**Lesions**	Thresholding	Median filtering	Artificial neural network	Sensitivity-0.987, Specificity-0.725
23	**Microaneurysms**	Radon Cliff operator	Radon Cliff operator	Support Vector Machine	Sensitivity-41%
34	**Exudates**	Morphological reconstruction	Fuzzy C-means clustering	Not used	Sensitivity-87.28%, Accuracy-99.11%, Specificity-99.24%, PPV- 42.77%, PLR- 224.26
38	**Exudates**	Region growing	Neural network system	Support vector machine	Accuracy-98.77%

4.3 LIMITATIONS OF THE CURRENT APPROACHES

The current study focuses on disease identification based on retinal grading using machine learning techniques like ANN (single and multi-stage) and SVM. To improve low contrast and dark images, pre-processing techniques such as homomorphic filtering and adaptive histogram equalization are used. The retinal images are segmented using grey level and moment invariant based features, 2D Gabor matched channel segmentation, and Fourier series power spectrum segmentation techniques. To improve the classifier's performance, the DWT-based PCA feature extraction approach is used. To improve classification performance, statistical parameters such as mean, mean square, standard deviation, variance, skew, and kurtosis are extracted from retinal images, as are grey level co-occurrence matrix parameters such as entropy, contrast, homogeneity, energy, correlation, and texture features extracted using local tetra pattern, and spectral features extracted using wavelet.

4.4 CONCLUSION

According to a review of the literature, all existing algorithms aim to improve the identification of abnormalities in retinal images through biomedical image processing using various techniques. As per existing research, the disadvantages of segmentation algorithms are that they are computationally complex and that the results do not produce a standard segmentation due to improper threshold choice, large intensity variation of the foreground and background, manual segmentation of the retinal blood vessels is difficult and time-consuming, and detail segmentation can be challenging if the complexity of the vascular network is too high. Although there are several methods for segmenting retinal images, there is still room for improvement because they do not work well for all types of retinal images. Because of the lack of clearly defined edges between adjacent retinal tissues, user selection of noisy seed leads to flawed segmentation that is very sensitive to noise, intensity variations, and over segmentation, identifying the initial curves can only provide superior results. These are significant drawbacks of existing algorithms, as outlined in the proposed research work. As a result, automatic segmentation techniques that are robust, simple, and suitable for real-time applications are now required.

In this study, five different segmentation algorithms are developed and implemented to detect DR using various techniques. The algorithms' complexities and segmentation performances differ. Different evaluation metrics are used to assess the performance of the developed algorithms.

4.5 FUTURE SCOPE

To implement the IoT-based framework for tracking the disease and storing the patient information in cloud which will be very useful to telemedicine practices. It is very useful for DR patients.

KEYWORDS

- **diabetic retinopathy**
- **fundus image**
- **hemorrhages**
- **exudates**
- **micro-aneurysms**
- **accuracy**

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CHAPTER 5

CRITICAL ANALYSIS OF VARIOUS SUPERVISED MACHINE LEARNING ALGORITHMS FOR DETECTING DIABETIC RETINOPATHY IN IMAGES

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ABSTRACT

Nowadays, it is very difficult to protect ourselves from several diseases. As we are growing technologically advanced, the population growth is also exponential. With the advancement in technology, it has become quite easy to identify any kind of disease, but not everyone can afford the expenses of the treatment or even for checkups. Thus, with the help of this technological advancement, we might find some easy and cost-effective solution, which not only helps in the identification of disease but also helps in providing cost-effective treatment by saving an enormous amount of time in the identification of infection. The objective of this research is to provide cost-effective and more accurate results in less time by less human intervention. The main challenge here is the pre-processing of the images, as the images are in a different format and may have different resolutions. Pre-processing may include the removal of noise and image enhancement, which helps to raise the accuracy of the outcome. Therefore, the motivation is to develop an approach that helps doctors and other practitioners

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to identify the disease well in time with more accuracy. This chapter talks about determining the disease from a standard image dataset using supervised machine learning algorithms and compares the accuracy obtained by different algorithms. After analyzing the accuracies, it is found that Random Forest Classifier performs better than others.

5.1 INTRODUCTION

This chapter talks about detecting people that might be suffering from diabetic retinopathy. Before jumping into the technical aspects of the work, let's understand what is diabetic retinopathy, its causes, how it can be treated, etc. The reason for choosing diabetic retinopathy is that the number of cases that are being recorded worldwide are increasing alarmingly. Many studies suggest that it is very uncommon that a healthy person may be suffering from this disease. Diabetic retinopathy, generally called diabetic eye sickness, is an infirmity wherein hurt hops out at the retina given diabetes mellitus. It is a fundamental wellspring of blindness.[1] Diabetic retinopathy influences up to 80% of the people, who have diabetes for quite a while or more. If timely treatment is done, then 90% of new cases could be reduced.

Persons having a long history of diabetes are more likely to develop diabetic retinopathy.[2] Each year in the United States diabetic retinopathy states 12% of each original example of graphical impedance. Glaucoma, the standard movement of fluid in the eye may get discouraged as crisp enlist the vessels structure. The blockage causes an increase in visual weight, or weight in the eye, thereby increasing the risk of optic nerve mischief and vision disaster. Figures 5.1a and 5.1b illustrate the normal retina and diabetic retinopathy-affected image.

Now here, we briefly examine the causes of diabetic retinopathy.

1. **Patent history:** It is one of the most prominent reasons that a person might be suffering from diabetic retinopathy. There might be a possibility that a person is having diabetes but is not prone to diabetic retinopathy. Here, the duration of diabetes in a patient matters. A person who is suffering from diabetes for a long time has more chances to encounter diabetic retinopathy. It means if a person is diabetic, he is more likely to have retinopathy at some point.[3]

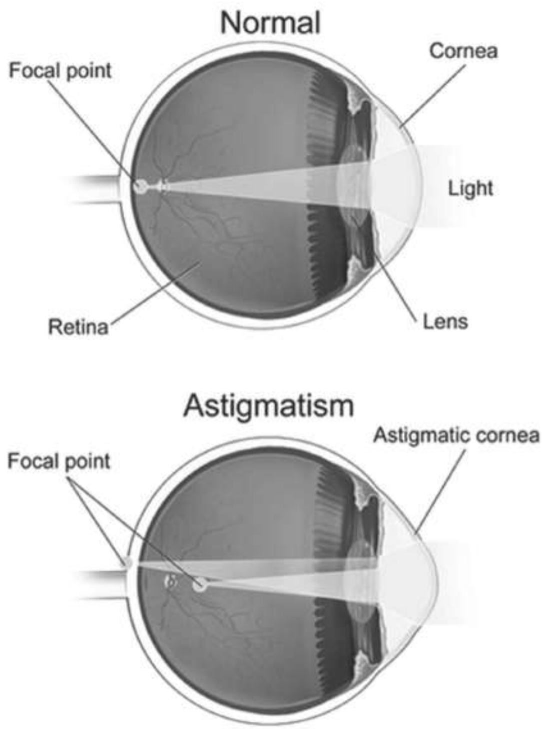

FIGURE 5.1 Normal retina image.

Photo credit: Interactive by BruceBlaus. https://storymd.com/journal/6j45zzkinw-astigmatism/page/xlrzo7hlzzg-how-does-astigmatism-occur

High blood pressure: It is the second most prominent reason. It has been imagined that the belongings of an enlarged bloodstream can harm retinal capillary endothelial cells in the eyes of diabetic patients. So, if a person is suffering from diabetes and high blood pressure, then there are high chances that he might suffer from diabetic retinopathy. Numerous clarifications from medical readings displayed a connotation among hypertension and the occurrence of severe retinopathy in people suffering from diabetes, in support of this hypothesis. Type 2 diabetes is non-insulin-dependent diabetes; retinal photos that were reserved at the diagnosis for six years shows that systolic blood pressure was meaningfully related with retinopathy occurrence In the Wisconsin Epidemiologic Study of Diabetic Retinopathy (WESDR), diastolic blood pressure was a significant predictor of progression of diabetic retinopathy to proliferative diabetic retinopathy over 14 years of follow up in patients with younger onset (type 1) diabetes mellitus, independent of glycosylated.Type 1 diabetes is insulin-independent of glycosylated hemoglobin occurrence of unrefined proteinuria and dependent diabetes mellitus.

2. **Raised cholesterol levels:** Lipid levels and high levels of serum cholesterol are very well-known factors of metabolic condition. Cardiovascular disease remains the leading cause of death in patients with diabetes. A big meta-analysis has shown that diabetics with macular edema are found to have higher total cholesterol levels. High cholesterol levels are also linked with advanced rates of stiff retinal exudates.
3. **Inflammatory markers:** It has been linked to high perils of cancer and cardiovascular diseases and are also linked through diabetic retinopathy. Quite high levels of C-reactive protein (CRP) are found in a person suffering from mild or severe diabetic retinopathy when compared to a person without diabetic retinopathy.[4]
4. **Sleep-disordered breathing:** It is also called obstructive sleep apnea (OSA). It is considered by recurrent upper airway obstruction which leads to blood oxygen desaturation and sleep distraction. OSA is closely related to obesity and is also considered a gradually recognized source of morbidity. A reading suggested that about 86% of the patient who is obese and suffering from diabetes meet the requirements for the diagnosis of OSA on overnight oximetry monitoring.[5]
5. **Pregnancy:** It is considered to be the main risk factor for the development of retinopathy and is prevalent and severe when compared to non-pregnant women suffering from diabetes. The microvascular changes in a pregnant woman suffering from diabetes are so poor that the physicians advise to terminate or avoid pregnancy. Many readings have revealed that the younger the age of the start of diabetes and longer the interval of diabetes, the higher is the peril of evolution of the illness. The longer the time of diabetes, the higher the chances of expansion of microvascular complications, so these patients are prone to severe form of starting-point retinopathy changes before pregnancy. The development is more substantial in women suffering from

severe-to-moderate forms of retinopathy associated with those with minor or no signs of retinopathy at the time of formation.[6]

In this survey paper, the lesion localization model has been designed through a deep network patch-based methodology. The authors focused to decrease the complexity of the model and also improve the performance of the model. They intended to use an efficient technique for selecting the training reinforcements. Authors have used the Standard Diabetic Retinopathy Database Calibration Level 1 (DIARETDB1) database to train the model and also verified it on several databases together with Messidor. Results range an expanse below the receiver operating representative arc of 0.911–95% CI (0.896–0.927) for DR broadcast, and a sensitivity of 0.941–95% (0.920–0.958).[7]

A directional multi-scale line detector method was used for the dissection of retinal vessels through the main emphasis on the small containers. Line detector has been used on images which progresses the correctness of the algorithm. Binarization operation is involved in the finishing step, which also improves the performance. In the proposed method, sensitivity is obtained at 0.8042, 0.8010, and 0.7973 for different datasets.[8]

Supervised technique was used for digital retinal images by using blood vessel detection. A neural network scheme has been used for pixel arrangement, which calculated a 7-D vector for pixel representation. DRIVE and STARE database has been used to evaluate the method. The performance of the suggested method is better than traditional approaches.[9]

The new method used was the Stationary Wavelet Transform with a multiscale Fully Convolutional Neural Network. The method was performed on three different databases and achieved correctness of 0.9575, 0.9693, and 0.9652 below the ROC arc of 0.9820, 0.9904, and 0.9854 on the CHASE DB1, DRIVE, and STARE databases, respectively.[10]

Cataract screening has been done efficiently using the proposed procedure. Authors emphasized on exploration constructed on consistency structures like uniformity, standard deviation, and intensity. Correspondingly, retinoblastoma cancer can be identified through the automatized recognition method for undeveloped cells in the retina. The authors' proposed idea was to compress an image processing algorithm that helped for recognition of white radius of the retina, through the help of canny edge detection, image filtering, and thresholding.[11]

A system for systematizing the manual effort done by an operator to censor the human error and aggregate the accuracy of the malaria diagnosis was developed. The proposed method uses a dataset of 80 images of a thin blood smear. The diseased cells are removed using HSV segmentation. The proposed method is useful for the rural regions, where there is a shortage of experts.[12]

The use of an image is not limited to medical and health care, rather it is used in many applications like monitoring, tracking, vision, surveillance, etc. Biochemical parameters have been analyzed in sugar cane crop by using the MODIS data.[13] An agricultural drought has been estimated by the fusion of satellite images.[14] Wetness in the strawberry plant has been identified using the color and thermal images,[15] to extract the land parameters with the help of machine learning approach.[16–18] The use of a drone has been increased; with the advancement in technology, researchers are using drone data in disaster management, rescue, etc. The rail track health is inspected by using the concept of vision;[19] fish plate detection has been identified using the feature-based template matching in rail tracks;[20] a drone is used in surveillance during disaster (drone-surveillance for search and rescue in a natural disaster), etc. The drone is also being used in fusion with the satellite data by researchers for agriculture monitoring, defence, and similar other applications.[21–24]

5.2 DATASET AND TESTING TOOL

5.2.1 DATASET

The dataset used is freely accessible. The Messidor database contains 1200 lossless images through 450 FOV and dissimilar determinations. A grading tally fluctuating from R0 to R3 is delivered from an individual image.[25] The dataset consists of features that are removed from the Messidor image set which is used to forecast whether an image encloses any sign of diabetic retinopathy or not. Entire features either signify a distinguished lesion, adescriptive feature of a functional part, or an image-level descriptor.

5.2.2 TESTING TOOL

We have tested our code using the Jupyter notebook. Adventure Jupyter is an altruistic affiliation completed to make open-source programming, open-measures, and organizations for instinctive enlisting through over many programming tongues. Spun-off from IPython in 2014 by Fernando Pérez, Project Jupyter reinforces implementation conditions in two or three dozen vernaculars. Adventure Jupyter's name is a position to the three focus programming vernaculars supported by Jupyter, which are Python, Julia, and R, and worship to Galileo's diaries cassette the revelation of the moons of Jupiter. Activity Jupyter has made and supported the instinctive enlisting things of Jupyter Notebook, Jupyter Hub, and Jupyter Lab, the bleeding edge version of Jupyter Notebook.

5.3 TECHNIQUES USED

In this section, we have discussed the techniques used in our chapter.

5.3.1 DATA CLEANING

Data cleaning infers the path toward perceiving a wrong, lacking, mistaken, immaterial, or missing bit of the data and a short time later changing, displacing, or eradicating it. Data cleaning is seen as an inborn segment of data science. Data is the most significant thing about analytics and Artificial Intelligence (AI). For authentic data, it is phenomenal that data can have lacking, clashing, or missing characteristics. In case the data is degenerative, it could upset the system or give wrong results. We have seen a couple of cases of the importance of data cleaning. Machine learning is a data-driven AI. In AI, if the data is irrelevant then it prompts the improvement of a wrong model. As much as you make your data clean, you can improve a model. Without the data quality, it is dumb to foresee a better than average result.

5.3.2 DATA VISUALIZATION

Data visualization is a graphical depiction of information and data. Using visual segments, for instance, graphs, diagrams, and maps, data portrayal instruments give a sensible strategy to see and get examples, features, and models in data. In the domain of enormous data, data discernment instruments, and headways are fundamental for dismembering wide data and choosing data-driven decisions.[26] We can rapidly portray red from blue and square from the circle. Our lifestyle is self- evident, together with everything from craftsmanship to advancing, to TV, and motion pictures. Data portrayal is an additional kind of graphic craftsmanship that snatches our eye and watches out for the message. If we are looking at a huge spreadsheet of data and cannot see a pattern, we understand how convincing the portrayal is. Figure 5.2 shows data visualization.

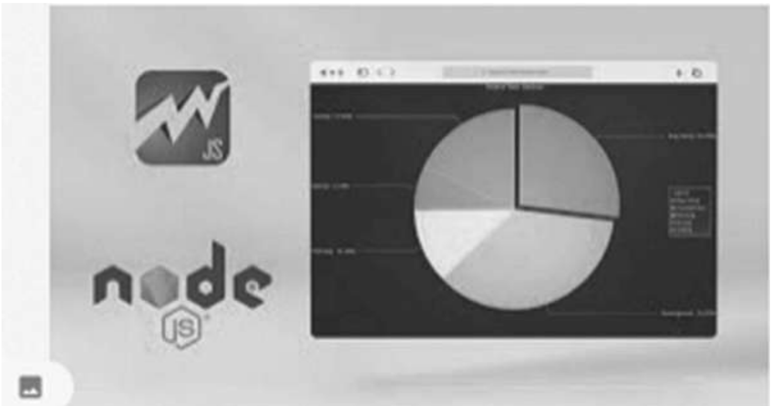

FIGURE 5.2 Data visualization.

5.3.3 MACHINE LEARNING

To decide AI in extremely straightforward terms, science must permit people to learn and work with machines. To choose AI is especially essential, science must allow machines to learn and work similarly to individuals. Computer-based intelligence is unquestionably not just another development. The estimations that run current applications for structure affirmation and AI have existed for quite a while. Regardless, AI models have begun to speak with continuously complex instructive records and are picking up from past checks and measures to convey reliable decisions and results. Assembling the right model will undoubtedly keep up a key good way from new risks and recognize beneficial open entryways over your business.[26] The calculations that run current applications for design acknowledgement and AI have existed for a long time. Assemble the correct model and you are bound to maintain a strategic distance from new dangers and recognize productive open doors over your business. Figure 5.3 shows the machine learning visualization.

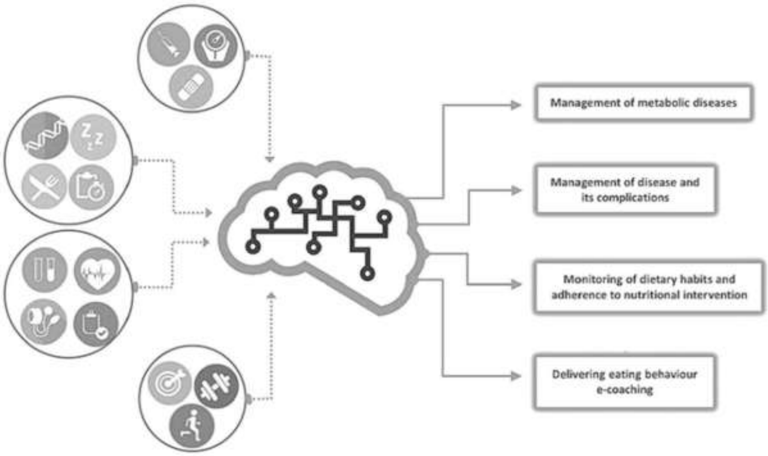

FIGURE 5.3 Machine learning visualization.

Source: Copyright © 2018 Francesco Pettini. https://creativecommons.org/licenses/by/4.0/

5.4 ALGORITHM USED

In this fragment, we are listing all the supervised machine learning algorithms which are reused to identify diabetic retinopathy in images.

5.4.1 KNN (K-NEAREST NEIGHBOR) ALGORITHM

KNN algorithm is used to solve the task in both classifications as well as regression. It is a supervised machine learning algorithm. If we see KNN as a classifier, then it works in majority voting principle, that is, if K = 5, the classes of five nearby pairs are tested. Similarly, KNN regression chooses the mean value of five nearby pairs.

Figure 5.4 shows that there are two clusters of data—class A (shown as red stars) and class B (shown as a green triangle). Here we have checked for the distances for the new value of K for both clusters of class A and class B. But here arises a question, do we need to calculate the distances of all the points of class A and class B? The answer is no because we have to specify the value of K. The K value will specify the number of variables or points in any cluster that is to be considered for the prediction. The point having the least distance will be treated as the class of the predicted class of the new variable. In Figure 5.4, we have the value of K as 3 so it is taking the three closest values of the cluster.[27]

KNN Pseudocode

1. Load the data.
2. Initialize a value of K (which is used for all predictions).
3. To get the predicted class, start the iteration from 1 and cover up to the total number of the training data sets.
4. Compute the distance among the test data concerning each row present in the training data. To calculate the distance, you can use any formula, but here we are using Euclidian's distance formula.
5. Based on the distance value, sort the data into either ascending or descending form.
6. From the sorted array, get the top K values
7. Get the class that is most frequently occurring in these rows.
8. Return the class that is predicted.

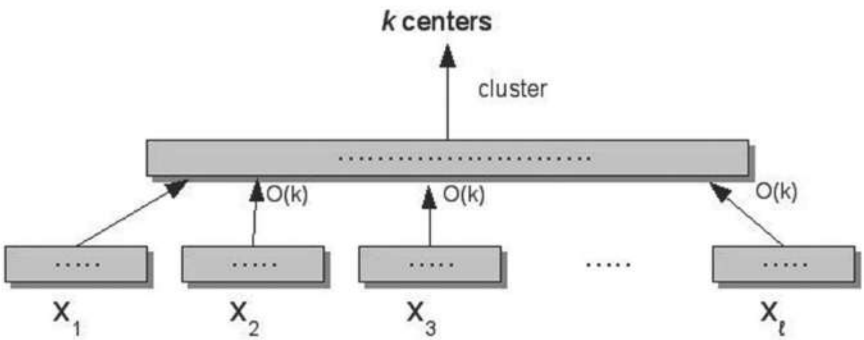

FIGURE 5.4 A cluster of data classes.

5.4.2 LOGISTIC REGRESSION

It is a supervised machine learning algorithm that is used to predict the possibility of an objective variable. The environment of the dependent or objective variable is divided into two parts, which means there will be merely two probable classes. In simple arguments, the objective variable is having information coded as moreover 1 or 0 or say binary. Statistically, a logistic regression model forecasts P(Y = 1) as a function of X. It is one of the simplest ML algorithms that can be used for numerous classification problems such as diabetes prediction, cancer detection spam detection, etc.[28]

5.4.3 SUPPORT VECTOR MACHINE (SVM)

The SVM is a learning control approach that can be used for classification, regression, and outlier detection. In the SVM count, we draw each datum point as a point in the *n*-dimensional space and the estimation of each part is the estimation of the particular orchestrate. The supporting features are only one-dimensional spectators. The SVM picture is the breaking point that best segregates two classes (hyper face/line).[29]

5.4.4 RANDOM FOREST CLASSIFIER

Decision trees are keen models that utilize a joined strategy of rules to ascertain the objective. Each tree is considered a key model with branches, packs, and leaves. Optional forests are a blend of tree wants, so each tree depends upon erratic model vectors with a relative dissipating in the forested zones. Another definition of "Erratic timberland" is a classifier, which contains a blend of formed masterminded trees self-assuredly passed on optional vectors. The most outstanding class while making the Random Forest chooses and hardenings two or three choice trees for continuously cautious and stability. Capricious forests make stirred up wants, if the vast majority of the key classifiers are confused. The general centrality of each property credit is incredibly simple to assess. In this way, it is more accurate than most different calculations.[30]

5.4.5 NAIVE BAYES

The Bayesian request relies upon the theory of Nice, with free hypotheses on the prophets. The Bayesian model is not hard to set up. It has a massive database with no critical straight mix and is incredibly useful. But essential, in the Naive Bayes classifier, much of the time bombs are additionally created and comprehensively used associations since it misses the mark with progressively refined request procedures. This supposition P (c | x) altogether is called class opportunity. P (x | c) is the probability of accentuation, the probability of hypothesizing for a particular class P (x) is the guess measure. There is anything but a practical substitution for the invalid model, we endeavour to find the best pointers of Bayesian organization self-ruling estimation of the prophets.[31]

5.4.6 GRADIENT BOOSTING

Gradient boosting is an AI technique for issues of appraisal and social event that makes insightful models for the most unprotected. Gradient boosting machines are a domestic of influential machine-learning methods that have revealed considerable achievement in an extensive range of real-world applications. They are extremely customizable to the specific needs of the application, like being learned concerning dissimilar loss functions. Hypothetical information is complemented with expressive examples and illustrations which shelter all the phases of the gradient boosting model design.[32]

5.5 RESULT AND DISCUSSION

This segment describes the performance of the above-discussed machine learning techniques by comparing their accuracies that have been applied to the same data set. The accuracy tells the correctly identified percentage of people who are suffering from diabetic retinopathy. We have divided the dataset into two parts for the training and testing. The training dataset is used to train the algorithm so that it gets familiar with the data and then we used the test dataset to see our prediction. After the prediction, we have calculated the accuracies of each algorithm and found which algorithm performs better among them with the highest accuracy. Table 5.1 describes the component that is considered for diabetic retinopathy. There are certain causes of diabetic retinopathy that are extracted from the dataset.

Figure 5.5 describes the details like the number of elements, mean value, standard deviation, minimum value, and the occurrences of values that are present in the dataset. We also checked for null values in the dataset and identified if there is any null value present in the dataset. This is important, as the presence of null values causes noise in the dataset,

TABLE 5.1 Component Considered for Diabetic Retinopathy.

	Pregnancies	Glucose	Blood pressure	Skin thickness	Insulin	BMI	Diabetes pedigree function	Age	Outcome
Count	2000.00	2000.00	2000.00	2000.00	2000.00	2000.00	2000.00	2000.00	2000.00
Mean	3.70	121.18	69.15	20.94	80.25	32.19	0.47	33.09	0.34
Std	3.31	32.07	19.19	16.10	111.18	8.15	0.32	11.79	0.47
Min	0.00	0.00	0.00	0.00	0.00	0.00	0.08	21.00	0.00
25%	1.00	99.00	63.50	0.00	0.00	27.38	0.24	24.00	0.00
50%	3.00	117.00	72.00	23.00	40.00	32.30	0.38	29.00	0.00
75%	6.00	141.00	80.00	32.00	130.00	36.80	0.62	40.00	1.00
Max	17.00	199.00	122.00	110.00	744.00	80.60	2.42	81.00	1.00

which affects the accuracy of the dataset and we are not able to take out the maximum potential from it and the algorithm as well.

All these parameters have proven to be the cause of diabetic retinopathy. There might be other possible parameters that can be used to identify the presence of diabetic retinopathy but for the time being, these are considered for the conclusion of the result. We have arbitrarily separated the dataset into training and test datasets in the ratio of 8:2, which means that 80% of the data is considered for training and 20% of the data are considered for testing. We have done it in this particular ratio because it is supposed to be the optimal division ratio to achieve the best accuracy. We have removed some outliers from the dataset. Outlier is a value that is in some other region and deviates significantly from the rest of the datasets in a particular pattern, which leads to error. The removal of these outliers is essential so that the model is trained accurately for predicting future events. We have around 80+ outliers from this dataset. We have analyzed the accuracy of different algorithms and itemized them in Table 5.2.

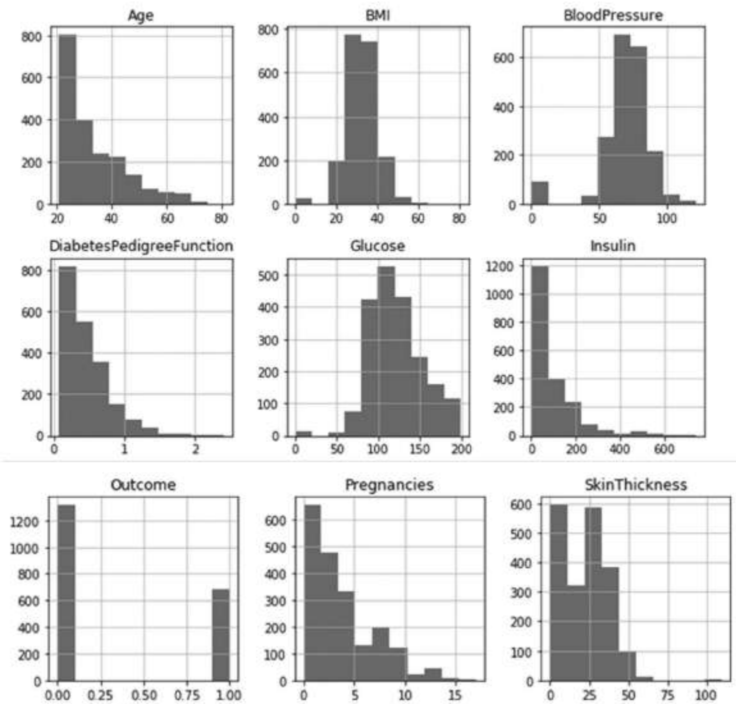

FIGURE 5.5 Graphical representation of diabetic retinopathy components.

TABLE 5.2 Accuracy Obtained by Different Algorithms.

Algorithms	Accuracy
SVM	0.7643
Logistic Regression	0.7734
KNN	0.87
Random Forest Classifier	0.9908
Naïve Bays Theorem	0.7311
Gradient Boosting Classifier	0.8942

Sklearn package has been used which has the SVM, logistic regression, KNN, Random Forest Classifier, Naïve Bays Theorem, and Gradient Boosting Classifier function which is used for the execution of all the listed algorithms on the dataset. We have fitted two training datasets and split them into *x* and *y* by using the train_test_split function from the sklearn and model_selection package. Similarly, the test dataset has been divided into *x* and *y* as well. For prediction, the test_x dataset has been used. To get the accuracy, we have the accuracy_score() function in which test_y and y_pred as parameters have passed. We analyzed the ROC_AUC (Receiver Operating Characteristics and Area Under the Curve) that was formed by using the roc_auc_score() function which takes test_y and y_pred as parameters and gave the score value. After that, cross-validation has been used by using the function cross_validation () where the train_x, train_y, scoring, and the cross_validation () value as parameters has been passed. Then finally results have been displayed which are obtained from the different algorithms. The graphical representation of the accuracy of different algorithms has been shown in Figure 5.6.

5.6 CONCLUSION AND FUTURE WORK

We have successfully achieved our aim and provided the most accurate algorithm that can be used among other algorithms to detect diabetic retinopathy from the image dataset. The dataset has been parsed most accurately by using the Random Forest Classifier algorithm with an accuracy of 99.09%. The main findings of this chapter are:

1. Pregnancy, age, blood pressure, insulin, and previous patient history are the key features for the identification of diabetic retinopathy in a person.

2. Random forest classifier gives the most perfect result in comparison to all other algorithms.
3. In the coming years, diabetic retinopathy can grow at an alarming rate as per people's lifestyles.

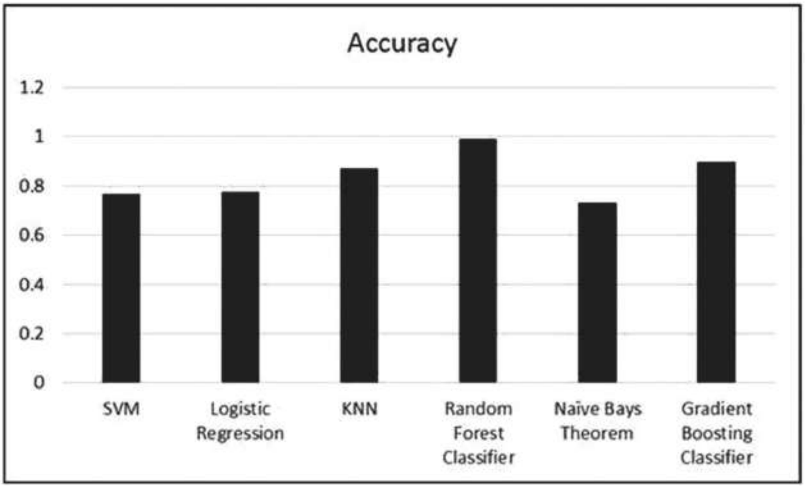

FIGURE 5.6 The accuracy obtained by different algorithms.

Recommendations work for prospects: There should be a test for retinopathy along with the diabetes test. The application of neural networks might make it easier to identify this problem. Identifying some other symptoms that might be relevant to diabetic retinopathy. Using some other dataset for detection with some other parameters can be helpful.

KEYWORDS

- **diabetic retinopathy**
- **image processing**
- **machine learning**
- **supervised approach**
- **glaucoma**

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PART II

Image and Video Processing

CHAPTER 6

ARTIFICIAL BEE COLONY OPTIMIZATION TECHNIQUE-BASED VIDEO COPYRIGHT PROTECTION IN DWT-PCA DOMAIN

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ABSTRACT

This paper addresses the issues in video copyright using local features of frames. The conventional algorithms assess various attacks and they do not envisage the redundancy of frames in the video. The proposed methodology focuses on the removal of redundant frames to reduce the processing time as well as engage the feature extraction algorithm to survive the geometric attacks. The keyframes are effectively identified by using entropy and the absolute mean difference of the video frames. The PCA technique is employed for lowering the source data's dimensionality and selecting the key features of the watermark. The speeded up robust features (SURF) feature selection technique is employed for extracting the important significant features of the keyframes. The Haar wavelet is adapted for decomposing the frames into two levels, the LH band is chosen for watermarking process. For the aim of improving the efficiency of the robustness and imperceptibility, the artificial bee colony optimization algorithm is applied to choose the optimal scaling factor. Based on

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Karm Veer Arya, Ciro Rodriguez Rodriguez, Saurabh Singh, & Abhishek Singhal (Eds.)

the results taken from the experiments, the proposed novel technique is resistant against image-oriented attacks, frame-based attacks, and also tenacities the false positive problem.

6.1 INTRODUCTION

Since the rapid expansion of the web, multimedia information such as images, videos, and audio are easily transmitted through the web. The digital multimedia data are passed on the internet without the owner's concern; hence unlimited duplicate copy is made by illegal users. For avoiding the above-said issues, the digital video watermarking technique is adapted. It is an effective technique adapted for maintaining security to the source data by entrenching either the digital image, audio, or any digital data. Various watermarking techniques have already been suggested by the researchers,[1,2] in connection with the performance of the DWT-PCA-based methodology provides more results in the metrics of imperceptibility and robustness.[3,5,9]

In recent years, a number of solutions for protecting video copyright have been created. Each strategy has advantages and disadvantages based on robustness and imperceptibility, which are represented in Table 6.1.

6.2 PROPOSED METHODOLOGY

The proposed methodology is intended to produce a one-of-a-kind video security system by combining the speeded up robust features (SURF) feature selection method with the DWT. The PCA technique is used for the dimensionality reduction of the secret image. The video is fragmented into groups of scenes based on the scene segmentation technique. The maximum energy level frames are identified in the individual scene based on the entropy level of the frame. Finally, the maximum entropy holding frame is selected as a Keyframe for the watermark embedding process. The scaling factor is the crucial factor to control the robustness and imperceptibility of the watermark as well as the video hence the artificial bee colony algorithm is applied to choose the optimal scaling factor. The PC of the scrambled watermark block is chosen to improve the security level of the watermark data. The SURF features are then retrieved, and the scrambled secret image blocks are inserted in the Keyframes' feature points.

TABLE 6.1 Literature Review.

Author	Technique	Limitations
Masoumi et al. [3]	Technique based on frequency domain-oriented temporal and spatial data hiding	Method survives the frame-based attacks and addition of noise
Karmakar et al. [4]	DCT-based video watermarking technique	Withstands rotation, collusion and frame-oriented attacks
Dutta and Gupta [5]	watermarking scheme for video coding	With stands various attacks like filtering, addition of noise, compression
Sake and Tirumala [6]	Bi-orthogonal-based wavelet transform and optimization method	With stands image processing attacks
Loganathan and Kaliyaperumal [7]	BAM neural networks and fuzzy system-based video watermarking system	Not robust against the frame dropping attack
Ejima and Miyazakia [8]	Wavelet-based watermarking system	Not robust for frame oriented, median filtering attacks
Da-Wen [9]	3D wavelet transform-based watermarking	Not with stands the filtering and specific image-oriented attacks
Thind and Jindal [10]	DWT- and SVD-based video watermarking	Not robust for video processing attacks
Raja et al. [11]	DWT-SCHUR-based video watermarking	Not robust against the frame dropping attack

The novelty of the proposed work

- The redundant frames are reduced by engaging the absolute mean difference between adjacent frames. Hence, the processing time is reduced for Keyframe selection and embedding region identification.
- The Keyframes are identified from the individual scenes based on the entropy.
- The SURF feature selection technique is employed to extract the suitable features from the Keyframes.
- The optimal scaling factor is selected by the ABC algorithm to improve the robustness and imperceptibility.
- The secret image is segmented into several blocks based on the various feature regions preferred in the scene. Hence, it avoids the occurrence of frame-dropping attacks in the video.

6.2.1 SCENE SEGMENTATION

The scene segmentation method is adapted to detect rapid changes in the scene that occurred in the source video by the metrics (ADFV) absolute difference of frame variance,[15,16] which is calculated by eq 6.1. The changed scenes of the Suzie video are shown in Figure 6.1.

$$ADFV'_n = | FV_n - FV_{n-1} | \tag{6.1}$$

FIGURE 6.1 Scene changes in Suzie video.

6.2.2 DISCRETE WAVELET TRANSFORM (DWT)

Haar wavelet is adapted to crumble the frames into four sub-bands like LL, HL, LH, and HH. The LH band is chosen to embed the watermark hence it is stable compared to other bands,[21,23–25] which is represented in Figure 6.2.

FIGURE 6.2 Two level decomposition of Haar wavelet.

6.2.3 ENTROPY

The entropy is used for measuring the randomness of image for the purpose of classification,[17] which is measured by eq 6.2.

$$\text{Entropy}\ (f_1) = \text{Sum}\ (P\ (f_1).{*}\log2\ (P\ (f_1))). \tag{6.2}$$

6.2.4 ABSOLUTE MEAN DIFFERENCE BETWEEN FRAMES (AMD)

The absolute mean difference metric is engaged to eliminate the duplicated frames in the video, which is accomplished by taking the AMD among the adjacent frames.[17,19,20]

If the difference is equivalent to zero, those frames are called redundant frames. Either one of the frames is eliminated for further processing. The formula for AMD is represented in eq 6.3.

$$\text{AMD}\ (f_i,f_{i+1}) = \text{Meanabs}\ (f\{i\} - f\{i+1\}) \tag{6.3}$$

6.2.5 FEATURE SELECTION ALGORITHM

Speeded up robust features (SURF) feature selection technique is applied for selecting the feature description of the Keyframes. This SURF transform provides important key features of a frame that are not exaggerated by numerous complications like rotation and scaling, illumination, and viewpoint.[20,26] Figure 6.3 shows the feature selection in the keyframes of Suzie video.

FIGURE 6.3 SURF feature selection for frame in Suzie video.

6.2.6 WATERMARK PREPROCESS

The gray scale image is used as a secret image. The image is decomposed into various blocks, that are same difference to the various feature discordant extracted in the Keyframes. The image block size depends on the feature set size. The Keyframes are selected by the following algorithm.

Step 1. Find the total frames in the scene.
Step 2. Identify the level of duplicated frames in the scene using AMD between the neighboring frames.
Step 3. Calculate the remaining available frames (RF) in scene by Remaining frames (RF) = Total frames– Number of duplicated frames.
Step 4. Select 25% of frames with maximum entropy and minimum Motion in each scene as key frames.

6.2.7 ABC (ARTIFICIAL BEE COLONY) OPTIMIZATION TECHNIQUE

ABC algorithm comprises three kinds of components such as employed bees, onlooker bees, and scout bees. In this algorithm, artificial bees search food sources in the multi-dimensional space. They fine tune their postures based on their knowledge of themselves and their nest members[22,23]

Step 1. Set N is the population of food sources of Bees.
Step 2. N = scaling factor lies between 0 and 1 in the interval of 0.01
Step 3. Choose the population range P = {i = 1,2,3…..N}
Step 4. Fitness= PSNR $\{F_1, F_2 \ldots\ldots\ldots\ldots F_n\}$
n = Number of frames in the video
Step 5. In each iteration
Employed bees calculate the position of fitness value for each frame
Onlooker bees update the fitness value $\{F1, F2, \ldots\ldots\ F_n\}$
Step 6. Optimal position = The position in which the onlooker bees got optimal PSNR value.

The employed bee's search the position of the scaling factor based on the fitness function PSNR. The onlooker bees record the food sources, when a food source is depleted the employed bee transforms into a scout and begins searching for new food sources. Finally, the ABC algorithm predicts a value of 0.45 as the optimal scaling factor.

6.2.8 WATERMARK EMBEDDING ALGORITHM

The process flow of watermark embedding and extraction algorithm is shown in Figures 6.4 and 6.5, respectively.

Step 1. The Input video and watermark image are chosen for processing.
Step 2. Input Video is divided into the frames and split toward separate scenes using the absolute difference of frame variance.
Step 3. In each scene $(s_1,s_2 \ldots s_n)$, calculate the entropy of individual frames (EF_i)and average entropy of scene (ES_i).
Step 4. The Redundant frame is removed by using absolute mean difference between frames.
Step 5. Select the Keyframes of each scene based on the following constraint.

IF   EFi > ESi Select the frame (F i) as Keyframe Else Reject the frame

Step 6. Apply DWT to the Keyframes then select the LH sub band for further processing.
Step 7. Apply SURF feature detection algorithm to the individual Keyframes in a scene for detecting base features of the frames.
Step 8. Based on the SURF feature extraction algorithm, select the smooth features for watermark embedding.
Step 9. The secret image (W) is preprocessed and reordered toward various blocks equivalent to the selected key features in the Keyframes.
Step 10. Apply PCA to each block of the secret image to obtain the principal component of the image block.
Step 11. The reordered image block is embedded into the key features of selected Keyframes in scene. $W_f = V_f + k * W_h$
k is a scaling factor selected by ABC algorithm.
Step 12. Perform inverse DWT to entire frames in the video.
Step 13. Combine all watermarked frames with other frames in the scene to produce output video (V_w).

6.2.9 WATERMARK EXTRACTION ALGORITHM

Step 1. The output video is processed and divided into different scenes.
Step 2. Identify the Keyframes of each scene based on the entropy constraint.
Step 3. Apply DWT to the Keyframes then select the LH sub band for extraction process.
Step 4. Extract PC of the watermark block from each key features in a scene (b_{h1}, b_{h2} ,… b_{hn}).
Step 5. Watermark blocks are converted into image block.
$W_b = (b_1, b_2 \ldots, b_n)$.
Step 6. Combine all extracted block to obtain the watermark image.

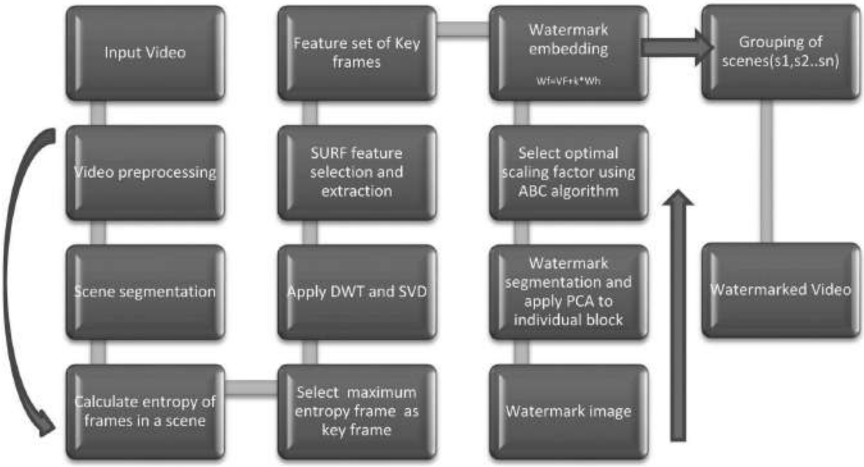

FIGURE 6.4 Process flow of watermark embedding.

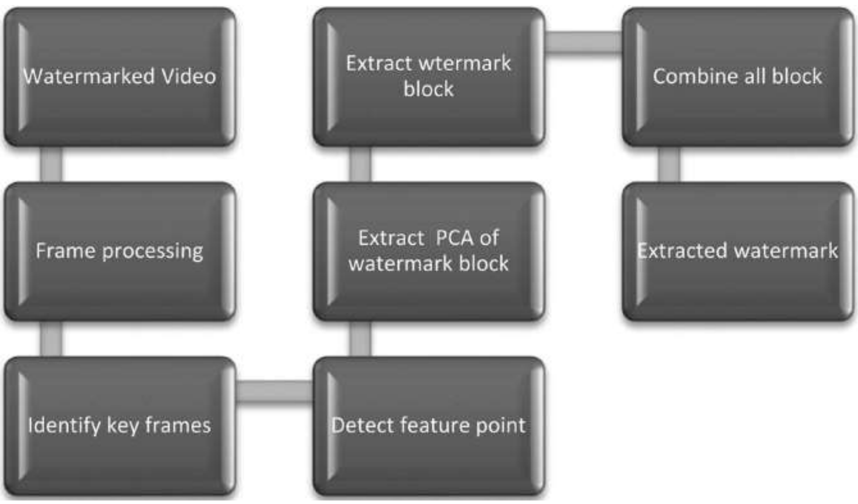

FIGURE 6.5 Process flow of watermark extraction.

6.3 RESULT AND DISCUSSION

To test the proposed artificial bee colony optimization technique-based video copyright protection in the DWT-PCA domain, various experiments are conducted. The chosen input video sequences for the experiments are

Suzie, Vipmen, News, and Foreman with 30 frames/s is presented in the standard database. The sample watermarks tried in the experiments are cameraman, Nature, and logo.

Several tests via MATLAB 2013 are conducted to estimate the robustness of the proposed technique. On the watermarked video, several image- and frame-oriented attacks mentioned in the results were applied. PSNR and NCC were used to assess the quality of the output video and retrieved watermark, the formula is represented in eqs 6.4 and 6.5.

$$PSNR(M,\hat{M}) = 10\ \mathrm{Log}_{10} \frac{255^2}{\frac{1}{LxK}\sum_{i=1}^{L}\sum_{j=1}^{k}(M_{ij}-\hat{M}_{ij})^2} \tag{6.4}$$

$$NC = \frac{\sum_i \sum_j M(i,j).M(i.j)}{\sum_i \sum_j [M(i,j)]^2} \tag{6.5}$$

TABLE 6.2 PSNR and NCC Values After Applying Various Attacks.

Type of attack	PSNR (dB) of watermarked video	PSNR of extracted watermark	NCC of extracted watermark
No attack	58.33	59.51	0.98
Gaussian noise	49.91	50.51	0.93
Poisson noise	48.38	52.32	0.90
Salt and pepper noise	47.42	49.78	0.88
Blur	57.31	53.65	0.96
Brighten	55.64	54.47	0.95
Frame averaging	57.23	45.83	0.96
Frame swapping	58.23	53.31	0.93

6.3.1 RESULT COMPARISONS WITH EXISTING ALGORITHMS

The robustness of watermarked video produced by the proposed technique is correlated with existing methods. The performance variation of the proposed technique in the results of PSNR and NCC values for different attacks is represented in Table 6.2. PSNR fluctuation of output video between the existing and proposed approach is represented in Figure 6.6. It is witnessed from Table 6.2 and Figure 6.6 that the proposed methods survive various image-oriented and frame processing attacks as well as produce better results compared with existing methodologies.

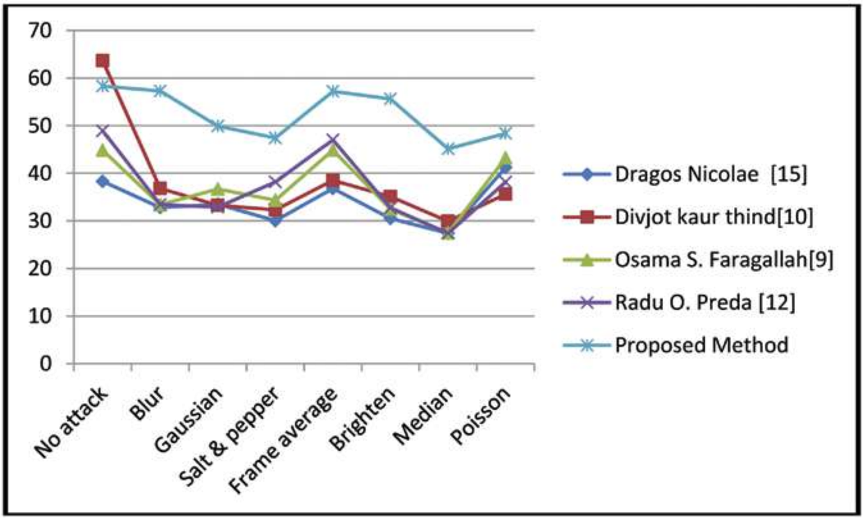

FIGURE 6.6 Watermarked video PSNR of existing vs. proposed approach.

TABLE 6.3 NCC for proposed Vs. Existing Methodology.

Attacks	Ejima [8]	Dawan [9]	Chetan [14]	Cruz-Ramos [13]	Raja [11]	Proposed method
Frame dropping	0.94	0.89	0.96	0.95	0.27	0.98
Frame averaging	0.64	0.81	0.77	0.93	0.99	0.96
Frame swapping	0.81	0.85	0.81	0.92	0.99	0.98
Addition of noise	0.7	0.91	0.9	0.98	0.97	0.96
Median filtering	0.53	0.8	0.74	0.9	0.9	0.93

6.4 CONCLUSION

In this paper, artificial bee colony optimization technique-based video copyright protection in the DWT-PCA domain is proposed. The PC of reordered watermark block is inserted into the LH subband of keyframes in the video. This methodology serves various advantages related to existing techniques. The aim of the proposed approach is the removal of duplicated frames for reducing the processing time. The security of the method is improved by introducing the entropy-based keyframe selection concept. The PCA algorithm is engaged to avoid the false positive problem. The imperceptibility and robustness of the proposed watermarking process is increased by choosing the optimum scaling factor through the ABC algorithm. The proposed methodology survives the geometric attacks by incorporating SURF feature extraction in the keyframes. This also resists specific kinds of image-oriented and frame processing attacks. The further enhancement will be introducing the multilevel authentication to the source video and watermark.

KEYWORDS

- **discrete wavelets transform**
- **feature extraction**
- **video watermark**
- **false positive**
- **SURF**
- **redundant frames**
- **artificial bee colony optimization algorithm**

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CHAPTER 7

GRAY TONE SPATIAL DEPENDENCE MATRIX: TEXTURE FEATURE FOR IMAGE CLASSIFICATION

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ABSTRACT

In any image processing activity, it is necessary to extract meaningful features from the raw image that defines the properties of the texture of an image. Texture patterns play a vital role in classifying the image type or the region of interest. Texture analysis based on extracted features is widely used in classification images, it could be medical, photomicrograph, satellite, or aerial images. Many researchers have proposed various feature extraction methods using raw images for efficient image classification and reduction of columns as compared to a raw image. The raw image is an intensity matrix of an image. This chapter gives a clear explanation of one state-of-art method, the gray-level co-occurrence matrix (GLCM) or gray tone spatial dependence matrix. This chapter also discussed advantages, disadvantages, examples, and applications of this feature extraction method.

7.1 INTRODUCTION

Nowadays, image processing[1,30] is one of the emerging fields in almost every area of research. With the fast processing computers, it has become

Artificial Intelligence and Machine Learning Techniques in Image Processing and Computer Vision. Karm Veer Arya, Ciro Rodriguez Rodriguez, Saurabh Singh, & Abhishek Singhal (Eds.)

feasible to process large image datasets for various purposes. Digital images are stored in the form of a matrix and each value of a matrix represents the gray tone of a resolution pixel in the image.[4,26] An image can be represented as $I : Lx * Ly \; \forall \; \in G$ where G is the set of gray tone value within some range and Lx and Ly are a number of rows and column in matrix.[10] Based on this 2D data (image), various analyses such as classification[19] and restoration of images[5] have been performed. In the past decade, a lot of work has been done in medical image analysis such as brain tumor, lung cancer, and bone fractures.[6,7,11,21,28,31,32]

In this chapter, the gray-tone dependence matrix (GTDM) feature extraction technique[10] is discussed in detail with example. The advantages, disadvantages, and applications of the technique are discussed. The technique is used for the classification of images such as medical images to classify whether the pixel belongs to cancerous cells or not. Moreover, the features extracted from the gray-level co-occurrence matrix (GLCM) can be used for reducing columns in the dataset. Features extracted from the GLCM matrix based on various mathematical formulas are used for different applications like predicting coarse textures, fine texture, and homogeneity in the image. In this technique, the pair of pixels are counted with some particular gray tone values and placed at the appropriate position in the resultant matrix. Therefore, the resultant matrix shows a sort of relationship between the pixels in the image. Now various features are calculated based on the resultant matrix. With the values of features, the type of image can be categorized as if the value of a certain feature is high we can conclude that the image is more homogeneous (e.g., image of water). The GTDM is based on the textural properties of an image. The texture of an image means the distribution of gray values (pixel value) over an image. Based on this gray tone distribution, textures can be analyzed as fine, homogeneous, broad textures, smooth or rough (coarse), and many other forms. With this tonal distribution, we can get information about the relationship of pixels with neighboring pixels and the texture pattern. And using this information, the classification of images can be done compared to other images. Therefore, texture information is widely used for extracting various texture features for image classification. We explained state-of-art work for developing various texture features based on the information about the gray tone distribution of the image. Here, GTDM are created in various directions $(0, 45, 90, 135)°$ with distance d = 1 between the neighboring pixels. Based on these matrices, various texture features are extracted which are used for image discrimination.

Many state-of-art methods have been developed using statistical image data for texture feature extraction. Some of the state-of-art methods are gray level run length matrix,[36] histogram of gradients magnitude,[33] and local binary pattern.[17] These methods are used in many applications such as medical imaging. The run-length matrix considers the maximum number of pixels with the same gray tone in the image in the same direction. In the local binary pattern, thresholding is done based on the gray level of the middle pixel with the neighboring pixels. And a binary value is calculated such that if the neighboring pixel p1 grey value is greater than the middle pixel m1, one will be assigned to p1 else one will be assigned. The decimal equivalent of the generated binary number is assigned to m1. This process continues for each pixel.

In this chapter, we discussed the procedure of calculating GTDM for (0, 45, 90, 135)° direction and distance d = 1 between the neighbor-hood pixels. Also, mention the advantages/disadvantages and application of discussed feature extraction method. The rest of the chapter is organized as follows, related work in Section 2, method of calculating GLCM in Section 3, advantages/disadvantages in Section 4, application in Section 5, and finally conclusion and references are mentioned at the end.

7.2 RELATED WORK

In this section, the work of various authors related to image feature extraction techniques has been discussed. State-of-art work from the year 1970s to 2021 has been investigated. In Ref. [5], the work is proposed to recover the original image from the distorted or noisy image. The intensity of pixels is compared with the neighboring pixels. The work is divided into majorly three parts; in the first part is the study of Markov Random Fields (MRF) to find the probability of the original image from the given distorted image. In the second and third parts, the technique to maximize the probability and spatial coherence model is developed. Tang et al.,[36] has proposed a novel feature extraction technique based on an eigenvector algorithm called a run-length algorithm which creates a run-length metric. In this approach, a run-length metric is calculated based on the run length of pixels with particular gray levels. This method provides important information about the images which prove to be beneficial in classification. In Ref. [26], the author has considered that calculating the variance, average, and entropy of co-occurrence metric can be considered as prominent texture features for the classification of poultry images. The different intimations of the images can be extracted based on the directions of the co-occurrence metrics. Patil et al.,[27] proposed an algorithm for the classification of lung cancer based on chest X-rays. The proposed work comprises pre-processing of the dataset, nodule detection, and feature extraction. Resizing and filtering is done in pre-processing of the dataset after that thresholding with region-based segmentation is applied over the filtered data for detecting nodules. GLCM is used for feature extraction to prepare a model for the classification of lung cancer. Sulochana et al.,[35] proposed a method of an image retrieval system based on GLCM. The method uses various features calculated using mean and standard deviation applied on each sub-band of framelet transform. In Ref. [23], Gabor and GLCM techniques are used for feature extraction to classify the satellite images. The major drawbacks of GLCM are more time-consuming and low accuracy in the boundary region. In the proposed work to overcome the drawback of GLCM, a fast GLCM method is proposed. Gabor filter is used for boundary regions to improve the accuracy of the combined model. Saleck et al.,[29] proposed a method of breast cancer detection using the breast X-ray using FCM. GLCM is used for feature extraction to set the threshold for making the boundary between two clusters. Zotin et al.,[38] proposed the method for detecting the irregularities in the lung boundary using chest X-rays, for the detection of some series diseases. GLCM is used for extracting 14 features and the probabilistic neural network (PNN) approach is used for the classification of lung boundaries as normal or abnormal. Shivhare et al.,[32] proposed a model having two stages for segmenting the brain tumor. In the first stage, the convex hull approach is used for estimating the coarse of brain tumors. Enhanced tumor, necrosis, and edema are segmented from the tumor and all results are combined for detection of brain tumor with high accuracy without any intervention of a human, in the second stage. Kanaguraj et al.,[16] proposed a model for detecting lung tumors using biopsy images. GLCM is used for extracting features and backpropagation approach of neural network for training the model. Finally, the biopsy images are classified as cancerous or non-cancerous. Yamunadevi et al.,[37] proposed a method for segmenting lung carcinoma using fuzzy GLCM. F-GLCM is used for extracting features and GoogLeNet CNN model is used for the classification of lung cancer. In Ref. [31], different segmentation methods are combined to propose a model for segmenting the brain tumor with higher accuracy.

TABLE 7.1 Summary of Related Work.

Author name and year	Approach	Proposed work
Geman et al. [5], 1984	Spatial coherence	Technique and model is proposed to recover the original image from the corrupted (noisy) image
Tang et al. [36], 1998	Multilevel eign vector calculation	Run length algorithm for feature extraction of an image
Park et al. [26], 1998	Co-occurrence metric	Variance, sum variance, average variance, and entropy of co-occurrence metric is used as image features for poultry image classification
Patil et al. [27], 2010	GLCM	Model for lung cancer classification based on GLCM
Sulochana et al. [35], 2013	Framelet transform with GLCM	Image retrieval system based on framelet transform and GLCM
Mirzapour et al. [23], 2015	Gabor and GLCM	Model to classify satellite images. Features are extracted using GLCM and Gabor for boundaries
Saleck et al. [29], 2017	FCM (C = 2) and GLCM	Model to classify breast cancer using X-rays. Features are extracted using GLCM
Zotin et al. [38], 2019	GLCM and PNN	Model to classify lung boundaries using chest X-rays
Shivhare et al. [32], 2019	Convex hull	Brain tumor segmentation model using MRIs
kanaguraj et al. [16], 2020	Back propagation neural network	Lung tumor detection model using biopsy images
Yamunadevi et al. [37], 2021	F-GLCM and GoogLeNet CNN	Model for lung cancer detection and classification
Shivhare et al. [31], 2021	Gray wolf, artificial electric field, and, spider monkey optimization algorithms	Brain tumor segmentation model by combining other segmentation algorithms

Gray wolf, artificial electric, and spider monkey optimization algorithms are used for learning multi-layer perceptron (MLP). Further, majority voting is used for result selection from these three algorithms.

7.3 METHOD OF CALCULATING GRAY TONE SPATIAL DEPENDENCE MATRIX

The image I is a collection of resolution cells (pixels) with some intensity value or gray level G. An image is represented as (X, Y), where X and Y are spatial domains. As every image is stored in computer memory is in the form of a matrix; therefore, an image I can be represented as $L_x * L_y$. Where $L_x = \{0, 1, ... N_x\}$ and $L_y = \{0, 1, ...N_y\}$, N_x and N_y are the number of rows and columns. Each value in the matrix represents a gray tone that is within the range of 0–255 but can be scaled within any range. Here, $G \in \{0, 2, 3, 4\}$, the gray level is scaled between 0 and 4. The pixels with some value $v \in G$ represent the gray tone distribution over the image I in some specific pattern. This pattern can be wide and narrow representing the broad and fine texture of an image. To calculate the relationship between the neighboring pixels in I, we calculate the co-occurrence matrices M_1 to M_4 of I in four directions. That is M_1 represents the relation of a pixel with the neighbor-hood pixel at 0° direction. Similarly, M_2 to M_4 represents the relation of a pixel with neighborhood pixel in $\{45, 90, 135\}°$ directions. Based on M_1 to M_4, various texture features are extracted for performing image classification. Considering 3 × 3 matrix (the portion of an image) having some values surrounding the center pixel #. Figure 7.1 show the direction of neighborhood pixel from the # of distance 1. Kindly note that the pixel values in the figure are not representing grey tone values.

Now, we will calculate the gray tone spatial dependence matrix in each direction for the nearest neighboring pixels. In more general way, let P_{ij} representing the number of occurrence of pair of pixels of distance d and direction r, where $r \in \{0, 45, 90, 135\}°$. Where i and j are the gray tone values of two pixels separated by d. The mathematical equations for finding the values of P_{ij} in each direction are shown below in eqs 7.1–7.4. Where (k, l) and (m, n) are the row-column indexes of neighborhood pixels in I. And I is considered as a 2D (row-column) representation of an image (in form of a matrix).

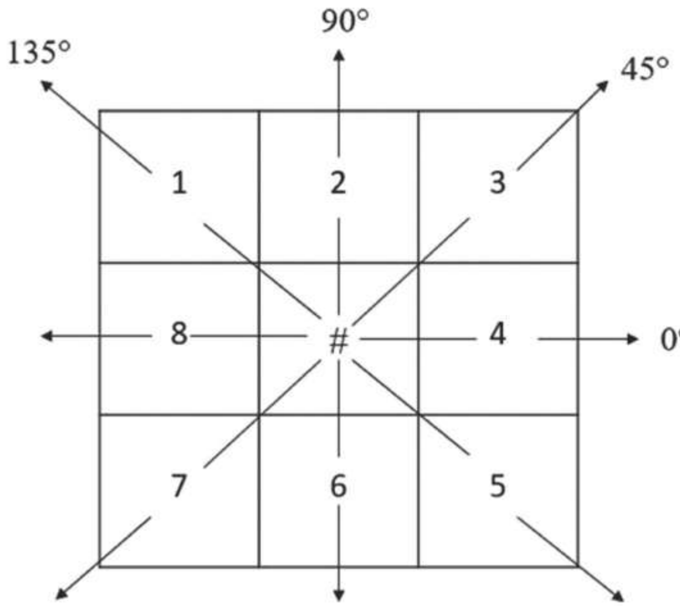

FIGURE 7.1 Pixels 8 and 4 are neatest neighbor of center pixel # in 0° direction similarly, 7 and 3 in 45°, 6 and 2 in 90°, 5 and 1 in 135°, respectively.

$$P\,(i, j, d, 0°) = *((k, l), (m, n)) \in I \text{ where } k - m = 0 \;\&\; |l - n| = 1 \quad (7.1)$$

$$P\,(i, j, d, 45°) = *((k, l), (m, n)) \in I \text{ where } k - m = 0 \;\&\; |l - n| = 1 \quad (7.2)$$

$$P\,(i, j, d, 90°) = *((k, l), (m, n)) \in I \text{ where } k - m = 0 \;\&\; |l - n| = 1 \quad (7.3)$$

$$P\,(i, j, d, 135°) = *((k, l), (m, n)) \in I \text{ where } k - m = 0 \;\&\; |l - n| = 1 \quad (7.4)$$

where * is representing frequency of particular pixel in I.

7.3.1 GRAY TONE SPATIAL DEPENDENCE MATRIX FOR DIRECTION = 0° AND DISTANCE D = 1

Considering the 4 × 4 matrix A representing a portion of an image with the gray tone values $v \in G$ as shown in Figure 7.2. For calculating P_{ij} for 0° and distance $d = 1$, R_H would be calculated. The procedure for calculating R_H is shown below in Figure 7.3 we consider a 4 × 4 matrix B, each value of the matrix is representing the row–column coordinates of that pixel in the matrix.

In Figure 7.4, R_H is calculated for the matrix that is the pair-wise representation of the neighboring (distance $d = 1$) pixels in 0° direction (horizontal). The rows from r_1 to r_4 in Figure 7.4 are showing the all pairwise combination of neighboring pixels of first row in the matrix. Now following the same procedure, we will calculate the R_H for 4 × 4 matrix shown if Figure 7.2. The R_{H0} for Figure 7.2 is shown in Figure 7.5 in which row $r1$ is pairwise combination of horizontal neighboring pixels of the matrix shown in Figure 7.2. Similarly, $r2$ to $r4$ are the pairwise combination of neighboring pixels in the matrix in horizontal direction that is 0°.

Now for calculating GTDM for direction 0° and distance $d = 1$ (P_{H0}), we will consider R_{H0} as shown in Figure 7.5. The first coordinate (0, 0) of P_{H0} will be the total count of (0, 0) in R_{H0}, as it is 6 times. So, the value of first coordinate (0, 0) of P_{H0} will be 6. In the same way, the value of (0, 1) will be calculated and total number of (0, 1) are counted in R_{H0} which is 1 in this case. Now, if we want to calculate the value of coordinate (2, 1) in P_{H0} than again we will count the occurrence of (2, 1) in R_{H0}.

$$A = \begin{array}{|c|c|c|c|}\hline 0 & 0 & 2 & 2 \\ \hline 0 & 0 & 2 & 3 \\ \hline 0 & 0 & 1 & 1 \\ \hline 3 & 2 & 2 & 2 \\ \hline\end{array}$$

FIGURE 7.2 4 × 4 matrix with gray tone values scaled in range of 0–3.

$$B = \begin{array}{c|c|c|c|c|} & 0 & 1 & 2 & 3 \\ \hline 0 & (0,0) & (0,1) & (0,2) & (0,3) \\ \hline 1 & (1,0) & (1,1) & (1,2) & (1,3) \\ \hline 2 & (2,0) & (2,1) & (2,2) & (2,3) \\ \hline 3 & (3,0) & (3,1) & (3,2) & (3,3) \\ \hline \end{array}$$

FIGURE 7.3 Co-ordinate representation of 4 × 4 matrix.

That is why matrix P_{H0} is also called as gray tone co-occurrence matrix in 0° direction and distance $d = 1$. The P_{H0} matrix is shown in Figure 7.6.

$$R_H = \left\{ \begin{array}{l} r1 = \{(0,0),(0,1)\}, \{(0,1),(0,0)\}, \{(0,1),(0,2)\}, \{(0,2),(0,1)\}, \\ \quad \{(0,2),(0,3)\}, \{(0,3), (0,2)\} \\ \\ r2 = \{(1,0),(1,1)\}, \{(1,1),(1,0)\}, \{(1,1),(1,2)\}, \{(1,2),(1,1)\}, \\ \quad \{(1,2),(1,3)\}, \{(1,3), (1,2)\} \\ \\ r3 = \{(2,0),(2,1)\}, \{(2,1),(2,0)\}, \{(2,1),(2,2)\}, \{(2,2),(2,1)\}, \\ \quad \{(2,2),(2,3)\}, \{(2,3), (2,2)\} \\ \\ r4 = \{(3,0),(3,1)\}, \{(3,1),(3,0)\}, \{(3,1),(3,2)\}, \{(3,2),(3,1)\}, \\ \quad \{(3,2),(3,3)\}, \{(3,3), (3,2)\} \end{array} \right\}$$

FIGURE 7.4 All combination of neighboring pixels of the matrix in with distance $d = 1$ and direction 0°.

$$R_H = \left\{ \begin{array}{l} r1 = \{(0,0),(0,0)\},\ \{(0,2),(2,0)\},\ \{(2,2),(2,2)\}, \\ r2 = \{(0,0),(0,0)\},\ \{(0,2),(2,0)\},\ \{(2,3),(3,2)\}, \\ r3 = \{(0,0),(0,0)\},\ \{(0,1),(1,0)\},\ \{(1,1),(1,1)\}, \\ r4 = \{(3,2),(2,3)\},\ \{(2,2),(2,2)\},\ \{(2,2),(2,2)\} \end{array} \right\}$$

FIGURE 7.5 R_H of Figure 7.2 with $d = 1$ and direction 0°.

7.3.2 GRAY TONE SPATIAL DEPENDENCE MATRIX FOR DIRECTION = 45° AND DISTANCE D = 1

In this section, we will calculate the co-occurrence matrix of neighboring pixels in 45° direction and distance among the pixels that $d = 1$. The same procedure will be followed as discussed in the previous section. In the Figure 7.7, we can see the procedure of selecting the pixels in 45° direction using the matrix 2. Now for calculating the gray tone dependence matrix for 45° direction and distance $d = 1$ (P_{45}), we will count the number of pairs with (0, 0) gray tone.

$$\begin{array}{c|cccc} & 0 & 1 & 2 & 3 \\ \hline 0 & *(0,0) & *(0,1) & *(0,2) & *(0,3) \\ 1 & *(1,0) & *(1,1) & *(1,2) & *(1,3) \\ 2 & *(2,0) & *(2,1) & *(2,2) & *(2,3) \\ 3 & *(3,0) & *(3,1) & *(3,2) & *(3,3) \end{array} \longrightarrow P_{H0} = \begin{bmatrix} 6 & 1 & 2 & 0 \\ 1 & 2 & 0 & 0 \\ 2 & 0 & 6 & 2 \\ 0 & 0 & 2 & 0 \end{bmatrix}$$

FIGURE 7.6 Number of occurrences of neighboring pixels in horizontal direction with distance 1.

$$A = \begin{array}{|c|c|c|c|} \hline 0 & 0 & 2 & 2 \\ \hline 0 & 0 & 2 & 3 \\ \hline 0 & 0 & 1 & 1 \\ \hline 3 & 2 & 2 & 2 \\ \hline \end{array}$$

FIGURE 7.7 Direction of pairing neighboring pixels with distance $d = 1$.

And assign the total number of pairs in the matrix P_{45} with gray tone values as (0, 0). Similarly, for calculating the value of coordinate (0, 1), the pair of pixels with gray tone (0, 1) will be counted and assigned to P_{45} at similar position. This procedure is followed to build the whole P_{45} matrix as shown in Figure 7.8. As shown in the Figure 7.8 the value of coordinate (2, 2) is 2 because the occurrence of pair (2, 2) gray tone values in image 7 is 2 times. Many people may have confusion that the occurrence of (2, 2) is only 1 time, then why are we reading it as 2? This is because the pairing is done in both the ways as shown for 0° in RH0. Similarly, here also coordinate (2, 2) will consider as (2, 2) (2, 2). Considering the coordinate (1, 3) of P_{45}, its value is 0 as there is no pair of gray tone (1, 3) in Figure 7.7.

7.3.3 GRAY TONE SPATIAL DEPENDENCE MATRIX FOR DIRECTION = 90° AND DISTANCE D = 1

The spatial dependence matrix with distance $d = 1$ and direction of 90° will be calculated. Taking the reference image as shown in Figure 7.2, we pair the gray tone values in vertical direction 9. And count the number of occurrences of particular pair of gray tone values in the image to create the dependence matrix (P_{90}). For example, to calculate the value of (0, 0) coordinate position in (P_{90}), we count pair of vertical pixels with gray tone values as (0, 0). In this case 8, so we assign the same value at (0, 0) position in (P_{90}). Figure 7.10 showing the process of pairing (0, 0) of gray tone value in the image. The number of arrows is showing the number of pairs of pixels with gray tone values as (0, 0). Let's take one more example to calculate the value of coordinate (1, 2) of the matrix P_{90}, again we will do pairing as shown in Figure 7.11. As the pair (1, 2) /= (2, 1) , therefore, only one direction arrow will be there. The total number of pairs with gray tone (1, 2) will be 3, so we assign the value 3 in P_{90} at the same coordinate position. In this way, all the coordinate positions in P_{90} will be assigned with the number of occurrences of vertical pair pixels with the same gray tone values as the coordinate position in P_{90}. The complete P_{90} matrix is shown in Figure 7.12. In this chapter, we are skipping the detailed procedure for calculating the spatial dependence matrix with direction 135° and distance $d = 1$, as the process is almost same as for direction 45°.

$P_{H45} =$

4	0	2	1
0	0	2	0
2	0	2	0
1	1	0	0

FIGURE 7.8 Gray tone co-occurrence matrix for direction 45° and $d = 1$.

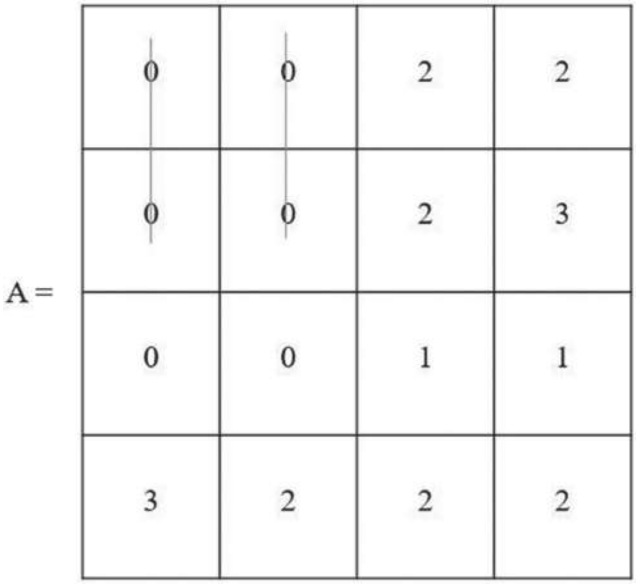

FIGURE 7.9 Direction of pairing neighboring pixels with distance $d = 1$.

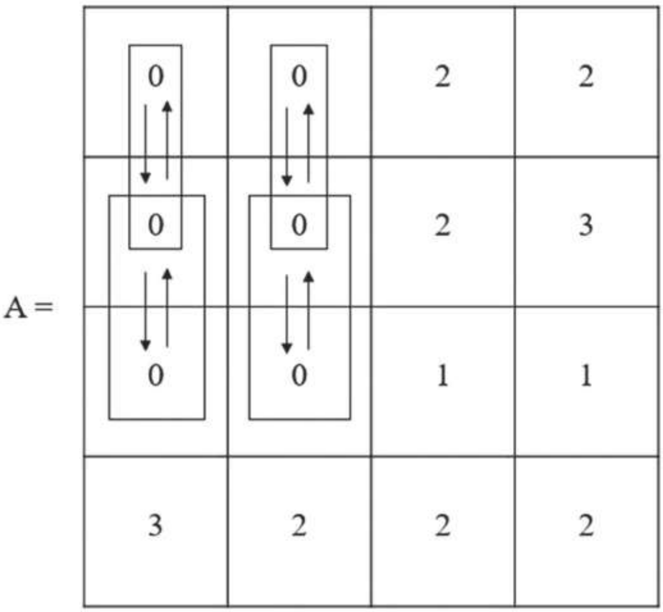

FIGURE 7.10 Vertical pairing of pixels with (0, 0) gray tone values with $d = 1$.

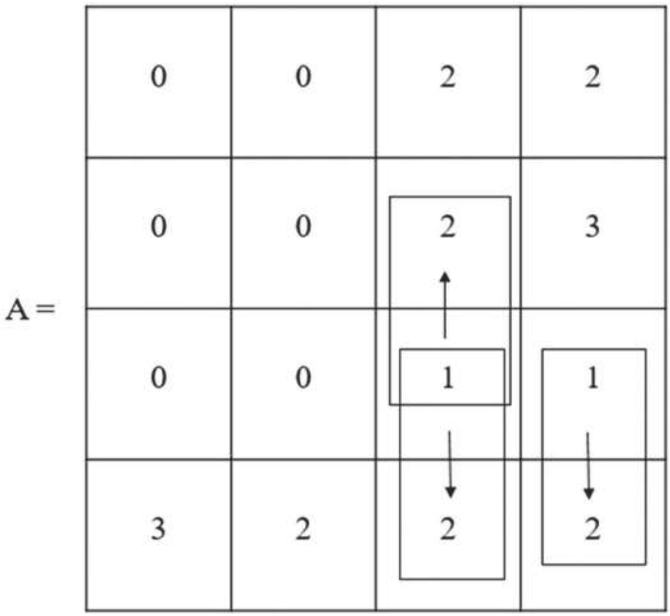

FIGURE 7.11 Vertical pairing of pixels with (1, 2) gray tone values with $d = 1$.

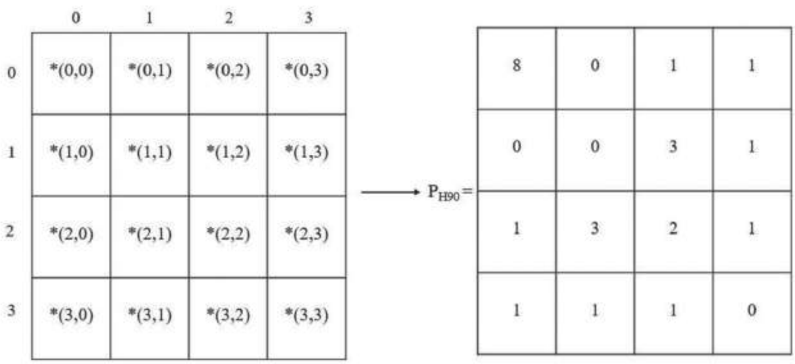

FIGURE 7.12 Gray tone co-occurrence matrix for direction 90° and $d = 1$.

7.4 ADVANTAGES AND DISADVANTAGES

GLCM is a good choice for the images with less noise like simple text images[35] and the method can be used for large amounts of image data.[25] This approach provides the relationship between the center pixel and its neighborhood pixels in almost all directions. The gray tone of the center pixel is compared with the gray tone of adjacent neighboring pixels. This pairing is done in all the directions that is 0, 45, 90, and 135° and the distance between the pixels is 1 ($d = 1$). In this way, we get four gray tone dependence matrices. Based on these matrices, various texture features are engineered. This approach of gray tone dependence matrix shows good results for the classification of simple images.[24] For images with a lot of noise, the algorithm does not show results with high accuracy.[20] Due to high dimensionality, GLCM is time-consuming[22,23] approach and also depends on the size of images used for texture feature extraction. In GLCM, selecting the value of d that is the distance between the neighboring pixels is important for its performance and speed of computation. The value of d should be selected such that the accuracy of image classification should not be compromised to enhance the speed of computation and vice versa.[9,18]

7.5 APPLICATIONS

GLCM or GTDM is one of the state-of-art methods for texture feature extraction that is used in image pattern analysis. Nowadays, GLCM is widely used in medical image classification. Texture features based on GLCM are calculated for brain tumor MR images. Brain tumor segmentation is one of the important areas in medical research. By the term brain tumor segmentation, we mean that identifying the tumor size in the brain based on magnetic resonance imaging (MRI). GLCM is used for classifying each pixel of the image, whether the pixel belongs to malign or benign tissues. Malignant tissues are the cancerous tissues that multiply or spread very fast whereas benign are non-cancerous tissues.[2] Automation in finding the tumor size with high accuracy is one of the challenging tasks. GLCM can be used for feature extraction of MRIs to segment the size of tumor.[8,13,34,39]

GLCM is used in automating the process of lung tumor detection (LTD)[12,15,29,16] and lung boundary detection (LBD).[27,37,38] Both for LTD and LBD, CXR (chest X-ray) is used for segmentation. GLCM is applied to CXR images to extract the textural features of the images. In the case of LBD, the boundary detection of the lungs using CXR depends on the experience of the radiologist. And doctor uses this primary report for further medication. As the report depends on the quality of CXR and the experience of a radiologist, therefore, these reports lack uniformity. So, it is very important to automate the process of LBD to maintain the uniformity of the report for better diagnosis by the doctor. GLCM texture features can be used to train the model for automating the process of detecting the lung boundary to determine the shape of lungs (normal/abnormal) using CXR data.[14] The same issue of uniformity is persisted with the segmentation of lung tumor size. To automate the process of LTD, GLCM-based extracted features can be used for training the model based on CT scans (computed tomography). As the task of segmenting, the lung tumor is a tedious task for radiologists to manually identify the tumor based on experience and knowledge. In case of bone fracture, doctors diagnose the X-rays based on the experience and knowledge to analyze the fractured and non-fractures bones. Various work has been done to automate the process of discriminating the fractured and non-fractured bones and classifying the type of bone fracture. In various approaches for automating the later process, GLCM textural features have been used Ref. [3].

7.6 CONCLUSION AND FUTURE SCOPE

In this chapter, we discuss one popular image feature texture extraction technique. Feature texture extraction plays a vital role in the reduction of columns in the dataset, improves efficiency and accuracy. The method of generating a GTDM or GLCM is discussed here with its advantages/disadvantages and major application areas. The procedure of generating the GLCM matrix is explained in detail with the help of an example, considering the distance between neighboring pixels as one ($d = 1$) in $(0, 45, 90, 135)°$ directions. The method was proposed by Haralick et al.,[10] in which the author has explained the procedure for generating a matrix with $d = 1$ and direction as 0°. In this chapter, the procedure for generating GTDM is explained for $d = 1$ and direction as $(0, 45, 90)°$ for better understanding of the concept. Moreover, readers of this chapter also get the idea about the application and importance of this technique.

KEYWORDS

- **image processing**
- **feature extraction**
- **texture pattern**
- **gray level feature analysis**
- **image classification**
- **signal processing**

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CHAPTER 8

IMAGE COLORIZATION AND RESTORATION USING DEEP LEARNING

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ABSTRACT

The growing use of machine learning and deep learning has encouraged many people to try and automate difficult tasks which when done manually it tends to be a tedious process and energy inefficient. We present a technique to offload this work onto computer resources. We are using CNN for visual recognition of objects in an image and using Resnet34 pre-trained on ImageNet to separate these objects into individual entities and train our model to guess the color of the objects. This also ends up restoring the picture to a certain extent. This method is a lot less tedious than using photo-shopping software which also requires certain technical skills.

8.1 INTRODUCTION

The subfield of machine learning in which model is inspired by the working of a real human brain is called deep learning. Supervised, semi-supervised,

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and unsupervised are the three ways in which learning can be done. Deep neural network, deep reinforcement, deep belief network, deep training, convolution neural networks, and recurrent neural networks have been deployed in the areas including natural language processing, computer vision, speech recognition, translation machine mechanisms, biological information, climate science, materials, and parlor game inspection, resulting in exceeding the results of human performance.

The process in which some useful information is extracted from an image or some techniques are applied to it to alter its properties is called image processing. Image processing belongs to that part of the family of signal processing in which the input fed to the model is an image and the output could be one of the properties of an image or it could be an image also. Over the last few years, image processing has shown excellent development and it is one of the rapidly growing technologies. It has remained a major focus in the field of research and development because of its increased efficiency and applications, and it is prevalent in many different fields of study. Digital image processing and analog image processing are the two methods in which image processing can be carried out. When an image processing task is performed on two-dimensional analog signals it is called analog image processing. When an algorithm is used to process a digital image with the help of a digital computer it is called digital image processing.

Preprocessing, enhancement and display, and information extraction, are the three main steps that data must undergo when it is processed using digital technology.

Gray scale images which are colorized impact various domains, for example, historical images are recreated and surveillance feeds are improved. Gray scale images have limited information, so to get a better insight into the semantics of an image gray scale images need to be converted into an RGB image.

As discussed by Albawi et al.,[1] that in the fields like pattern recognition; from image processing to voice recognition, over the past decades convolutional neural networks (CNNs) have had neoteric results.

Using CNN for high-resolution images on a large scale is very expensive, despite its incomparable features and limited efficiency of its local architecture.[2]

In the development field, CNNs have shown observable results but not resulting it to be used often.[2] According to our research, unpopularity of convolutional neural networks is because of its complexity. The frequently viable use of convolutional networks is in segmentation tasks, in which the developed image is a single-phase label. The necessary outcome includes localization in bio-medical image processing, that is, per pixel class label should be specified.[3]

This means that the model must provide each pixel of the input gray scale image with the rgb component rather than just giving an output of the selfsame size as the fed input image. We facilitate a complete convolutional model architecture using the regression loss as a criterion and then extending this idea to adversarial networks.

FIGURE 8.1 Image colorization.

Source: Reprinted from here: https://hotpot.ai/colorize-picture. © 2023 Panabee, LLC.

The process of delineation of multi-scale input to a multi-scale output is called image coloring. This could be viewed as a pixel-to-pixel regression problem where the input structure is strongly matched to the output structure.

Generative adversarial networks (GANs), much like CNNs use deep learning methods as an approach to generative modeling. Generative modeling is an unsupervised learning operation in ML that involves self-learning the discovery of homogeneity or pattern in input data, so that a model can be used to create or infer new samples that can be drawn from the original dataset. There have been many innovations in image classification thanks to deep convolutional neural networks.[4–7] Deep networks innately amalgamate attributes[6] and classifiers in an all-around multi-layered approach, and feature "layers" which can be augmented by the number of stacked layers. According to recent data,[8,9] network depth is important, and the main results[8–11] for the complex ImageNet dataset[12] use a "very deep[8]" model with a depth of sixteen[8] to thirty.[11] Various insignificant visual perception applications[13–17] also benefited significantly from very deep models. GANs are a smart way to train generative models by fashioning the problem as a supervised learning problem and framing the problem into two sub-models: real or fake (generated). The two models are trained together in a zero-sum adversarial game until the discriminator model tricks it half the time. This means that the generator model produces credible samples.

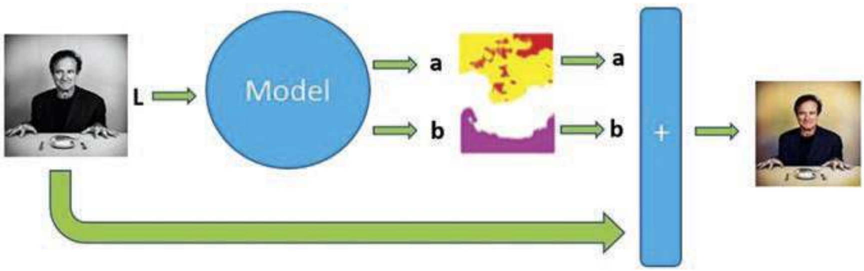

FIGURE 8.2 Stages of image colorization model.

GANs are the cutting-edge and fast-changing field that ascertains the possibilities of generative models in their ability to generate realistic examples in the variety of disciplines. Primarily, this involves image-to-image conversion operations like converting summer pictures into winter or day to night, producing realistic images of scenes, objects, and people that cannot be distinguished from fakes even by humans.

Built on the top of conventional CNN, U-Nets were first developed and applied for bio-medical image processing in 2015.[18]

Because typical CNNs focus their work on image classification where image is the input and single label is the output; in bio-medical, it is necessary to localize the region of anomaly as well as whether there is a disease or not. U-Net is able to do image localization by predicting the image pixel by pixel. There are many applications of image segmentation using U-Net.

8.2 LITERATURE SURVEY

U-Net is a structure for proper segregation.[19] It contains the contract method and the comprehensive method. The contract method follows the normal formation of a convolutional network. In addition, thousands of training images are often inaccessible to bio-medical activities.

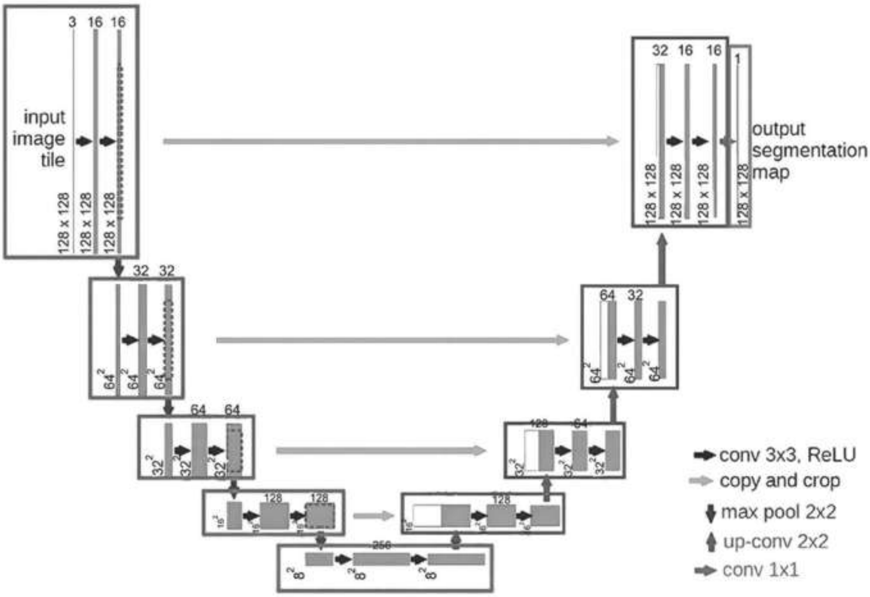

FIGURE 8.3 Working of UNet.

Source: Reprinted from Ref. [32]. © 2022 by the authors. Licensee MDPI, Basel, Switzerland. (https://creativecommons.org/licenses/by/4.0/).

U-Net enhances CNN's standard configuration by adding a comprehensive compatible method, also known as a decoder for the purpose of generating a semantic prediction with full correction. In other words, to produce segment images that highlight certain features and elements observed in the picture.

Architecture of the Generator and Critic is illustrated in Figure 8.4 below.

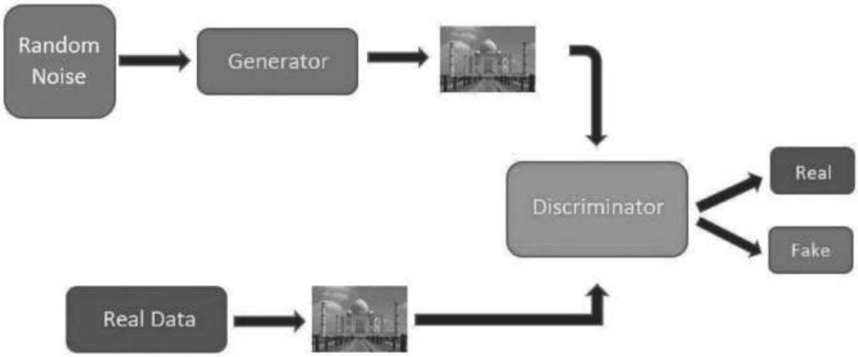

FIGURE 8.4 Generator and critic.

Contracting path is placed on the left-hand side, and an extensive path can be found on the other half. CNN architecture is followed by the contracting path. Two 3 × 3 convolutions are reapplied, each being accompanied by a modified Rectified Linear Unit (ReLU) and two steps of a 2 × 2 maximum pooling operation for down sampling.[19] One important change to the structure is that in terms of up sampling, there are many functional channels that allow the network to propagate contextual information to the high-definition layer.

As a result, the scaled approach is almost symmetrical to the contract approach and has a productive *U*-shaped structure. The network does not involve fully connected layers, rather the valid part of each convolution is utilized. That is, pixels are contained by partition map only and whole text is embedded in an input image.[20]

GAN training has been successful but is deemed to be troublesome at times and is delicated to hyperparameter selection. Several papers in the past gave a shot to stabilize GAN learning dynamics and even tried to improve sample diversity by developing novel network architectures (Che et al., 2017; Zhao et al., 2017; Jolicoeur Martineau, 2019; Karras et al., Arjovsky et al., 2017; Salimans et al., 2018; Metz et al., 2017; Radford et al., 2016; Zhang et al., 2017; Zhao et al., 2017), addition of regularization methods (Gulrajani et al., 2017; Miyato et al., 2018) and (Salimans et al., 2016; Azadi et al., 2018; Odena et al., 2017). Recently, Miyato et al. (2018) pitched to limit the spectral norm of the weight matrix in the discriminator to restraint the Lipschitz constant of the discriminator function.

A projection-based discriminator when combined with spectral normalization model (Miyato & Koyama, 2018) significantly improves image generation using conditional classes from ImageNet. In image recognition, VLAD (vector of linearly aggregated descriptors)[20] is a residual vector encoded representation of the glossary, and Fisher vector[21] which can be produced as an anticipated version of VLAD.[20]

Both are found to be powerful superficial depictions for image search and categorization.[22,23] It has been found that coding of residual vectors[24] is more efficient than coding of original vectors in vector quantization. The multigrid method,[25] widely used for solving partial differential equations (PDE) in short-level vision and computer graphics, redesigning the system as a subproblem between two scales namely coarser and finer scales.

Variables that showcase residual vectors forming a bridge between two scales is hierarchical basis preconditioning[26,27] which is an alternative to multigrid method. Standard solvers that are not aware of the residual properties of the solution, converge much slower than these solvers.[25–27] To make optimization easier, the processes that we are suggesting is good reconstruction or pre-processing.

As discussed by Nazeri et al.,[30] an image-to-image transformation operation that maps a multidimensional input to a multidimensional output, is called as image coloring.

This could be viewed as a pixel-to-pixel regression problem where the input model is strongly matched to the output model. It shows that the network should also provide the color details to every pixel of the input gray scale image along with the output of the similar spatial dimensions as the input.

An output having the same spatial dimensions as the input, must be produced by the network and we need to give color details to every pixel in the input gray scale image.

The convolutional model architecture that we are providing uses regression loss as the foundation that goes on to extending our research to adversarial nets. L*a*b* color space is being utilized for colorization task. The reason we are using the L*a*b* color space as it consists of a dedicated channel to represent the brightness of an image and color details are fully encoded in the other two channels. It prevents sharp changes in brightness and color due to small perturbations of intensity values occurring in the RGB (Red Green Blue) baseline and extends this idea to adversarial nets.

8.3 METHODOLOGY

We have used two CNN models to achieve our result, namely Generator and Critic. Generator is based on the unit which has the ability to recognize different objects in an image. The Generator also acts as the backbone in determining colors to be used and hence, outputs all this to a colored image. The Generator basically makes a mirror image of the black and white input to get a colored output. The Critic is a simple convolutional network based on deep convolutional generative adversarial networks (DC-GAN). This model utilizes a Resnet34 backbone on a U-Net with a focus on multiple layers on the decoder side. And with No GAN, a new type of GAN training, we train five important iterations of the pre-training/GAN cycle.

It facilitates the advantages of GAN training by investing a minimal amount of time in GAN training itself. Instead, we spent most of our time separately, training the creators and critics using the simpler, faster, and more reliable traditional method. The critic basically determines how realistic the output image looks.

FIGURE 8.5 Discolored output.

Generator and the Critic constantly exchange their output to make sure the output produced by Generator is as realistic as possible. This training of models among the others is an example of GAN. GANs[28] are effective in generating art. As mentioned by Vincent James, GAN-generated images are turning out to be a paragon of AI-based modern art.[29] We have used the “fastai” library because of its extensible graphics processing unit (GPU) optimized computer vision in pure Python. High-grade units used by practitioners for generating up-to-date outputs instantly and in an effortless manner are facilitated by a library called fastai in the deep learning domain. Low-grade units used by researchers to mix and match and generate new results courtship.

Utility, adaptability and efficiency are not compromised when both high-grade and low-grade units are put to use. Past endeavors to extend GANs using CNNs for image modeling have not been successful. We also faced difficulties when trying to extend GANs using CNN architectures commonly used in the supervised learning. However, after thorough model exploration, we developed an architecture that provides reliable training on different datasets and can train higher resolution and deeper generative models.

8.4 SIMULATION RESULTS

We have used a colored image as a base and processed it to turn it into a gray scale image. Then, we have used this gray scale image as an input for our model. The model does add colors to the input as advertised but the resulting image is not true to the original-colored input. This is due to the use of UNet[18] that was used for object segmentation. As UNets were originally built to perform bio-image segmentation which does not require any color.

8.5 COMPARISON WITH OTHER RESEARCH METHODS

We have used fastai library with the Resnet34 model pre-trained on ImageNet. Nazeri et al.[30] proposed training on datasets CIFAR-10 and Places 365 to obtain their result. They also used DCGAN to colorize their inputs. CIFAR-10 produced better looking images than the U-Net, used in our model. U-Nets have a brownish hue due to the loss function. Federico Baldassarre et al.,[31] proposed the use of inception ResNet-V2 models which is also a pre- trained model. In this method, objects were well recognized but certain objects were not well colored.

FIGURE 8.6 (a) Ground truth.

In this image (Ground Truth) colors are vibrant with no discoloration and high resolution. Quality of the image is high, since no image processing operations have been performed on it.

FIGURE 8.6 (b) Gray scale.

In a gray scale image, the value of each pixel represents only the intensity information of the light. Hence, there is no color.

FIGURE 8.6 (c) Result.

This is the output image. Colors are not as vibrant as ground truth; image is rather dull.

FIGURE 8.7 (a) Ground truth.

In this image (Ground Truth), colors are vibrant with no discoloration and high resolution. Quality of the image is high, since no image processing operations have been performed on it.

FIGURE 8.7 (b) Gray scale.

In the gray scale image, there is a drop in quality.

FIGURE 8.7 (c) Result.

The colors again are dull in the output and the image is not as sharp as ground truth.

TABLE 8.1 Comparison With Other Research Paper.

Parameters Authors	Title of paper	Methodology used	Network	Limitation
Kamyar Nazeri, Eric Ng, and Mehran Ebrahimi	Image Colorization using Generative Adversarial Networks (May 16, 2018)	CIFAR-10 Dataset	GAN	Miscoloration ofcertain images.
Federico Baldassarre, Diego González Morín, Lucas Rodés- Guirao	Deep Koalarization: ImageColorization using CNNs and Inception- Resnet-v2 (December 9, 2017)	Inception Resnet-V2	Visual Geometry Group (VGG)	Discoloration ofobjects.
Ours	Image Colorization and Restoration Using DeepLearning	Resnet34	UNet and DCGAN	Dullness in outputimages.

8.6 CONCLUSION

In this research paper, technique of image colorization and restoration is successfully executed. The findings from this paper, based on the output, were that the gray scale images which were given as an input got colorized methodically. Also, the level of colorization of our model was on par with online software. However, the sharpness of an image got curtailed and the same can be resolved by further refining of the code, hence increasing the efficiency of the model. We found that there is a scope for improvizations hereafter.

KEYWORDS

- **deep learning**
- **colorization**
- **convolutional neural network (CNN)**
- **Resnet34**
- **ImageNet**

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CHAPTER 9

DETERMINING IMAGE SCALE IN REAL-WORLD UNITS USING NATURAL OBJECTS PRESENT IN IMAGE

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ABSTRACT

The photograph is a 2D representation of intensity values in columns and rows which are stored in a digital computer. Defining the physical size of article in actual world units (e.g., mm, cm) is a challenging task. Some images have a real-world unit scale that helps individuals to assess the size of the objects present in the image. This scale is introduced at the acquisition time of the picture by the metalware infrastructure. Inopportunely, the scale is not present in a lot of visual descriptions (images), and it poses a problem to define the genuine size of the objects captured in an illustration. Thus, the determination of the size of objects in an image became the main goal of this research proposal.

The reason that this segment of the study is yet to be explored to its fullest was the incentive behind the work. In this research endeavor, our proposition is to find image scale (size in real-world unit per pixel) using the common size of the objects existing in the visual description. These regular objects could be people, cars, bikes, signposts, etc., depending

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upon the location, traffic, and time when the image was taken. The dataset which is used for the investigation is attained from KITTI and the RGB-D images of the dataset have been taken into account. It is then wielded in the derivation of the mathematical function to co-relate the depth of the concerned entity with the object in real-world unit per pixel. In this way, the desired outcome is achieved.

9.1 INTRODUCTION

An image comprises rows and columns and is stored in a digital computer in a digitized arrangement. The picture elements are an important aspect of a photograph as it opens the path for numerous dimensions of research. The pixels help in understanding the image and also provide a way to interpret it. By the means of existing tools and image-refining techniques, an image can be enhanced, filtered, cropped, stretched, sharpened, etc., and studied in various forms.

An image is attained by a camcorder in the proximity of sunlight or any varied source of illumination, followed by pre-processing techniques which vary relying upon the individual or the problem account. The illustration is of miscellaneous kinds. The types of images that this account has closely dealt with are monocular and stereo images.

FIGURE 9.1 Image.

$$\begin{array}{c c}
 & \begin{array}{cccc} 1 & 2 & \cdots & n \end{array} \\
\begin{array}{c} 1 \\ 2 \\ 3 \\ \vdots \\ m \end{array} &
\begin{bmatrix}
a_{11} & a_{12} & \cdots & a_{1n} \\
a_{21} & a_{22} & \cdots & a_{2n} \\
a_{31} & a_{32} & \cdots & a_{3n} \\
\vdots & \vdots & \vdots & \vdots \\
a_{m1} & a_{m2} & \cdots & a_{mn}
\end{bmatrix}
\end{array}$$

FIGURE 9.2 Image representation.

Monocular image incorporates a single image whereas stereo images deal with two images. The contradiction between these two is that the monocular vision ascertains articles which are categorized like cars and pedestrians whereas stereo vision detects even those objects which are yet to be categorized. It takes every object which appears in the field of vision into consideration, unlike mono-vision. When mono-vision is considered, it heavily relies on what it has been skilled or experienced on. These entities are grouped into the predefined classes which it was trained on. Whenever a new object pops in its field of vision, it simply ignores it.

This approach leads to many mistakes, some of which have been fatal. A common example of this mistake is the cars manufactured with a mono-vision system. On the other hand, stereo vision overcomes this shortcoming very easily. It uses the inherent 3D capabilities and views all the entities that are in the field of vision. It has a broad field of sight as well and takes appropriate measures and avoids casualty. Therefore, stereo images are commonly used and commercially preferred as well. Table 9.1 elaborates the points more clearly.

The next important aspect is the size of the objects. It determines the real-world characteristics of the entities and in turn, assists the individuals to understand the image well. The height of the article, the expanse from the camcorder and size in physical world unit all help in understanding the image.

The size assessment aspect has been majorly used in[3,16,17,29,31,35] which is of utmost importance for this work. It has helped in understanding the appropriate time for the harvest of fruits and vegetables,[3,16] for precise calorie intake,[17,29] for dietary estimation[31] and food serving size assessment.[35] It is being widely used in various categories and has ever since proven helpful.

TABLE 9.1 Mono Camera System vs. Stereo Camera System.[18]

Comparison parameter	Mono camera system	Stereo camera system
Number of image sensors, lenses, and assembly	1	2
Physical size of the system	Small (6” × 4” × 1”)	Two small assemblies separated by ~25–30 cm distance
Frame rate	30–60 frame/s	30 frames/s
Image processing requirements	Medium	High
Reliability of detecting obstacles and emergency braking decisions	Medium	High
System is reliable for	Object detection (lanes, pedestrians, traffic signs)	Object detection “AND” calculate distance to object
System cost	1×	1.5×
Software and algorithm complexity	High	Medium

The novelty of the work is determination of scale of an image. Size estimation is still not widely explored and through this work it will be highlighted and experimentally scrutinized for achieving the motive of the work. Therefore, determining size estimation of articles in an image will surely help in producing the scale in relation to the physical world which will assist in understanding the ambiguous descriptions with more confidence.

9.2 LITERATURE REVIEW

According to Zhao et al.,[1] depth information is predominant for autonomous systems in order to perceive the atmosphere and to proximate the phase. Conventional depth assessment methods are based on feature correspondence from several perspectives. Meanwhile, depth maps predicted by research are rare thus inferring depth information from an image (i.e., estimating monocular depth) is a serious problem. The deep learning-assisted monocular depth estimation has been extensively studied. In terms of precision, it has already performed very promisingly. In the meantime, the dense depth maps from individual images are estimated by a deep neural network in a format that various forms of network frames, loss functions, and strategies are then proposed to increase the accuracy of the depth estimation. Therefore, during this review, the researchers examined these deep learning-assisted monocular depth estimation methods.

According to Fernandes et al.,[2] the research paper provides an accurate process for computing the scale of the boxes directly from perspective projection images acquired by conventional cameras. The approach is dependent on projective geometry and calculates the dimensions of the box by using the data. The facts, figures, and data are extracted from the silhouette of the box with the prediction of two parallel laser beams on one of the faces of the box. To identify the silhouette of the object, a statistical model is developed for homogeneous background color removal that works with a moving camera. A voting scheme for the Hough transform is incorporated in the archetype that ascertains the collinear pixels' groups. The efficiency of the proposed approach is reflected when the dimensions of real boxes are calculated using a scanner prototype that implements the algorithms and the approaches described within the paper.

According to Wang et al.,[3] in-field mango fruit sizing is beneficial for estimating the fruit maturation and size distribution, as it suggests harvesting, produce resourcing (e.g., tray insert sizes), and marketing. In-field, machine vision imaging is used for fruit count, and now it is being used for the analysis of fruit sizes from images. The low-cost examples of three technologies for estimating the distance of a camera to fruit is assessed. The RGB-D camera is used due to cost and performance sanctioned, but it operated poorly under direct sunlight. For detection of fruits, a cascade detection with the histogram of oriented gradients (HOG) feature is used, then Otsu's method, followed by color thresholding application to get rid of the background objects. Finally, fruit lineal dimensions are premeditated using the RGB-D depth statistics, fruit image extent, and the thin lens formulation. The authors believe that this work signifies the leading practical execution of machine vision fruit sizing infield, with realism evaluated in terms of capital and ease of operation.

According to Standley et al.,[4] a successful robotic influence of real-world entities necessitates thoughtful adaptability of the physical properties of these objects. The authors have proposed a model which estimates, pass, from the image of an object. They have compared a variety of baseline replicas for an image-to-mass problem then they were trained on this dataset. The authors have also characterized the performance of a human on this problem. Finally, a model is presented which interprets the 3D shape of the object.

According to Zhu et al.,[6] in their work, the advantages of two common computer vision tasks: semantic segmentation and self-supervised depth estimation from images are studied. The authors proposed a technique to measure the regularity of border explicitly between depth and segmentation and diminish it greedily by iteratively administering the network toward a sectional optimal solution. Through expansive examinations, this suggested approach advanced to the state of the art on unsupervised monocular depth estimation on the KITTI dataset.

According to Liu et al.,[7] the matter of depth approximation from solitary monocular images is addressed. Estimating depth in monocular images is challenging as compared to estimating depth using numerous images. Earlier work focuses on making the most use of additional sources of data. The authors have put forward a deep architecture learning strategy that acquires the pairwise and unary potentials of continuous conditional random field in amalgamated deep CNN framework. Then further the proposal of an equally effective model based on fully convolutional networks and a novel super pixel pooling method is made. This method is used for estimating basic scenes with no geometric priors or any extra information injected.

According to Ma et al.,[9] the authors have considered the prediction of dense depth from a sparse set of dimensions of depth and an RGB image. Since the estimation of depth from monocular visual descriptions is intrinsically ambiguous and unreliable, to achieve a better level of robustness and accuracy, additional sparse depth samples were introduced, which are obtained by visual simultaneous localization and mapping (SLAM) algorithms. The experiments conducted by the team highlights that in comparison to the usage of RGB images, the addition of 100 spatially random depth samples reduces the prediction of root-mean-square error (RMSE) by 50% on the NYU-Depth-v2 indoor dataset. It boosts the section of reliable prediction from 59 to 92% on the KITTI dataset.

According to Li et al.,[13] self-supervised depth estimation has shown significant possibilities in hypothesizing 3D structures using entirely un-annotated images. However, the performance significantly decreases when the trained on images with varying brightness/moving objects. In this paper, the academicians have accredited this matter by intensifying the strength of the self-supervised criterion using a set of image-based/ geometry-based constraints. Firstly, a gradient-based robust photometric loss framework is proposed. Secondly, the irregular regions are cleared by leveraging the inter-loss consistency and the loss gradient consistency.

Thirdly, the motion estimation was repressed to generate across-frame consistent motions via proposing a triplet-based cycle consistency constraint. Expansive examinations directed on KITTI, Cityscape, and Make3D datasets indicated the supremacy of this method.

According to Eigen et al.,[15] the calculating depth is an indispensable constituent in realizing the 3D geometry depicted in a certain scene. Furthermore, the chore is characteristically abstruse, with a great source of indecision from the inclusive scale. In this paper, the journalists have conferred a novel technique that discusses this assignment by retaining two deep network stacks: making an abrasive universal estimation established on the grounds of entire image and additional that enhances this prediction locally.

According to Ponce et al.,[16] fruit grading is a necessary post-harvest chore in the olive industry. The combination is based on size, assists, and mass in the processing of high-quality table olives. The study presents a technique fixated on assisting olive grading by computer vision techniques and feature modeling. The sum total of 3600 olive fruits from nine variations was captured, stochastically distributing the individuals on the scene, using an improvised imaging chamber. Then, an image analysis algorithm was invented to divide olives and extract descriptive characteristics to evaluate their minor and major axes; and their mass. Determining the accurate performance for the individual division of the olive fruits, the algorithm was proven through 117 captures containing 11,606 fruits, producing only six fruit-segmentation mistakes.

According to Ege et al.,[17] analysis of the estimation of images of food for precise food calorie estimation comprises three prevailing techniques with two new methods: (1) CalorieCam—it evaluates real food size on the foundation of a reference object, (2) Region Segmentation—it employs food calorie estimation, (3) AR DeepCalorieCamV2—it is about visual-inertial odometry built-in the iOS ARKit library, (4) DepthCalorieCam—it is based on camera (stereo) such as iPhone X/XS, and (5) RiceCalorieCam—it utilizes rice grains as reference articles. The last two approaches attained 10% or less estimation error, which is good for estimating food calorie.

9.3 METHODOLOGY

The proposed methodology adopted is described in Figure 9.1:

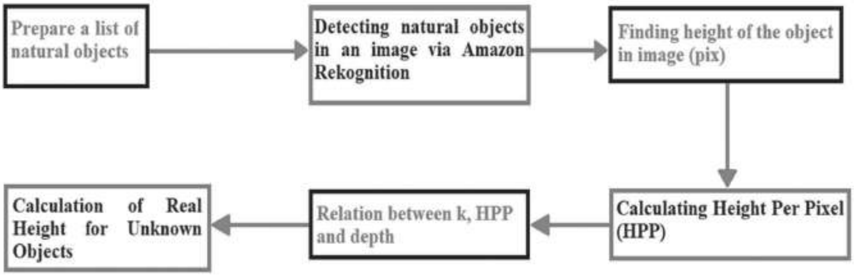

FIGURE 9.3 Proposed methodology.

9.3.1 DATASET

The dataset has been attained by Karlsruhe Institute of Technology. They have used autonomous driving platform known as Annieway for the development of unique and inspiring real-world computer vision benchmarks. Annieway is equipped with two high-resolution color and grayscale video cameras on its hood. The accurate and precise ground truth is supplied by a Velodyne laser scanner and a GPS localization system. All the datasets are captured by driving around the mid-size city of Karlsruhe, Germany.

FIGURE 9.4 Autonomous driving platform for KITTI.

Source: Reprinted with permission from Ref. [19].

The info set used for this work is Data Scene Flow 2015/Stereo 2015 which has been downloaded from KITTI Dataset and consists of 200 training scenes and 200 test scenes (four color images per scene, saved in loss less PNG format). The estimation server calculates the proportion of bad pixels intermediated over all ground truth pixels of all 200 test images. In the dataset, the images contain up to 15 cars and 30 pedestrians.

Sample imageries from the dataset are as follows:

FIGURE 9.5 (i) Sample image 1.

FIGURE 9.5 (ii) Sample image 2.

FIGURE 9.5 (iii) Sample image 3.

9.3.2 ALGORITHM

1. Creation of table of natural objects in the image.
2. Detecting natural objects in an image.
3. Once the objects are detected the table is prepared.
4. The subsequent stage is to calculate HPP which is height per pixel. Its unit is cm/pix. This can be calculated by applying $= \dfrac{\textbf{\textit{Real Height}}}{\textbf{\textit{No. of Pixels}}}$.
5. k is calculated by applying the formula $\boldsymbol{k} = \dfrac{\boldsymbol{HPP}}{\boldsymbol{depth}}$.
6. Then ***Real Height = k * Depth * Pixel Height*** is calculated which is Predicted Height.

9.3.3 STEPS

1. **Preparing the list of natural objects:** The images were perused and the natural objects were noted. After that, the standard height of these objects was appropriated from reliable web sources like websites of construction companies, Wikipedia, car manufacturing websites. [20,23–25] For trees, the average height was taken from research paper.[21] Then the list was processed along with the mean height of the objects.

TABLE 9.2 List of Natural Objects.

Sr no.	Natural objects	Mean height (cm)
01	Train	402
02	Car	170
03	Person	172
04	Traffic signal	210
05	Truck	280
06	Wheel	83

The mean height is taken in cm.

2. **Detecting natural objects in an image:** Amazon Rekognition tool is brought in usage to detect objects in the KITTI images. It is provided by AWS.

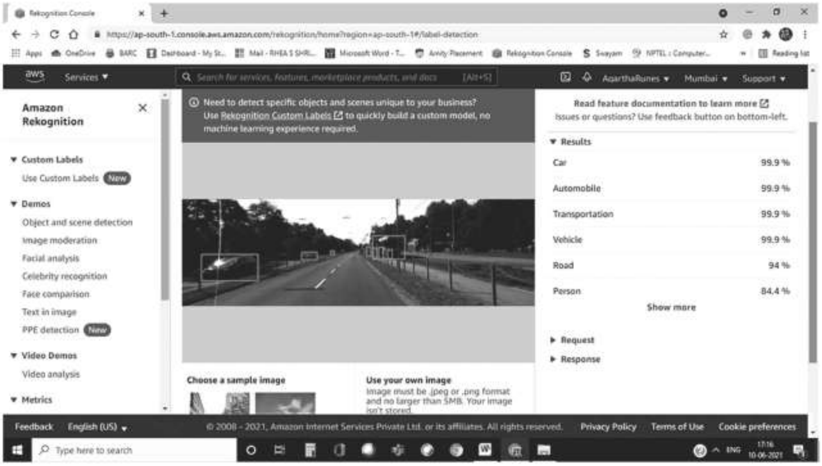

FIGURE 9.6 Object detection working on image from dataset.

From Figure 9.6, it is evident that it is detecting articles predominant in KITTI dataset image. The bounding box appears on some items for determining the image scale.

3. **Finding height of the object in image (in pix):** This step is carried out by the help of bounding box. When AWS recognition is used, the responses include numerical data pertaining to width, height, left, and top.[22,32]

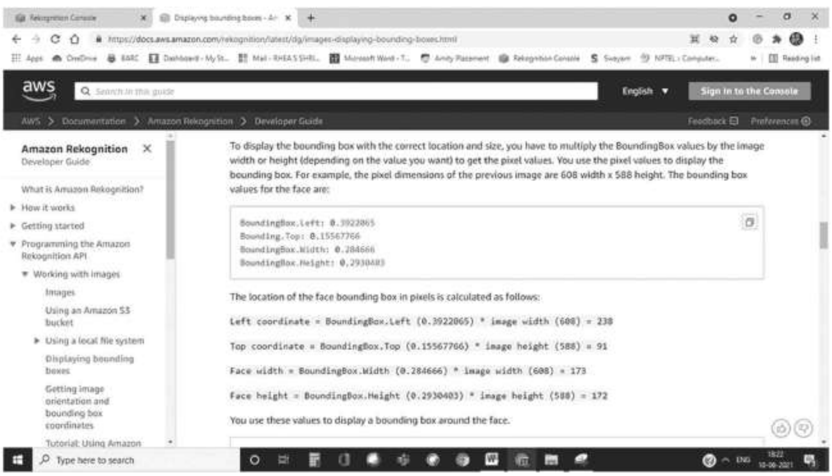

FIGURE 9.7 Snapshot of the AWS developers guide explaining bounding box calculations.[22]
Source: Adapted from Ref. [22]

It is to procure the needed information. The bounding box returns the quantity of image elements to express an object. Only height will be considered for further calculations as it indicates the ratio of overall image height which is the requirement of this stage.


```
{
    "Labels": [
        {
            "Name": "Car",
            "Confidence": 99.9004135131836,
            "Instances": [
                {
                    "BoundingBox": {
                        "Width":
0.1577451378107071,
                        "Height":
0.2390778809785843,
                        "Left":
0.05361174792051315,
                        "Top":
0.5120320916175842
                    },
                    "Confidence":
99.9004135131836
                },
```


FIGURE 9.8 The response after image (in Figure 9.5(i)) was uploaded in Amazon Rekognition Object Detection.

The real height of the object will remain same irrespective of its position in the illustration which is understandable. The formula which is applicable in this step is as follows:

- $$HPP = \frac{Real\ Height}{No.\ of\ Pixels} \tag{9.1}$$
  Real height is taken from the table created in Step 1.
- $HPP \propto \frac{1}{No.\ of\ Pixels}$. This indicates that if HPP (height per pixel) is increased then no. of pixels will decrease and vice versa.

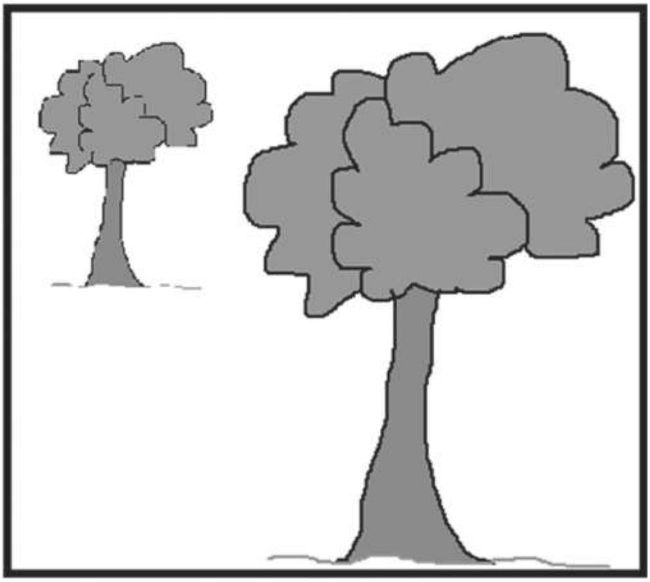

FIGURE 9.9 Understanding the real height of the object.

Irrespective of the location of the article in the photograph, its real height or natural height remains same. The only modification is the pixel representation of the interested object. In the above Figure 9.7, it's unmistakably noticeable that the closer the object is to the camera more number of pixels will be used to represent it, hence the height per pixel will decrease and vice versa.

Relation between k, HPP, and depth: Through step 4 and from eq 9.1, we can say that,

- $\boldsymbol{HPP \propto depth}$ (9.2)
- $\boldsymbol{HPP = k * depth;}$ k is a constant which is to be calculated.
- $\boldsymbol{k = \frac{HPP}{depth}}$ (9.3)

According to eq 9.2, height per pixel will increase with increase in depth and will decrease with decrement in depth. This is done to evaluate k which is an important component of this whole research.[37] k remains constant throughout the image and is different for different images. Also, through availability of k, the calculation of the scale of unknown substances in the image has become easy.

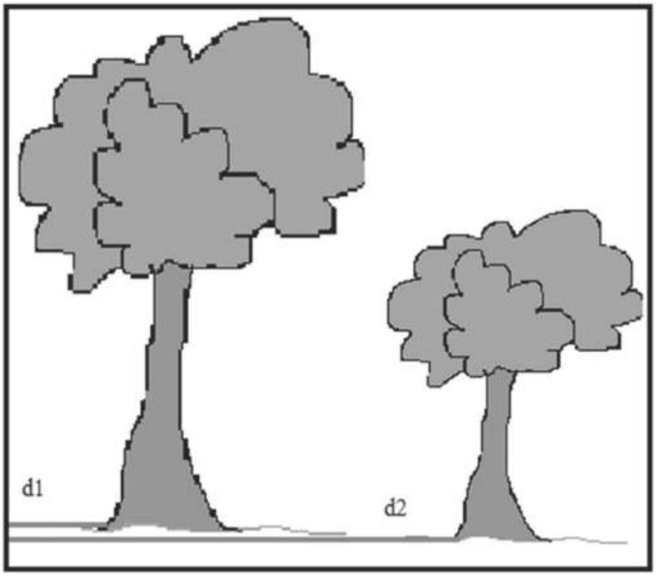

FIGURE 9.10 Pictorial representation of how depth is related to no. of pixels to represent an object in the image.

Calculation of real height for unknown objects: The last step uses all the acquired details from above steps and substitutes them in eq 9.4 to get the results.

$$\textit{Real Height} = k * \textit{Depth} * \textit{Pixel Height} \tag{9.4}$$

In this equation, k attained for the entire image will be substituted along with the depth of the entity (step 5) and pixel height (step 3). This provides us with the desired outcome.

9.4 IMPLEMENTATION

Step 1

First step to implement the methodology (Figure 9.3) is to create a table (Table 9.2) for natural objects with their mean height. This is thoroughly explored and the dataset being used is from Karlsruhe, Germany. Therefore, the estimations and calculations have been done keeping this in the forefront of the research.

Step 2:

After the list was prepared, the next task was to identify the articles in the images. This was done with the assistance of Amazon Rekognition.

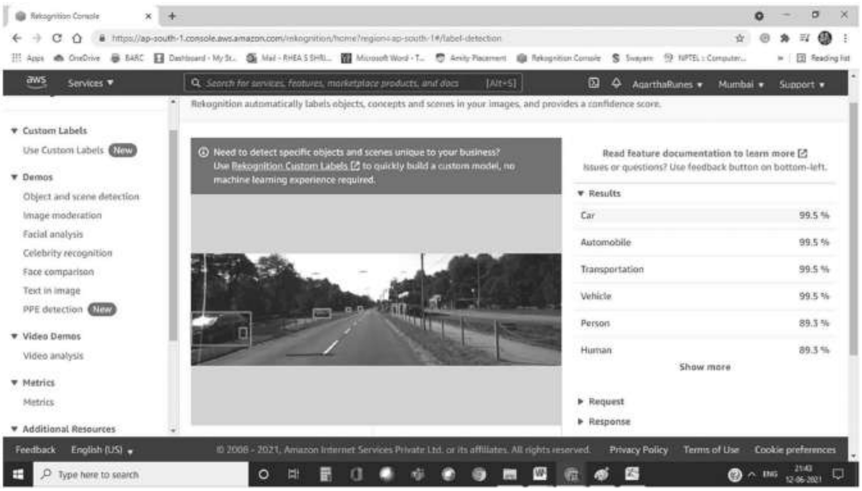

FIGURE 9.11 Amazon Rekognition recognizing the objects in the KITTI dataset image.

Step 3

After the objects are detected the table is prepared using Microsoft Excel as follows:

The column "*Natural Objects*" and "*Height Ratio*" has been generated by the results from Amazon's software. The "*Dimension Of Image*" is already known to us. The column with heading "*Height Ratio* × *375 (pix) pix value*" is indicative of step 3. Here, the values are in pixel units: pix and define the extent of pixels that define an object restrained by bounding box.

Step 4

The subsequent stage is to calculate HPP which is height per pixel. Its unit is cm/pix. This can be calculated by applying eq 9.1. Once it's done the resultant column is achieved as follows:

ImageID	Natural Objects	Height Ratio	Dimension Of Image	Height Ratio x 375 (pix) pix value
000000_10	Car 1	0.1181	1242x375	44.2875
000000_10	Car 2	0.1261	1242x375	47.2875
000000_10	Car 3	0.0857	1242x375	32.1375
000000_11	Car 1	0.1416	1242x375	53.1
000000_11	Car 2	0.0937	1242x375	35.1375
000000_11	Car 3	0.1164	1242x375	43.65
000000_11	Car 4	0.1883	1242x375	70.6125
000000_11	Car 5	0.0586	1242x375	21.975
000003_10	Car 1	0.239	1242x375	89.625
000003_10	Car 2	0.0656	1242x375	24.6
000003_10	Car 3	0.0281	1242x375	10.5375
000003_10	Person 1	0.1136	1242x375	42.6
000003_10	Person 2	0.0994	1242x375	37.275
000003_10	Person 3	0.0851	1242x375	31.9125
000003_10	Person 4	0.077	1242x375	28.875
000003_10	Train	0.214	1242x375	80.25
000003_11	Car 1	0.2851	1242x375	106.9125
000003_11	Car 2	0.0742	1242x375	27.825
000003_11	Car 3	0.0273	1242x375	10.2375
000003_11	Car 4	0.0355	1242x375	13.3125
000003_11	Person 1	0.1025	1242x375	38.4375
000003_11	Person 2	0.1027	1242x375	38.5125
000003_11	Wheel	0.0827	1242x375	31.0125

FIGURE 9.12 Table depicting the values garnered from Amazon Rekognition Software.

HPP (cm/pix)
3.8385
3.595
5.2897
3.201
4.8381
3.8946
2.4075
7.736
1.8967
6.9105
16.1328
4.0375
4.6143
5.3897
5.9567
5.0093
1.59
6.1096
16.6056
12.7699
4.4747
4.466
2.6763

FIGURE 9.13 HPP column.

Step 5

Upon estimating HPP, we calculate *k*. This is done with the help of eq 9.3. The depth is already given and the required details are substituted in the formulae. This effect in generation of column *k* which is showed in Figure 9.14 below:

Depth (cm)	k=HPP/Depth	k_Mean
150	0.022666667	
300	0.023333333	
1500	0.024	
750	0.022933333	0.026966
80	0.0215	
50	0.022933333	
38	0.022631579	
40	0.052894737	
39	0.063173541	
375	0.004533333	
975	0.005128205	
1025	0.004738676	0.005826
375	0.008493827	
385	0.013139801	
160	0.014409722	

FIGURE 9.14 Depth and constant *k*.

Step 6

Then eq 9.4 is used and the column "Predicted Real Height" is generated. These values are then paralleled with the natural heights of real world objects which provide insight as to how much the predicted height is varying from real height.

9.5 RESULTS AND CONCLUSION

The attached screenshot below describes the requirements for calculation.

Pred H=k*depth*pixh	Actual Height (cm)
170	170
175	175
180	180
172	172
172	172
172	172
172	172
402	402
170	170
170	170
170	170
170	170
172	172
172	172
83	83

FIGURE 9.15 Natural height vs. predicted real height.

Image	Object	Actual Height (cm)	k=HPP/Depth	k_Mean	Depth (cm)	Pixel Height	Pred H=k*depth*pixh
000003_10	Car 1	170	0.022666667		150	50	170
000003_10	Car 2	175	0.023333333		300	25	175
000003_10	Car 3	180	0.024		1500	5	180
000003_10	Person 1	172	0.022933333	0.026966	750	10	172
000003_10	Person 2	172	0.0215		80	100	172
000003_10	Person 3	172	0.022933333		50	150	172
000003_10	Person 4	172	0.022631579		38	200	172
000003_10	Train	402	0.052894737		40	190	402
000003_11	Car 1	170	0.063173541		39	69	170
000003_11	Car 2	170	0.004533333		375	100	170
000003_11	Car 3	170	0.005128205		975	34	170
000003_11	Car 4	170	0.004738676	0.005826	1025	35	170
000003_11	Person 1	172	0.008493827		375	54	172
000003_11	Person 2	172	0.013139801		385	34	172
000003_11	Wheel	83	0.014409722		160	36	83

FIGURE 9.16 Necessary columns for calculation.

According to preceding steps, *k* was calculated and its mean was taken. Now, for every image there is a constant *k*. This *k* when substituted in the formulae gives the desired outcome which is the predicted size. All articles in the photograph are at a certain depth from the camera's focal point so there's a requirement of taking mean depth which is illustrated in the figure below. The mean depth will aid in estimating the height of the unknown objects in the image.

Image	Mean Depth
000003_10	363.5
000003_11	476.28571

FIGURE 9.17 Mean distance of images.

Lastly, mean squared error of the images is calculated. The formula for MSE is as follows:

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y - y')^2$$

Where, MSE is mean squared error, n = number of data points, y = observed value, and y′ = predicted value. This results in the values as mentioned in Figure 9.18:

Image	K	Mean depth	Mean squared error
000003_10	0.026966	363.5	0
000003_11	0.005826	476.2857143	0

FIGURE 9.18 Mean squared error of the images.

The data so far calculated help in estimating the size of the unknown objects in the image. Further, calculation of MSE is carried out and it is depicted below. The predicted height is subtracted from the actual height.

It is understandable that MSE governs the correctness and accuracy of the model, system. It is a loss of calculation the amount of error. It is said that MSE and accuracy are inversely related to one another as MSE's ideal value 0 corresponds to utmost accuracy. According to Wikipedia,[27] "The MSE is a measure of the quality of an estimator. As it is derived from the square of Euclidean distance is constantly a positive value with the error decreasing as the miscalculation approaches zero." That can be seen from the image. Since MSE is 0, the algorithm works perfectly well with the dataset. Therefore, it can be said that the methodology and the algorithm are working with full efficiency and desired outcome is achieved. This study has been very beneficial as the visual descriptions without scale are now interpreted with ease which is helpful in understanding them, whatever may be the purpose. This is going to aid future researchers and their analysis in comprehending the true essence of varied photographs for experimentation. It is further going to open new horizons into undiscovered or less explored terrain of research.

Image	Object	Actual Height (cm)	Pred H=k*depth*pixh	Error = Actual - Predicted	Sq(Error)
000003_10	Car 1	170	170	0	0
000003_10	Car 2	175	175	0	0
000003_10	Car 3	180	180	0	0
000003_10	Person 1	172	172	0	0
000003_10	Person 2	172	172	0	0
000003_10	Person 3	172	172	0	0
000003_10	Person 4	172	172	0	0
000003_10	Train	402	402	0	0
000003_11	Car 1	170	170	0	0
000003_11	Car 2	170	170	0	0
000003_11	Car 3	170	170	0	0
000003_11	Car 4	170	170	0	0
000003_11	Person 1	172	172	0	0
000003_11	Person 2	172	172	0	0
000003_11	Wheel	83	83	0	0
				Mean Square Error	0

FIGURE 9.19 Calculation of MSE for all objects in images.

9.6 FUTURE SCOPE

Size estimation of material world objects through images is a progressive research community and within a span of few years it is anticipated to experience extensive scrutiny. It is indicative of a breakthrough in a multitudinous of fields ranging from computer vision, image processing, artificial intelligence, military science, medical research, criminology, food technology, space research, oceanography, volcanology, and many more. It has introduced a way for novel and generative ideas which results in progressive research. The scheme of ideas pertaining to future research entails carrying out the research on larger dataset and on a variety of images unlike the executed one. Considering the effort in this field has just begun, there exists a colossal scope and potential in it. It will be truly expansive and skyrocketing years from now.

KEYWORDS

- **image processing**
- **machine learning**
- **size estimation**
- **size determination**
- **KITTI**
- **stereo images**
- **natural objects**

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CHAPTER 10

IMAGE SEGMENTATION USING METAHEURISTIC

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ABSTRACT

Image segmentation is an important task in image processing. Its purpose for segmentation is to rearrange the description of an image into anything a lot simpler to understand. Metaheuristic methods are global optimization that finds the global solution and avoids stagnation. These algorithms are mostly nature-inspired, and they are turning out to be exceptionally incredible in solving global optimization problems. This paper will review various existing metaheuristics-based methods regarding image segmentation. Particle swarm optimization (PSO) algorithm, artificial bee colony optimization (ABC), genetic algorithm (GA), and simulated annealing (SA) are some of the metaheuristic algorithms reviewed here. Further, the freely accessible benchmark datasets for segmentation are also reviewed.

10.1 INTRODUCTION

Image segmentation (IS)[1] is a process of dividing an image into several segments. It is an important phase in image processing. In machine learning, it is a pre-preparing period of many picture-based applications, such as medical imaging, object detection, biometrics authentication,

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and pattern detection.[2] We use image segmentation so that we can group certain pixels based on certain criteria.

- **Medical imaging:** Medical imaging assists the diagnosis using images and does the treatment by using these images of patients. Segmentation helps the medical field in several ways from analysis to procedures. The influence of medical images in healthcare is constantly growing, so is the demand for better image segmentation algorithms. Examples include segmentation of tumors,[3] tissue image segmentation,[4] etc.
- **Object Detection:** Object detection is a process to locate a region of interest, and to show where a region is in an image then object recognition is used. Here, detection may be of a human face, tree, cell, etc.
- **Video Surveillance:** It is the process of detecting a moving object using a camera.[5] This capturing of the object helps us in the seeing of the activity being acted in video or for the traffic, to know the different objects present there, etc.
- **Biometric Identification:** It simply means the measurement of the human body's physical or behavioral attributes. It analyzes attributes distinct to each person to validate their individuality. Examples include Fingerprint identification,[6] Iris Segmentation,[7] Face Recognition, Finger Vein Identification, etc.

The region for these algorithms' popularity is that they mimic natural phenomena, and the efficiency of metaheuristic algorithms can be because they mimic the best elements, particularly the choice of the best in an organic manner which has developed by regular determination more than a long period.[8]

Several challenges may affect the efficiency of an image segmentation method. Some of them are mentioned here:

- **Intra-class variation:** Issues in this is that the region of interest is in several different forms which means different images of the same type showing variation. Which frequently makes the segmentation technique troublesome. That is the reason a segmentation strategy ought to be invariant to such sorts of varieties. Examples include tables that are shown in different shapes, person face detection, etc.
- **Inter-class similarities:** Another main issue in the field of segmentation is image similarity occurs between different images of a

different class, which is referred to as inter-class similarity. Which often makes the process difficult to handle. Therefore, it should be invariant to such sorts of changes. Examples include modality-based medical image classification, etc.

- **Fuzzy background:** Image segmentation for the picture with fuzzy background is viewed as a major test. Segmenting a picture for the necessary district of interest might blend with the complex environment and limitations. Examples include fishes with a boggling foundation in the water thinking about a major test, a picture that compares to H&E-stained breast cancer, etc.
- **Uneven lighting condition:** It is among the main sources that can influence the division result. It is regularly created during the catching of a picture and affects pixels. The essential drivers for the aggravation of uneven lighting are the ROI cannot be disconnected from the shadows of different objects, the light may be temperamental, and the object to be detected is very large which therefore creates an uneven light distribution. It may therefore sometimes generate a false segmentation result.

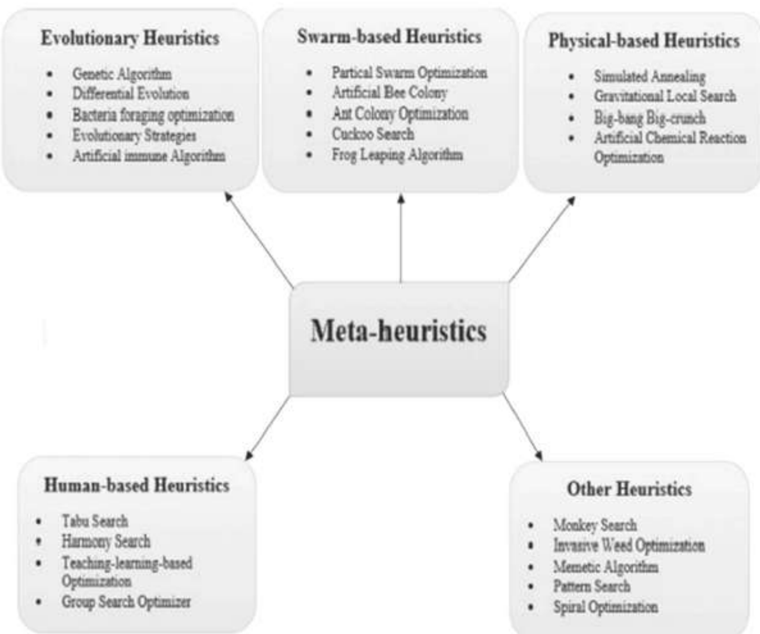

FIGURE 10.1 Different metaheuristic algorithm.

10.2 LITERATURE REVIEW

Image segmentation[1] is a process of dividing a picture into several parts, it is an essential phase of image processing in which the required region of interest is separated from a picture that can go from discovering objects while moving to distinguish irregularities in a picture. Segmentation expects to make a basic type of picture which can be analyzed easier. It is the exploration region that relates to the parceling of a picture into its constituent items. Also, Image segmentation can dole out a class for every pixel in the advanced picture, where the pixels with a similar class have similar properties. Image segmentation can locate lines or curves in the picture which will help to find objects and boundaries in digital images. Image segmentation seems to have a bright future as it makes complex problems easy such as region of interest recognition, limit assessment inside the movement, picture pressure, picture altering, or picture data set turn upward, and more else.[31]

Laporte and Osman[10] characterized a metaheuristic as: "An iterative generation measure which directs a subordinate heuristic by consolidating shrewdly unique two significant part sents of metaheuristic algorithms exploration and exploitation. Exploration means creating diverse solutions for the global search, while exploitation means focusing on the local region. A decent mix of these two significant parts will guarantee great learning methodologies which used to structure data to track down the effective worldwide ideal arrangement."

10.2.1 GENETIC ALGORITHM (GA)

According to Man et al.,[11] "The GA can be useful for making various design tools for industrial Working engineers." Here they have explained why and where the GA could be used for industrial engineers as an optimal tool.

According to Abdel-Khalek et al.,[13] "Utilized a clever two-dimensional picture segmentation approach based on Tsallis and Renyi entropies and the genetic algorithm. The utilization of the entropy here is for the data contained in the 2D histogram. The genetic algorithm is utilized to boost the entropy to the parts made proficiently, the picture into required items and foundation. They tentatively show that their methodology amplifies the entropy effectively and produces better picture segmentation for the given picture compared with the other classical thresholding techniques."

According to Maulik,[14] "Genetic algorithms have given very desirable output for the medical segmentation and diseases prediction. The problem can be made as one of search space. The difficulties which are faced by most segmentation in clinical picture segmentation happen given helpless picture differentiation and relics that bring about absent or diffuse limits might prompt helpless forecast for a future picture. Thus, the subsequent space is regularly uproarious with numerous neighborhood optima. Genetic algorithmic was observed to be proficient from emerging from neighborhood optima, and it likewise upgraded division methodology improved with significant adaptability. In this paper, he has reviewed the various applications of Genetic Algorithms, especially for medical image segmentation."

According to Ahmed et al.,[39] "Their principal space for work was in Object identification and acknowledgment which is a powerful and crucial method for knowing articles precisely in complex scenes. Item examination has grabbed a lot of the eyes of analysts throughout the decade to investigate and utilize the parts of article location and acknowledgment-related issues in the advances, such as horticulture, mechanical technology, clinical, observation, and showcasing. Right off the bat, they have utilized the bunching of comparable tones and districts which they have accomplished by applying the K-mean clustering algorithm. Besides, they have done segmentation by blending the recently accomplished clusters, which are comparable and connected. Thirdly, they have Generalized Hough transform. At long last, a Genetic algorithm is applied as a recognizer to perceive the notable objects for various natural conditions. For really looking at the exactness of their proposed work, a benchmark dataset has been utilized."

According to Ahmed et al.,[40] "Their domain of was Agriculture for this paper. According to them the inspection of fruit diseases time-taking and we can use metaheuristic methods to control this problem. They have three techniques to follow. In the initial step, they have spots of the leaf are improved by utilizing a combination of 3-dimension box filtering, de-correlation, 3D-Gaussian, and 3D-Median. Then, at that point, the spots are portioned by a solid connection-based strategy. At last, the color, color histogram, and (LBP) highlights are joined by an examination-based equal combination. The necessary region is then utilized genetic algorithm for improvement and afterward characterized utilizing One-vs-All M-SVM. The yield of the analyses is performed on the Plant Village dataset. They then, at that point, applied their proposition for various kinds of sicknesses. In addition, they finish up by saying that very much put pre-handling consistently delivered highlights that accomplished huge grouping precision for the given information."

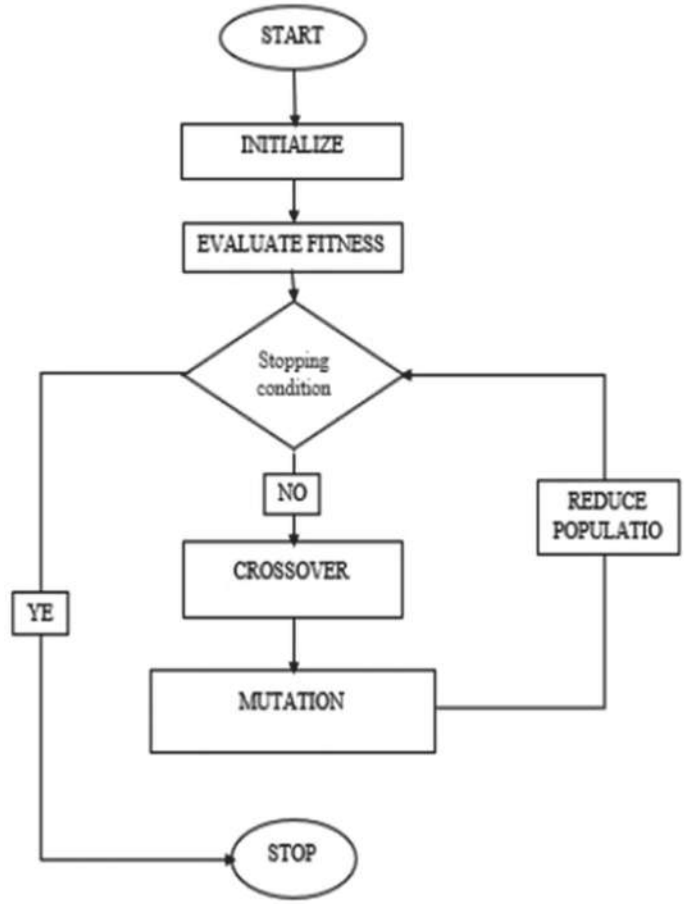

FIGURE 10.2 Flow chart for genetic algorithm.

10.2.2 SIMULATED ANNEALING (SA)

Simulated annealing (SA) was developed in 1983, by Kirkpatrick et al., by going through the annealing process. SA is a direction-based calculation

TABLE 10.1 Review of Genetic Algorithm in Image Segmentation Domain.

Author	Application	Advantages	Reference
John Holland (the1960s and 1970s)	Data mining,classification problem	Normally equivalent, reliably a reply;reliably gives indications of progress depends on the period, odds of getting an ideal course of action are more	
Man et al.	Boundary and System Identification, Robotics, Pattern and Speech Recognition,Engineering Designs,Planning and Scheduling, ClassifierSystem	Easily handle problem constraints. The multi-objective problem can be addressed. Solve multimodal, nondifferentiable, non-continuous, oreven NP-complete problems. Givesan instrument to advance the geography or the design in corresponding with the boundaries of the arrangement. It is an extremely straightforward strategy with very little math. Can without much of a stretch interface to existing reproductions and models	[11]
Hammouche et al.	Image segmentation	Enables to decide the proper number of edges, just as the sufficient edge esteems. Handles problem-dependent characteristics	[12]
Abdel-Khalek et al.	Segmentation	The approach effectively boosts the entropy and produces better picture division quality contrasted with the old technique	[13]
Maulik	Medical imagesegmentation	Equipped for dealing with colossal, convoluted, and multimodal search spaces. The joining of area information of images brings considerable adaptability in the segmentation	[14]
Ahmed et al.	Sports scenesunderstanding, monitoring, and traffic recognition	The proposed method producedprevalent article acknowledgment consequences of 86.1% when contrasted with other methods	[39]
Khan et al.	Segmentation and classification	Better execution as far as exactnessand execution time	[40]

with an underlying estimate arrangement at a high temperature at first, and afterward steadily chilling off. A new solution by this algorithm is acknowledged whether it is helpful; else, it is utilized as a likelihood-based methodology, which makes it conceivable to escape from any nearby arrangement. It is perceived that when the framework is chilled off leisurely enough that the worldwide yield can be produced by this method.[8,15]

According to Liu and Yang[16] "Image segmentation can partition a unique picture into some homogeneous districts. In this paper, they have proposed a multiresolution color image segmentation (MCIS) algorithm. The methodology is an unwinding cycle that combines the MAP (maximum *a posteriori*) gauge of the division. For the multiresolution framework, they have utilized the quadtree structure, and for the parting and combining, they have utilized a SA procedure to limit an energy capacity and which thusly, augments the MAP gauge. The multiresolution plot helps them to empower the utilization of various divergence measures at various goal levels. Their proposed calculation is noise resistant. Furthermore, for the worldwide bunching data of the picture, the scale-space filter (SSF) as the initial step is utilized. Exploratory consequences of the blended and genuine pictures as exceptionally viable and valuable. Eventually, the paper has another assessment rule which is proposed and created by the creator."

According to Bhandarkar and Zhang[17] "Image segmentation means it is depicted as a course of heterogeneous and homogeneous by which a crude info picture is divided into various locales which are homogeneous and any two neighboring areas are heterogeneous by the association.

A divided picture is the most elevated area autonomous reflection of a given picture which will likewise be our district of interest. They have taken the picture division issue as one of the combinatorial advancement. Genetic algorithm, here combined with SA. A hybrid evolutionary combination has shown better results as compared with canonical genetic algorithm."

10.2.3 PARTICLE SWARM OPTIMIZATION

According to Kennedy and Eberhart[20] "In this paper, strategy is presented for improvement. This strategy was found through a social model; accordingly, the calculation remains without figurative help in this paper.

Here they have portrayed as far as its antecedents, then, at that point, they have momentarily assessed the stages. Then, at that point of final execution, one paradigm is talked about in further detail, and tests are performed effectively. Then, at that point, they have expounded on more clear ties which are with its artificial life in general, and with bird flocking fish schooling and swarming theory. The proposed calculation is likewise identified with developmental calculation. Then, at that point, their connections are evaluated here without limit."

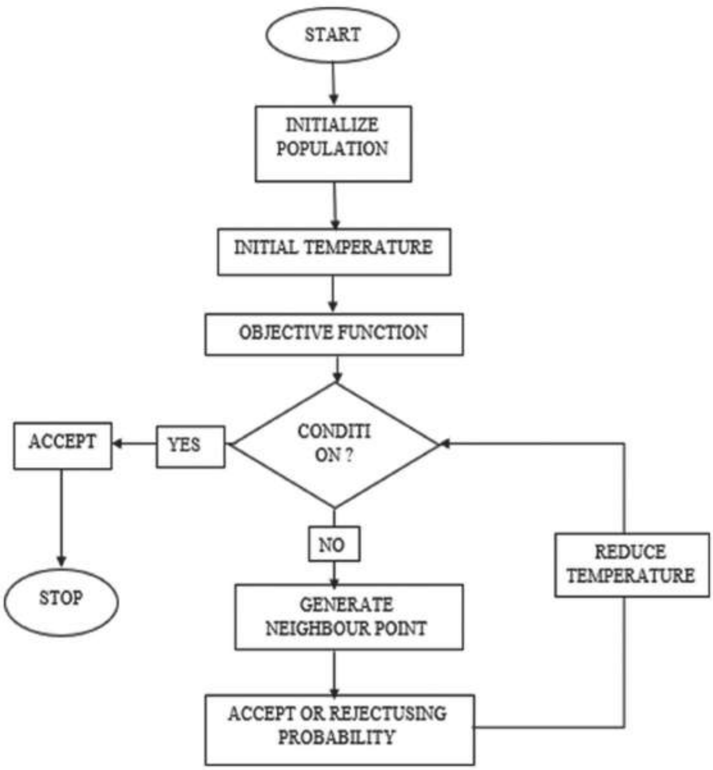

FIGURE 10.3 Flow chart for simulated annealing.

According to Eberhart and Kennedy,[19] "A new enhancer, the algorithm looks at that what changes in the worldview mean for the number of iterations required an error criterion and analyzes the recurrence with which models relentlessly a nonglobal arrangement."

TABLE 10.2 Review of Simulated Annealing in Image Segmentation Domain.

Author	Application	Advantages	Reference
Kirkpatrick et al.	Problem Optimization,Statistical Mechanics, Designof Computers, Placement problem, wiring problem	Straightforward, easilyextended to new problems, gives a structure to the enhancement of the huge and complex system	[15]
	Traveling Salesmen problem	Associationwith statistical perspectives new data and gives a new point of view on old optimization questions andstrategies	
Liu and Yang	Segmentation	Need not bother with deduced data of the picture and is verycommotion safe. Performs well on both synthesized and genuine pictures	[16]
Bhandarkar andZhang	Image segmentation	A hybrid algorithm works much better than the GA (genetic algorithm) in visual quality and execution time	[17]
Cordeiro et al.	Biomedical image	Ready to manage with complex non-characterized boundaries, it does not need extra human intercession once dubious sore regions areas of now clinically, better qualitative segmentation	[18]

According to Ghamisi et al.,[21] “Have gone through a thresholding method which multilevel is presented for the division of multispectral pictures. The technique depends on fractional order Darwinian particle swarm optimization (FODPSO) which goes through the many multitudes of test arrangements that might exist for a period being. Likewise, they have additionally delivered another characterization approach utilizing SVM and FODPSO. Furthermore, the results checks the technique can enhance results got with SVM as far as accuracies.”

According to Ghamisi et al.,[22] “The other molecule particle swarm optimization-based adaptive interval type-2 intuitionistic fuzzy C-means clustering algorithm (A-PSO-IT2IFCM) delivered by them and afterward they applied this proposed strategy to shading picture division. To manage the vulnerability, a target work is built by the intuitionistic fuzzy information. Then, at that point, they additionally plan another PSO plan to improve fuzzified and bunch focuses then again.”

According to Anderson,[23] “In this paper, the author is focused upon segmentation utilizing PSO-improved deep and clustering troupe models with enhanced hyper-parameters and cluster centroids. The proposed PSO model by this writer utilizes SA, Levy distribution, as well as helix, particle swarm optimization (PSO), and DE activities with twisting pursuit boundaries to build broadening which helps in various conditions by this methodology, henceforth increment its adaptability. Here each molecule in activities follows two remote leaders as portrayed by the creator at the same time in a winding manner to keep away from stagnation.”

According to Ahilan et al.,[24] “Author of this paper used (PSO) and its variations like Darwinian PSO and Fractional Order Darwinian PSO for the staggered thresholding of clinical pictures. They have proposed order and mixing expectation lossless calculation for picture arrangement in the clinical field. Their proposed approach beats the issues of the traditional Otsu calculation. FODPSO gives productive outcomes for fitness, PSNR, and MSE esteems from others.”

10.2.4 ARTIFICIAL BEE COLONY (ABC)

This Algorithm was developed in 2005 by Dervis Kara Boga. It is a foraging behavior of a honeybee swarm. And from this, a new algorithm simulating the foraging behavior of real honeybees is described, which is observed to be exceptionally helpful in tackling multidimensional and multimodal improvement issues. ABC algorithm comprises employed bees, on lookers bees, and scouts bees groups. Its benefits are straightforwardness, versatility, power, capability to examine neighborhood game plans, capability to deal with target cost. The main half is of the artificial bees and the second 50% of the onlookers.[25]

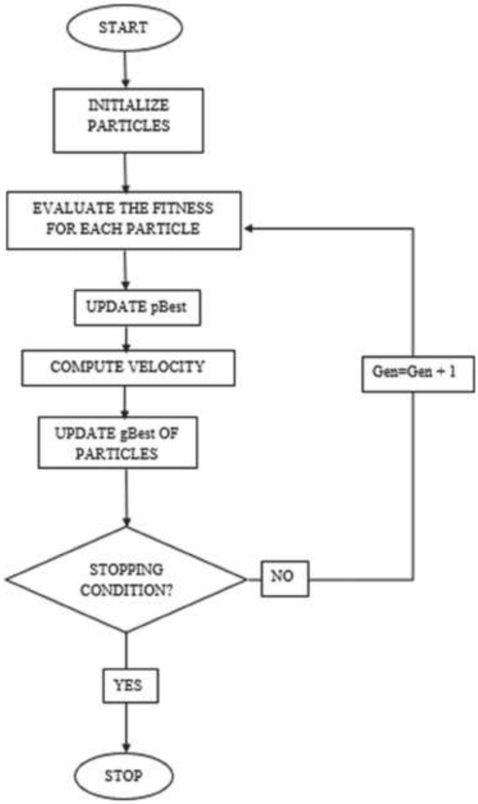

FIGURE 10.4 Flow chart for particle swarm optimization.

According to Horng,[26] "In this paper, the creator has fostered another multilevel maximum entropy thresholding dependent on the ABC algorithm: the most extreme entropy-based artificial bee colony thresholding (MEABCT) method. Then, at that point, further, he has contrasted four unique techniques with his proposed one. The PSO, algorithm (HCOCLPSO), the Fast Otsu's technique, and the honeybee mating optimization (HBMO) were the calculations utilized for examination.

TABLE 10.3 Review of PSO in Image Segmentation Domain.

Author	Application	Advantages	Reference
Kennedy and Eberhart	Communication, information extraction,combinatorial issues,signals handling, andenergy structures	PSO is reliant upon the bits of knowledge that should be dispatched on the scientific examination and the designing portion, estimation in the PSO,which is considered at its fundamentals	[19]
Kennedy and Eberhart	Application in social psychologist and electrical engineering	Simple and robust requires just crude numerical administrators and is computationally economical for memory prerequisites and speed	[20]
Ghamisi et al.	Segmentation for the real-time deployment and distributedrestriction of sensor hubs, disastermonitoring, and combat zoneobservation	The approach is more robust andhas a higher potential for tracking down the ideal arrangement of limits with additional between-class difference significantly quicker,especially for higher division levels and for pictures with a wide arrangement of forces. The new approach with SVM is proposed to improve the SVM as far as arrangement correctness. due to the low computational complexity	[21]
Tan et al.	Image segmentationapplications	Shows impressive performance concerning hyper-boundaryadjusting and bunch centroid improvement for base portions produces assorted base evaluators with extraordinary variety	[23]
Ahilan et al	Telemedicine applications, segmentation inremote sensing, segmentation of ultrasound images, multilevel thresholdingapplication in abdomen CT medicalimages	Efficient for fitness, peak signal-to- noise ratio and mean square error esteems, fractional coefficient favors a more significant level of investigation subsequentlyguaranteeing the worldwide arrangement	[24]
Zhao et al.	Color imagesegmentation	Can adaptively decide the bunchhabitats and acquire great division output	[22]

Also, the outcomes verifications that model was utilized to look for to search for multiple thresholds and were the calculation utilized for examination. Also, the outcomes verifications that model was utilized to look for a MEABCT calculation is more limited than that of the other four techniques."

According to Ma et al.,[27] "Finding optimal Threshold is a search process, and they have utilized the ABC algorithm for enhancing the arrangement. Artificial bee colonies (ABCs) are utilized for the quick segmentation speed and discovered to be an exceptional assembly execution. The division nature of their methodology can be better utilized by two-dimensional gray entropy, by which noise vanishes, and the most needed data of edge and surface are yet unchanged. After exploratory, it was discovered that their strategy is better than GA or Artificial Fish Swarm-based techniques as far as precision, time, and assembly speed is a concern."

According to Bose and Mali[28] "A segmentation algorithm, where they join the course of ABC and the well-known (FCM) and named it FABC. In FABC, the fuzzy C-means participation capacity to look through the ideal group communities utilizing enhancement calculation ABC. FABC is observed to be a more effective enhancement procedure. FABC as a combo of ABC and FCM defeats the disadvantages of famous fuzzy C-means (FCM) concerning this it does not rely upon introductory group habitats and it performs a lot of well as far as convergency, time intricacy, strength, and exactness. What's more, FABC turns out to be more productive and considerably more successful as it enjoys the benefit of randomized qualities of ABC for the of the group communities."

According to Menon and Ramakrishnan[29] "The MRI Brain Image division strategy on ABC calculation and fuzzy C-means (FCM) calculation is observed to be exceptionally compelling. To build the division nature of their technique they utilized two-dimensional gray entropy, by which noise totally vanishes, and the most valuable data about edge and surface is as yet unchanged. The worth is looked through utilizing a limit strategy. Then, at that point, for the ideal edge, the ABC calculation is utilized by them. To get effective wellness work, the first picture is disintegrated by discrete wavelet transforms. Then, at that point, they have played out a commotion decrease to the estimation picture, which delivers a sifted picture remade with low-recurrence parts. The FCM calculation proposed here is utilized for the bunching of the sectioned picture to recognize the tumor."

A blended metaheuristic methodology division by joining both the ABC calculation and sine-cosine calculation (SCA) called ABCSCA. ABCSCA technique uses ABC for enhancement and to confine the pursuit region. Starting there, the SCA calculation uses the yield of ABC to get the global solution, which tends to the ideal worth. It beats various calculations as far as execution measures, such as PSNR and the SSIM.[30]

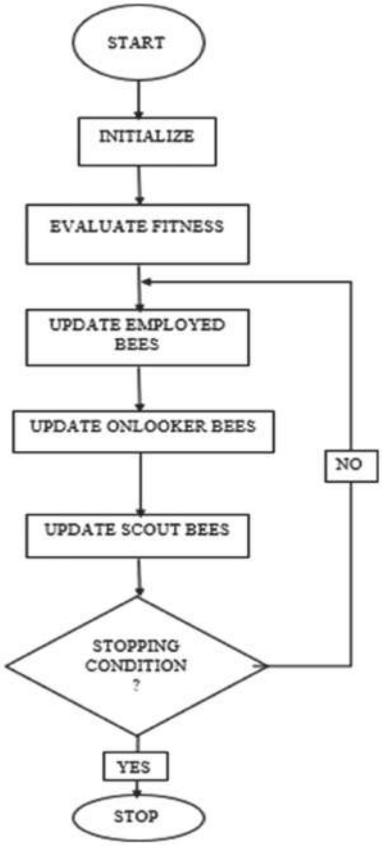

FIGURE 10.5 Flow chart for artificial bee colony (ABC).

10.3 DATASET

Selecting a dataset is a very important part of verifying any algorithm or method. A good dataset increases method validity and its output. Some of

TABLE 10.4 Review of Artificial Bee Colony in Image Segmentation Domain.

Author	Application	Advantages	Reference
KaraBoga (2005)	Optimization, Scheduling, Bioinformatics, Clustering of the picture, pictureprocessing, and division, and Engineeringapplications	Straightforwardness, flexibility, power,capable to examine neighborhood,capable to deal with target cost	[25]
Horng	Segmenting the Images	Can search for multiple thresholds, computation time is shorter	[26]
Ma et al.	Synthetic apertureradar segmentation	Quick division technique for SAR pictures, this strategy is better than old algorithms-based strategies as far as division precision, time, and convergence speed is considered	[27]
Bose and Mani	Gray imagessegmentation	It does not rely upon the decision of beginning cluster centers, and it performs better as far as convergence, time, robustness, and correctness are concerned. The adequacy of FABC (fuzzy-based ABC) is shown by both quantitative and subjective measures	[28]
Menon and Ramakrishnan	Tumor deductionfrom MRI images	Fast segmentation of MRI, effective fitness function to develop the segmentation, low noise content. This method detects the tumor and gives its intensity	[29]
Ewees et al.	Image segmentation	The algorithm gives the best outcomes in the low values of the thresholds based on PSNR and SSIM values, beaten different techniques by 41% and 39% as far as PSNR and SSIM, are concerned	[30]

the common challenges faced by segmentation are intra-class variation, inter-class similarities, fuzzy backgrounds, and uneven lighting conditions. Benchmark datasets contain a variety of images and are always good choices for analysis.

Some of the challenges are mentioned here: Intra-class variation:

- **Intra-class variation:** Issues in this is that the region of interest is in several different forms which means different images of the same type showing variation, which frequently makes the segmentation technique troublesome. That is the reason why a segmentation strategy ought to be invariant to such sorts of varieties. Examples include tables that are shown in different shapes, person face detection, etc.
- **Inter-class similarities:** Another main issue in the field of segmentation is image similarity that occurs between different images of a different class, which is referred to as inter-class similarity, which often makes the process difficult to handle. Therefore, it should be invariant to such sorts of changes. Examples include modality-based medical image classification, etc.
- **Fuzzy background:** Image segmentation for the picture with fuzzy background is viewed as a major test. Segmenting a picture for the necessary district of interest might blend with the complex environment and limitations. Examples include fishes with a boggling foundation in water thought about a major test, picture which compares to H&E-stained breast cancer, etc.
- **Uneven lighting condition:** It is among the main sources that can influence the division result. It is regularly created during the catching of a picture and affects pixels. The essential drivers for the aggravation of uneven lighting are the ROI cannot be disconnected from the shadows of different objects, the light may be temperamental, and the object to be detected is very large which therefore creates an uneven light distribution. It may therefore sometimes generate a false segmentation result.

Some open-source datasets:

- **MRI dataset:** This dataset provides us with a segmented image of MRI and it also has manual FLAIR abnormality data masks.[36]
- **Chest X-ray-pneumonia:** This dataset is consisting of 5863 images and is divided its dataset into test training and Val folders.[32]

- **Cell-images-for-detecting-malaria:** This dataset consists of 27,558 images and is divided into two folders parasitized and uninfected.[33]
- **Face-detection-in-images:** This dataset consists of 500 images with 1100 faces physically labeled via a bounding box to identify them. Faces are marked with boxes.[34]
- **Flower-color-images:** It is a collection of fragments of photos of flowers. Here images reflect the flowering features of plant species.[35]
- **Sky open-source dataset:** It comprises pictures taken from the Airplanes Side. Altogether, there are 60 pictures alongside ground truth.[37]

10.4 CONCLUSION

In conclusion, four nature-inspired metaheuristic algorithms were reviewed: genetic algorithm (GA), SA, PSO, and ABC optimization (ABC). Metaheuristic tries to improve the constraints of old segmentation algorithms. Metaheuristic is a global optimizer that tries to get the global value and avoid stagnation by using exploration and exploitation property. This improves the performance for segmentation and improves the quality. The GA has been inspired by evolutionary heuristics, PSO, and ABC have been inspired by swarm intelligence heuristics, and SA has been inspired by physical-based heuristics. However, the existing model fails to maintain a balance between exploitation and exploration.

KEYWORDS

- **image segmentation**
- **metaheuristic**
- **genetic algorithm**
- **simulated annealing**
- **particle swarm algorithm**
- **artificial bee colony optimization**
- **benchmark datasets**

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PART III

Advanced Machine Learning

CHAPTER 11

A COMPUTER VISION USE CASE: DETECTING THE CHANGES IN AMAZON RAINFOREST OVER TIME

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ABSTRACT

The widespread availability of high-resolution satellite images and recent advances in the field of computer vision has opened a lot of possibilities in terms of monitoring and managing climate risk. In this chapter, we demonstrate how computer vision can be used to identify the changes in Amazon Rainforest in Brazil using satellite imagery at two different points in time using the metrics of the Structural Similarity Index Measure (SSIM). As a result, from the SSIM index of 0.7368 and by analyzing the resultant difference image it can be concluded that there has been a significant shift in the Amazon Rainforest from year 2000 to 2021 that needs to be closely looked at to avoid any future damage. This study demonstrates how using a simple implementation of principles of computer vision and efficient open-source libraries can be used to detect the difference between two images in the most efficient way. This implementation with some modifications and enhancements can be applied in many domains such as fraud detection, medical diagnosis, assessing the impact of natural disasters, increased activities on borders, changes

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in land classification, and many more. The application in monitoring deforestation as demonstrated in this chapter alone has multiple utilities including but not limited to avoiding/controlling wildfires, identifying the areas for new plantations, avoiding some serious climate risks, preventing floods and droughts in the long term, and saving some wildlife from extinction.

11.1 INTRODUCTION

While the idea of computer vision has been around for a long time, it never got much traction due to a lot of constraints in terms of data availability as well as tools used for analyzing image data. However, with advances in technology, the availability of large amounts of data and development of the state-of-the-art data analysis tools, interest in the field of computer vision is growing across industries in recent years. People and organizations alike have realized the potential of computer vision in solving modern world problems with minimum resources. One such area where computer vision is supposed to play a pivotal role is in managing climate risk. The rapid pace of change in land use and in turn deforestation are identified as the key reasons behind the drastic climate changes we are experiencing or will experience in the future. Due to a lack of proper monitoring and accountability at international levels, it is often a case that the activities of deforestation go unnoticed until they reach advanced stages and cause some serious damage.

If we want to control such activities, it is very important to actively track our large forests and identify any major activities of deforestation by people, organizations, and governments across the world as soon as possible and take corrective action to stop them. The large amount of satellite imagery made available to the public in recent years amalgamated with advances in computer vision provides promising prospects to actively monitor the forests and track changes in them using various change detection techniques.

Amazon Rainforest is one of the largest forests in the world and is popularly called as “lungs of the earth” as by some estimates it produces nearly 20% of the world’s oxygen. This use case demonstrates how computer vision can be used to identify the changes in Amazon Rainforest in Brazil using satellite imagery at two different points in time.

11.2 LITERATURE SURVEY

Analyzing the earth's surface using the high spectral imagery from satellites has created a lot of traction in recent years. The satellites such as Sentinel and Landsat continuously monitor and take pictures of almost every corner of the earth every few days. What is more exciting is that this data is made available to the public almost free of cost in most circumstances. This enables a large number of people and more importantly, the researchers to use these images in creative ways that were never thought of before. There is a lot of research done in the domain of remote sensing and computer vision to use aerial/satellite images for effective land use and land cover classification.[2,3,9,10] A recent research paper[14] aptly summarizes various techniques used for change detection and challenges in their implementation. Advanced algorithms such as Convolutional Neural Networks with their many forms provide extremely high accuracy[11,22] in identifying the correct land use category or image surface texture analysis[1] and in many other remote sensing applications.

Computer vision and related models have been applied successfully in image change detection too. Goyette et al.[7] demonstrates how computer vision can be effectively applied in change detection in a video dataset. Radke et al.[15] has conducted a survey of different image change detection algorithms and their efficacy. Bosc et al.[4] has developed a system for automatic change detection in MRI scans to study the evolution of diseases across time. Fang et al.[5] has adopted advanced neural networks in detecting various changes in the driving environment to assist the drivers. Gu et al.[8] has developed a change detection method based on Markov random field specific to remote sensing images. There are multiple approaches adopted by researchers to identify the pixel that significantly differs in the images of the same objects taken at different times. The approaches include but are not limited to a simple differencing model,[17] shadow and shading models,[20] fuzzy clustering[6], and an unsupervised learning model.[13]

11.3 EXPERIMENTAL METHODS AND MATERIALS

11.3.1 DATA COLLECTION

For the purpose of this use case, we downloaded two images of the exact same location of the Amazon Rainforest taken in the years 2000 and 2021.

The images are downloaded from Google Earth Pro which is available free of cost for public use. To ensure high image quality, the images with a 4K UHD resolution of 3840 × 2160 were selected for the study. The location coordinates of the image are as listed in Table 11.1.

TABLE 11.1 Location Coordinates of the Image.

Location coordinates of the image (degrees, minutes, and seconds (DMS) scale)	
South	S 3°45′ to S 3° 57′
West	W 62° 33′ to W 61° 33′

Note that the area chosen is intentionally from the bordering area of Amazon Rainforest because the probability of deforestation is high in the bordering area of the forest.

The final images obtained are shown in Figures 11.1 and 11.2:

FIGURE 11.1 Amazon Rainforest–Year 2000.

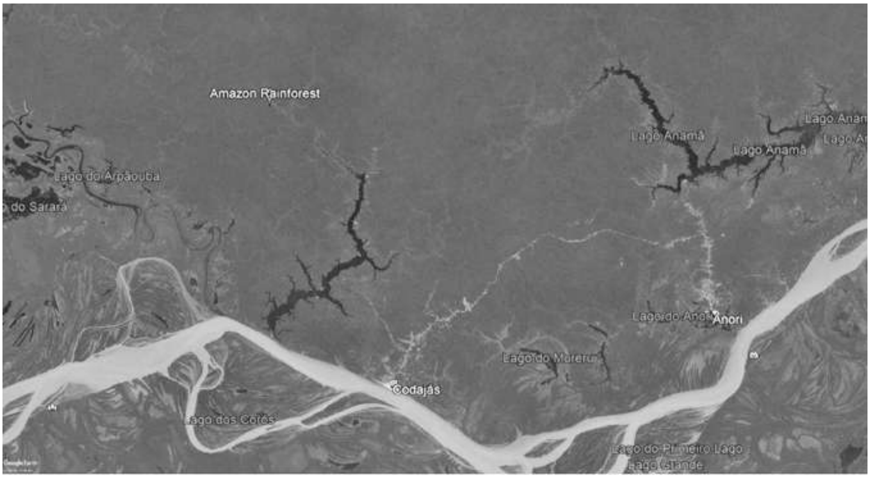

FIGURE 11.2 Amazon Rainforest – Year 2021.

11.3.2 TOOLS USED

The entire change detection process was performed in Python. OpenCV's cv2 package was extensively used for all the analysis. OpenCV (Open source Computer Vision) is a collection of methods mainly developed for computer vision-related tasks. OpenCV is extremely popular in the computer vision community and is being used for many pathbreaking research activities. One of the biggest advantages of using OpenCV methods is its support for NumPy operations. This makes communicating with OpenCV extremely smooth and efficient.

11.3.3 DATA PROCESSING

The images downloaded for analysis were read into Python environment using the image read (imread) method of OpenCV's cv2 package. The imread () function returns the image in the form of a Numpy array with structure (height × width × colors) in Blue–Green–Red (BGR) format. The dimensions of the images read are 2160 × 3840 × 3.

The images are then converted into grayscale images using the COLOR_BGR2GRAY option in the cvtColor method of the cv2 package. Doing this converts all the colors in the image to different shades of gray which simplifies the image comparison process in terms of noise reduction, processing time, and resources required for comparison without losing any information required for our purpose. The grayscale images generated are shown in Figures 11.3 and 11.4.

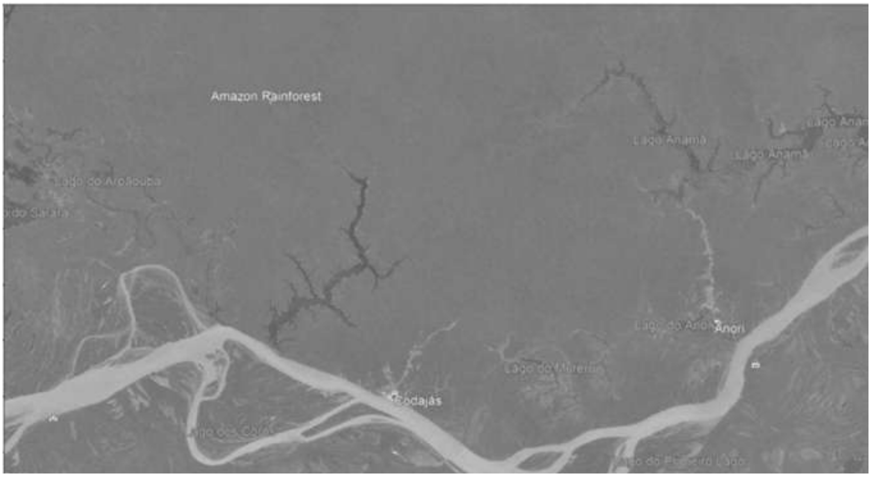

FIGURE 11.3 Amazon Rrainforest: Grayscale Image – Year 2000.

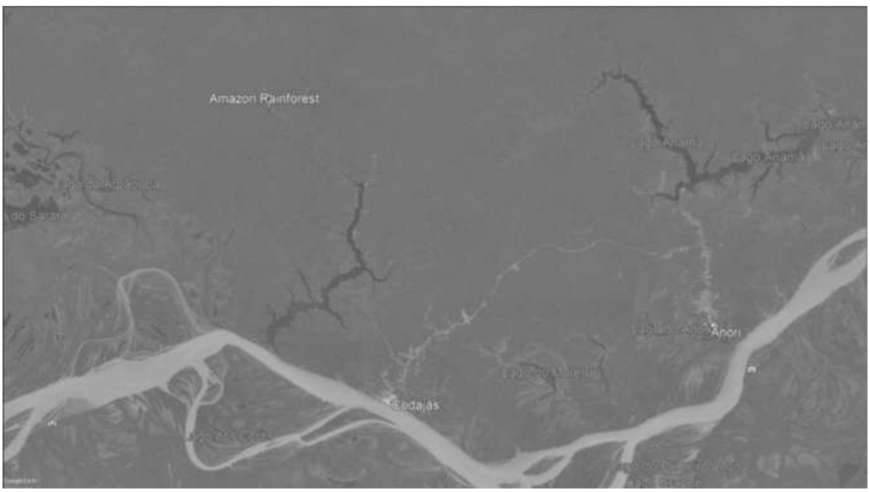

FIGURE 11.4 Amazon Rrainforest: Grayscale Image – Year 2021.

11.3.4 APPROACH USED FOR CHANGE DETECTION

To identify the difference between the two images, Structural Similarity Index Measure (SSIM) is used. The SSIM was introduced in the research paper named "Image quality assessment: from error visibility to structural similarity".[19] Since then, it has become a popular index in the image processing and computer vision community with many research papers using it for image comparison. It is cited by more than 15,000 papers and 250 patents.

Historically, researchers have explored many techniques of change detection such as differencing,[18] rationing,[16] principle component analysis,[12] Slow Feature Analsis[21] and many more. The scope of this chapter is to demonstrate change detection using SSIM only.

If we break down the idea of change detection, in most granular and simplistic terms we are trying to find if the pixel in an image differs from the same pixel in the image with which it is being compared. In the more complex implementation, we take this idea a little further to check holistically if the pixel along with the group of pixels around it follows the same pattern/behavior based on the context of underlying images. The pixels in both images could differ in multiple ways depending on features such as brightness, luminance, angle, contrast, intensity, reflections, shadows, and many more. The severity of the difference between the pixel values could vary on the scale between between "extremely small" difference and "extremely large" difference. Most methods of change detection acknowledge this subjectivity and allow manual setting of thresholds to decide whether the change is significant or not. Depending upon the context in which change detection is being performed, the changes can be highlighted or ignored. In this case, we will be showing all the structural differences between the images irrespective of their severity.

SSIM follows the principle of looking at the group of pixels for the structural similarity between two images. By doing so, it inherently takes care of the problem of noise due to small differences in the images up to a large extent. SSIM measures the similarity between two images based on three key parameters:

1. Luminance
2. Contrast
3. Structure

Suppose we have two images, A and B, the formula for SSIM as proposed in the study by Wang et al.[19] is as shown below:

$$SSIM(A,B) = \frac{(2 * \mu_A * \mu_B + C_1)(2 * \sigma_{AB} + C_2)}{(\mu_A^2 + \mu_B^2 + C_1)(\sigma_A^2 + \sigma_B^2 + C_2)}$$

Where,

μ_A is the average value of A

μ_B is the average value of B

σ_A^2 is the variance of A

σ_B^2 is the variance of B

σ_{AB} is the covariance of A and B

$C_1 = (K_1 * L)^2$ and $C_2 = (K_2 * L)^2$ are ratio stabilizers

L is the range of pixel values (2 bits per pixel –1)

$K_1 = 0.01$ and $K_2 = 0.03$

The derivation of this equation is beyond the scope of this chapter. However, readers can find more details on it in an original research paper by Wang et al.[19]

The value of SSIM can lie between 0 and 1 where 0 indicates no structural similarity between the images at all whereas 1 indicates the exact same images. The Scikit-image library in Python already has a method named "structural_similarity" that computes the SSIM value given two images as input. The same is used for the change detection in the images of the Amazon Rainforest selected during the data collection step. The structural_similarity function returns the SSIM score as well as the image highlighting the difference in the input images.

11.3.5 CODE

```
import numpy as np
from skimage.metrics import structural_similarity
import cv2
import matplotlib.pyplot as plt
def plot_image(image, factor=1):
 figure = plt.subplots(nrows=1, ncols=1, figsize=(15, 15))
 if np.issubdtype(image.dtype, np.floating):
 plt.imshow(image * factor)
 else:
 plt.imshow(image)
# load input images
image2000 = cv2.imread('C:/Users/patil/Desktop/GANESH/Book Chapter/
Change detector/Images/Amazon Rainforest/Amazon rainforest 2000.jpg')
plot_image(image2000)
print(image2000.shape)
image2021 = cv2.imread('C:/Users/patil/Desktop/GANESH/Book Chapter/
Change detector/Images/Amazon Rainforest/Amazon rainforest 2021.jpg')
plot_image(image2021)
print(image2021.shape)
# converting images to grayscale
grayA = cv2.cvtColor(image2000, cv2.COLOR_BGR2GRAY)
grayB = cv2.cvtColor(image2021, cv2.COLOR_BGR2GRAY)
# Find structural similarity using structural_similarity method
(score, diff) = structural_similarity(grayA, grayB, full=True)
diff = (diff * 255).astype("uint8")
print(“SSIM:”, score)
# Find threshold difference image
thresh = cv2.threshold(diff, 0, 255, cv2.THRESH_BINARY_INV | cv2.
THRESH_OTSU)[1]
thresh1 = cv2.cvtColor(thresh, cv2.COLOR_BGR2RGB)
plot_image(thresh1)
```

11.4 RESULTS AND DISCUSSION

The SSIM and the difference image obtained using the above approach are as shown in Figure 11.5 and Table 11.2.

TABLE 11.2 SSIM Final Result.

Structural Similarity Index (SSIM)	0.7368

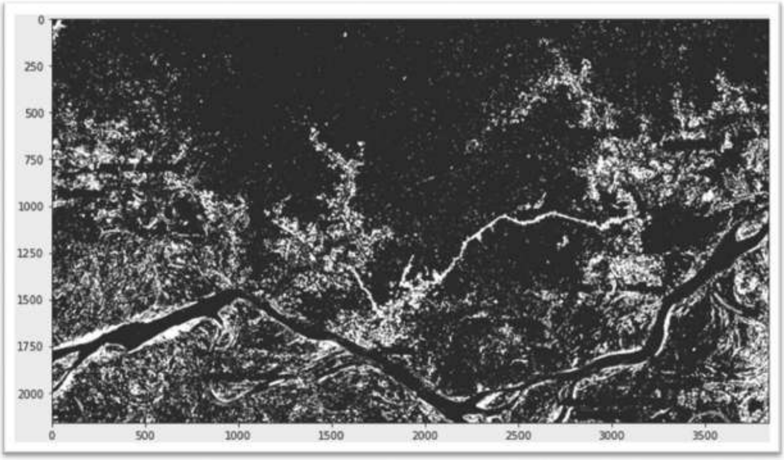

FIGURE 11.5 Difference between two input images of Amazon Rainforest.

The SSIM of 0.7368 indicates that there is a significant difference between the two images. The difference image provides the areas of difference in the form of white areas on the black canvas. Visually inspecting both the input images confirms that the difference image very efficiently captures the key areas of difference as shown in Figure 11.5.

11.5 CONCLUSIONS

In this chapter, we presented a case study of applications of computer vision in image change detection to identify the deforestation of the Amazon Rainforest in Brazil. To start with we discussed the concerns around proper monitoring of forests and its impact. Identifying this as a severe concern and opportunities introduced by recent advancements in the area of computer vision, an approach for using computer vision in identifying deforestation was explored. A detailed literature survey introduces readers to the depth and breadth of the past and present research being conducted in the areas of using satellite imagery and computer vision to solve some real-world problems including the pathbreaking research in the area of image change detection.

Subsequently, we introduced the concept of the SSIM used widely in the industry to measure the similarity between the two images by analyzing the structural similarities between them. By implementing a Python code that uses OpenCV's cv2 library and structural similarities method from the Scikit-image library we found out the difference in two satellite images of Amazon Rainforest in 2000 and 2021.

As a result, from the SSIM index of 0.7368 and by analyzing the difference image it can be concluded that there has been a significant shift in the Amazon Rainforest from year 2000 to 2021 that needs to be closely looked at to avoid any future damage. This demonstrates how using a simple implementation of principles of computer vision and efficient open-source libraries could help us in identifying and resolving some critical issues at a speed and accuracy that was never thought of before.

The use case discussed in this chapter demonstrates how computer vision can be used to detect the difference between two images in the most efficient way. This implementation with some modifications and enhancements can be applied in many domains such as fraud detection, medical diagnosis, assessing the impact of natural disasters, increased activities on borders, changes in land classification, and many more. The application in monitoring deforestation as demonstrated in this chapter alone has multiple utilities including but not limited to avoiding/controlling wildfires, identifying the areas for new plantations, avoiding some serious climate risks, preventing floods and droughts in the long term, and saving some wildlife from extinction.

KEYWORDS

- **computer vision**
- **Amazon Rainforest**
- **change detection**
- **structural similarity index**

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CHAPTER 12

USING CNN AND IMAGE PROCESSING APPROACHES IN THE PRESERVATION OF SEA TURTLES

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ABSTRACT

The concern to restore and maintain a balance of the Earth with respect to its natural ecosystems has been enhanced in this last decade. Some of these concerns have focused on giving specific monitoring to animal species, to have and conserve them in good biological condition, as well as establishing support schemes to preserve the habitat and environment of animal species. Therefore, it is extremely important to monitor any change in natural ecosystems, no matter how small, to prevent it from having repercussions on human beings and the different forms of life. To maintain monitoring that affects the control of natural ecosystems and their animal species, it is necessary to manage a high scale of data information, as well as develop digital tools based on artificial intelligence, which allow carrying out the work of classifying the type of animal species and identify the animal individually by converging space-time information from diverse ecosystems. This chapter describes a new biometric recognition model which is based on a proposed convolutional neural network (CNN) using biometric information from sea turtle images, which makes it a

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non-invasive model and allows the classification of six species of sea turtles that have arrived in Mexico. Likewise, the biometric information obtained for each of the six species of sea turtle is described, through the optimized mathematical algorithms. In addition, the creation of a database with images is presented, whose content represents the phylogenetic information of the sea turtles of the different species that arrive in Mexico. The results generated for the recognition of sea turtles with the proposed biometric model, specifically for the six species that reach the coast of Mexico, are presented in the final part of the chapter, which has an accuracy of 87.90% in validation and 86.95% in testing. This means that the proposed model outperforms the artificial intelligence technique that is implemented with a red–green–blue (RGB) neural network architecture—AlexNet—which has an accuracy of 73% in validation and 74% in testing.

12.1 INTRODUCTION

Sea turtles are animal species considered to be bioindicators. Indeed, their study shows information on the state of ocean pollution because different chelonians are sensitive to various pollutants.[1,2] The Mexican coasts concentrate six of the seven species of sea turtles, unfortunately for various reasons such as incidental fishing, poaching, pollution, and climate,[3,4] three of these species are listed as critically endangered and the others as endangered (Official Mexican Standard NOM-059-SEMARNAT-2010). Therefore, measures are needed to conserve, repopulate, and preserve sea turtle species. Sea turtles also have an indirect impact on Mexico's economy, as they promote the sustainable development of commercially valuable marine species such as tuna, and marine crustaceans such as shrimp and lobster. In Mexico, the National Commission of Natural Protected Areas (CONANP) has categorized as Natural Protected Areas (NPA) several beaches where these animals nest and spawn.[5] One of the technological fields that has recently supported conservation and natural preservation is artificial intelligence.

Positively, artificial intelligence has solved complex problems such as the classification of species within a population in a relatively easy way. For example, work has been reported on the classification of animals based on phenotypic characteristics,[6] facial recognition in lemurs,[7] macaques,[8] gorillas, and chimpanzees.[9] These works demonstrate the advances of a new field known as animal biometrics[10] whose primary source of information is images and video acquired with digital cameras. Information in the form of biometric patterns is extracted from this multimedia content, which is defined as a non-invasive study technique for identification and classification. With the incentive to continue with the support in the activity of preservation of sea turtles and especially of the six species that arrive in Mexico and proposing novel solutions based on artificial intelligence, this research work presents a *technological development of non-invasive classification for sea turtle species*, this implementation and development is based on convolutional neural network (CNN) models using *images with patterns of morphological* body *characteristics* of these animals. This development allows increasing accuracy in the classification of sea turtle species.

12.2 DESCRIPTION OF SEA TURTLE SPECIES

The changes that sea turtles have undergone in their physical configuration over time have been minimal. Their main characteristic is that their body is protected by a rigid shell,[11] and the dorsal part of the shell is known as the carapace. These species that currently inhabit the seas are covered with keratinous scales of different thicknesses specifically on the face and on the carapace present an arrangement of scutes. The leatherback sea turtle (*Dermochelys coriacea*) has only minimal keratin coverage on the jaws and the carapace has small white spots and dorsal ridges all along the carapace. The current classification of sea turtles based on Carr, Pritchard, and Marquez[12–14] catalogs them in a superfamily called *chelonioidea* that groups the families *cheloniidae* (turtles with scutes) and the family *dermochelyidae* (turtles without scutes). This chapter describes the research work that contemplates the six species that are present in Mexico, while ignoring the flat sea turtle (*Natator depressus*) that is endemic to Australia, and the black sea turtle which is classified as a subspecies.

12.2.1 MORPHOLOGICAL TRAITS OF THE SIX SPECIES

The combination of different features such as the shape of the head, the number of scutes, and the configuration of the carapace provide the necessary information to know the species of any sea turtle, with the advantage that this is done visually. The carapace is made up of plates that form a generally symmetrical pattern. The dorsal sections are called vertebral scutes, where each is attached to two plates called lateral scutes. The perimetral edge of the carapace is divided into marginal scutes. The configuration of the pattern formed on the carapace is characteristic of each species, varying the number of scutes, their shape, and symmetry. Figure 12.1 shows how to number the scutes and their respective location.[13]

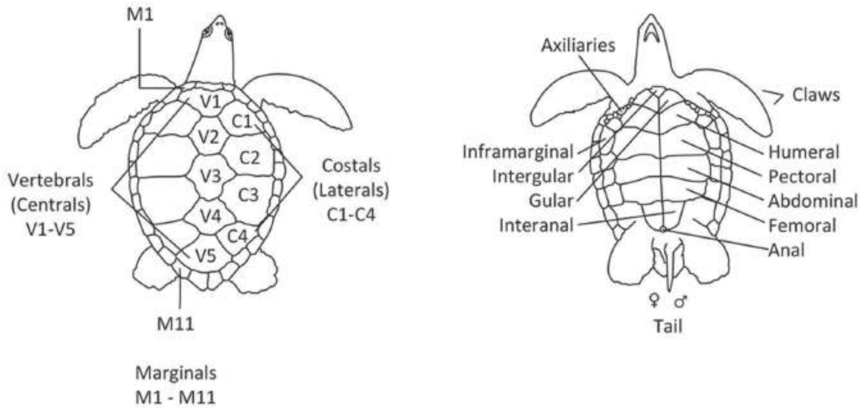

FIGURE 12.1 Morphological features of the carapace and plastron (ventral part).
Source: Adapted from Ref. [28].

Figure 12.2 shows a top view of the carapace configuration and at the same time an illustration of the head of the different chelonid species.[13]

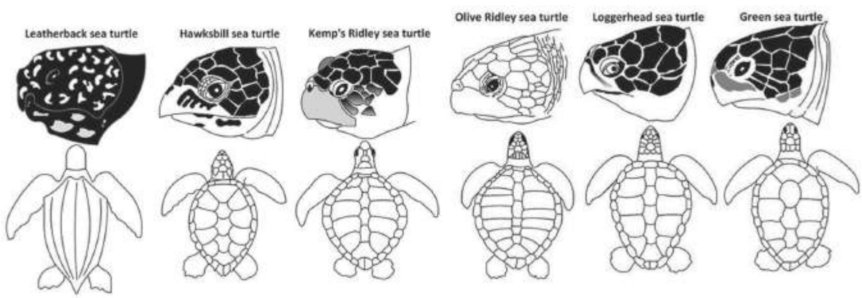

FIGURE 12.2 Sea turtles: visual description of six species.
Source: Adapted from Ref. [28].

The phylogenetic relationships between the species of the family *Cheloniidae* are very close, that is, the structure of the body of sea turtles, both the olive ridley and that of the Kemp's Ridley present a very high similarity. In the same way as sea turtles such as the hawksbill and the green, this feature of great similarity increases the complexity of the classification problem and can cause confusion in differentiating between these species, since visually they are very similar. In the opposite case is the leatherback sea turtle, which has unique characteristics that allow it to be classified without any problem.

12.2.2 DISTRIBUTION OF SEA TURTLES IN MEXICO

Mexico, like other countries, receives several species of sea turtles along its coastal areas, but it is the only one that receives the Kemp's Ridley species in mass, which is why conservation and preservation efforts are a necessity. Figure 12.3 shows the nesting beaches and common areas of concentration. Some of these areas are protected by the government and have been granted NPA status.

FIGURE 12.3 Map of sea turtle distribution in Mexico.

12.3 PROPOSED BIOMETRIC RECOGNITION MODEL

The term biometrics means a science that involves a statistical analysis of biological characteristics.[15] The various studies applied to humans led to the birth of animal biometrics which is defined as a set of quantitative information intended to represent and detect species, individuals, behaviors or morphological traits based on phenotypic characteristics.[10] That is, through the biometric information which comes from an animal, and which is contained in the coat, morphology, behavior, etc., it is possible to know the species to which they belong or to identify the individuals that make up a population. Biometric digital images can store all this information which contain the patterns of the carapace, face, and head.

Considering these particularities, animal biometrics systems started to use images mainly with biometric patterns coming from animals to classify and identify species or individuals. In 2013, Kuhl and Burghardt[10] showed a synthesis of the main components of an animal biometric system, as well as the flow of information.

This chapter presents the work where the following biometric recognition model is proposed, constituted with deep learning architectures to carry out the indexing and classification processes of sea turtles.

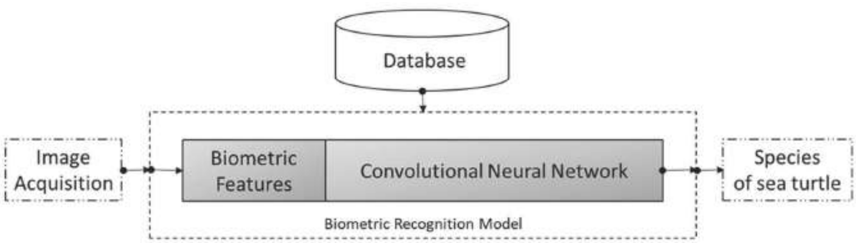

FIGURE 12.4 Proposed biometric recognition model.

12.3.1 FEATURE EXTRACTION IN BIOMETRIC IMAGES

According to the proposed biometric recognition system (Figure 12.4), in the first phase the first block is the extraction of features from digital images, both for storage in the database and for classification of a completely new image. A function defines a feature, said function is based on one or more measures, at the same time these measures represent a specific and quantifiable property that is related to the object.[16] These are usually visual characteristics such as shape, color, and texture of biometric patterns. These characteristics are integrated into a vector called a feature vector or descriptor, which is nothing more than organized data.

To obtain the biometric features of sea turtles, mathematical methods for computer vision tasks have been applied to this work. One of them is known as the *scale invariant feature transform* (SIFT). This SIFT method extracts characteristic points of an image and information around the coordinates of the points and then constructs a vector with this information. The commercial software "NaturePatternMatch" is based on this transform and has been used for sea turtle recognition.[17] Another mathematical method, *local binary pattern* (LBP), can create a very efficient local descriptor of the image content, where units of texture are obtained from the image.[18] It provides texture units that are invariant to luminosity. Regarding the extraction of contours from biometric pattern images, the descriptor was generated by the method called *histogram of oriented gradients* (HOG).[19] Regarding color, there is a color model based on the so-called CIE 1976 L*a*b* (this color space was created by the *Comission Internationale de L'Éclairage*) was used, which is very often cited in the literature because it achieves an approximation of the way humans perceive colors.[20] This biometric feature extraction information was applied to the sea turtle images and added to the abstract feature extraction phase proper to the mathematical operations of the CNN of the proposed biometric recognition system. The use of all these techniques provides a robust characterization to an image, as it supplies multiple descriptors to the same feature.

12.3.1.1 SCALE INVARIANT FEATURE TRANSFORM

Invariant features were calculated using the SIFT method,[21] which is constructed from scaling and rotation invariant minutiae in images and includes the information of the vicinity surrounding that minutiae. The SIFT transform in its first phase creates a scale–space by applying different Gaussian blurs using the convolution operation (eq 12.1):

$$L(x, y, \sigma) = G(x, y, \sigma) * I(x, y)$$

$$G(x, y, \sigma) = \frac{1}{2\pi\sigma^2} e^{\frac{-(x^2+y^2)}{2\sigma^2}} \tag{12.1}$$

where $I(x, y)$ is the original image and $G(x, y, \sigma)$ is a Gaussian function in two dimensions.

Another key aspect for scale invariance is the subsampling applied to consecutive octaves creating a pyramidal scale-space of Gaussian images (eq 12.2):

$$\begin{aligned} D(x,\ y,\ \sigma) &= L(x,\ y,\ k\sigma) - L(x,\ y,\ \sigma) \\ &= \left(G(x,\ y,\ k\sigma) - G(x,\ y,\ \sigma)\right) * I(x,\ y) \end{aligned} \tag{12.2}$$

Figure 12.5 shows the key points obtained in an image containing a hawksbill sea turtle, which form a pattern of edges and contours of the head and carapace. Despite changes to which the image may be exposed such as noise or geometric transformations, the coordinates of the points point to the same region of the image.

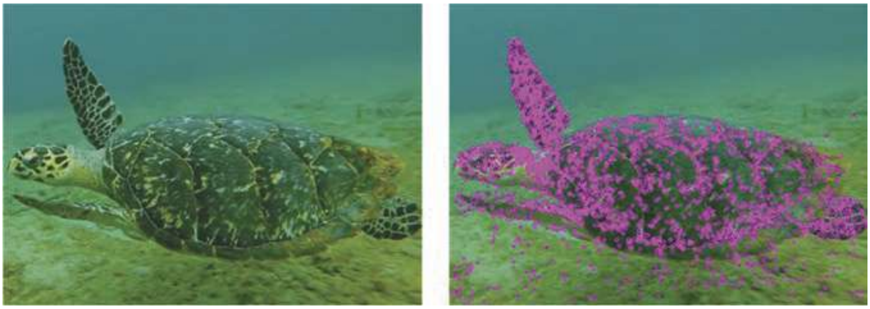

FIGURE 12.5 Distribution of points obtained with the SIFT transform.

The descriptor of a key point requires the calculation of its magnitude and orientation. The eqs 12.3 and 12.4 calculate these values using the smoothed images of the scale space. Each histogram has eight cells so that the final descriptor of each point acquires a length of 128 values.

$$m(x,y) = \sqrt{\left(L(x+1,y) - L(x-1,y)\right)^2 + \left(L(x,y+1) - L(x,y-1)\right)^2} \tag{12.3}$$

$$\theta(x,y) = arc\tan\left(\frac{L(x,y+1) - L(x,y-1)}{L(x+1,y) - L(x-1,y)}\right) \tag{12.4}$$

12.3.1.2 LOCAL BINARY PATTERN

The LBP method[22] uses a binary system to encode the texture characteristics present in a digital image. Texture relates to how the surface composition of an object is perceived. This texture information is represented with specific numerical values or texture units that are concentrated in the form of a feature vector for later use in object classification or semantic segmentation.

The coding of the intensities is done with the following equation:

$$y = \sum_{p=0}^{7} s\left(gp - gc\right)2^{p} \tag{12.5}$$

where

$$s\left(x\right) = \begin{cases} 1 & si\ x \geq 0 \\ 0 & otro \end{cases} \tag{12.6}$$

Figure 12.6 shows the texture feature map that gives the sensation of relief and different sections can be appreciated according to the semantic areas of the image: water, head, carapace, fin, sand, etc.

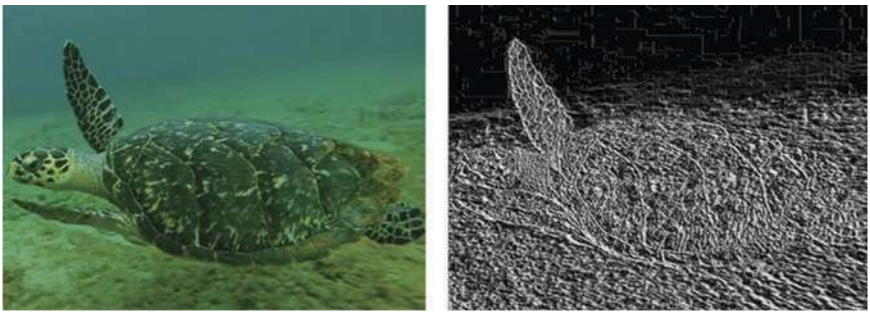

FIGURE 12.6 Feature map with texture units.

12.3.1.3 HISTOGRAM OF ORIENTED GRADIENTS

In the work developed by the authors Dalal and Triggs,[23] they established the mathematical method called HOG. This method extracts the gradient distribution in an image, the gradient involves the partial derivatives of a function as shown in eq 12.7:

$$\nabla I\left(x, y\right) = \begin{bmatrix} gx \\ gy \end{bmatrix} = \begin{bmatrix} \dfrac{\partial I\left(x, y\right)}{\partial x} \\ \dfrac{\partial I\left(x, y\right)}{\partial y} \end{bmatrix} \tag{12.7}$$

From them, a vector with magnitude and direction is obtained as follows:

$$m(x,y) = \sqrt{gx^2 + gy^2}$$

$$\theta(x,y) = arc\tan\frac{gy}{gx} \tag{12.8}$$

The map containing the magnitudes of the original image, as well as their respective angle, is shown in Figure 12.7. Note how it creates a perfectly defined outline of the body lines as well as the animal outline.

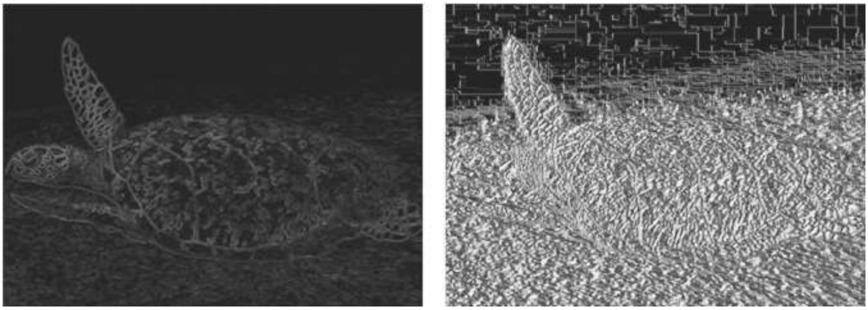

FIGURE 12.7 Gradient magnitudes and angles.

*12.3.1.4 COLOR SPACE: CIE 1976 L*A*B**

The representation of human visual perception is done with color spaces. One of the color spaces that is frequently used is the CIE 1976 L*a*b*, its main characteristic is based on the fact that the numerical values assigned to the colors present a consistent correlation with the colors that are perceived through of human visual perception.

12.3.2 CONVOLUTIONAL NEURAL NETWORK

The work carried out on the visual cortex by the authors Hubel and Wiesel has been used to support the understanding of the bioinspired model in which CNNs are built.[24] The potential of CNN networks was glimpsed in the year 2012 during the challenge: *ImageNet Large Scale Visual Recognition Challenge dataset*, known as (*ILSVRC*), for this challenge where the work presented by Alex Krizhevsky whit his network called

AlexNet, was tested with the ILSVRC-2012 database. AlexNet obtained the lowest classification test error for top-5 error rate with 16.4%. What caught the attention was the marked difference compared to the second place in the competition that used the average predictions of some Fisher vector classification methods computed from different types of densely sampled features and obtained 26.2%. In CNN networks, convolution operations are found in the first layers and are responsible for extracting visual abstract features automatically by means of trainable filters. The convolution masks and their parameters are initiated randomly and as the network learns it adjusts their values. The convolution layers are separated by subsampling layers called *pooling*, which take a region of the neighborhood and replace it with a statistical value. This helps ensure invariance to shifting, scaling, and distortion of the feature map. The last layers of the network are a multilayer perceptron classifier, called fully connected layer (FCL).[25]

In the architecture of an CNN network, a multi-input neuron performs the mathematical operation called dot product, which consists of performing the multiplication between each of the input values and its corresponding value *w* called weight. That is, the weights of the neuron are the filter parameters, and the input comes from a region of the image. The subregion of the image that reaches the inputs of the neuron is called the field of view, thus in the training stage the parameters are modified. It is important to mention that the entries are made up of more than one plane, which adds depth dimension to the operation. Figure 12.8 shows this process.

The output of the neuron that generates the new feature map is defined by the following equation:

$$a_s^m = f\left(\sum_{r=1}^{R} a_r^{m-1} * w_{r,s}^m + b_s^m\right) \tag{12.9}$$

where a represents the value of the feature of the output for the neuron marked with the index s, the index m represents the number of the layer, R is the number of inputs, which at the same time represent the outputs of what are called feature maps that correspond to the previous layer, and w represents the synaptic weights of the neuron between layers $m - 1$ and m. The last term b corresponds to the bias of the neuron. Regarding the activation function, it depends on the problem; however, the one that is widely used is the ReLU function. Figure 12.8 schematically represents the essential parts of a CNN.

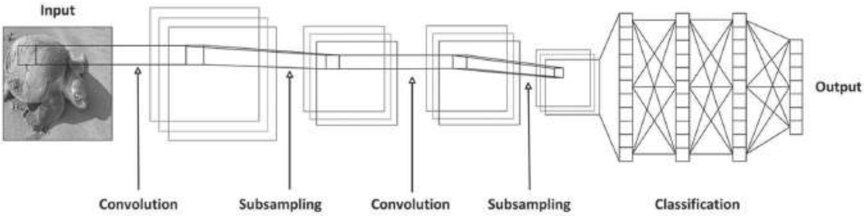

FIGURE 12.8 Convolutional neural network.

12.4 SEA TURTLE SPECIES DATABASE

The database is shown in Table 12.1, each class is made up of 1000 images and only the six species that are present in Mexico were considered. Another distinctive feature is the addition of an extra class with images from the CALTECH256 database that do not have any content related to sea turtles. The format of the images is Portable Network Graphics (PNG) with a resolution of 227 × 227 pixels.

TABLE 12.1 Sea Turtle Species Database.

	Class	Images
0	Loggerhead sea turtle	1000
1	Hawksbill sea turtle	1000
2	Olive Ridley sea Turtle	1000
3	Leatherback sea turtle	1000
4	Kemp's Ridley sea turtle	1000
5	Green sea turtle	1000
7	CALTECH256	1000
	Total	7000

The databases were divided as shown in Table 12.2.

TABLE 12.2 Training Specifications: Sea Turtle Species Database.

Training	Validation	Test
4090	1050	1050
700 per class	150 per class	150 per class

From some of the images found in the database are presented in Figure 12.9.

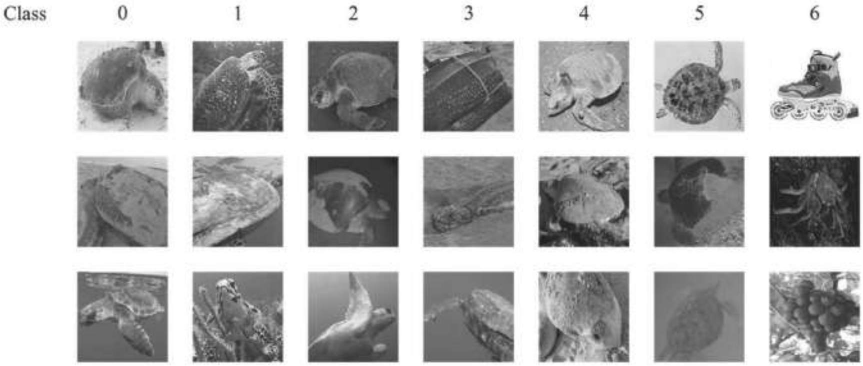

FIGURE 12.9 Samples of images held in the sea turtle species database.

12.5 PROPOSED SEA TURTLE SPECIES CLASSIFIER MODEL

With respect to the model for the classification of sea turtle species, this model is built with the specifications required for a CNN architecture, it is considered, by its nature, that the information presented at the input is made up of two dimensional.

The information of features obtained through the first block of the proposed classification model is composed of layers with different depths; in these layers, the extraction of features is carried out (see Figure 12.4), corresponding images were taken from the database of sea turtle species (Table 12.1). The different characteristics were extracted from each one using the mathematical methods described in Section 3.3.1.

The information obtained from the SIFT transform is based on the coordinates of the points of interest. Regarding the LBP features, the RGB images are transformed to images in grayscale space. Texture units are obtained according to eqs 12.5 and 12.6. Regarding the use of the CIE color space L*a*b* 1976, the decomposition of the RGB image into its R, G, and B channels is performed to obtain the new L*, a* and b* values. Referring to the HOG method, the magnitude features, and their respective angle in each channel of the image are generated with eq 12.7 and 12.8, obtaining six feature maps, two for each channel of the original image (magnitude and angle). The obtained arrangement is shown in Figure 12.10.

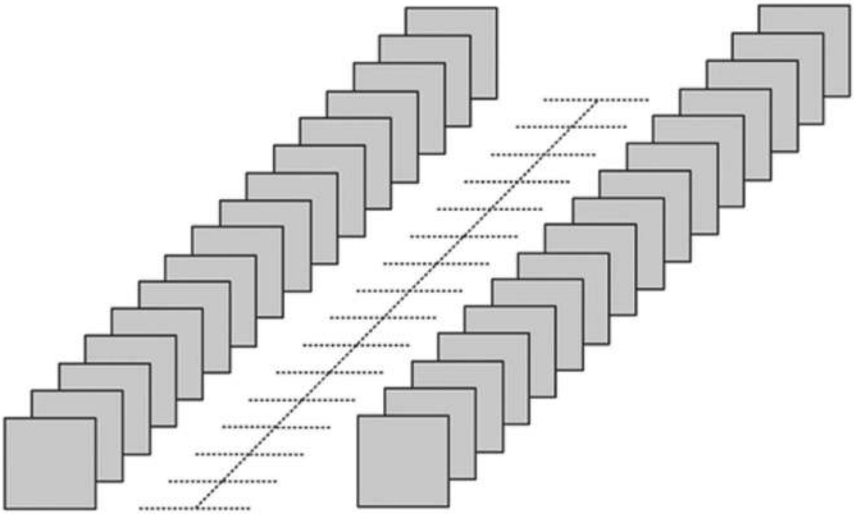

FIGURE 12.10 Plan-based feature arrangement.

Figure 12.11 presents the architecture of the proposed CNN. This architecture has five layers to perform the convolution operation, three layers are added that are in charge of carrying out the pooling operation, as well as two normalization layers, finally there are two layers that prepare the detection and classification operations and their neurons are fully connected. The first convolution layer has 11 × 11 convolution filters applied with a stride of four pixels. After this operation, 96 feature maps are obtained, which are subsampled with an operation that extracts the maximum value from a pooling size of 3 × 3 pixel. The first normalization layer is applied at stabilizing the dynamics of the hidden state in the neural architecture. Convolution 2 uses 5 × 5 filters for the weight values, these filters are applied to the convolution operation with a stride set to one pixel. A total of 256 outputs are generated in this layer and again maximum value subsampling is applied to them. The output of pooling layer two has the second normalization layer applied to it. Next, convolution layers 3 and 4 produce 384 feature maps each with a stride that has been set to one pixel. Finally, 256 feature maps are generated in convolution layer 5 with a stride that has been set to 1 pixel. These last three layers have 3 × 3 filters and the subsampling used in the previous layers is used at the output of convolution layer 5.

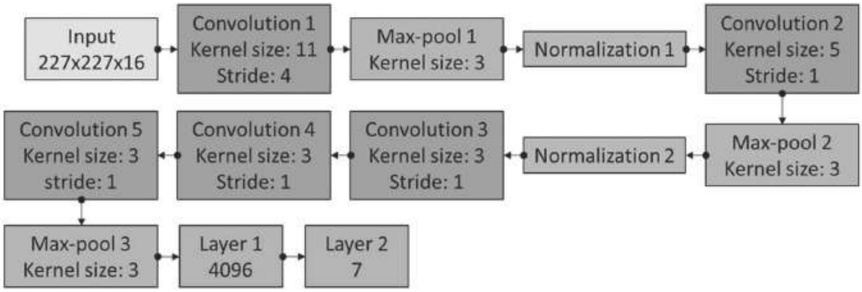

FIGURE 12.11 Proposed CNN network architecture.

The convolution layers represent the abstract feature extraction proper to the mathematical operations of the CNN. These features are delivered to a first FCL that has 4096 neurons, as can be seen in Figure 12.11, for the second FCL has seven neurons representing to each of the six species of the sea turtle and the class CALCTECH256. The activation function used is ReLU. For the training of the proposed RNC network, the following parameters are used:

$$\text{Learning ratio} = 10 \times 10^{-6}$$
$$\text{Weight decay} = 10 \times 10^{-6}$$

During training, batches of 32 images are used, which is the same number of images used for validation. The Adam optimizer, which is an extension of gradient descent, was used to adjust the weights.

12.6 RESULTS

In this work, the cases that are concentrated in Table 12.3 were evaluated. In case 1, the original AlexNet network architecture was trained using RGB images and all the convolution filter weights of the network are initialized randomly. In case 2, which is the architecture proposed in this chapter, convolution filter weights are obtained with a pre-trained model.

Figure 12.12 presents the accuracy obtained in the validation operation for the two implemented architectures: on the one hand, the AlexNet architecture (random initialization), which in the literature has performed well in deep learning and has become a benchmark, on the other hand, the architecture of the proposed model. Accuracy is the metric used, which defines the relationship between the images that have had a correct classification and the total images used to generate the network model during training and performing the validation operation with the data set for validation. Figure 12.12 presents on the horizontal axis the iterations to generate the model. The accuracy of the model whose maximum value is 1 is represented on the vertical axis.

TABLE 12.3 Cases Evaluated in Species Classification.

Case	Architecture	Input information	
1	AlexNet	RGB channels (227 × 227 × 3)	Random
2	Proposal	RGB channels + Mathematical features (227 × 227 × 16)	Pre-training

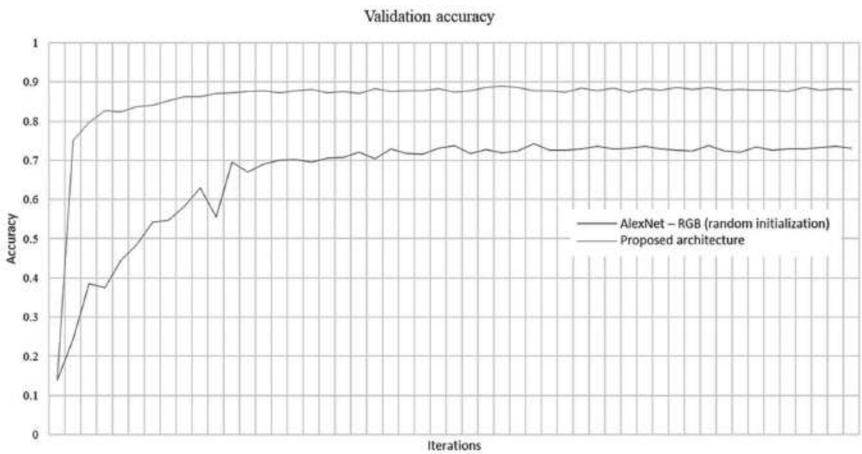

FIGURE 12.12 Validation accuracy for model architectures.

Table 12.4 shows the average validation accuracy value for the two evaluated architectures, with the proposed neural network architecture obtaining the highest validation accuracy value.

TABLE 12.4 Validation Accuracy Values.

Case	Architecture	Validation %
1	AlexNet—RGB (random initialization)	73.00
2	Proposal	87.90

Figure 12.13 presents the cumulative error of the validation set for the two evaluated architectures. An increase in this error during training indicates an overfitting of the parameters, as observed in the AlexNet neural network model. The vertical axis represents the error value and the horizontal axis the iterations.

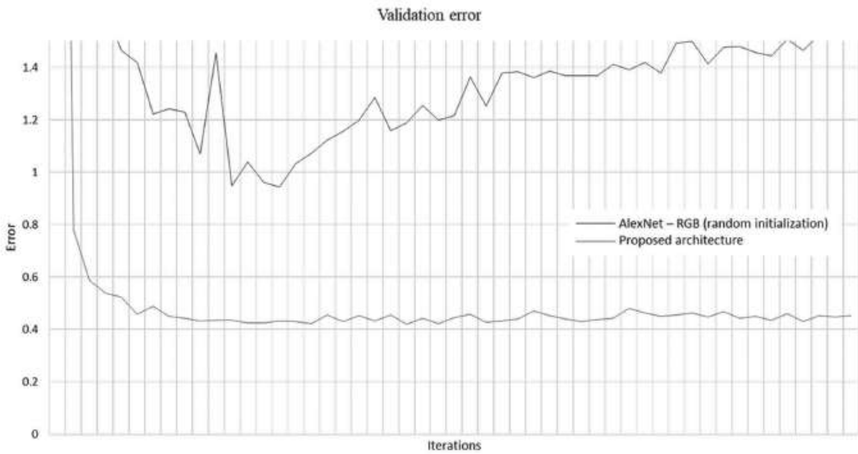

FIGURE 12.13 Validation error for model architectures.

The set of test images was applied to the two cases and the accuracy, as presented in Table 12.5, was obtained. Table 12.5, the best result is again that of the proposed neural network architecture.

TABLE 12.5 Accuracy of the Test Set.

Case	Architecture	Test %
1	AlexNet—RGB (random initialization)	74.00
2	Proposal	86.95

From case 2, which is the proposed neural network architecture and the one that provided the best results, the confusion matrix presented in Figure 12.14 was generated; the main diagonal is the number of correctly classified images.

Known \ Prediction	Loggerhead	Hawksbill	Olive ridley	Leatherback	Kemp's ridley	Green	Object
Loggerhead	125	4	6	1	8	6	0
Hawksbill	4	130	4	4	1	6	1
Olive ridley	5	1	124	4	14	2	0
Leatherback	0	4	3	143	0	0	0
Kemp's ridley	8	2	10	2	126	2	0
Green	9	12	6	4	4	115	0
Object	0	0	0	0	0	0	150

FIGURE 12.14 Model confusion matrix based on the proposed neural network architecture: Case 2.

12.6.1 ANALYSIS OF RESULTS

There is a marked difference in accuracy between the AlexNet model with random initialization and the proposed model (see Table 12.4). The proposed system has an accuracy of 87.90% in validation and 86.95% in test, these values being higher than those of the AlexNet model. The proposed model and its results are the product of the addition of biometric information obtained mathematically; thus, the network extracts abstract features that involve those previously extracted mathematically. In this way, emphasis is obtained in the biometric characteristics in areas of contours and points of interest that allow differentiating the distinct sea turtle species. From the confusion matrix (Figure 12.14) it is observed that the proposed model can discern between images containing an object different from a sea turtle: the class corresponding to the leatherback sea turtle obtains 143 hits, being the highest classification value among distinct species. This is confirmed by the fact that this sea turtle species has morphological characteristics that are very distinct from the rest of the species. Similarly, the confusion matrix shows the close relationship between the different sea turtles' species according to the phylogenetic diagram based on their morphology[12–14] (Figure 12.2). Hawksbill sea turtles are mainly mistaken for Green sea turtles and vice versa, which coincides with the descriptions presented in Section 3.2. Similarly, there is a fairly close relationship between the Olive Ridley and Kemp's Ridley species. This is again reflected in the values of the confusion matrix: 14 images are incorrectly classified as Kemp's Ridley and 10 as Olive Ridley. Regarding the cumulative error of the validation (Figure 12.13), an overfitting of the parameters in the original AlexNet architecture and the RGB images is appreciated. In the case of the proposed model, the error remains stationary as the iterations progress, resulting in that the lowest value of the cumulative error is obtained for the proposed neural network model.

12.6.1.1 THE PROPOSED NEURAL NETWORK MODEL AND ITS RELATIONSHIP WITH THE STATE OF THE ART

There is no properly specific research on sea turtle species classification using an artificial intelligence approach within its two main areas which are machine learning and deep learning. Recently, some works have been reported that do animal species classification employing deep learning architectures. For example, in 2019 Miao and coworkers[26] proposed a model capable of classifying 20 animal species in the wild with an accuracy of 87.50%, in tests using different architectures and images in uncontrolled environments. In another paper presented in 2020, Ahmed and coworkers[27] report classification accuracies of 73.00 and 48.90% for different databases consisting of 10 and 20 species, respectively. In addition, due to the uncontrolled environment, the images have a high degradation and face the problem that the database has mislabeled images. Although it is not possible to make a direct comparison between these investigations and the research work presented here, there are points in common such as the classification of species and images of animals in the wild. Similar results are observed between the first work of Miao and the model proposed in this research work for the classification of sea turtles. With due respect to the work of Ahmed and his collaborators, the model proposed here has superior results. A particularity of the present research work is that it deals with a classification of species belonging to the same family, unlike the aforementioned works, which classify animals with biometric characteristics that are quite different from each other because they belong to different families.

12.7 CONCLUSIONS

Adding mathematically derived biometric features from color, contours, and textures to an image improves accuracy in automatic extraction and classification tasks.

It is possible to perform a classification of species using the deep learning philosophy; therefore, it is necessary to build a database that contains the necessary information in digital images of sea turtles for each of the six species analyzed, which allows the adjustment of the parameters specifically for the task of extracting features present in the morphology of sea turtles. The same difficulties a person faces when classifying very similar sea turtle species using visual features were learned by the system as it emulates the visual system and the decisions made. To minimize the confusion error in the model presented in this chapter, it is necessary to give specific and timely information concerning each species.

ACKNOWLEDGMENTS

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KEYWORDS

- **CNN**
- **image processing**
- **image classification**
- **preservation of see turtles**
- **see life preservation**
- **green technology**

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CHAPTER 13

DEEP LEARNING-BASED SEMANTIC SEGMENTATION TECHNIQUES AND THEIR APPLICATIONS IN REMOTE SENSING

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ABSTRACT

The recent few years have witnessed advancements in the field of computer vision owing to the development of more precise deep learning algorithms, the availability of abundant data, and more computing power. This has led to newer developments in various fields related to computer vision, and remote sensing is one of them. Satellite imagery is a vast source of information about the earth and its environment, but it is also inherently challenging to draw useful conclusions from it owing to the presence of complex spectral and spatial details. Satellite image analysis demands pixel-level accuracy as there is a semantic meaning associated with each pixel in the image. But most of the work on semantic segmentation has been performed on natural images. This chapter presents the state-of-the-art techniques for semantic segmentation based on deep learning and their applications in satellite imagery. The various challenges faced by the researchers in the application of these techniques to satellite imagery are also discussed.

Artificial Intelligence and Machine Learning Techniques in Image Processing and Computer Vision.
Karm Veer Arya, Ciro Rodriguez Rodriguez, Saurabh Singh, & Abhishek Singhal (Eds.)

13.1 INTRODUCTION

Due to the explosive growth in the subject area of deep learning during the recent few years, there has been an acceleration in the fields of computer vision[1] and natural language processing.[2] Advances in deep learning,[3] as well as access to large volumes of data and high computation power, have led to the achievement of state-of-the-art results in the field of computer vision. Deep learning-based convolutional neural networks (CNNs)[4] have been able to attain near-human performance in the image classification task on the ImageNet dataset.[5] Significant improvements have been noticed in other image-based tasks such as image localization, object detection, object segmentation, image style transfer, image reconstruction, super resolution, etc.

The application of deep learning is penetrating newer areas, including remote sensing. Satellite images are a source of a vast amount of data, which can be explored to get very useful insights into various fields such as urban development,[6–8] precision agriculture,[9–12] environmental monitoring detection,[13–16] biodiversity studies,[17] land cover mapping,[18–20] natural disaster management,[21–23] etc. This domain of monitoring the earth's physical, chemical, and biological systems from space is generally known as earth observation.[24] Because high-resolution satellite imagery is abundant and freely available, it has been possible to explore the applicability of machine learning algorithms to draw useful conclusions about the earth's surface and its environment. Easily accessible satellite images provide large spatial coverage as well as a high temporal revisit frequency. This makes them particularly helpful for regional data collection in near real time.[25]

Processing satellite imagery is an inherently complex task owing to the fine spatial and spectral details present in the images. Satellite images consist of several other bands rather than just red, green, and blue. Furthermore, the enormous number of channels present in satellite data makes applying existing deep learning frameworks for semantic segmentation challenging, if not impossible. Because each pixel in an image has semantic importance, satellite image processing necessitates pixel-level precision. When compared to RGB image data, this makes satellite imagery analysis challenging. Lack of ground truth data and labeled training images is another obstacle in satellite imagery. Deep learning architectures' performance is dependent on the availability of a large number of training samples. Moreover, these models are very deep and complex and require a lot of computational power to train.

This chapter aims to develop a better understanding of semantic segmentation and its application in satellite imagery. In this chapter, state-of-the-art semantic segmentation architectures are presented and reviewed.

The rest of this chapter is laid out in the following manner: Section 13.2 discusses satellite imagery, its unique properties, and the challenges faced in processing it. Section 13.3 provides an overview of CNNs, which are the backbone of computer vision tasks using deep learning. Various state-of-the-art semantic segmentation architectures are discussed in Section 13.4. Section 13.5 discusses developments in the field of semantic segmentation of satellite images. The chapter concludes with pointers to future research in Section 13.6.

13.2 SATELLITE IMAGERY

Satellite data serve as a reliable source of information that is globally available. Landsat-1 was the first civil earth surveillance satellite launched in 1972. Researchers have been researching satellite imagery applications since then. At present, more than 4500 satellites are orbiting the earth, with more than 700 being imagery satellites for earth observation.

13.2.1 PROPERTIES OF SATELLITE IMAGERY

- *Spatial resolution:* It refers to the visible details in pixel space. A spatial resolution of 15 m, for example, means that each pixel on the image corresponds to a $15 \times 15\ m^2$ area on the ground.
- *Temporal resolution:* It is dependent on the satellite's revisit period.
- *Spectral resolution:* It refers to the number of electromagnetic (EM) bands present in an image. Instead of only red, blue, and green, satellite photography includes microwaves (radar), infrared (IR), near-IR (NIR), mid-IR (MIR), ultraviolet, and other EM bands, as shown in Figure 13.1. Depending on the spectral resolution, satellite imagery can be classified as multispectral and hyperspectral. Hyperspectral images are made up of a large number

of very small EM bands with wavelengths ranging from 10 to 20 nm. One such example is NASA's Hyperion imaging spectrometer, which produces 30 m resolution images in 220 spectral bands. The multispectral images obtained from the Landsat-8 sensors are similar, although they have fewer and larger bands.

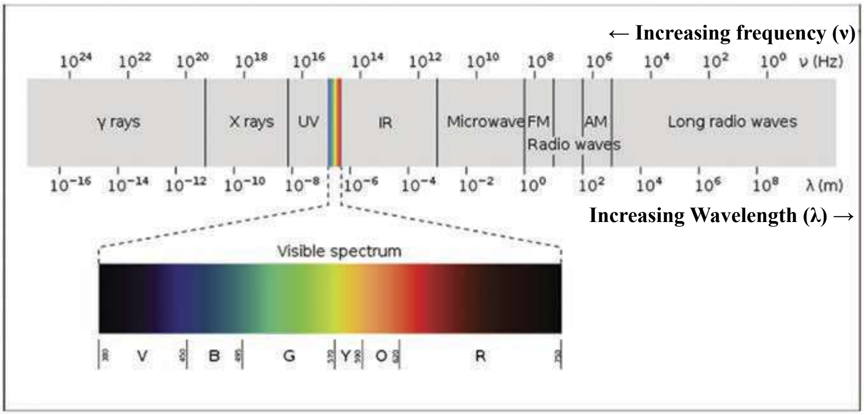

FIGURE 13.1 The EM spectrum.

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- *Radiometric resolution:* It is defined as the ability of the sensor to discriminate between differences in energy.

Current state-of-the-art satellites have a resolution of 25 cm per pixel, which means that a person takes 3 pixels on an image. The images are captured every 14 days by the American Landsat satellite, which has a resolution of 30 m per pixel. The Sentinel satellites Sentinel-1, Sentinel-2, Sentinel-3, Sentinel-4, Sentinel-5, and Sentinel-5P are part of the European Space Agency's Copernicus Program. Sentinel-1 and Sentinel-2 are used for monitoring land and water, while the others are meant for purposes such as air quality monitoring, ozone monitoring, UV radiation monitoring, etc. The Sentinel-2 satellite mission involves two satellites with a phase of 180° and a resolution of 10 m per pixel, which take images every 5 days. MODIS captures images 46 times per year with a resolution of 500 m per pixel. Table 13.1 lists the satellite missions providing free imagery.

TABLE 13.1 A List of Satellite Missions Providing Free Imagery.

Satellite	Instruments	Spatial resolution	Temporal resolution	Radiometric resolution
Sentinel-1	Radar	5–40 m (depends on acquisition mode)	6 days (12 days, per week with two satellites)	1 dB
Sentinel-2	Visible, NIR, SWIR, and coastal	10 m (visible, NIR), 20 m (SWIR), 60 m (coastal)	5 days (10 days, per week with two satellites)	12-bit
Landsat-8	Panchromatic, visible, NIR, SWIR, coastal, and thermal	15 m (panchromatic), 20 m (visible, NIR, SWIR, and coastal), and 100 m (thermal)	16 days	12-bit

13.2.2 CHALLENGES INVOLVED IN PROCESSING SATELLITE IMAGERY

The RGB images, which have only three channels used in general computer vision tasks, are relatively smaller as compared to satellite images with multiple bands ranging from tens to thousands. Furthermore, gathering ground truth data is time-consuming and costly. Similar spectral and feature values frequently correlate to different objects at various locations, making classification a difficult task.

13.3 CNNS

CNNs are dedicated networks that handle image-related tasks. A simple linear classifier is of the form:

$$yi = W_i x_i + b_i \tag{13.1}$$

where x_i is the input (in the case of images, x_i can be thought of as a flattened image vector), W_i is the weight matrix, b_i is the bias, and y_i is the output. In the case of semantic segmentation, y_i represents the class label predicted for a particular pixel x_i in the image. A nonlinear activation function, such as sigmoid, tan h, or ReLU (Rectified Linear Unit), can be fed the output of eq 13.1. Each layer in the neural network consists of multiple such nodes (or neurons). Multiple similar layers are placed on top of each other to form a multilayered neural network.

To learn patterns in images, CNNs (Figure 13.2) use convolutional filters that traverse the image from left to right and top to bottom. Learning is performed layerwise through weight sharing in an end-to-end manner. The first layers learn fine features from the image, such as lines, curves, and edges, whereas the latter layers learn more abstract elements. Pooling layers are used in conjunction with convolution filters to broaden the receptive field at each deeper layer.

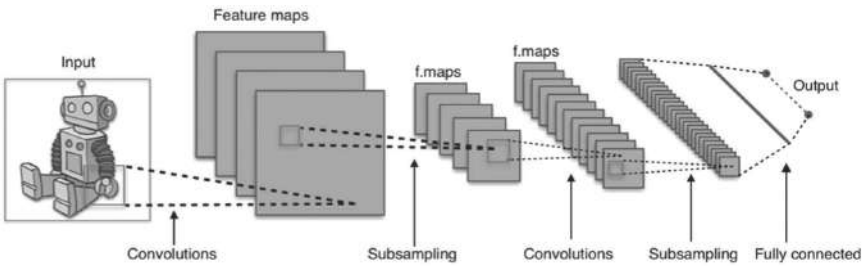

FIGURE 13.2 A typical CNN architecture.

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Following are the various steps performed in a typical CNN:

- *Convolutions:* Each convolutional layer is made up of $h*w*d$ kernels (or filters), where h and w are the kernel's height and width, respectively, and d is the depth, which denotes the number of channels. While sliding across the width and height of the image, a kernel is convolved with a corresponding volume of the image. The sum of the elementwise dot product of the neurons in a kernel with the corresponding values in the input is the output of the convolution process.
- *Nonlinearity functions:* Nonlinearity activation functions, such as sigmoid, tan h, or ReLU, determine whether a neuron fires or not. For the most part, ReLU has been determined to be the most effective.[1] As shown in eq 13.2 below, ReLU squashes negative input values to zero while allowing positive values to pass unchanged.

$$f(z) = \max(0, z) \tag{13.2}$$

where z is the ReLU unit's input.

- *Pooling layers:* The goal of pooling layers is to spatially downsample the input information. Maxpooling, average pooling, the L2 norm of a rectangle neighborhood, or a weighted average based on the distance to the central pixel are all examples of pooling.[26]
- *Fully connected layers:* These are the final set of layers in a CNN.
- *Classifier:* At the final output layer, the softmax function is used to estimate the input data's class from many mutually exclusive classes. It assigns probabilities to different classes whose sum equals one. The cross-entropy loss function, which is the softmax function's negative log-likelihood, is used here (eq 13.3):

$$p(y = k|x_i) = e^{(sk)}/\sum_j e^{(sj)} \tag{13.3}$$

where "*s*" is the total score for a certain class.
- *Regularization:* Regularization is used in neural networks to prevent the model from overfitting the training data. Overfitting reduces the model's capability to generalize well to new and unseen problem instances. Dropout[27] is a commonly used regularization strategy wherein certain neurons are dropped randomly based on a probability. L^2 regularization is another widely used regularization approach.
- *Training:* A forward computation produces an output at the last layer during neural network training, and then an error or loss is backpropagated through the network to alter the network parameters to lessen the difference between the output produced and the actual output. This forward-pass and backpropagation operation is repeated until convergence or a set number of epochs are reached.
- *Hyperparameters:* The learning method, learning rate, minibatch size, weight initialization, regularization, and number of epochs are just a few of the hyperparameters that can be tweaked in a neural network.

The most prominent CNN models for image classification are AlexNet,[1] VGGNet,[28] and GoogleNet.[29] They've served as the foundation for many semantic segmentation architectures.

13.4 SEMANTIC SEGMENTATION

Computer vision has been used to solve a variety of image classification, object recognition, and semantic segmentation challenges. The work of image classification entailed creating labels for the objects shown in a photograph. Object detection not only identifies but also locates each object in an image by producing bounding boxes around it. Object detection is suitable in a scenario where the objects in a scene are clearly distinguishable, such that bounding boxes can be placed around them. But in the case of satellite imagery, image segmentation serves a better fit. Image segmentation deals with segmenting the objects on a pixel-by-pixel basis rather than just creating bounding boxes. Semantic segmentation and instance segmentation are two types of image segmentation. Each pixel in an image is assigned a class label using semantic segmentation.[30,31] It recognizes the objects and their locations during this process. Instance segmentation is similar to semantic segmentation except that it distinguishes between different instances of the same object.

Most of the research on semantic segmentation has been performed on natural images. Semantic segmentation in satellite images involves the extraction of roads, buildings, vehicles, trees, etc. Semantic segmentation is a complex job since it must account for both the image's overall context and the context surrounding each individual pixel. Blurry class boundaries make this task even harder.

Before the era of deep learning, semantic segmentation was performed using traditional approaches.[32] Patch classification[33] was a prominent early deep learning-based approach to semantic segmentation that determined the class label to be assigned to a pixel by examining a patch of the image surrounding that pixel.

The last decade has witnessed enormous success in image processing tasks due to breakthroughs in CNNs.[1] CNNs have now become a prominent approach to image classification.[34]

In order to assign a label to each pixel in the image, semantic segmentation must examine small details in the image. Similar to image classification, CNN-based techniques for semantic segmentation have become the industry standard. Fine structural features are lost during the abstraction process for picture classification, which is undesirable if pure CNNs are to be used for image segmentation.[30] Pooling layers are one of the biggest issues with using CNNs for segmentation, aside from fully connected layers. Pooling layers expand the field of view and can aggregate the context while eliminating the "where" data. However, semantic segmentation demands the precise alignment of class maps, necessitating the preservation of "where" information. To address this issue, two separate groups of architectures have emerged in the literature. The encoder–decoder architecture was adopted by the first class. U-Net[35] is a well-known architecture in this category. The encoder's job is to use pooling layers to gradually reduce spatial resolution, while the decoder gathers segmentation information and gradually upsamples the spatial resolution using learned features. Shortcut links between the encoder and decoder layers are generally present for better segmentation. Dilated or atrous convolutions were employed in the second class, as described in Refs. [36, 37].

13.4.1 FULLY CONVOLUTIONAL NETWORKS

Long et al.[38] proposed fully convolutional networks (FCNs), wherein they adapted the CNN models (VGGNet model,[28] AlexNet,[1] and GoogLeNet[29]) created for image classification to image segmentation. The fully connected layers of a CNN were replaced with convolutional layers using 1*1 convolutions by the authors. The transposed convolutions were then used to upsample the image. Segmentation might be applied to variable size images using this method. It was also a lot faster than patch classification. Although FCNs achieved state-of-the-art results on the PASCAL VOC 2012 dataset,[39] the segmentation maps produced were coarse. To deal with this, the FCNs also used skip connections[40] for the later layers that feed information from the earlier layers operating at a finer resolution. Among the three variants of FCNs, that is, FCN8, FCN16, and FCN32, FCN8 and FCN16 use skip connections. Skip connections have been further refined in U-Nets.[35]

13.4.2 SEGNET

Despite using upsampling and shortcut connections, FCN produced coarse segmentation maps. To overcome this issue, instead of replicating the encoder features, the SegNet architecture[41] (Fig. 13.3) added extra shortcut connections and used the maxpooling indices of the associated encoder layers for upsampling; hence, it does not use any learnable parameters, unlike FCN. Although the model size of SegNet is smaller than FCN with a much lower memory requirement, it is comparatively slower due to its complex decoder architecture.

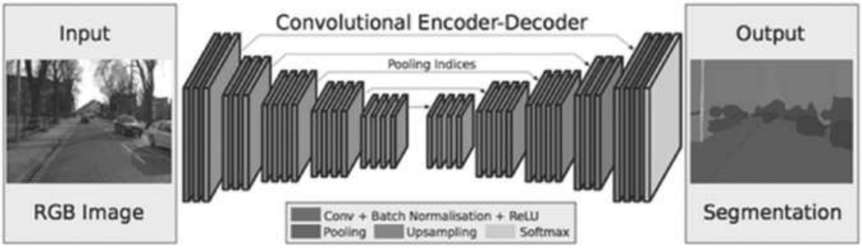

FIGURE 13.3 The SegNet architecture.

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13.4.3 DECONVNET

Noh et. al.[42] introduced the deconvolution layers, which take the convolutional transpose of the input. A deconvolution network is stacked on top of a CNN. The authors also proposed a network ensemble approach by combining the deconvolution network and the FCN[38] to improve the performance further. The model achieved state-of-the-art results on the PASCAL VOC 2012 segmentation benchmark dataset.[39] The major drawback of DeconvNet is the presence of fully connected layers, which make it slower.

13.4.4 U-NET

U-Net[35] is a well-known encoder–decoder-based image segmentation framework. It was originally developed for the medical domain, which is able to yield more accurate segmentation masks with much less training data. There are two symmetrical pathways, one contractive and one expansive, with the same number of layers in each. The contractive path employs a sequence of two 3*3 filters followed by a 2*2 maxpooling layer to gradually downsample the spatial resolution at each layer. The expansive paths have a similar structure, but upconvolutions replace the maxpooling layers. Skip connections are used to connect the corresponding layers in the contractive and expansive routes.

The original U-Net architecture suffers from the problem of loss of localization precision. Several modifications to the underlying U-Net architecture have been proposed in the literature to address this. Zhou et al.[43] proposed U-Net++, which connects the encoder and decoder subnetworks via a set of nested, dense skip routes. Iglovikov and Shvets[44] tried to improve the accuracy of U-Net by using a pretrained VGG11 encoder. Li et al.[45] showed that a deeper U-Net architecture yields better results by allowing the network to create much more detailed segmentation maps. Jegou et al.[46] tried to improve the basic U-Net by introducing residual blocks to train a deeper network. The residual blocks added residual connections inside each block. They inserted two dense blocks in the contractive as well as the expansive paths, which are composed of multiple convolution layers stacked together. The residual connections transport all of the information inside a block's layers to the block's output. As a result, an output with both low-level and high-level features is produced. Avenash and Viswanath[47] proposed U-HarDNet, which suggested a novel activation function, the Hard-Swish activation function.

13.4.5 DILATED CONVOLUTIONS

Yu and Koltun[36] experimented with dilated (atrous) convolutions to enhance the receptive field without sacrificing spatial resolution. In a convolutional filter, the authors utilized a dilation factor l, which reflects the space between entries. This increases the receptive field and allows for the extraction of contextual information at many scales without sacrificing spatial resolution. Due to the systematic structure of the created filters, this strategy similarly achieved state-of-the-art performance but had the drawback of producing gridding artifacts.

13.4.6 DEEPLAB

DeepLab (v1 and v2)[48,49] further improved the results by employing atrous spatial pyramid pooling (ASPP) modules along with atrous convolutions. ASPP allows the application of multiple atrous convolutions with different sampling rates simultaneously. The output feature maps are then fused together. ASPP helps in dealing with objects belonging to the same class but having different scales. Conditional random fields (CRFs)[50] were employed as a postprocessing step, which they claim produces better outlines of objects. CRFs' smooth segmentation results are based on the underlying image intensities, assuming that pixels with comparable intensities belong to the same class. CRFs can improve accuracy by 1–2%. Chen et al.[37] modified the ResNet model to incorporate dilated and atrous convolutions and improved ASPP to propose DeepLabv3. They employed the encoder–decoder architecture to capture sharper object boundaries and depthwise separable convolution to increase computational efficiency. DeepLabv3+[51] further improved on DeepLabv3 by incorporating a decoder module to enhance the segmentation outcomes.

13.4.7 RefineNet

Because they were applied to a large number of high-resolution feature maps, dilated convolutions were computationally highly expensive. Lin et al.[52] proposed RefineNet, an encoder made up of ResNet-101 blocks, to solve this problem. The decoder is made up of RefineNet blocks that combine high-resolution information from the relevant encoder block with low-resolution features from the previous RefineNet block to form the decoder. By upsampling lower-resolution features, a RefineNet block concatenates multiresolution features.

13.4.8 PSPNet

PSPNet (Pyramid scene parsing network)[53] learns a better global context representation of a scene by capturing the smaller objects using the features pooled at a high resolution and detecting the larger objects using the features pooled at a lower resolution. PSPNet, in particular, uses a pretrained CNN with dilated convolutions to extract the feature map of the size of 1/8th of the input image. Then, to cover the entire, half, and small areas of the image, a pyramid pooling module is used, which uses a four-level pyramid with varied-sized pooling kernels. Each pooling kernel's feature maps are then concatenated to form the global prior, which is then concatenated with the original feature map and a convolution layer to create the final segmentation map.

13.4.9 FastFCN

Wu et al.[54] developed FastFCN (Figure 13.4), where they used FCNs with the latest improvements in the initial layers, but in the encoding route, replaced dilated convolutions by joint pyramid upsampling (JPU). JPU generates high-resolution segmentation maps from the low-resolution output images by transferring the structure and details of a guidance image. Their method was shown to be more efficient in terms of memory and time consumption while also generating good outcomes.

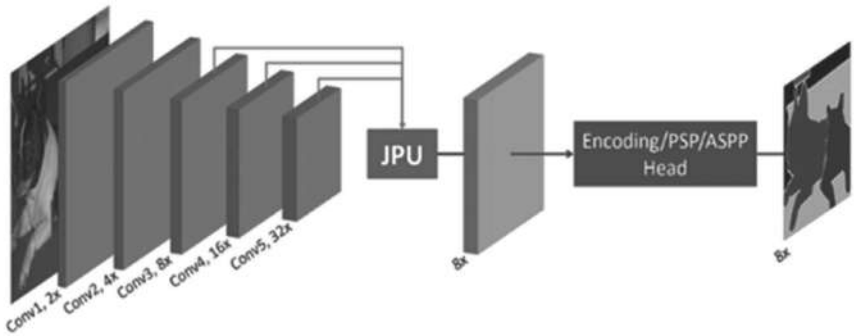

FIGURE 13.4 FastFCN architecture.

Source: Reprinted from Ref. [54] https://creativecommons.org/licenses/by-sa/4.0

13.4.10 KERNEL-SHARING ATROUS CONVOLUTION

Though the ASPP used in the DeepLab family[37,48,49,51] improved the segmentation results to a great extent, there is no information sharing between the parallel layers, which lowers the ability of the kernels in each layer to generalize. In addition, many parallel branches of kernels with varied receptive fields are used to capture the semantics of objects of various sizes, altering total generalizability. Furthermore, the number of parameters in the network grows linearly with the number of parallel branches, resulting in overfitting. To address these issues, Huang et al.[55] proposed KSAC (kernel-sharing atrous convolution). Instead of having a different kernel for each parallel branch, like in ASPP, it uses a single kernel shared across layers. This improves the network's generalization capability. Using dilation rates of 6, 12, and 18 in KSAC saves 62% of the parameters as compared to ASPP. Furthermore, the number of parameters is independent of the number of dilation rates utilized, allowing for the addition of as many rates as possible without expanding the model. With rates 6, 12, and 18, ASPP produces the greatest outcomes, but accuracy declines with rates 6, 12, 18, and 24, indicating possible overfitting. However, KSAC accuracy improves significantly, showing improved generalization potential. Because the same kernel is used across several rates, this kernel sharing technique can also be viewed as a feature space augmentation.

The performance of the state-of-the-art image segmentation models presented in this section on the PASCAL VOC 2012 benchmark dataset[39] is summarized in Table 13.2.

TABLE 13.2 Performance Comparisons of the State-of-the-Art Semantic Segmentation Algorithms on PASCAL VOC 2012 Benchmark Dataset.[39]

References	Semantic segmentation architecture	Benchmark score (mean IoU) (%)
[38]	FCN	62.2
[41]	SegNet	59.9
[42]	DeconvNet	69.6
[36]	Dilated convolutions	75.3
[49]	DeepLab	71.6
[48]	DeepLabv2	79.7
[37]	DeepLabv3	85.7
[51]	DeepLabv3+	89.0
[52]	RefineNet	84.2
[53]	PSPNet	85.4
[55]	KSAC	88.1

13.5 SEMANTIC SEGMENTATION IN SATELLITE IMAGERY

In the pre deep learning era, machine learning methods such as random forests,[56] support vector machines,[57–60] decision trees,[61,62] maximum likelihood,[63] artificial neural networks,[64,65] etc. were used for semantic segmentation. The quality of the handcrafted elements was crucial in these approaches. The problem here is that handcrafting the best feature representations in multiband satellite imagery is nearly impossible.

Deep learning-based methods have recently emerged as the preferred method for remote sensing research such as crop type detection,[66] crop counting, natural disaster monitoring, land cover classification, etc.

Striking the balance between strong downsampling, which allows richer feature extraction, and precise boundary detection, which depends on fine details, has been a challenge for semantic segmentation of remote-sensing images. In the satellite imagery domain, four main strategies have been applied for semantic segmentation: (i) Developing no-downsampling encoder networks using atrous convolutions[67] or combining features from multiple resolutions[68–70]; (ii) constructing symmetric unconvoluted layers and skip connections to improve the decoder network[40,71,72]; (iii) combining various networks with distinct initializations or architectures[70,73]; and (iv) using a probabilistic graph model to postprocess the semantic segmentation results[67,71,74] by fusing segments produced by unsupervised segmentation[74] by using an overlay strategy[69,71] or by using a filtering method.[75]

Mnih and Hinton[76] produced cutting-edge results when labeling aerial images from existing maps. The authors also created benchmark datasets for deep learning with satellite imagery including the Massachusetts roads dataset. Zhao and Du[77] proposed the use of a multiscale CNN (MCNN) to learn deep spatial correlation features. Längkvist et al.[78] employed four CNNs in unison, each focusing on a different size of context. Liu et al.[79] proposed a self-cascaded network employing dilated convolutions[36] to simultaneously handle both large man-made and fine-structured objects for multiscale representation on the last layer of the encoder. Wu et al.[80] proposed a multiconstrained FCN built on the top of FCN for the task of automatic building segmentation in aerial imagery.

For road extraction from high-resolution satellite images, Zhou et al.[81] proposed D-LinkNet with a pretrained encoder and dilated convolution. Attention dilation-LinkNet (AD-LinkNet)[82] is a neural network for semantic segmentation that uses encoder–decoder structure, serial-parallel combination dilated convolution, channelwise attention mechanism, and a pretrained encoder. In the CVPR 2018 DeepGlobe road extraction competition, AD-LinkNet defeated the D-LinkNet algorithm and took first place.[83] Audebert et al.[84] improved SegNet by creating three parallel convolution layers that apply filters with kernel sizes of 3 × 3, 5 × 5, and 7 × 7. Averaging the receptive cells at different scales aggregates the predictions from these layers. For semantic segmentation of slums from satellite images, Wurm et al.[85] used FCNs with transfer learning.

Originally developed for biomedical image segmentation, U-Net has become a common alternative for satellite image segmentation. Attributed to its ability to merge low-level feature maps with higher-level ones, enabling exact localization, all of the top submissions in the 2017 "DSTL Satellite Imagery Feature Detection" challenge on Kaggle used some flavor of U-Net.[86] Zhang et al.[87] used deep residual U-Net for road extraction. Building extraction was automated using U-Net-based semantic segmentation approaches.[7,8] Rui et al.[88] proposed MACU-Net with the help of an asymmetric convolution block to aggregate multiscale features generated by multiple layers of U-Net for semantic segmentation of high-resolution remote-sensing imagery. Su et al.[89] suggested an end-to-end deep CNN (DCNN) for semantic segmentation of satellite images that combines the benefits of DenseNet, U-Net, dilated convolution, and DeconvNet. Mohanty et al.[30] experimented with different modifications to the U-Net architecture and with Mask R-CNN for semantic segmentation of high-resolution satellite images to detect buildings. They used the Spacenet dataset and used IoU as the evaluation metric. They showed that Mask R-CNN outperformed the U-Net-based architectures and achieved an average precision of 0.937 and an average recall of 0.959, with IoU $\geq$ 0.5 being considered a true detection.

U-Net and its variants have been applied in numerous tasks related to satellite imagery such as identifying landslide scars,[21] building extraction,[7,8] forest cover mapping,[17] land cover mapping,[18–20] cloud detection,[13] natural disaster damage mapping,[22] etc.

13.6 CONCLUSIONS AND FUTURE RESEARCH DIRECTIONS

This chapter presented the state-of-the-art deep learning-based semantic segmentation techniques and their applications in satellite imagery. Although there exist pretrained semantic segmentation models like FCN ResNet101 and DeepLabV3 ResNet101 available in deep learning frameworks such as Keras, PyTorch, and Fast.ai, the problem here is that these models have pretrained weights learned on RGB images, while the satellite images consist of several other bands rather than just red, green, and blue.

Furthermore, the enormous number of channels present in satellite data makes applying existing deep learning frameworks for semantic segmentation hard, if not impossible. Because each pixel in an image has semantic importance, satellite image processing needs pixel-level precision. When compared to RGB image data, this makes satellite imagery analysis challenging.

TABLE 13.3 A Summary of Deep Learning-Based Semantic Segmentation Methods Used for Satellite Imagery.

References	Type of application	Dataset used	DL model	Results
[66]	Land cover and crop type classification	Multitemporal Landsat-8 and Sentinel-1A satellite imagery	Fully connected multilayer perceptron (MLP), random forest, and CNN	For all main crops, accuracy is greater than 85% (wheat, maize, sunflower, soybeans, and sugar beet).
[67]	Labeling high-resolution aerial imagery	ISPRS Vaihingen and Potsdam datasets[90]	Deep FCN with no downsampling	90.3% accuracy
[68]	Semantic segmentation in a continuous domain	Close-range images (PASCAL VOC dataset[39] and CamVid dataset[91]) and remote sensing images (ISPRS Vaihingen dataset[90] and EvLab-SS dataset)	In the deep learning architecture, a dual multiscale manifold ranking (DMSMR) network combines dilated, multiscale techniques with the single stream MR optimization method.	88.7% overall accuracy
[69]	Labeling high-resolution aerial imagery	ISPRS Vaihingen and Potsdam datasets[90]	Shuffling CNNs with field-of-view (FoV) enhancement	85.78% overall precision
[70]	Labeling high-resolution aerial imagery	UC Merced Land Use benchmark dataset[92]	CNN with transfer learning	92.4% overall accuracy
[40]	Labeling high-resolution aerial imagery	ISPRS Vaihingen and Potsdam datasets[90]	Deep FCNs with shortcut blocks	91.2% overall precision
[71]	Labeling high-resolution aerial imagery	ISPRS Vaihingen and Potsdam datasets[90]	An hourglass-shaped network (HSN) design structured into encoding and decoding stages	89.42% overall accuracy

TABLE 13.3 *(Continued)*

References	Type of application	Dataset used	DL model	Results
[72]	Labeling high-resolution aerial imagery	RIT-18 dataset[72]	Pretrained FCN	59.8% mean-class accuracy
[73]	Labeling high-resolution aerial imagery	CamVid dataset[91] and ISPRS Potsdam dataset[90]	Conditional least squares generative adversarial networks	69.8% overall accuracy
[8]	Automatic extraction of building footprints	SpaceNet building dataset in the DeepGlobe satellite challenge[83]	U-Net-based semantic segmentation method	70.4% total F1-score
[74]	Semantic segmentation of very high resolution (VHR) satellite imagery	ISPRS Vaihingen dataset[90] and satellite images of Beijing, China	CNN and a CRF model	82–96% overall accuracy
[75]	Automatic building extraction	ISPRS Vaihingen and Potsdam datasets[90]	Deep neural network with a guided filter	97.71% F1-score
[77]	Labeling high-resolution aerial imagery	Hyperspectral digital imagery of Pavia center, scene of the University of Pavia, and part scene of Beijing City, China	Multiscale CNN (MCNN)	91.12% overall accuracy
[78]	Labeling high-resolution aerial imagery	Full city map in northern Sweden	Four CNNs in parallel	94.49% overall accuracy
[79]	Semantic labeling for VHR images in urban areas	Massachusetts building dataset,[93] ISPRS Vaihingen and Potsdam datasets[90]	Self-cascaded network using dilated convolutions	85.58–93.31% mean F1-score

TABLE 13.3 *(Continued)*

References	Type of application	Dataset used	DL model	Results
[80]	Automatic building segmentation in aerial imagery	Aerial image dataset and the corresponding building outlines downloaded from the website of Land Information of New Zealand	Multiconstrained FCN	83.3% IoU score
[81]	Road extraction from high-resolution satellite imagery	CVPR DeepGlobe 2018 road extraction challenge dataset[83]	D-LinkNet with a pretrained encoder and dilated convolution	63.42% IoU score
[82]	Road extraction from high-resolution satellite imagery	CVPR DeepGlobe 2018 road extraction challenge dataset[83]	Attention dilation-LinkNet (AD-LinkNet)	64.79% IoU score
[84]	Labeling high-resolution aerial imagery	Various datasets including ISPRS Vaihingen and Potsdam datasets[90]	FCN trained in a multitask setting by learning jointly classification and distance regression	92.22% overall accuracy on ISPRS datasets
[85]	Semantic segmentation of slums from satellite images	VHR satellite imagery from QuickBird transferred to Sentinel-2 and TerraSAR-X data	FCNs using transfer learning	86–88% accuracy
[87]	Road extraction	Massachusetts roads dataset[93]	Deep residual U-Net	91.87% F1-score
[7]	Automatic building extraction from VHR satellite imagery	Aerial images and the corresponding building outlines collected from a public source (https://data.linz.govt.nz)	Deep ResU-Net	97.09% overall accuracy

TABLE 13.3 *(Continued)*

References	Type of application	Dataset used	DL model	Results
[88]	Semantic segmentation of high-resolution remote-sensing imagery	Wuhan dense labeling dataset[94,95] and Gaofen image dataset[96]	MACU-Net based on U-Net and asymmetric convolution block	84.062% overall accuracy
[89]	Semantic segmentation of satellite images	ISPRS Potsdam dataset[90]	DL model combining DenseNet, U-Net, dilated convolution, and DeconvNet	62.6% mean IoU score
[30]	Automatic building detection	SpaceNet building dataset in the DeepGlobe satellite challenge[83]	U-Net architecture and Mask R-CNN	93.7% average precision, 95.9% average recall
[21]	Identifying landslide scars	Scenes from the Landsat-8 satellite of a region of Nepal	U-Net architecture	74% F1-score
[17]	Forest cover mapping	10-m resolution imagery from the Sentinel-2 satellite covering portions of the legal Amazon region	U-Net architecture	94.7% accuracy
[18]	Land cover mapping	BigEarthNet satellite image archive	Modified U-Net architecture	74.9% overall F1-score
[19]	Land cover mapping	Land cover classification task of the DeepGlobe challenge[83]	U-Net architecture with the Lovasz-Softmax loss	64.1% mean IoU score
[20]	Land cover mapping	Worldview-2 image of a Brazilian conservation unit	U-Net architecture	87% overall accuracy

TABLE 13.3 *(Continued)*

References	Type of application	Dataset used	DL model	Results
[13]	Cloud detection	A dataset derived from Landsat-8 satellite data	Attention based U-Net architecture	88.72% IoU score
[22]	Natural disaster damage mapping	Four-band multispectral high-resolution Worldview-2 images collected before and after the 2011 Tohoku earthquake-Tsunami	U-Net architecture	70.9% overall accuracy

Another barrier to satellite imagery is the lack of ground truth data and annotated training images. Deep learning architectures' performance is dependent on the availability of a large number of training samples. Moreover, these models are very deep and complex, which require a lot of computational power to train.

All of the aforementioned issues open up a lot of possibilities for future study in the field of satellite image processing and its real-world applications. Satellite images, for example, are accessible at a variety of resolutions and scales. The fusion of these images and the creation of a generalized classifier model are an open research problems. Also, creating ground truth masks for satellite images is a laborious and time-consuming task. There is a lack of image variation, diversity, a limited number of classes, and an accuracy saturation problem in many existing remote-sensing datasets. This is an open research question about whether the features learned by a deep learning algorithm on everyday objects generalize well to the satellite imagery domain.

What makes satellite imagery different from scenic imagery consisting of everyday objects is that a semantic meaning is ascribed to each pixel in a satellite image. Satellite imagery contains various semantically meaningful objects of interest, such as roads, buildings, water bodies, crop fields, land use patterns, clouds, etc. As the satellite images contain objects at a much lower resolution than normal RGB images, accurate delineation of objects, especially at the boundaries, is of utmost importance for semantic segmentation.

Change detection is another area that has not been researched enough in satellite imagery. It is an automatic method of monitoring changes in a scene over a period of time. Satellite imagery automatically lends itself to change detection by providing multitemporal images of the same geographical areas.

KEYWORDS

- **semantic segmentation**
- **satellite imagery**
- **convolutional neural networks**
- **U-Net**
- **dilated convolutions**

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CHAPTER 14

DEEP CONVOLUTIONAL NEURAL NETWORK-BASED SINGLE IMAGE SUPERRESOLUTION

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ABSTRACT

Recently, deep learning-based convolutional neural networks have achieved remarkable performance in the field of image and computer vision tasks. In this chapter, we explore recent deep convolutional neural networks (CNNs) related to image superresolution (SR) algorithms. Furthermore, we discuss the basic terminology that is used in image SR reconstruction. Additionally, image datasets play a vital role in the area of deep learning, so in this regard, we give information on the training as well as testing datasets that were used in the deep CNN model. Finally, we present the quantitative results in terms of PSNR/SSIM and the number of parameters as compared to other state-of-the-art methods. The extensive evaluation discussed in this chapter shows that the race for deep learning-based image SR methods still does not achieve the target and has more space available for researchers.

Artificial Intelligence and Machine Learning Techniques in Image Processing and Computer Vision. Karm Veer Arya, Ciro Rodriguez Rodriguez, Saurabh Singh, & Abhishek Singhal (Eds.)

14.1 INTRODUCTION

Single image superresolution (SR) has achieved remarkable performance in the field of image and computer vision tasks. The main target of image SR is to convert the low-resolution (LR) input image into a high-resolution (HR) output image, as shown in Figure 14.1. Image SR is also known as image upsampling, image enhancement, image enlargement, image upscaling, or image zooming.

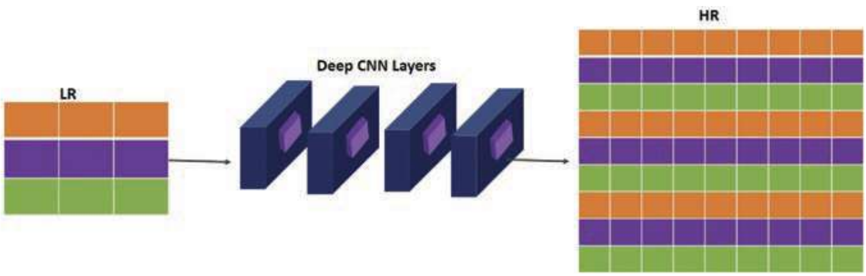

FIGURE 14.1 General overview of image superresolution.

Image SR is the process of reconstructing the visually pleasing high-quality or HR output image from the degraded version or low-quality input, which is a challenging task for many hardware applications, such as medical imaging devices,[1] forensics,[2] video security surveillance,[3] and object detection.[4] The main approaches of image SR include traditional approaches, frequency-based approaches (including Fourier transform, wavelet transform, and second-generation wavelet transform), spatial domain-based approaches (such as projection onto convex sets, Kalman filter, and least means square algorithms), direct methods (including median filter, weighted median filter, or SVD-based filters), probabilistic-based approaches (including maximum likelihood method, maximum a posteriori estimation, and Markov random fields), and deep learning-based approaches. In this chapter, we focus on deep learning-based image SR using a convolutional neural network (CNN).

In summary, the main contributions of our chapter are as follows:

- We explore the basic knowledge of recent deep CNN methods used in image SR methods.
- We discuss the datasets used in the deep CNN during the training as well as testing the image SR methods.

- Finally, we discuss some quantitative results in terms of several peak signal-to-noise ratios (PSNR) and SSIM.

The remaining part of the chapter is organized as follows: Section 14.2 discusses some conventional and deep learning-based image SR methods. Sections 14.3 and 14.4 explain the different training and testing datasets involved in the deep CNN-based image SR and their quality matrix. In Section 14.5, we discuss some quantitative results in terms of PSNR, SSIM, and number k parameters. Finally, we present the conclusion with future suggestions.

14.2 CONVENTIONAL AND DEEP LEARNING-BASED IMAGE SUPERRESOLUTION METHODS

Single image SR has been reviewed by many studies, including conventional as well as deep learning-based approaches. Here initially, we discuss the basic concept of the conventional approach that involved the deep CNN for upscaling the LR image into an HR image.

14.2.1 INTERPOLATION-BASED UPSCALING APPROACH FOR IMAGE SUPERRESOLUTION

Image interpolation is a method that changes a LR image into a HR image. The design of interpolation-based methods is very simple, and implementation is very easy. The applications of interpolation in various areas include space imagery, military applications, medical image enhancement, and image decompression. The main categories of interpolation-based techniques are nearest neighbor, bilinear, and bicubic interpolation-based methods.

14.2.1.1 NEAREST-NEIGHBOR INTERPOLATION-BASED IMAGE SUPERRESOLUTION METHOD

The simple and basic method of interpolation is the nearest-neighbor interpolation-based method. In this method, rather than calculating an average value using weighting criteria or generating an intermediate value using complex methods, this method simply finds the “nearest” neighboring pixel and assumes its intensity value. The basic operation of nearest-neighbor interpolation is shown in Figure 14.2.

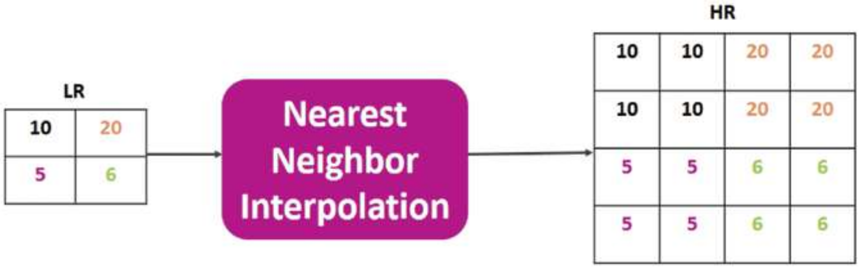

FIGURE 14.2 A general architecture of the nearest-neighbor interpolation-based image SR method.

14.2.1.2 BILINEAR INTERPOLATION-BASED IMAGE SUPERRESOLUTION METHOD

The bilinear operation also has the same function to upscale the LR image into a HR one. This approach works in both directions. The detailed operation of this approach is to take the weighted average of the four closest neighbors to generate the output. The numerical resultant output of bilinear interpolation is shown in Figure 14.3.

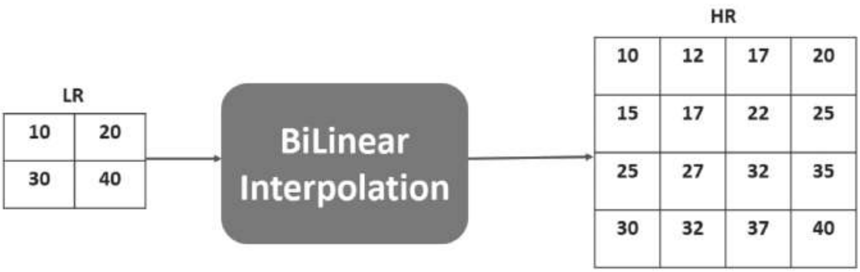

FIGURE 14.3 A general architecture of the bilinear interpolation image SR method.

14.2.1.3 BICUBIC INTERPOLATION-BASED IMAGE SUPERRESOLUTION METHOD

Bicubic interpolation is different from the previous two approaches. Bicubic interpolation uses a 4 × 4 neighborhood, and it generates a sharper image as compared to previous approaches. It also balances processing time and output quality. The main mathematical operation of this method is shown in Figure 14.4.

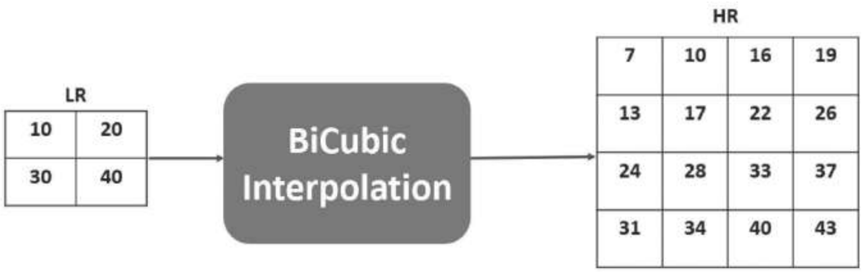

FIGURE 14.4 A general architecture of the bicubic interpolation image SR method.

14.2.2 POSITION OF UPSAMPLING USED IN CNN-BASED IMAGE SUPERRESOLUTION METHOD

There are many ways to shift the position of upsampling operations to reconstruct the high-level features from the low-level features. The main positions available for upsampling in CNN-based image SR are as follows:

14.2.2.1 EARLY UPSAMPLING POSITION

This type of strategy depends on the preprocessing step to upscale the LR image into an HR one. The resultant output used by the CNN is shown in Figure 14.5. This approach was first proposed by Dong et al.[5] in the image SR method, known as SRCNN. In this approach, authors initially upscale the LR image using bicubic interpolation, and the resultant image is fed into the CNN layers. To further increase the early upsampling position, Kim et al.[6] used bicubic interpolation as a preprocessing step to upscale the LR image into an HR image. Additionally, DnCNN[7] and IRCNN[8] also employed this strategy.

14.2.2.2 LATE UPSAMPLING POSITION

The early upsampling approach is more computationally expensive and cannot achieve satisfactory performance. Researchers shift the position of upsampling to the later end layer, which means that all of an image's features are retrieved in the convolutional layers before the final upsampling layer, as shown in Figure 14.6. This method has become more popular due to its lower computational cost and improved performance as compared to previous methods. This approach includes recent deep CNN methods, such as FSRCNN[10] and ESPCN.[11]

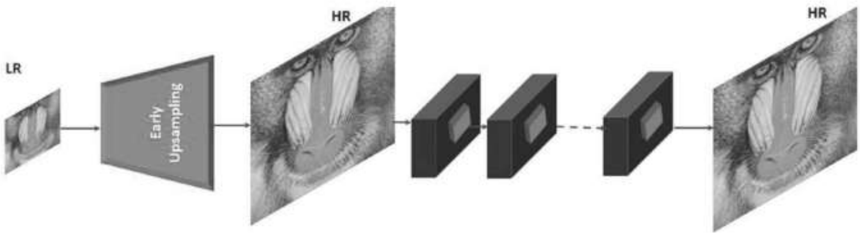

FIGURE 14.5 Early upsampling in image SR method.

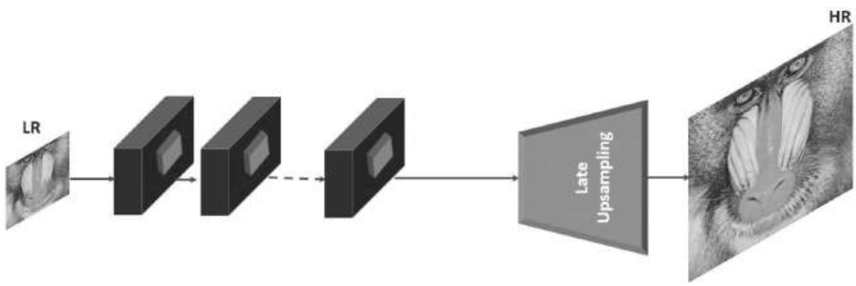

FIGURE 14.6 Late upsampling in image SR method.

14.2.2.3 PROGRESSIVE UPSAMPLING POSITION

Naturally, the 2x enlargement factor is the most common upsampling factor used by everyone to reconstruct the HR image. For example, image size 32 × 32 is to enlarge the scale factor 2x, which is equal to 64 × 64 pixels. In case we want to upscale this image up to scale 4x or 8x, then the network has a challenging task to alter directly in one step to enlargement factor 8x, but we can solve this issue by multiple upscaling steps to reconstruct the higher scale factor HR image, as shown in Figure 14.7. In this approach, an upscale operation is performed in many steps to reconstruct the HR image.

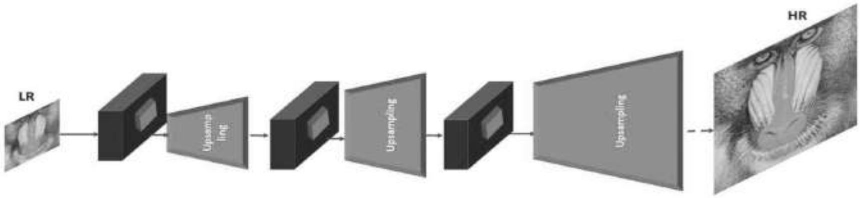

FIGURE 14.7 Progressive upsampling in image SR method.

14.2.2.4 ITERATIVE UP-AND-DOWN SAMPLING POSITION

Iterative up-and-down sampling is one of the most important techniques. Currently, most frameworks employ this technique to iteratively add upsampling and downsampling at various stages of the network, as shown in Figure 14.8. The recently published articles include RBPN,[12] SRFBN,[13] DBPN,[14] and others.

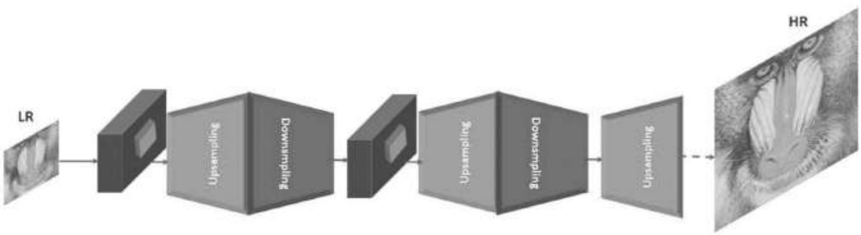

FIGURE 14.8 Iterative up-and-down upsampling in image SR method.

14.2.3 DEEP LEARNING-BASED UPSCALING APPROACH FOR IMAGE SUPERRESOLUTION

The deep learning-based approach has achieved remarkable performance and reduced the computational cost in terms of network parameters. In this subsection, we discuss the upscaling technique involved in deep CNN for converting the LR image into an HR image.

14.2.3.1 DECONVOLUTION LAYER-BASED IMAGE SUPERRESOLUTION METHOD

The deconvolution operation is obtained from the convolution operation, and it works as an inverse operation to enlarge the LR features into HR features. The basic operation of deconvolution is shown in Figures 14.9 and 14.10. The main differences between convolution and deconvolution operations are compared in Table 14.1.

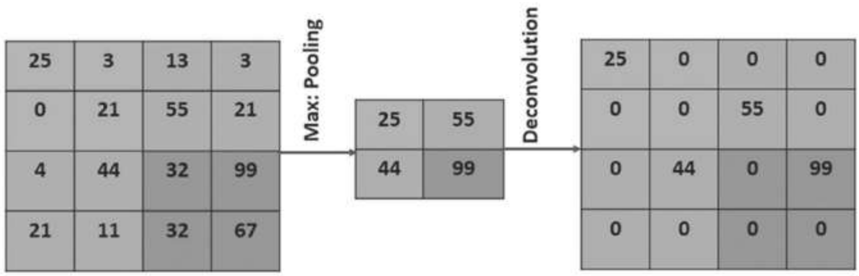

FIGURE 14.9 Basic principle operation of deconvolution in image SR method.

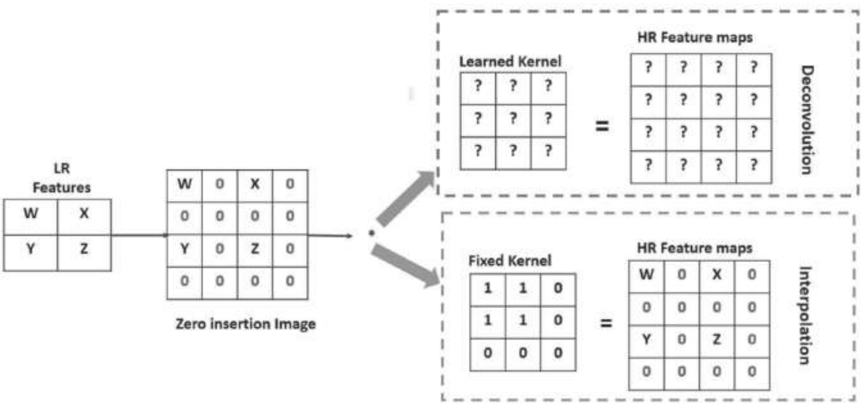

FIGURE 14.10 Upscaling operation in deconvolution and nearest-neighbor interpolation.

14.2.3.2 SUBPIXEL CONVOLUTION LAYER-BASED IMAGE SUPERRESOLUTION METHOD

Subpixel convolution, often known as pixel shuffle, is one of the most important notions introduced in the chapter.[11] Its main function is to increase the LR input image into a HR one, as shown in Figure 14.11. The important benefit of subpixel convolution is that it reduces the checkerboard artifact.

TABLE 14.1 Differences Between Convolution and Deconvolution Operations.

Convolution operation	Deconvolution operation
• Its main function is to extract main features' information from the original low-quality input image. • Convolution operation reduces the dimension of the input image at each stage. • Convolution operation spreads the many pixels information into one pixel.	• Deconvolution is an inverse operation of convolution. • It is also known as transposed convolution. • The main function of the deconvolution operation is to convert the feature maps back into the original image. • Deconvolution operation spreads the information present in one pixel to many pixels.

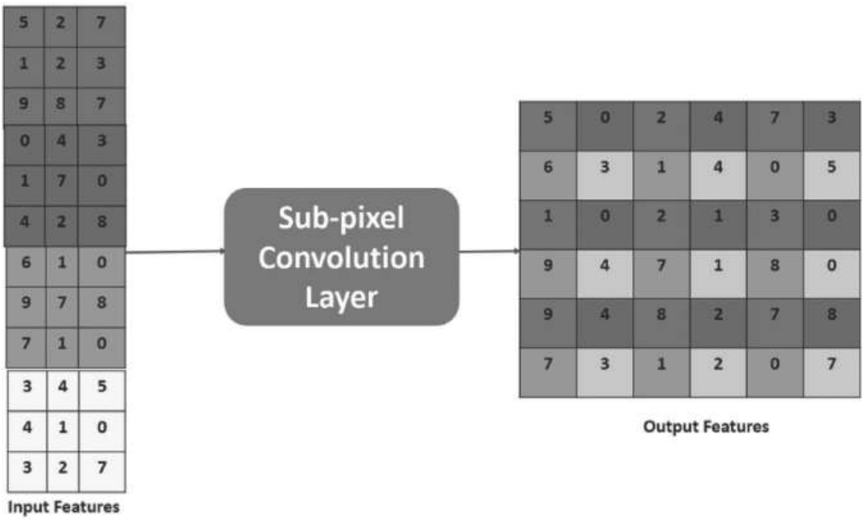

FIGURE 14.11 Upscaling operation in subpixel convolution layer.

14.2.3.3 UPSAMPLING CONVOLUTION LAYER-BASED IMAGE SUPERRESOLUTION METHOD

The upsampling layer is also used to upscale the LR features into HR features. It is commonly used in upsampling blocks. The upsampling increases in both directions in terms of width and height. Figure 14.12 shows how many elements in the input array would be replicated in two directions. After upsampling, the resolution would always be 2x the original input LR image.

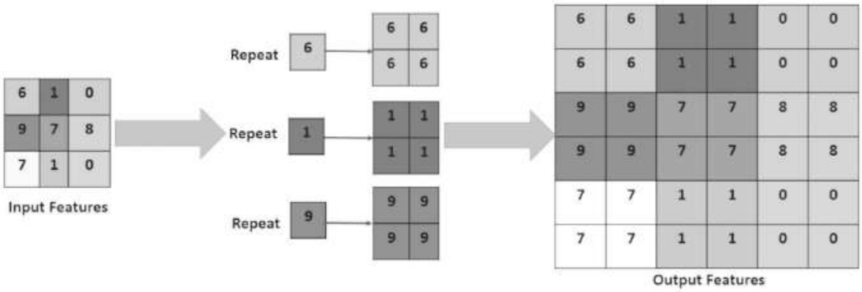

FIGURE 14.12 Upsampling layer operation.

14.2.3.4 SUPERRESOLUTION CONVOLUTIONAL NEURAL NETWORK

SRCNN[5] is the first successful attempt at SR utilizing only convolutional layers. This effort is appropriately regarded as groundbreaking work in deep learning-based image SR. There were a few more attempts at this approach after that. The SRCNN structure is simple. It comprises solely of a convolutional layer followed by a rectified linear unit (ReLU) activation function. Three convolutional layers and two ReLU layers are stacked linearly, as shown in Figure 14.13. The performance of SRCNN is improved compared to previous methods, but there are still some issues, such as authors using bicubic interpolation as a preprocessing step to upscale the LR input image into the HR output image. Bicubic interpolation is not designed for this purpose, and due to this, it introduces extra new noises in the model.

14.2.3.5 FAST SUPERRESOLUTION CONVOLUTIONAL NEURAL NETWORK

The improved version of SRCNN was proposed by the same authors with minor variations and is known as the fast superresolution convolutional neural network (FSRCNN). It increases the resolution as well as the processing time compared to its predecessors. In this approach, the authors replace the bicubic interpolation with a deconvolution layer, and early upsampling is revised with late upsampling to reconstruct the HR output image, as shown in Figure 14.14. FSRCNN includes five stages such as feature extraction, shrinking, nonlinear mapping, expanding, and finally deconvolution.

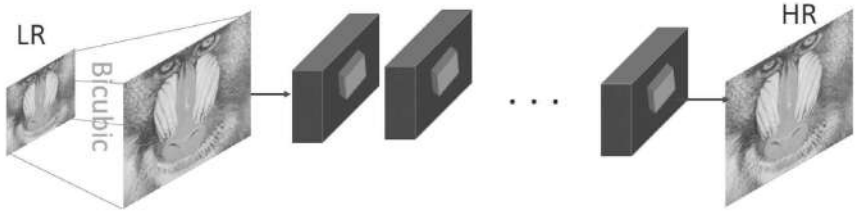

FIGURE 14.13 The basic architecture of SRCNN for CNN-based image superresolution.

FIGURE 14.14 The basic architecture of FSRCNN for CNN-based image superresolution.

14.2.3.6 VERY DEEP SUPERRESOLUTION

The first deep model proposed for image SR is a very deep superresolution (VDSR) having a 20-weight layer, as shown in Figure 14.15. VDSR[6] reconstructs the high-quality or HR (HR) output image from the degraded version of the low-quality or LR input image. VDSR extracts the features from the interpolated upscale version of the HR image and feeds them to the convolution layer, followed by the ReLU activation function. VDSR adds the original information through the residual skip connection to obtain a high-quality HR output image. Furthermore, VDSR followed the VGGNet architecture and used a filter of 64 with a kernel size of 3 × 3 in all layers.

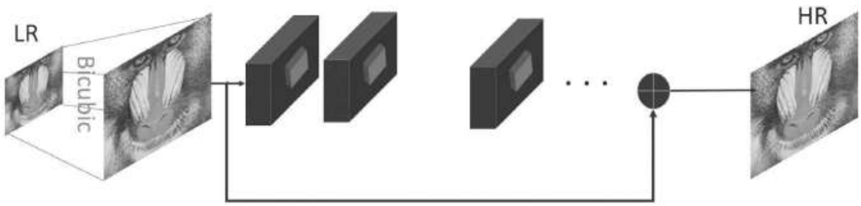

FIGURE 14.15 The basic architecture of VDSR for CNN-based image superresolution.

14.2.3.7 EFFICIENT SUBPIXEL CONVOLUTIONAL NEURAL NETWORK

Earlier approaches such as SRCNN, FSRCNN, and VDSR, the CNN-based approaches, obtained improved results but still had some limitations. These approaches used the interpolation and deconvolution layers to upscale the LR image into an HR image. Interpolation, which is not designed for this purpose, also introduces the ringing, jagged artifacts on the reconstructed image. The deconvolution layer introduces the checkboard effect on the HR image as well. To address these issues, the novel ESPCNN[11] strategy has been presented, which involves adding an efficient subpixel convolutional layer to the CNN network. ESPCNN improves the network's resolution toward the end. The upscaling process is accomplished by the last layer of ESPCNN, which means that the smaller LR image is used, as shown in Figure 14.16. ESPCNN used three CNN layers with one subpixel shuffle. The first layer used 64 filters with a kernel size of 5 × 5, followed by a tan h activation function. The second layer used 32 filters with a small kernel size of the order 3 × 3, followed by the same tan h activation function. The third layer used a fixed number of input channels with the same kernel size (3 × 3). Finally, ESPCNN applied a subpixel shuffle function to upscale the reconstructed HR output image, followed by the sigmoid activation function.

14.2.3.8 DEEPLY-RECURSIVE CONVOLUTIONAL NETWORK

DRCN also follows the principles of SRCNN and VDSR to use bicubic interpolation as an early upsampling technique to reconstruct the HR output image. The architecture of DRCN[18] consists of three basic subnetworks, such as embedding, inference, and reconstruction-type networks, as shown in Figure 14.17. This upsampling image goes through an embedding stage to be converted into a set of feature maps. The output of the embedded network is used by one recursive layer, followed by the ReLU activation function (inference stage). Finally, the reconstruction network converts the output feature maps of the inference network into the original HR output image space.

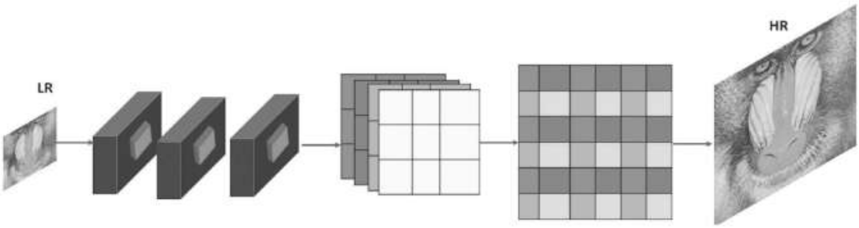

FIGURE 14.16 The basic architecture of ESPCNN for CNN-based image superresolution.

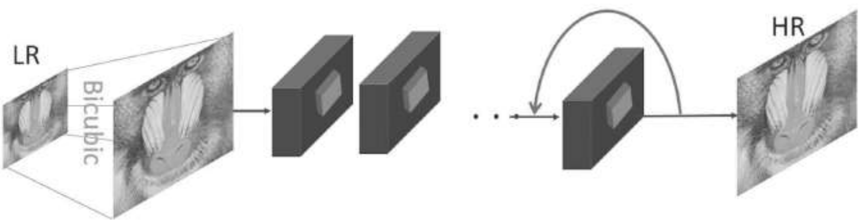

FIGURE 14.17 The basic architecture of DRCN for CNN-based image superresolution.

14.2.3.9 DEEP LAPLACIAN PYRAMID SUPERRESOLUTION NETWORK

In contrast to one-step upsampling, the authors[19] used a progressively pyramidal approach to reconstruct the residuals of subbands of HR images at multiple levels up to an enlargement scale factor of 8x. The first, second, and last subnetworks are residues of 2x, 4x, and 8x. The cumulative sum of these residual images gives the final reconstructed HR output image. In this framework, the authors used three types of layers, convolution, leaky ReLU, and deconvolution layers, as shown in Figure 14.18. The leaky ReLU used the negative slope up to 0.2. LapSRN first used new loss functions, such as the Charbonier loss function, which can resolve outlier problems. The number of fitters and kernel size are 64, 3×3 in the convolution layer, and 4 × 4 in the deconvolution layer, respectively. Yang et al. used the LapSRN and the BSDS200 datasets for training.

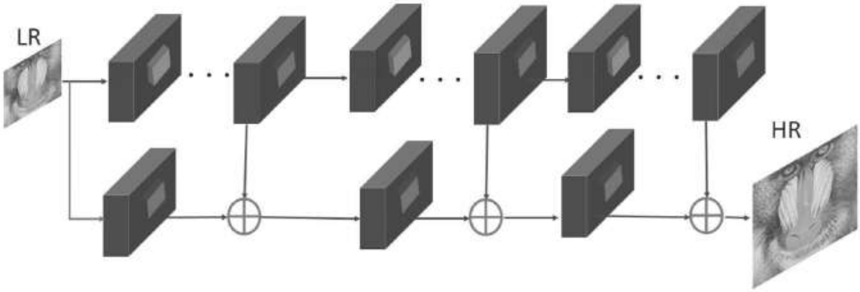

FIGURE 14.18 The basic architecture of LapSRN for CNN-based image superresolution.

14.2.3.10 A PERSISTENT MEMORY NETWORK FOR IMAGE RESTORATION

A very deep persistent memory network, also known as MemNet,[20] was presented at the 2017 ICCV conference for image denoising and SR. The MemNet architecture consists of FENet (feature extraction net), memory blocks, and reconnect (reconstruction net), as shown in Figure 14.19. The FENet uses the CNN layer to extract the features' information from the degraded or blurry input image. The memory blocks use recursive and gated units. The recursive unit learns the features' information at multiple levels. The recursively extracted features' information is sent to the gate unit. The main function of the gate unit is to control the number of previous states that should be reserved and how much of the current state should be stored. MemNet used the mean squared error (MSE) as a loss function.

14.2.3.11 BALANCED TWO-STAGE RESIDUAL NETWORKS FOR IMAGE SUPERRESOLUTION

Fan et al. proposed the concept of balanced two-stage residual networks (BTSRN)[21] for image SR. In this architecture, the authors used two-stage resolution, one called the LR stage and the other the HR stage, as shown in Figure 14.20. In the LR stage, input patch size and feature maps are the same. For upsampling purposes, they used the deconvolution layer with nearest-neighbor interpolation technique to convert LR features into HR features. The reconstructed upsampled feature maps are fed to the HR stage. Residual blocks are used for the LR and HR stages, known as projected convolution. The LR stage employed six blocks (residuals).

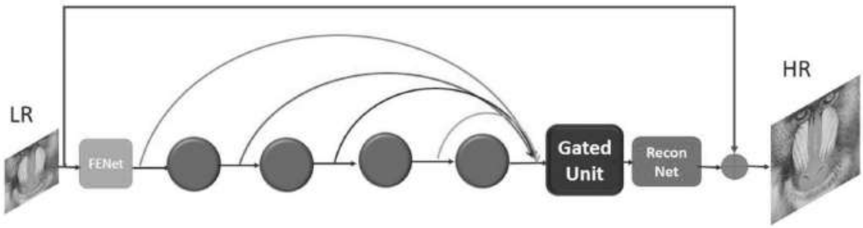

FIGURE 14.19 The basic architecture of MemNet for CNN-based image superresolution.

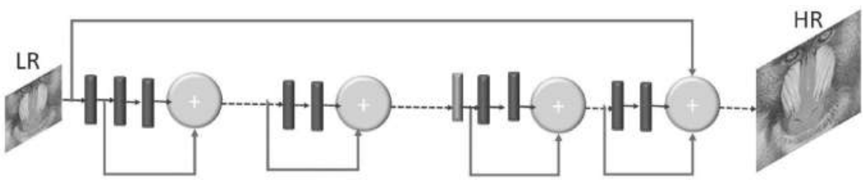

FIGURE 14.20 The basic architecture of BTSRN for CNN-based image superresolution.

14.2.3.12 IMAGE SUPERRESOLUTION BASED ON FUSING MULTIPLE CONVOLUTION NEURAL NETWORKS

Ren et al.[22] suggested a new architecture known as fusing multiple CNNs for image SR. The authors assigned a name as CNN network contextwise network fusion (CNF), in which each type of SRCNN architecture is stacked with a different number of CNN weight layers, as shown in Figure 14.21. The resultant output of each SRCNN is fed to a single CNN layer and finally fused using a sum-pooling operation.

14.3 EXPERIMENTAL CALCULATIONS

In this section, we explain the most commonly used datasets, quantitative evaluations, and computational costs in terms of PSNR and SSIM. There are many datasets available for the training of image SR using a deep CNN model. These training datasets include DIV2K,[23] ImageNet,[24] General 200, Yang91,[25] and Flickr[26] image datasets. For testing purposes, the research community of image SR evaluates the performance on Set5,[27] Set14,[28] BSDS100,[29] Urban100,[30] and Manga109[31] test datasets.

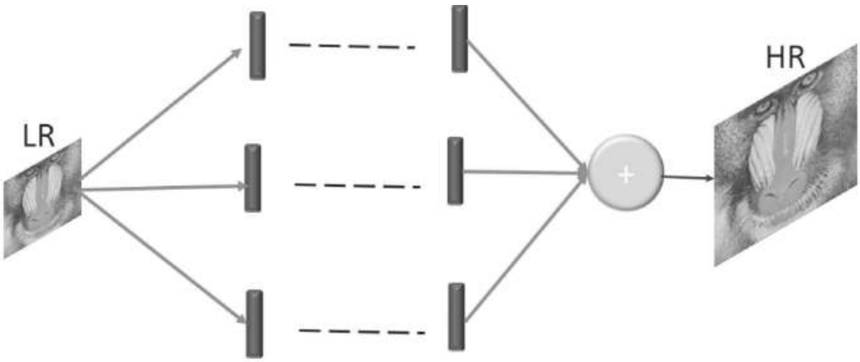

FIGURE 14.21 The basic architecture of CNF for CNN-based image superresolution.

14.3.1 EVALUATIONS QUALITY METRICS

The main benchmark quality metrics to evaluate the performance of image SR are the peak signal-to-noise ratio (PSNR) and the structural similarity index measures (SSIM).[32] The PSNR results of state-of-the-art methods are presented in Tables 14.2–14.5 for 2x, 3x, 4x, and 8x enlargement factors, respectively. Furthermore, Tables 14.6–14.9 present the SSIM of all enlargement scale factors as 2x, 3x, 4x, and 8x. Finally, Table 14.10 presents information about the number, parameters, types of input and output, network depths, and types of loss function. All quantitative results used are from the latest research work in Refs. [33, 34].

14.4 CONCLUSIONS AND FUTURE WORK

Recently, deep convolutional neural network-based image SR solutions have achieved remarkable improvements over the previous methods. Furthermore, image SR plays a vital role in real-time applications such as medical images, security surveillance images, satellite images, and astronomical imaging. Earlier methods of image SR depended on

TABLE 14.2 Average PSNRs of Enlargement Factor 2x in Recent Benchmark Image Superresolution Methods.

Algorithm	Enlargement factor	SET5	SET14	BSDS100	URBAN100	MANGA109	DIV2K
		PSNR	PSNR	PSNR	PSNR	PSNR	PSNR
Bicubic	2x	33.68	30.24	29.56	26.88	31.05	32.45
SRCNN [5]	2x	36.66	32.45	31.36	29.51	34.59	35.72
FSRCNN [10]	2x	36.98	32.62	31.50	29.85	36.62	34.74
VDSR [6]	2x	37.53	33.05	31.90	30.77	37.16	35.43
DRCN [18]	2x	37.63	33.06	31.85	30.76	37.57	35.45
LapSRN [19]	2x	37.52	32.99	31.80	30.41	37.53	35.31
DRRN [35]	2x	37.74	33.23	32.05	31.23	37.92	35.63
DnCNN [7]	2x	37.58	33.03	31.90	30.74	N.A	N.A
IDN [36]	2x	37.83	33.30	32.08	31.27	38.02	N.A
CNF [22]	2x	37.66	33.38	31.91	N.A	N.A	N.A
BTSRN [21]	2x	37.75	33.20	32.05	31.63	N.A	N.A
MemNet [20]	2x	37.78	33.28	32.08	31.31	37.72	N.A

TABLE 14.3 Average PSNRs of Enlargement Factor 3x in Recent Benchmark Image Superresolution Methods.

Algorithm	**Enlargement factor**	**SET5**	**SET14**	**BSDS100**	**URBAN100**	**MANGA109**	**DIV2K**
		PSNR	**PSNR**	**PSNR**	**PSNR**	**PSNR**	**PSNR**
Bicubic	3x	30.40	27.54	27.21	24.46	26.95	29.66
SRCNN [5]	3x	32.75	29.29	28.41	26.24	30.48	31.11
FSRCNN [10]	3x	33.16	29.42	28.52	26.41	31.10	31.25
VDSR [6]	3x	33.66	29.78	28.83	27.14	32.01	31.76
DRCN [18]	3x	33.82	29.77	28.80	27.15	32.31	31.79
LapSRN [19]	3x	33.82	29.79	33.82	27.07	31.22	32.21
DRRN [35]	3x	34.03	29.96	28.95	27.53	32.74	31.96
DnCNN [7]	3x	33.75	29.81	28.85	27.15	N.A	N.A
IDN [36]	3x	34.11	29.99	28.95	27.42	32.69	N.A
CNF [22]	3x	33.74	29.90	28.82	N.A	N.A	N.A
BTSRN [21]	3x	34.03	29.90	28.97	27.75	N.A	N.A
MemNet [20]	3x	34.09	30.00	28.96	27.56	32.51	N.A

TABLE 14.4 Average PSNRs of Enlargement Factor 4x in Recent Benchmark Image Superresolution Methods.

Algorithm	Enlargement factor	SET5	SET14	BSDS100	URBAN100	MANGA109	DIV2K
		PSNR	PSNR	PSNR	PSNR	PSNR	PSNR
Bicubic	4x	28.43	26.00	25.96	23.14	25.15	28.11
SRCNN [5]	4x	30.48	27.50	26.90	24.52	27.66	29.33
FSRCNN [10]	4x	30.70	27.59	26.96	24.60	27.89	29.36
VDSR [6]	4x	31.35	28.02	27.29	25.18	28.82	29.82
DRCN [18]	4x	31.53	28.03	27.24	25.14	28.97	29.83
LapSRN [19]	4x	31.54	28.09	27.32	25.21	29.09	29.88
DRRN [35]	4x	31.68	28.21	27.38	25.44	29.46	29.98
DnCNN [7]	4x	31.40	28.04	27.29	25.20	N.A	N.A
IDN [36]	4x	31.82	28.25	27.41	25.41	29.40	N.A
CNF [22]	4x	31.55	28.15	27.32	N.A	N.A	N.A
BTSRN [21]	4x	31.82	28.25	27.41	25.41	N.A	N.A
MemNet [20]	4x	31.74	28.26	27.40	25.50	29.42	

TABLE 14.5 Average PSNRs of Enlargement Factor 8x in Recent Benchmark Image Superresolution Methods.

Algorithm	Enlargement factor	SET5	SET14	BSDS100	URBAN100	MANGA109
		PSNR	PSNR	PSNR	PSNR	PSNR
Bicubic	8x	24.39	23.19	23.67	20.74	21.47
SRCNN [5]	8x	25.33	23.85	24.13	21.29	22.37
FSRCNN [10]	8x	25.41	23.93	24.21	21.32	22.39
VDSR [6]	8x	25.72	24.21	24.37	21.54	22.83
DRCN [18]	8x	25.93	24.25	24.49	21.71	23.20
LapSRN [19]	8x	26.15	24.35	24.54	21.81	23.39
DRRN [35]	8x	26.18	24.42	24.59	21.88	23.60

TABLE 14.6 Average SSIMs of Enlargement Factor 2x in Recent Benchmark Image Superresolution Methods.

Algorithm	Enlargement factor	SET5	SET14	BSDS100	URBAN100	MANGA109
		SSIM	SSIM	SSIM	SSIM	SSIM
Bicubic	2x	0.931	0.870	0.844	0.841	0.936
SRCNN [5]	2x	0.955	0.908	0.889	0.896	0.968
FSRCNN [10]	2x	0.956	0.909	0.892	0.902	0.971
VDSR [6]	2x	0.959	0.913	0.896	0.914	0.975
DRCN [18]	2x	0.959	0.912	0.895	0.914	0.974
LapSRN [19]	2x	0.959	0.913	0.895	0.910	0.974
DRRN [35]	2x	0.959	0.914	0.897	0.919	0.976
DnCNN [7]	2x	0.959	0.913	0.896	0.914	N.A
IDN [36]	2x	0.960	0.915	0.898	0.919	0.975
CNF [22]	2x	0.959	0.914	0.896	N.A	N.A
SRMDNF [37]	2x	0.960	0.916	0.898	0.920	0.976
MemNet [20]	2x	0.959	0.914	0.898	0.919	0.974
CARN [38]	2x	0.959	0.916	0.898	0.925	0.976

TABLE 14.7 Average SSIMs of Enlargement Factor 3x in Recent Benchmark Image Superresolution Methods.

Algorithm	Enlargement factor	SET5	SET14	BSDS100	URBAN100	MANGA109
		SSIM	SSIM	SSIM	SSIM	SSIM
Bicubic	3x	0.869	0.775	0.741	0.737	0.859
SRCNN [5]	3x	0.909	0.823	0.788	0.801	0.914
FSRCNN [10]	3x	0.914	0.824	0.791	0.808	0.921
VDSR [6]	3x	0.921	0.832	0.799	0.829	0.934
DRCN [18]	3x	0.922	0.832	0.797	0.828	0.936
LapSRN [19]	3x	0.922	0.832	0.798	0.828	0.935
DRRN [35]	3x	**0.924**	0.835	0.800	0.764	**0.939**
DnCNN [7]	3x	0.922	0.832	0.798	0.827	N.A
IDN [36]	3x	0.925	0.835	0.801	0.836	0.938
CNF [22]	3x	0.923	0.832	0.798	N.A	N.A
SRMDNF [37]	3x	0.925	0.838	0.802	0.839	0.940
MemNet [20]	3x	0.925	0.835	0.800	0.838	0.937
CARN [38]	3x	0.925	0.841	0.803	0.849	0.944

TABLE 14.8 Average SSIMs of Enlargement Factor 4x in Recent Benchmark Image Superresolution Methods.

Algorithm	Enlargement factor	SET5	SET14	BSDS100	URBAN100	MANGA109
		SSIM	SSIM	SSIM	SSIM	SSIM
Bicubic	4x	0.811	0.704	0.670	0.660	0.790
SRCNN [5]	4x	0.863	0.753	0.712	0.725	0.859
FSRCNN [10]	4x	0.866	0.755	0.715	0.728	0.861
VDSR [6]	4x	0.883	0.768	0.726	0.754	0.887
DRCN [18]	4x	0.884	0.768	0.725	0.752	0.887
LapSRN [19]	4x	0.885	0.772	0.727	0.756	0.890
DRRN [35]	4x	0.888	0.772	0.728	0.764	0.896
DnCNN [7]	4x	0.885	0.767	0.725	0.752	N.A
IDN [36]	4x	0.890	0.773	0.730	0.763	0.894
CNF [22]	4x	0.886	0.768	0.725	N.A	N.A
SRMDNF [37]	4x	0.892	0.778	0.734	0.773	0.902
MemNet [20]	4x	0.889	0.772	0.728	0.763	0.894
CARN [38]	4x	0.894	0.781	0.735	0.784	0.908

TABLE 14.9 Average SSIMs of Enlargement Factor 8x in Recent Benchmark Image Superresolution Methods.

Algorithm	Enlargement factor	SET5	SET14	BSDS100	URBAN100	MANGA109
		SSIM	SSIM	SSIM	SSIM	SSIM
Bicubic	8x	0.658	0.566	0.548	0.516	0.650
SRCNN [5]	8x	0.690	0.591	0.566	0.544	0.695
FSRCNN [10]	8x	0.697	0.599	0.572	0.550	0.692
VDSR [6]	8x	0.724	0.614	0.583	0.571	0.725
DRCN [18]	8x	0.723	0.614	0.582	0.571	0.724
LapSRN [19]	8x	0.738	0.620	0.586	0.581	0.735
DRRN [35]	8x	0.738	0.622	0.587	0.583	0.742

TABLE 14.10 Information About the Number of Parameters, Types of Input, Types of Output, Network Depths, and Types of Loss Function.

Algorithm	Enlargement factor	Parameters	Input	Output	Network depth	Type of loss function
SRCNN [5]	2x	57k	Bicubic	Direct	3	l_2
FSRCNN [10]	2x	12k	LR	Direct	8	l_2
VDSR [6]	2x	665k	Bicubic	Direct	20	l_2
DRCN [18]	2x	1775k	Bicubic	Direct	20	l_2
LapSRN [19]	2x	812k	LR	Progressive	24	l_1
DRRN [35]	2x	297k	Bicubic	Direct	52	l_2
DnCNN [7]	2x	566k	Bicubic	Direct	17	l_2
IDN [36]	2x	796k	LR	Direct	31	I_1,l_2
CNF [22]	2x	337K	Bicubic	Direct	15	l_2
BTSRN [21]	2x	410K	LR	Direct	22	l_2
MemNet [20]	2x	677k	Bicubic	Direct	80	l_2

hand-designed preprocessing techniques, which introduced new noises in the reconstructed images and increased memory consumption during the testing process. This chapter provides comprehensive knowledge of deep CNN image SR methods. We have noted that the performance of the image SR is increasing day by day, but we have not yet achieved the target results. In this regard, the recent deep CNN model still faces the problems of computational complexity, an inadequate number of quality metrics, and taking more processing time. We hope that this chapter will inspire fresh efforts by researchers to address these critical issues. In the future, we will highlight these issues as a separate book chapter for readers, especially in the image SR research area. Additionally, we will also discuss all quantitative metrics in detail, with some practical results.

KEYWORDS

- **deep learning**
- **CNN**
- **super resolution**
- **image processing**
- **signal processing**
- **image enhancement**

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CHAPTER 15

A REVIEW OF MACHINE LEARNING TECHNIQUES FOR VISION-ESTABLISHED HUMAN ACTION RECOGNITION

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ABSTRACT

Observing human behavior is an important area of computer vision. Monitoring systems that involve interactions between people and electronic devices, such as human–computer interfaces, are among their applications. These applications require automatic identification of irregular or conflicting action states caused by individuals' direct acts. A study is being conducted on how to construct accurate data representations with high-level abstraction from large-scale noiseless video data, which is one of the most relevant issues in the human action recognition process. The majority of recent active research in this field has focused on machine learning. In the field of image detection, machine learning methods have outperformed other approaches. The authors first investigate the role of machine learning in image and video processing and recognition in this chapter. Because of the various machine learning approaches, the authors

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compare and contrast them. This chapter also discusses open datasets for testing accreditation procedures to compare the outcomes of various approaches in these databases. This research looks at approaches to basic human actions as well as approaches to abnormal activity levels. The interests and constraints of each system are observed and used to classify these methods. The chapter also discusses several methods for representing space-time module approaches and actions, as well as perceiving such action datasets directly from images.

15.1 INTRODUCTION

In the field of computer vision, human activity recognition (HAR) is a major topic of discussion. This is attributed to the exponential growth of video recordings and the large number of applications that use scheduled video testing, such as visual surveillance, human–machine interfaces, video interrogation, and video recovery. HAR, especially abnormal state action recognition, is one of the most appealing of these applications. An activity is a continuation of the human body's growth that may include several body parts at the same time. Action recognition is the computer's incorporation of an idea (such as a video) into precategorized types. After that, it is sent to an action-style label. Human movements are often divided into four phases in terms of diversity: gestures, behavior, interactions, and group activities,[1] as shown in Figure 15.1. The essential movement of physical body parts is described as a gesture. Some examples include handshaking and facial expressions. Within the defined forms, a gesture usually takes a really short time and features a very low level of complexity. An action is a type of action that an individual engages in. In reality, it's a mash-up of various gestures (nuclear actions). Running, biking, jogging, and boxing are some examples of actions. Communication is a type of interaction in which two characters interact. It may be a human or object encounter. Human interaction includes fights, handshakes, and embraces between two people, while object communication includes an ATM user, a computer user, and a bag thief. The most complicated form of function is the team function. It can, of course, be a mix of movements, acts, and interactions. It includes more than two people as well as one or more things. Team action may take the form of a protest, a game between two teams, or a team meeting, and much study has been done following the basic development of human motion recognition. Feature extraction, process learning, classification, process identification, and division are all important aspects of such a structure.[2]

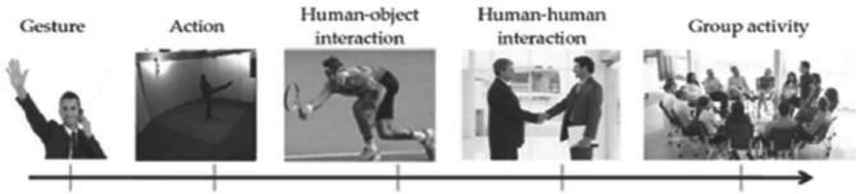

FIGURE 15.1 Categorization for different levels of activities.
Source: Reprinted from Ref. [93]. https://creativecommons.org/licenses/by/4.0/

Systems for action representation, division, and recognition were studied by Wayneland et al.[3] Durga and her companions[4] found that the accreditation issue was identified as a process with unpredictable outcomes and controlled procedures that demonstrated the capacity to manage fluctuating levels of multiple standards. Other classification criteria can be found.[1,5,6]

Over the past decade, several machine learning approaches have been used in the field of computer vision; however, there is currently no defined framework for machine learning methods. Approaches to machine learning[8,9] investigate a few of them. Furthermore, they were looked at in the areas of time-series data[7] and HAR.[9] These databases analyzed important works using machine learning techniques and implemented the most recent human activity authentication databases. In this regard, this chapter makes an attempt to provide a structural analysis based on commonly used learning models. So, those are the final thoughts on computers from both a theoretical and practical standpoint. This chapter looked at the various machine learning methods that have been proposed over the years and then rated them based on some key functional activities. The best-performing machine learning methods for HAR are then classified, and the most interesting recent approaches in this field are discussed.

The remaining section is organized as follows: The background issues are discussed in Section 15.2. Also discussed is the importance of using computer vision in machine learning research. Section 15.3 discusses the major techniques of machine learning in computer vision. We examine categorized machine learning algorithms for person recognition in Section 15.4. Finally, conclusions and future work are reviewed in Section 15.5.

15.2 MACHINE LEARNING FOR RECOGNITION OF HUMAN ACTIVITY: OPPORTUNITIES AND CHALLENGES

In recent years, one of the toughest problems in computer vision has been automatic or functional recognition. It is critical for a variety of AI applications, including video monitoring, computer games, robotics, and human interactions. Physical movement of a person, human–object interaction, playing musical instruments, human–human interaction, and play are some samples of the five sorts of activities displayed in Figure 15.2. Figure 15.3 shows the three key steps in an action authentication system: feature extraction, action representation, and classification.

FIGURE 15.2 Action in UCF101[10] dataset.[11]
Source: Adapted from Ref. [10] and [11]

As a result, each step is critical to achieving the correct recognition ratio. Sample inputs include source data for a feature vector, feature extraction, and process representation. The result can be greatly influenced by proper feature extraction and classification. Machine learning methods for learning high-level representations directly from source video data have recently been implemented.

15.3 VISION-ESTABLISHED ACTION RECOGNITION APPROACHES

Vision-established action recognition approaches are often divided into two divisions.

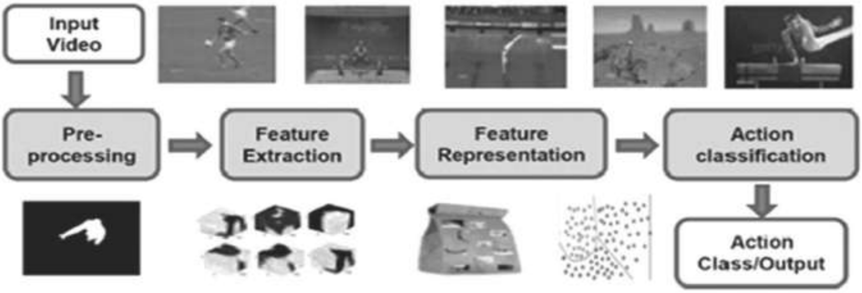

FIGURE 15.3 Schematic diagram of a typical activity recognition system.

i) Scale-invariant features: Hesian-3-D, elevated fast-up functions, transforms, and histograms of oriented gradients within the traditional handicraft representation-established entirely approach, LBPs, as well as carefully crafted distinctive inventors, are covered. Figure 15.4 shows the breakdown of well-known education for motion popularity.

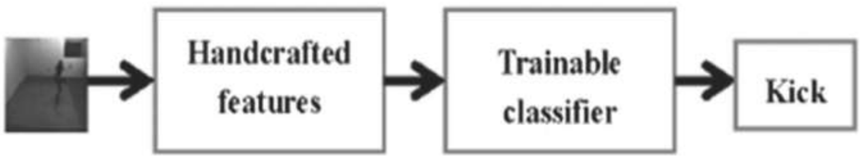

FIGURE 15.4 Example of a kicking action using handcrafted representation-based approach.

A learning-established representative method is a contemporary approach that allows for automatic learning of characteristics from source data. For action representation, this reduces the need for craft feature inventors and illustrators. As shown in Figure 15.5, it employs the principle of trainable feature extraction, followed by trainable classification.

15.3.1 HANDCRAFTED REPRESENTATION-ESTABLISHED APPROACH

The downstream technique for HAR is primarily focused on craft representations. Feature-established representation with crafts is the standard

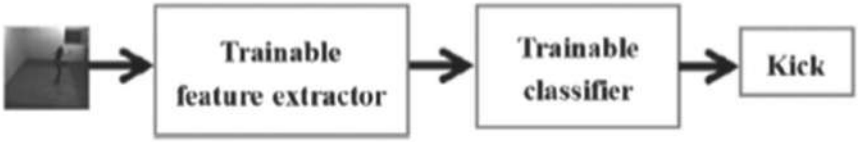

FIGURE 15.5 Example of a kicking action using learning-based representation approach.

method for functional recognition. Key features are extracted from the series frames using feature-designed feature detectors and descriptors, resulting in the development of the feature descriptor. After that, a common classifier, similar to the support vector machine (SVM),[12] is used to finish the classification. This approach is separated into three stages. Figure 15.6 shows frontal identification, craft feature extraction, visualization, and categorization. It includes space-time, appearance-established local binary patterns, and obscure logic-established techniques. It is divided into three stages: anterior identification, handcraft feature extraction, and representation and classification. A large number of survey documents focused on handcraft representation have been published at various stages of HAR processes, with various taxonomies being used to discuss HAR approaches. According to their survey, Agarwal and Ryu[3] divide functional recognition approaches into two types: single-layer techniques and hierarchical approaches. Simple functions are identified by single-layer approaches from a video series, while more complex functions are identified by breaking them down into simple functions using hierarchical approaches (subevents). Based on the feature illustration and classification methods used for authentication, these are divided into special modules and directions. Wang and Schmid[13] present a detailed survey of segmentation strategies, outlining the problems, tools, repositories, and public databases used for segmentation. Another research[14] looked at three different levels of HAR, including the main infrastructure, HAR structures, and applications. As described in Ref. [13], problems such as cover, anthropology, implementation pace, background clutter, and camera movement have a major impact on functional authentication systems. The analysis classified current methods according to their ability to address these issues and defined new research areas. Methods for recognizing human action established on feature representation and classification were addressed in Ref. [14]. Wayneland et al.[15] looked at HAR mechanisms and classified them into three categories: segregation, feature representation, and classification. R. Bergevin and Ziaeefard proposed a research paper on established semantic human activity classification methods[16] as well as powerful state-of-the-art functional recognition algorithms established on semantic characteristics. For the purpose of identifying actions, various handcrafted feature extraction and representation methods have been proposed.[17–22]

15.3.1.1 SPACE-TIME ESTABLISHED APPROACHES

The four fundamental factors of space-time established techniques are space-time interest point identification, language development, function characteristics, and identification.[23] Space-time attentiveness point identification inventors might be either dense or distributed. To detect points of "interest, V-Fast, Hessian Detector, and Dense Model" intensively cover all video footage, whereas Cupid Detector Harris 3-D[24] and indirect spatiotemporal structure model employ the distributed (local) subgroup of this content. STIP inventors have come from a variety of sources.[25,26] Lighting shifts, phase shifts, and video speed variations are all possible. Local and global feature descriptions are two types of feature descriptions. Local descriptors, such as cuboids, enhanced speeded-up robust features, and N-jet, give local information such as texture, color, and pose, whereas global descriptors employ global information such as illumination shifts, phase changes, and video speed variation.

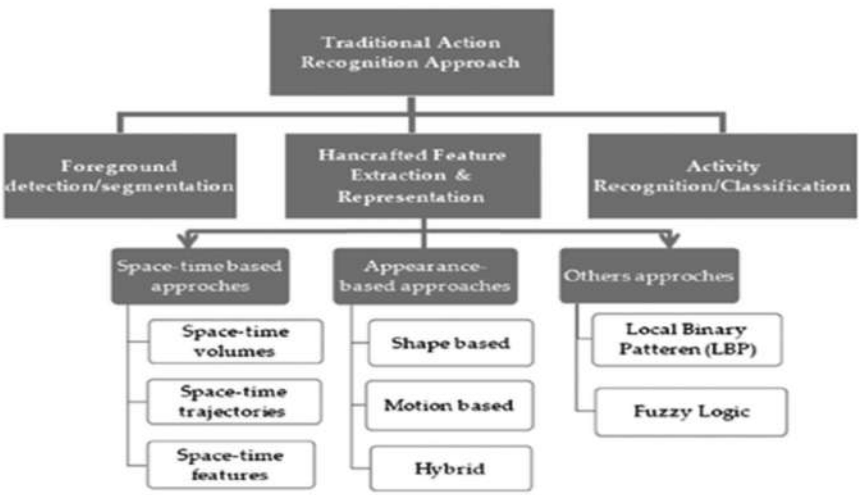

FIGURE 15.6 The traditional approach to action representation and recognition.

15.3.1.1.1 Space-Time Volumes

Three-dimensional spatiotemporal cubics, also known as space-time modules, are the characteristics of space-time volumes. The unification process between the two modules for action authentication is central to STV-established methods. Instead of using real-time modules, Bobby and Davis[27] introduced a process authentication scheme based on template matching. They used templates with power and movement in a two-dimensional binary movement history image instead of real-time modules. For action authentication, a basic prototype matching technique was used. Hu et al.[28] introduced MHI and two appearance-established features, the anterior image and the "histogram of the oriented gradients (HOG)," which were used to describe actions in this study, and the learning "support vector machine (SMILE-SVM)" was used to simulate several events for action classification. Row et al.[29] suggested a method that used a module motion template to expand the work of Bobby and Davis[28] from 2-D to 3-D space to find autonomous human action recognition. Table 15.1 compares and contrasts the various strategies.

15.3.1.1.2 Space-Time Trajectory

A process is defined as a collection of space-time trajectories in trajectory-established approaches. In these methods, an individual is represented by two-dimensional XY or three-dimensional XYZ points that correspond to his or her body's limb locations. When an individual performs an action, the essence of the action causes certain changes in his or her collective state. These shifts are captured as space-time trajectories, which are then used to create a three-dimensional XYZ or four-dimensional XYZT representation of the action. To distinguish between various types of actions, space-time trajectories monitor the joint location of the body. Several approaches to action detection established on trajectory methods have been proposed based on this concept.[30–32] The principle of dense trajectory from video to action recognition was introduced,[33] which was inspired by the dense paradigm of image classification. The authors used displacement information from the dense optical flow field to sample the dense dots from each frame and track them. These trajectories provide motion information and are effective in detecting unusual motion changes.[34] In Ref. [33], it has been suggested that camera movement-related performance be improved. The researchers used a robust function descriptor and dense optical flow to assess camera movement. Motion-established interpreters, such as the optical flow histogram and motion boundary histograms, performed substantially better as a result of this. When high density is combined with the trajectory in the video, however, the computational cost rises. Several attempts to reduce the computational cost of dense trajectory-established approaches have been made. Wick et al.[35] used a diagram to isolate key areas within the picture frame for this reason. The efficiency of trajectory-established methods can be improved by discarding a large number of dense trajectories established on the emission map. Human action authentication using deep images captured by the Kinect sensor has recently been proposed.[36] The complex skeleton of the human body is represented as a trajectory. HAR in uncontrolled videos is a difficult problem, and some approaches have been suggested in this conclusion. A human action recognition framework established on the explicit operating model[37] has been proposed for this reason. For action representation, this method used visual code words created from dense trajectories without the use of pre-background separation.

15.3.1.1.3 Space-Time Features

For human action recognition, space-time feature established methods extract features from spatial modules or space-time trajectories. In general, they have distinct action categories. These characteristics can be distinguished based on the existence of the space-time modes and trajectories as dispersed and dense. Harris three-dimensional[25] and Dollar,[54] two feature innovators established on points of interest, are rarely considered, as are feature inventors established on optical flow. These points of interest serve as the foundation for the inventors' recently proposed methods. Points of interest were described using Harris three-dimensional,[25] which formed a feature definition established on these points and used PCA (primary component analysis) SVM for classification in the work of Thi et al.[55] The Bag of-Visual-Words model[56,57] or its variants[58,59] are the most common representation methods in this group. Feature extraction, code-book construction, encryption/pooling, and normalization are the four steps in the BoVW model. They extracted local features from the video, learned the visual dictionary using the Gaussian compound model or a tutorial that included K-average clustering, encryption, and pool features, and then marked the video as the default pool vectors, followed by a general classifier for action authentication. Low-level features like denser trajectory features,[34] coding methods like Fisher vectors,[59] and space-time parallel event descriptions,[41] all contribute to the BoVW model's high efficiency. In terms of space-time features, improved density trajectory (iDT)[34] offers better performance in many public databases.

15.3.1.2 APPEARANCE-ESTABLISHED APPROACH

Two-dimensional XY and three-dimensional XYZ deep image-established methods are discussed in this chapter, which utilize a combination of effective shape, motion, or form, and motion features. For action representation, two-dimensional shape-established approaches[60] use shape-and-edge-established features, whereas motion-established approaches for action representation and recognition combine form and kinetic features.[61] A human body model is designed for action representation in three-dimensionally established approaches; this model is established on a visual hull consisting of cones, ellipses, silhouettes, or surface meshes. Three-dimensional optical flow,[62] shape histogram,[63] kinetic history scale,[64] and 3-D body skeleton[65] are some examples of these approaches.

15.3.1.2.1 Shape-Established Approach

A human form or shadow is captured using shape-established methods.[66] Using front parting techniques, these methods first extracted the front shadow from a picture frame. They then separated features between the canvas and the human body from the shadow (positive space) or the area around the shadow (negative space).[67] The edges, regional features, and geometric features are some essential features that differentiate the shadow from the background. Regional Human Rights Recognition System[68] is a proposal. This approach categorizes the human image into a small number of phases and cells and recognizes action using a hybrid classifier called support vector machine and nearest neighbor (SVM-NN). Centered on the marginal points with a set of multiview keys for action representation, a pose-established noninvasive human action recognition method has been discussed.[69] Chaaraoui and Flrez-Revuelton[70] suggested an expansion of this approach. As a classifier for the marginal points of the human shadow and radial scheme, this method employs the functional representation and support vector machine. By extracting features (negative space) from the surrounding areas of the human silhouette, a regionally established descriptor for human action representation was created.[71] Another approach is to provide action authentication information. The shadow-scale features are taken from the shadow first, and then these features are combined to form the main features in this mode. Finally, a weighted voting scheme was used to classify the objects.

15.3.1.2.2 Motion-Established Approach

Motion capabilities are used for movement representation, observed through a trendy classifier for movement reputation in movement-primarily established total movement authentication methods. For multiview movement representation, a brand new movement narrator has been counseled.[72] The help vector engine is used to categorize this movement descriptor, which is primarily established at the movement course and histogram of movement strength. Murtaza et al. counseled some other method based primarily on 2-D movement models.

15.3.1.2.3 Hybrid Approach

For motion representation, those strategies integrate form-primarily established functionality and movement-primarily established functionality.[73] For visual-invariant motion recognition, optical glide and shadow-primarily established totally form functions had been used, accompanied with the aid of using number one issue analysis (PCA) to lessen the data's dimension. Other motion authentication strategies based entirely on layout and running information have been proposed.[74,75] Pelican et al. used tough shadow functions, radial grid functions, and movement functions for multiview motion recognition.[75] Jiang et al.[61] used form-movement prototype timber to categorize human actions. In the form-running space, the authors marked an operation as a sequence of prototypes and used distance measurements for collection fit.

TABLE 15.1 Comparison of Space-Time-Based Approaches for HAR.

Author name and year	Feature type	Accuracy (%)
	KTH[15]	
Sadanand and Corso (2012) [38]	Space-time volumes	98.2
Wu et al. (2011) [39]	Space-time volumes	94.5
Ikizler and Duygulu (2009) [40]	Space-time volumes	89.4
Peng et al. (2013) [41]	Features	95.6
Liu et al. (2011) (42)	Features (attributes)	91.59
Chen et al. (2015) [43]	Features (mid-level)	97.41
	UCF Sports[16,17]	
Sadanand and Corso (2012) [44]	Space-time volumes	95
Wu et al. (2011) [39]	Space-time volumes	91.3
Ma et al. (2015) [45]	Space-time volumes	89.4
Chen et al. (2015) [46]	Features (mid-level)	92.67
Wang et al. (2013) [30]	Features (pose-based)	90
	HDME.51	
Wang and Schmid (2013) [34]	Dense trajectory	57.2
Jiang et al. (2012) [47]	Trajectory	40.7
Wang et al. (2011) [33]	Dense trajectory	46.6
Kliper et al. (2012) [48]	Space-time volumes, bag-of-visual-words	29.2
Sadanand and Corso (2012) [44]	Space-time volumes	26.9
Kuehne et al. (2011) [49]	Features	23
Wang et al. (2013) [30]	Features (mid-level)	33.7
Jain et al. (2013) [51]	Features	52.1
Fernando et al. (2015) [52]	Features (Video Darwin)	63.7
Wang et al. (2013) [30]	Microsoft Research Action3-D	90.22
Amor et al. (2016) [36]	Features (pose-based)	89
Zanfir et al. (2013) [53]	Trajectory 3-D pose	91.7
Wang et al. (2011) [33]	YouTube action dataset Dense trajectory	84.1
Peng et al. (2014) [41]	Features (FV+ SFV)	93.38

This method has been tested on five distinct public datasets and yielded the expected results. Eweiwi et al.[76] proposed an action authentication system that used action keyboards as a version of motion energy pictures and gait energy images for action representation, together with a simple near-neighbor extractor.

15.3.1.3 OTHER APPROACHES

Two key techniques are explored that do not fall within the boundaries of the above groups. Local binary patterns and fuzzy logic-established methods are two of these approaches.

15.3.1.3.1 Local Binary Pattern

The visual depiction of system classification is the local binary system.[77] Since its beginning, several improved versions of this description for jobs relating to distinct classification in computer view have been presented.[78–80] Combining appearance modification with patch matching, a human action authentication system established on LPP has been developed.[81] This method has been illustrated for action authentication in a number of public databases. Using the LBP-TOP concept,[82] another way for functional authentication was proposed. In this mode, the action module is separated into submodules, and a feature histogram is formed by merging the submodule histograms. They encrypted movement at three separate levels using this representation: pixel level, regional level, and globe level. Multiview human activity authentication has also been done using LPP-established approaches. Based on margin-established pose characteristics and uniform rotation-invariant LPP, a multivisual HAR system has been suggested. For multiview action authentication, a new operating system called motion binary pattern was recently introduced.[83]

15.3.1.3.2 Fuzzy Logic

Traditional vision primarily established totally act reputation algorithms for spatial or temporal data, observed with the aid of frequent categorization for motion illustration and classification. However, quantifying these strategies to affect the ambiguity and difficulty of applications in the actual world is difficult. Fuzzy-established approaches are thought to be the simplest option for handling these challenges. For action representation, a fuzzy-established framework supported by fuzzy recording polar histograms and temporal self-identities was presented, followed by SVM for action categorization.[83] The tactic proposed within the two public databases was evaluated and found to be very accurate and suitable for actual world applications. Another method supporting symbolic logic[84] is developed, which uses shadow particles and motion velocity data as input to a fuzzy system and obtains membership functionality for the proposed system employing a fuzzy C-object clustering algorithm. The results showed that the proposed fuzzy system was more accurate than other fuzzy systems for an equivalent general database. The majority of act recognition systems are visual and may recognize an action from one point of view. The real-time act authentication system, on the other hand, can recognize the action from any angle. Many state-of-the-art techniques do that by combining data from many cameras. This is often not a practical option, however, because calibrating many cameras in real-world scenarios is extremely challenging. The last word solution for vision-unchanging action identification should be the utilization of one camera. A hazy logic-established solution for visually-unchanging action identification with one camera was developed in this way.[85]

15.4 DATASET

This section discusses and illustrates datasets used after 2009. Aggarwal and Ryoo's work[86] goes into greater detail about datasets used before 2009. We concentrate on the fresh datasets we obtain and motivate them to break them down and consider them from many angles.

15.4.1 WEIZMANN ACTION DATASET

Running, walking, skipping, bounce jack, two-legged forward jumping, two-legged setting, lateral jogging, two-handed waving, one-handed waving, and twisting are among the 10 natural activities within the database.[87] There are a total of 93 sequels within the series. All of the layouts are captured using a normal camera with a 25 fps edge rate and a spatial resolution of 180,144 pixels. The database is additionally viewed by 10 additional groups, each of which is captured from a special perspective, switching between[66,67] in reference to the image plane.

15.4.2 KTH HUMAN ACTION DATASET

The current database comprises 25 persons performing five activities (walking, running, boxing, waving, and clapping) in four distinct scenarios: outside, outside with various scale types, outside with various gear, and indoors. There are a total of 2391 groups. A standard camera with a 25-frame-per-second edge rate and a spatial resolution of 160 × 120 pixels is included in all packages. In the original publication,[88] the arrangements were classified as a training package (eight people), an acceptance package (eight people), and a test package (eight people). Shadow samples and exteriors that have been deleted are not included in the database. Sadanand and Corso's[44] technique achieved 98.2% accuracy in this database, the highest accuracy ever reported.

15.4.3 IXMAS DATASET

IXMAS[41] is an INRIA Xmas Motion Acquisition Sequence. In 2006, a multiview dataset was created to check the approaches to visually unchanging human interest recognition. There are 13 everyday existence performances within the database, each carried out in three instances with the aid of 11 specific actors. Those actions include folding hands, extending the head, sitting, watching the time, having to get up, walking, coming, striking, hitting, clapping, picking, shouting, and hurling. Four aspect cameras and five scaled cameras, which include one pinnacle camera, were used to film those actions. For testing, photos of the video cloth that has been extracted are also provided. In general, varieties of multiview motion authentication techniques were proposed, with 2-D and 3-D primarily established techniques. In this database, 3-D-primarily established total techniques have better accuracy than 2-D-primarily established total methods, but at a higher computational cost. Holte et al.[89] received the finest accuracy on this database through the use of 3-D movement descriptors (HOF3-D descriptors and 3-D spatial pyramids (SP)).

TABLE 15.2 Human Action Dataset.

Dataset	Challenges	Year	No. of actions	Accuracy	Class
KTH	Homogeneous backgrounds with a static camera	2004	6	97.6	General purpose action recognition
Weizmann	Partial occlusions, nonrigid deformations, significant changes in scale and viewpoint, high irregularities in the performance of an action and low-quality video	2005	9	100	General purpose action recognition
IXMAS	Multiview dataset for view of invariant human actions	2006	13	89	Motion acquisition
CMU MoBo	Human gait	2001	10	78	Motion capture
HOHA	Unconstrained videos	2008	12	56	Movie
HOHA-2	Comprehensive benchmark for human action recognition	2009	11	58	Movie
Human Eva	Synchronized video and ground truth 3-D motion	2009	51	84.3	Pose estimation and motion tracking
CMU MoCap	3-D marker positions and skeleton movement	2006	101	100	Motion capture
UCF sports	Wide range of scenes and viewpoints	2008	200	93	Sports action
UCF YouTube	Unconstrained videos	2008	200	84.2	Sports action
i3D Post multiview	Synchronized uncompressed HD 8 view image sequences	2009		0	Motion acquisition

15.4.4 YOUTUBE ACTION DATASET

In 2009, the YouTube Action Database[92] was established. Due to camera movement, visual fluctuations, lighting situations, and loud backdrops, this database is difficult to work with. Bike riding, diving, basketball shooting, horseback riding, swinging, soccer juggling, trampoline leaping, volleyball spiking, golf swinging, tennis swinging, and on foot playing with a canine are a number of the 11 pastime categories. Peng et al.[50] used FV and SFV to reach the best accuracy ever in this database, that is, 93.38%.

15.5 CONCLUSIONS

In computer vision, learning direct representations from source video data is a major topic. In this study, machine learning was found to be an excellent option for constructing high-level conceptual representations with appropriate summaries of complex data. This study includes a comprehensive review of the literature on human functional representation and recognition techniques, such as handcrafts and learning-established representations. These methods have generated significant results in a variety of publicly available databases. Local dense sample interpretations are the most successful handcraft representation approaches, but they are computationally expensive. Using personal feature detectors and descriptors, relevant features are retrieved from a sequence of frames to produce a feature vector. After that, a generic predictor is instructed to do the cognate identification. Space-time, appearance-based, structural feature-based, known categorization patterns, and fuzzy logic-based techniques are among these strategies. Handcraft representation methodologies are still widely employed due to the computational difficulties and the necessity for a database of machine learning processes for functional recognition. To provide a fuller understanding of the issue, certain well-known public datasets for interest identification are provided for testing and assessment of HAR procedures. KTH dataset, Weizmann dataset, IXMAS dataset, UCF Sports dataset, Hollywood 2, YouTube dataset, HMDB-51 dataset, UCF-101 dataset, and Activity Net dataset are among the datasets used.

KEYWORDS

- **computer vision**
- **human–computer interfaces**
- **noiseless video data**
- **human action recognition**
- **machine learning approaches**
- **classify**

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